ZionEX Suite

The central intelligence of business

Installation & Setup Guide

Requirements · Definitions · Every installation scenario · Modules · Devices · Daily running · Troubleshooting

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About this guide

This guide takes you from an empty computer to a working shop, restaurant, pharmacy or chain of branches running **ZionEX**. It is written for someone who is **not** an IT professional. Every step says **what** to do, and where it helps, **why** it is needed.

Who to contact

ZionEX is built by **SES Labs**, the product division of **SyncLytics**, which licenses and supports it (receipts read *ZionEX · Powered by SES Labs*):

Hotline (support)	024 563 6889
Office	055 668 6975 · 050 693 8589
E-mail	info@synclyticsenterprisesystems.com
Website	synclyticsenterprisesystems.com
Address	Puff Cl, Race Course St, Tabora, Accra (GhanaPost GPS GC-027-0884)

How to read it

1. Read **Chapter 1 (Words used in this guide)** once. It explains terms such as *server*, *till*, *branch*, *module* and *licence*.
2. Use **Chapter 2 (Choose your setup)** to pick the installation that fits your business. Each setup lists the chapters you need, in order.
3. Check **Chapter 3 (What you need)** before buying or preparing anything.
4. Follow the installation chapter for your setup **in order**. Do not skip steps. Each step ends with a **Check** that tells you how to know it worked.
5. Keep the guide for daily running: backups, updates and troubleshooting are at the end.

Boxes used in this guide

Why: explains the reason for a step, so you understand what you are doing.

Note: extra information that is useful but not required.

Tip: a shortcut or a good habit.

Warning: something that can lose data, money or security if ignored. Read these carefully.

Check: how to confirm that a step worked before you move on.

Typing commands

Some steps ask you to type a **command**. A command is a line of text you type into a black window called a **terminal** and confirm with **Enter**.

- On **Windows** the terminal is **PowerShell**. Open it like this: click **Start**, type `PowerShell`, then click **Windows PowerShell**. When a step says *as administrator*, right-click it and choose **Run as administrator**, then click **Yes**.
- On **Ubuntu Linux** the terminal is called **Terminal**, or you reach the server through **SSH** (explained in Chapter 5).

Commands are shown in a dark box, like this:

```
php --version
```

Type them exactly as shown (you can copy and paste). Text in angle brackets, such as `<your-password>`, is a placeholder: replace the whole thing, brackets included, with your own value.

> **Tip:** In PowerShell you can paste with a right-click. Press the **Up arrow** key to bring back the last command.

What is ready today and what comes later

The software is built in phases. This guide always describes the software **as it is now**, and is updated with every release.

Area	Today	Later
Server installation	Step-by-step manual installation (this guide) on Windows or Ubuntu Linux	One-click installers for every setup (Phase 13)
Till app	Runs in the Chrome or Edge browser and can be installed as an app; works offline	Optional desktop app with printer and cash-drawer drivers (Phase 13)
Backups	Manual and scheduled database backups (Chapter 12)	Encrypted automatic backups with a tested restore (Phase 13)
Industry modules	Grocery, Pharmacy, Restaurant, Electronics, Fashion, Services, Sales & Distribution, Manufacturing, Online Storefront	Logistics, Assets, Projects and Helpdesk
Insights and AI	Forecasts, stock health, customers, pricing, cash flow, loss prevention and alerts (no AI needed); optional local AI assistant (Chapter 15)	Scheduled reports by e-mail (Phase 12)

> **Warning:** Use the software with test data until your accountant has checked the tax set-up (Chapter 7, step 7) and you have practised a backup and a restore (Chapter 12).

1. Words used in this guide

You do not need to remember these. Come back to this page whenever a word is unclear.

The parts of the system

Word	What it means
Server	The computer that keeps all the data (products, prices, stock, sales, customers, accounts) and runs the software. Every other device talks to it. It can be a PC in your shop, a computer in your office, or a rented computer on the internet (*cloud*).
Back office	The management screens, opened in a web browser at `http://<server>/admin`. Managers use it to set up products and prices, receive stock, see reports, run payroll and so on.
Till (POS, point of sale)	The selling screen used by cashiers. It runs in the Chrome or Edge browser at `http://<server>/pos/` on a PC, laptop or tablet, and keeps working when the server or the internet is down.
Device	Any till, handheld scanner, kitchen screen or self-service screen. Each one is *registered* in the back office and *enrolled* once with a one-time code, so only your own devices can sell.
Enrolment code	A one-time code (like `ABCD-EFGH-JKMN`) that the back office shows when you register a device. You type it on the device once.
Handheld	A phone or rugged scanner used in the warehouse to receive goods, put them away and count stock.
Kitchen screen (KDS)	A screen in the kitchen or bar that shows the orders to prepare (Restaurant module).
Online store / channel	A website that sells your products (the built-in shop, WooCommerce or Shopify), or a delivery app. It receives prices and stock automatically and sends its orders back.

Your business in the system

Word	What it means
Company	Your business. One installation normally holds one company.
Legal entity	The registered company, with its TIN and VAT number, printed on VAT invoices.
Branch	One shop, restaurant, pharmacy or warehouse location. One branch is the **head office** (HQ).
Warehouse	A place where stock is kept. Every branch has a main warehouse; you can add more (store room, cold room, van).
Till (in the set-up)	A selling point at a branch, for example *Till 1*. A device is assigned to one till.
Branch code	Up to 6 letters or digits, made by the system from the branch name (*Pharmacy Osu* → `PHAOSU`). It appears in receipt numbers and never changes.
User	A person who signs in. Back-office users sign in with email, password and a second factor. Cashiers sign in on the till with a **username and PIN**.

Word	What it means
Role	A set of permissions, such as *Cashier*, *Supervisor*, *Branch Manager* or *Accountant*. You give people roles.
Scope	Where a role applies: the whole company, one branch, one warehouse or one till. A cashier with a till scope can sell only on that till.
Security level	A ceiling (L1 to L6) on what a person can ever be given. It stops anyone from handing out more power than they have.
Two-factor sign-in (2FA)	A second proof of identity after the password: a 6-digit code from an authenticator app on your phone, a code sent by WhatsApp or e-mail, or a passkey. Required for managers.
Superadmin	The SyncLytics support account (level L7), added on the server by SyncLytics. Always uses the authenticator app; the owner is told each time it signs in.
Passkey	A sign-in using your device's fingerprint, face or PIN instead of a code. Optional.

How the software is organised

Word	What it means
Core	The part every business has: company, branches, users, products, prices, tax, stock, tills, customers, purchasing, sales, finance and HR (if licensed).
Module	An extra package for a type of business, switched on **per branch**: Grocery, Pharmacy, Restaurant, Electronics, Fashion, Services, Online Storefront, and more later. A branch can use several.
Licence	The key from SyncLytics that allows the software to run. It sets how many branches, devices and users you may have and which modules and features (Finance, HR & payroll) you bought.
Installation code	A code starting with `SYNREQ-` that your server shows on the Licence page. You send it to SyncLytics to receive your licence key (starting with `SYN1-`).
Read-only	What happens when a licence expires or is withdrawn: you can still see everything, but nothing can be changed or sold. Nothing is deleted.

Working offline and with several sites

Word	What it means
Offline	Without a connection to the server. Tills keep selling offline and send their sales later.
Offline sign-in (grant)	Permission for a cashier to sign in on a till without the server. It is given when they sign in on that till while it is online, and lasts 72 hours by default.
Queue	The list of sales a till made offline that are waiting to be sent. They are sent automatically.
Review queue	A list in the back office of things a manager should check, such as a sale made offline at an old price.
Node	One server running the software. A company can have several.
HQ server (head office)	The main server. It owns the master data (products, prices, users) and receives everything from the branches.

Word	What it means
Branch server	An optional server **at** a branch. The branch keeps working fully (tills, back office, stock) when the internet to head office is down, and exchanges changes with HQ when it returns.
Standalone	A single server for a single site, with no other servers.
Sync	Sending changes between servers (branch ⇄ HQ) and from tills to their server. It is automatic.

Technical words you will meet during installation

Word	What it means
Operating system	The basic software of a computer: Windows 10/11, or Ubuntu Linux on servers.
PHP	The programming language the server is written in. You install it so the server can run.
Composer	A tool that downloads the PHP building blocks (libraries) the server needs.
Node.js / npm	A tool used once during installation to build the till app.
PostgreSQL	The database: the program that stores all your data safely.
Database	The organised store of all your data inside PostgreSQL. It is called `erp`.
`.env` **file**	A small settings file on the server holding passwords and keys. It must stay private and be backed up.
APP_KEY / ERP_AUDIT_KEY	Secret keys in `.env`. APP_KEY protects sessions and stored secrets; ERP_AUDIT_KEY seals the audit log. Lose them and some data cannot be read or verified, so **back them up**.
Scheduler	A background timer that runs the software's routine jobs every minute: sending changes to online stores, syncing branches, posting to the accounts, reminders. It **must** run all the time.
Live updates (Reverb)	An optional service that pushes price and stock changes to tills and screens instantly. Without it, tills check every 60 seconds.
IP address	The number of a computer on the network, such as `192.168.1.10`. Tills use it to find the server.
LAN	Your shop's local network (cables or Wi-Fi through your router).
Port	A numbered "door" on a computer. The server uses the standard web ports: 443 for https, and 80 only to send visitors on to https (and, on the shop network, to hand out the server's certificate).
Certificate	The proof a web server shows a browser that it is really that server, which also encrypts everything sent between them (the padlock in the address bar).
HTTPS / SSL certificate	Encrypted web connection (the padlock in the browser). Needed when tills reach the server over the internet, and for some browser features.
Domain	A web address you own, such as `erp.yourshop.com`. Needed for a cloud server.
Firewall	Protection that blocks unwanted network connections. You open only the ports you need.
Service	A program that starts automatically with the computer and runs in the background.
Backup	A copy of the database and the `.env` file kept somewhere else, so you can recover after a theft, fire or disk failure.

2. Choose your setup

There are five ways to run ZionEX. They use the **same software**; only where the server sits changes. You can start small and grow later (for example, start with one shop and add a cloud head office when you open a second branch).

The five setups at a glance

	Setup	Best for	Server	Where tills connect
A	**One computer**	A small shop with one till	The till PC itself (Windows)	The same PC (`https://localhost/pos/`)
B	**One shop, several tills**	A shop, pharmacy or restaurant with 2–20 tills and devices	One PC or small server in the shop (Windows or Ubuntu)	Over the shop's network (LAN / Wi-Fi)
C	**Central server for several branches**	Branches joined by a private network (VPN) or in one building	A server at head office (Ubuntu recommended)	Over the private network
D	**Cloud head office**	Several branches in different towns, owner wants to see everything from anywhere	A rented cloud server with a domain name and HTTPS (Ubuntu)	Over the internet
E	**Cloud or central HQ + branch servers**	Branches with unreliable internet, busy restaurants, supermarkets	HQ as in C or D, plus a branch server at each such branch	Tills use their **branch** server over the LAN; branch servers sync with HQ

> **Why:** a till can sell offline in every setup, but the back office, kitchen screens, handhelds, shared restaurant bills and stock look-ups need the server. If the server is in the cloud and the internet is down, those stop until it returns. A branch server (setup E) keeps them working on the local network.

Which one should I choose?

Answer these questions in order:

1. **Do you have only one location?**
   - One till only → **A**.
   - Several tills, kitchen screens or handhelds → **B**.
2. **Several locations?**
   - Do all branches have reliable internet (or a private network)? → **D** (cloud) or **C** (central server with VPN).
   - Do some branches have unreliable internet, or need kitchen screens, handhelds or shared restaurant bills to keep working during outages? → **E** (add a branch server at those branches).

> **Tip:** Most businesses with 2–5 branches in Ghana do well with **D** for head office plus a branch server (**E**) only at branches with weak internet or a restaurant.

What each setup needs (and the chapters to follow)

A — One computer

- One Windows 10/11 PC (the till) with a receipt printer and barcode scanner.
- Chapters: 3 → **4 (Windows)** → 6 → 7 → 8 → 9 → 12.
- The till opens at `https://localhost/pos/` on the same PC, so no extra network set-up is needed.

B — One shop, several tills

- One PC or small server that stays switched on during trading hours, with a **fixed IP address** on the shop network.
- Till PCs/tablets on the same network (cable or Wi-Fi).
- Chapters: 3 → **4 (Windows)** or **5 (Ubuntu)** → 6 → 7 → 8 → 9 (including 9.3, *Trust the server's network address*, on every device) → 12.

> **Note:** With the server on Windows today, the web server built into PHP answers one request at a time. That is fine for about 1–4 tills. For more tills, kitchen screens and handhelds, use an Ubuntu server (Chapter 5) until the Phase 13 Windows installer arrives.

C — Central server for several branches

- One Ubuntu server at head office, reachable from the branches through a VPN (your network provider can set this up) or within one building.
- Chapters: 3 → **5 (Ubuntu)** → 6 → 7 → 8 → 9 → 10 (optional branch servers) → 12.

D — Cloud head office

- A cloud server (VPS) running Ubuntu 24.04, a **domain name** (for example `erp.yourshop.com`) and a free HTTPS certificate (set up in Chapter 5).
- Chapters: 3 → **5 (Ubuntu, including HTTPS)** → 6 → 7 → 8 → 9 → 11 (online stores) → 12.

E — HQ plus branch servers

- HQ as in C or D.
- At each chosen branch: a PC or small server installed like setup B, then **paired** with HQ.
- Chapters: first set up HQ (C or D). Then, for each branch server: 4 or 5 → **10 (Branch servers)** → 9 → 12.

Moving between setups later

From → To	What to do
A → B	Give the PC a fixed IP address and set up the other tills (Chapter 9). Nothing is reinstalled.
A/B → D	Install a cloud server, then move the database to it (Chapter 12, *Moving to a new server*). The licence must be re-activated on the new server.
D → E	Install a branch server at the branch and pair it (Chapter 10). Re-enrol that branch's tills on the branch server.

3. What you need

Hardware

The server

Setup	Minimum	Recommended
A (one PC, also the till)	Windows 10/11 64-bit, 4-core CPU, 8 GB RAM, 256 GB SSD	16 GB RAM
B (shop server)	Windows 10/11 Pro or Ubuntu 24.04, 4 cores, 8 GB RAM, 256 GB SSD	16 GB RAM, Ubuntu for more than 4 devices
C / D (HQ, central or cloud)	Ubuntu 24.04, 2 vCPU, 4 GB RAM, 80 GB SSD	4 vCPU, 8 GB RAM, 160 GB SSD; scale up with branches
E (branch server)	as B	as B

> **Why:** the database works best on an **SSD** (not a spinning hard disk). Memory lets the database keep frequently used data in RAM, which keeps the tills fast.

Sizing by business size (head office or shop server; the SSD matters more than the processor):

Business	Example	Memory	Processor	Disk
Small	1 shop, up to 3 tills, under 5,000 products, under 500 receipts a day	8 GB	4 cores	256 GB SSD
Medium	2–5 branches, up to 15 tills, under 20,000 products	16 GB	4–6 cores	512 GB SSD
Large	6+ branches or 30+ tills, or several industries with separate books	32 GB	8 cores	1 TB NVMe SSD

Branch servers (Setup E) need the *Small* size. If **System** → **Health** → **Slow pages** keeps filling up, add memory first.

Also needed for an on-site server:

- A **UPS** (battery backup) so a power cut does not switch the server off mid-write. At least 15 minutes, with enough time to shut down safely.
- The **correct date and time**, set automatically from the internet. Wrong clocks cause sync refusals.
- A place where it will not be unplugged by accident, and is not open to customers.

Tills and other devices

Device	Requirement
Till PC	Windows 10/11 with Google Chrome or Microsoft Edge (current version), 4 GB RAM, screen at least 1280 × 720 (touch screen optional)
Till tablet	Android 10 or later with Chrome, 10" screen recommended

Device	Requirement
Receipt printer	80 mm thermal receipt printer installed as a normal Windows/Android printer (USB, network or Bluetooth). Receipts are printed through the browser's print window. A4 printers also work.
Cash drawer	One that opens from the receipt printer (RJ11 cable to the printer), with the printer driver set to *open drawer* when printing
Barcode scanner	Any USB or Bluetooth scanner that acts as a **keyboard** (most do). It must send **Enter** after each code.
Handheld (warehouse)	Android phone or rugged scanner with Chrome and a built-in scanner or camera
Kitchen screen	Any PC, tablet or TV stick with Chrome; touch screen helpful
Label scale (grocery)	A scale that prints EAN-13 labels with prefix 21 (see Chapter 8, Grocery)

Network

- **Setup A:** no network needed for selling. Internet is needed for the daily licence check-in (it tolerates 30 days without).
- **Setup B/C/E:** a router with cable or reliable Wi-Fi. Give the server a **fixed (reserved) IP address** in the router so tills always find it.
- **Setup D:** a reliable internet connection at every branch (4G backup router recommended). Tills keep selling during outages.
- **Internet for head office:** needed once a day for the licence check-in, and for online stores, SMS and delivery apps.

Software

All software below is free. Download only from the official addresses listed.

Software	Version	Needed on	Official download
PostgreSQL	17	Server	`https://www.postgresql.org/download/` (Windows: the EDB installer)
PHP	8.4 (8.3 minimum), 64-bit, *Non Thread Safe* on Windows	Server	`https://windows.php.net/download/` (Windows); Ubuntu: Ondřej Surý's PPA, see Chapter 5
PHP extensions	`pdo_pgsql`, `pgsql`, `intl`, `bcmath`, `sodium`, `mbstring`, `openssl`, `curl`, `fileinfo`, `zip`, `gd`	Server	Included with PHP
Composer	2.x	Server (installation and updates)	`https://getcomposer.org/download/`

Software	Version	Needed on	Official download
Node.js (with npm)	22 LTS or later	Server (installation and updates only)	`https://nodejs.org/`
Git	any recent	Server (to download and update the software), optional	`https://git-scm.com/`
Google Chrome or Microsoft Edge	current	Every till, handheld, kitchen screen, and your own PC for the back office	`https://www.google.com/chrome/`
Authenticator app	any (Google Authenticator, Microsoft Authenticator, Authy…)	Phones of managers	Phone app store
Caddy (web server, https certificates)	from caddyserver.com	All servers	installed by the installer or the script (Chapters 4 and 5)
Ollama (optional)	current	Server, only for the local AI assistant	`https://ollama.com/download` — see Chapter 15 (needs 16 GB RAM or more)

Why Node.js on the server? It is used once during installation (and after each update) to build the till app. It does not run while you trade.

Information to prepare

Write these down before you start (keep the sheet in a safe place, not on the server):

Item	Example	Used in
Company trading name	Kofi Stores	Chapter 7
Registered name, TIN, VAT number	Kofi Stores Ltd, C0012345678	Chapter 7
Branches: code, name, number of tills, modules	HQ · Head Office · 2 tills · Grocery	Chapter 7
Owner: full name, username, email, password (12+ characters)		Chapter 4/5
Database password (long and random)		Chapter 4/5
Server IP address or domain	192.168.1.10 / erp.kofistores.com	Chapters 4, 5, 9
Your tax set-up, checked by your accountant	VAT 15 %, NHIL 2.5 %, GETFund 2.5 %	Chapter 7
Products list with prices, barcodes and opening stock		Chapter 7

Item	Example	Used in
Staff list with roles		Chapter 7

Warning: Never use real customer data for practice or testing. Use made-up names and numbers until you go live.

4. Installing the server on Windows (setups A, B and branch servers)

Follow every step in order. Allow about two hours the first time. You need an internet connection during installation.

In this chapter the software goes in `C:\SESLabs` and PHP in `C:\php`. If you choose other folders, change them in every command.

4.0 The quick way: the installer (recommended)

SyncLytics gives you `SESLabs-server-setup-<version>.exe` with your release. It does everything in 4.2–4.9 for you.

1. Do 4.1 (prepare the computer) first.
2. Double-click the setup file, click **Yes** when Windows asks for permission (it must install a database and services).
3. Choose **Setup A** (this PC is also the till) or **Setup B** (shop server for tills on the network). Leave the server id as **HQ** unless SyncLytics gave you a branch server id (Chapter 10).
4. **Reached from the internet?** Leave both boxes empty for a server used only in the shop (the usual case). Only if this server has an internet name pointing to it (for example `shop.example.com`, with ports 80 and 443 forwarded to it by your router), type that name and an e-mail address. **Why:** with an internet name, a free certificate from Let's Encrypt is fetched and renewed automatically. Without one, the server makes its own certificate for the shop network (4.9).
5. A blue window shows each step. It downloads PostgreSQL, PHP, Composer and the **Caddy** web server from their official sites (about 450 MB; keep the internet on) and checks each download against its published checksum. It then creates the database with its own random password, the settings, the keys, the tables, and the start-up tasks.
6. At the end it shows the **backup passphrase** — write it on paper and keep it away from the server (Chapter 12) — and asks for your company name and the owner's account.

> **Why:** the installer follows exactly the manual steps below, so you can still read them to understand what was done, or do them by hand on an unusual PC.

> **Note:** the passwords the installer created (database) are in `C:\SESLabs\installer-secrets.txt`, readable by administrators only. Running the installer again (a newer version) updates the software and keeps your data, settings and keys.

Advanced: the same steps without the setup file: `powershell -ExecutionPolicy Bypass -File install.ps1 -Release <release.zip> -Setup B` (as administrator).

The server is then reached over **https** only: `https://localhost/admin` (setup A), or `https://<server-address>/admin` (setup B, for example `https://192.168.1.10/admin`). For setup B, every till, tablet and other PC must trust the server once: see **4.9**.

If you used the installer, do 4.9 (setup B), then continue with Chapter 6 (licence). The rest of this chapter is the manual way.

4.1 Prepare the computer

1. **Install all Windows updates** (Start → Settings → Windows Update), restart, and repeat until none are left. **Why:** updates fix security holes and avoid restarts in the middle of trading later.
2. **Set the clock to update automatically:** Settings → Time & language → Date & time → switch on *Set time automatically* and *Set time zone automatically* (or choose *(UTC+00:00) Monrovia, Reykjavik* for Ghana). **Why:** receipts, shifts and sync depend on the correct time. Tills more than 10 minutes ahead of the server are refused.
3. **Stop the PC from sleeping:** Settings → System → Power → *Screen and sleep* → set *When plugged in, put my device to sleep after* to **Never**. **Why:** a sleeping server cannot answer the tills.
4. **Setup B only — give the PC a fixed address on the network.** In your router's settings, find *DHCP reservation* (sometimes *Address reservation* or *Static lease*) and reserve an address for this PC, for example `192.168.1.10`. Ask whoever installed your router if unsure. Write the address down. **Why:** tills find the server by its address. If the address changes after a restart, the tills cannot connect.

> **Check:** open PowerShell and type `ipconfig`. The *IPv4 Address* of your network adapter shows the address (for setup B it must be the reserved one).

4.2 Install PostgreSQL (the database)

1. Download the **PostgreSQL 17 Windows installer** from `https://www.postgresql.org/download/windows/` (the *Download the installer* link goes to EDB).
2. Run it and accept the defaults, with these choices:
    - **Components:** keep *PostgreSQL Server* and *Command Line Tools*. *pgAdmin 4* is optional (a graphical tool). Untick *Stack Builder*.
    - **Password:** type a strong password for the *postgres* administrator account. **Write it down** in your safe place.
    - **Port:** keep `5432`.
    - **Locale:** keep *Default locale*.
3. Finish. PostgreSQL now runs as a Windows service and starts with the computer.

> **Why:** PostgreSQL is where every sale, product and account is stored. It is designed never to lose committed data, even in a power cut.

Now create the database and its own user. Open **PowerShell** and type (replace `<db-password>` with a new long random password, different from the *postgres* one, and write it down):

```
& "C:\Program Files\PostgreSQL\17\bin\psql.exe" -U postgres -c "CREATE ROLE erp LOGIN PASSWORD '<db-password>';"
& "C:\Program Files\PostgreSQL\17\bin\psql.exe" -U postgres -c "CREATE DATABASE erp OWNER erp ENCODING 'UTF8';"
```

Each command asks for the *postgres* password you chose in step 2.

> **Why:** the software uses its own database user (`erp`) that can only reach its own database, not the administrator account.

> **Check:** both commands answer `CREATE ROLE` and `CREATE DATABASE`.

4.3 Install PHP

1. Install the **Microsoft Visual C++ Redistributable (x64)** for Visual Studio 2015–2022 from Microsoft's website (`https://aka.ms/vs/17/release/vc_redist.x64.exe`). **Why:** PHP for Windows is built with it and will not start without it.
2. Go to `https://windows.php.net/download/` and download **PHP 8.4, VS17 x64 Non Thread Safe, Zip**.
3. Create the folder `C:\php` and extract the ZIP **into** it, so that `C:\php\php.exe` exists.
4. In `C:\php`, copy `php.ini-production` and rename the copy to `php.ini`.
5. Open `php.ini` in Notepad and change these lines (search with Ctrl+F; remove the `;` at the start of a line to switch it on):

```
extension_dir = "ext"
extension=curl
extension=fileinfo
extension=gd
extension=intl
extension=mbstring
extension=openssl
extension=pdo_pgsql
extension=pgsql
extension=sodium
extension=zip
memory_limit = 512M
upload_max_filesize = 20M
post_max_size = 25M
date.timezone = Africa/Accra
upload_tmp_dir = "C:\SESLabs\tmp"
sys_temp_dir = "C:\SESLabs\tmp"
```

Create the folder `C:\SESLabs\tmp` (it holds files while they are being uploaded, e.g. CSV imports).

Why: these extensions let PHP talk to PostgreSQL (`pdo_pgsql`), handle money and text in every language (`bcmath`, which is built in, `intl`, `mbstring`), and protect secrets (`sodium`, `openssl`). 6. Add PHP to the system path, so Windows finds `php` from any folder: Start → type `environment` → **Edit the system environment variables** → **Environment Variables...** → under *System variables* select **Path** → **Edit** → **New** → type `C:\php` → **OK** three times. 7. Close PowerShell and open it again (it reads the new path only when it starts).

> **Check:** type `php --version`. It shows `PHP 8.4…`. Then type `php -m` and make sure the list contains `bcmath`, `intl`, `pdo_pgsql`, `pgsql` and `sodium`.

4.4 Install Composer and Node.js

1. Download and run **Composer-Setup.exe** from `https://getcomposer.org/download/`. When it asks for PHP, choose `C:\php\php.exe`. Leave the proxy empty.
2. Download and run the **Node.js LTS** Windows installer (`.msi`) from `https://nodejs.org/`. Accept the defaults. You do not need the *Tools for native modules* option.
3. Close PowerShell and open it again.

> **Check:** `composer --version` shows `Composer version 2…`, and `node --version` shows `v22…` or later.

> **Why:** Composer downloads the server's building blocks; Node.js builds the till app. Both are only used during installation and updates.

4.5 Put the software in place

SyncLytics gives you the software as a ZIP file named like `ses-labs-<your-company>-<build>.zip`.

1. Create the folder `C:\SESLabs`.
2. Extract the ZIP into it, so that `C:\SESLabs\server\artisan` exists.

> **Note:** If you were given the **source code** (for example through Git) instead of a release ZIP, place it so that `C:\SESLabs\server` and `C:\SESLabs\pos` exist, and also do step 4.7 (*Build the till app*).

4.6 Configure the server

Open PowerShell and go to the server folder:

```
cd C:\SESLabs\server
```

1. **Download the server's libraries:**

   ```
   composer install --no-dev --optimize-autoloader
   ```

   > **Why:** the release contains only ZionEX's own code; this downloads the open-source libraries it builds on, at the exact versions it was tested with. `--no-dev` leaves out testing tools.

2. **Create the settings file** from the example:

   ```
   copy .env.example .env
   notepad .env
   ```

Change these lines and save:

```
APP_ENV=production
APP_DEBUG=false
APP_URL=https://<server-address>
SESSION_SECURE_COOKIE=true
DB_PASSWORD=<db-password>
ERP_NODE_ROLE=hq
ERP_NODE_ID=HQ
```

  - `<server-address>`: `localhost` for setup A; the fixed address from 4.1 (e.g. `192.168.1.10`) for setup B.
  - For a **branch server**, use `ERP_NODE_ROLE=branch` and the branch server's id (e.g. `B01`) as given in Chapter 10.

> **Why:** `.env` tells the software where its database is, what address it is reached on, and that it is a live system (`APP_DEBUG=false` hides technical error details from users). `SESSION_SECURE_COOKIE=true` makes browsers send the sign-in cookie only over https.

> **Warning:** `.env` will hold your secret keys. Never e-mail it, never put it on a shared drive, and include it in your backups (Chapter 12).

3. **Create the secret keys:**

```
php artisan key:generate
php artisan erp:audit-key
```

> **Why:** `key:generate` creates `APP_KEY`, which encrypts sessions and stored secrets. `erp:audit-key` creates `ERP_AUDIT_KEY`, which seals the audit log so changes to history can be detected. **Never change either key after this.**

4. **Create the database tables and the standard data** (roles, permissions, units, tax templates, chart of accounts):

```
php artisan migrate --seed --force
```

> **Why:** this builds the empty structure of the database. `--force` confirms you are on a live system.

> **Check:** it ends with lines saying `DONE` and no red errors.

5. **Create your company and the owner account:**

```
php artisan erp:install
```

It asks for the company name, the owner's full name, a username (for till sign-in), email and a password of at least 12 characters.

> **Why:** the owner is the first person who can sign in; they create everyone else.

> **Note:** Skip this step on a **branch server**: its company and people come from head office (Chapter 10).

6. **Build the files for the online shop's live updates** (optional, but harmless):

```
npm install
npm run build
```

7. **Speed up the server** (run again after every change to `.env`):

```
php artisan optimize
```

4.7 Build the till app (source code only)

Skip this if your release ZIP already contains `C:\SESLabs\server\public\pos\index.html`.

```
cd C:\SESLabs\pos
npm install
npm run build:server
```

> **Why:** this turns the till's source code into the app the server hands out at `/pos/`, including the offline files that let tills keep selling without the server.

> **Check:** the file `C:\SESLabs\server\public\pos\index.html` now exists.

4.8 Start the server automatically

The server needs these programs running all the time:

- the **web server (Caddy)**: the only program the tills and browsers talk to. It serves everything over **https**, so passwords, PINs and sales cannot be read by anyone else on the network;
- **PHP workers**: they run ZionEX itself, behind the web server. Each one serves one request at a time, so there are four;
- the **scheduler**: it runs the routine jobs every minute (sending to online stores, syncing branches, posting to the accounts, reminders, the daily licence check-in, the nightly insights and alerts);
- **live updates (Reverb)**: they tell tills about price and stock changes within seconds. They are reached through the web server, so no extra port is open.

We register them with Windows **Task Scheduler**, so they start with the computer even when nobody is signed in, and restart if they stop.

Step 1: get Caddy. Download `caddy_2.10.0_windows_amd64.zip` from the official releases page `https://github.com/caddyserver/caddy/releases` and unzip it to `C:\SESLabs\caddy`.

> **Why:** Caddy is a widely used web server that gets and renews https certificates by itself.

Step 2: protect its data folder. In PowerShell as administrator:

```
New-Item -ItemType Directory -Force C:\SESLabs\caddy\data
icacls C:\SESLabs\caddy\data /inheritance:r /grant:r "Administrators:(OI)(CI)F" "SYSTEM:(OI)(CI)F"
```

> **Why:** on the shop network Caddy acts as its own small certificate authority, and its secret key lives in this folder. Only administrators may read it.

Step 3: the settings file. Create `C:\SESLabs\Caddyfile` with Notepad. This example is for setup B with address `192.168.1.10` and computer name `shopserver`; change both. For setup A, use `localhost` as the address, remove the whole `http://…` block, and add the line `default_bind 127.0.0.1` under `admin off`.

```
{
	admin off
	skip_install_trust
	storage file_system "C:/SESLabs/caddy/data"
	pki {
		ca local {
			name "SES Labs on shopserver"
			root_cn "SES Labs shopserver Root"
			intermediate_cn "SES Labs shopserver Intermediate"
		}
	}
}

http://192.168.1.10, http://shopserver {
	handle /ca.crt {
		root * "C:/SESLabs/caddy/data/pki/authorities/local"
		rewrite * /root.crt
		header Content-Type application/x-x509-ca-cert
		file_server
	}
	handle {
		redir https://{host}{uri} permanent
	}
}

192.168.1.10, shopserver {
	tls internal
	root * "C:/SESLabs/server/public"
	encode zstd gzip
	request_body {
		max_size 25MB
	}
	header -Server
	@hidden {
		path_regexp /\.
		not path /.well-known/*
	}
	handle @hidden {
		respond 404
	}
	@live path /app/* /apps/*
	handle @live {
		reverse_proxy 127.0.0.1:8080
	}
	handle /pos/* {
		header /pos/sw.js Cache-Control no-cache
```

```
                try_files {path} /pos/index.html
                file_server
        }
        handle {
                php_fastcgi 127.0.0.1:9001 127.0.0.1:9002 127.0.0.1:9003 127.0.0.1:9004 {
                        lb_policy least_conn
                        lb_try_duration 10s
                        read_timeout 300s
                }
                file_server
        }
}
```

> **Why:**

- *Plain http* only hands out the server's certificate (4.9) and sends everything else to https.
- *Hidden files* such as `.env` are never served.
- *Live updates* use the same https address.
- *The till app* falls back to its own start page, so it works offline.
- *Everything else* goes to the PHP workers.

For a server with an **internet name**:

- remove the `http://…` block and the `tls internal` line;
- write the name instead of the address;
- add `email you@example.com` under `admin off`.

Caddy then gets a Let's Encrypt certificate by itself.

> **Check:** `C:\SESLabs\caddy\caddy.exe validate --config C:\SESLabs\Caddyfile --adapter caddyfile` ends with *Valid configuration*.

Step 4: the PHP worker. Create `C:\SESLabs\php-worker.cmd`:

```
@echo off
set PHP_FCGI_MAX_REQUESTS=0
:run
"C:\php\php-cgi.exe" -b 127.0.0.1:%1
ping -n 3 127.0.0.1 >nul
goto run
```

> **Why:** PHP on Windows stops a worker after 500 requests unless told otherwise. The loop restarts it if it ever stops.

Step 5: in `.env`, add or change these lines (with your address), then run `php artisan optimize`:

```
REVERB_APP_ID=<random-1>
REVERB_APP_KEY=<random-2>
REVERB_APP_SECRET=<random-3>
REVERB_HOST=192.168.1.10
REVERB_PORT=443
REVERB_SCHEME=https
REVERB_PUBLISH_HOST=127.0.0.1
REVERB_PUBLISH_PORT=8080
REVERB_PUBLISH_SCHEME=http
```

Use three random values of your own (letters and digits, at least 20 characters each).

> **Why:** tills connect to live updates through the https address. The server itself sends to them directly on the same PC.

Step 6: the start-up tasks. Open **PowerShell as administrator** and paste this whole block:

```
$app = "C:\SESLabs\server"; $php = "C:\php\php.exe"
$settings = New-ScheduledTaskSettingsSet -ExecutionTimeLimit ([TimeSpan]::Zero) -RestartCount 999 -RestartInterval (New-TimeSpan -Minutes 1) -AllowStartIfOnBatteries -DontStopIfGoingOnBatteries
$principal = New-ScheduledTaskPrincipal -UserId "SYSTEM" -LogonType ServiceAccount -RunLevel Highest
$trigger = New-ScheduledTaskTrigger -AtStartup
Register-ScheduledTask -TaskName "SES Labs Scheduler" -Trigger $trigger -Settings $settings -Principal $principal -Action (New-ScheduledTaskAction -Execute $php -Argument "artisan schedule:work" -WorkingDirectory $app)
Register-ScheduledTask -TaskName "SES Labs Live" -Trigger $trigger -Settings $settings -Principal $principal -Action (New-ScheduledTaskAction -Execute $php -Argument "artisan reverb:start --host=127.0.0.1 --port=8080" -WorkingDirectory $app)
foreach ($i in 1..4) { Register-ScheduledTask -TaskName "SES Labs PHP $i" -Trigger $trigger -Settings $settings -Principal $principal -Action (New-ScheduledTaskAction -Execute "cmd.exe" -Argument "/c `"`"C:\SESLabs\php-worker.cmd`" $(9000 + $i)`"" -WorkingDirectory $app) }
Register-ScheduledTask -TaskName "SES Labs Web" -Trigger $trigger -Settings $settings -Principal $principal -Action (New-ScheduledTaskAction -Execute "C:\SESLabs\caddy\caddy.exe" -Argument "run --config C:\SESLabs\Caddyfile --adapter caddyfile" -WorkingDirectory "C:\SESLabs\caddy")
Get-ScheduledTask -TaskName "SES Labs *" | Start-ScheduledTask
```

> **Why:** `ExecutionTimeLimit 0` stops Windows from ending the tasks after three days (its default). `SYSTEM` runs them without anyone signed in.

Step 7: trust the server on this PC. Wait 10 seconds, then:

```
Import-Certificate -FilePath C:\SESLabs\caddy\data\pki\authorities\local\root.crt -CertStoreLocation Cert:\LocalMachine\Root
```

> **Why:** the server's own browser then accepts its certificate (other devices: 4.9). Skip this step for a server with an internet name.

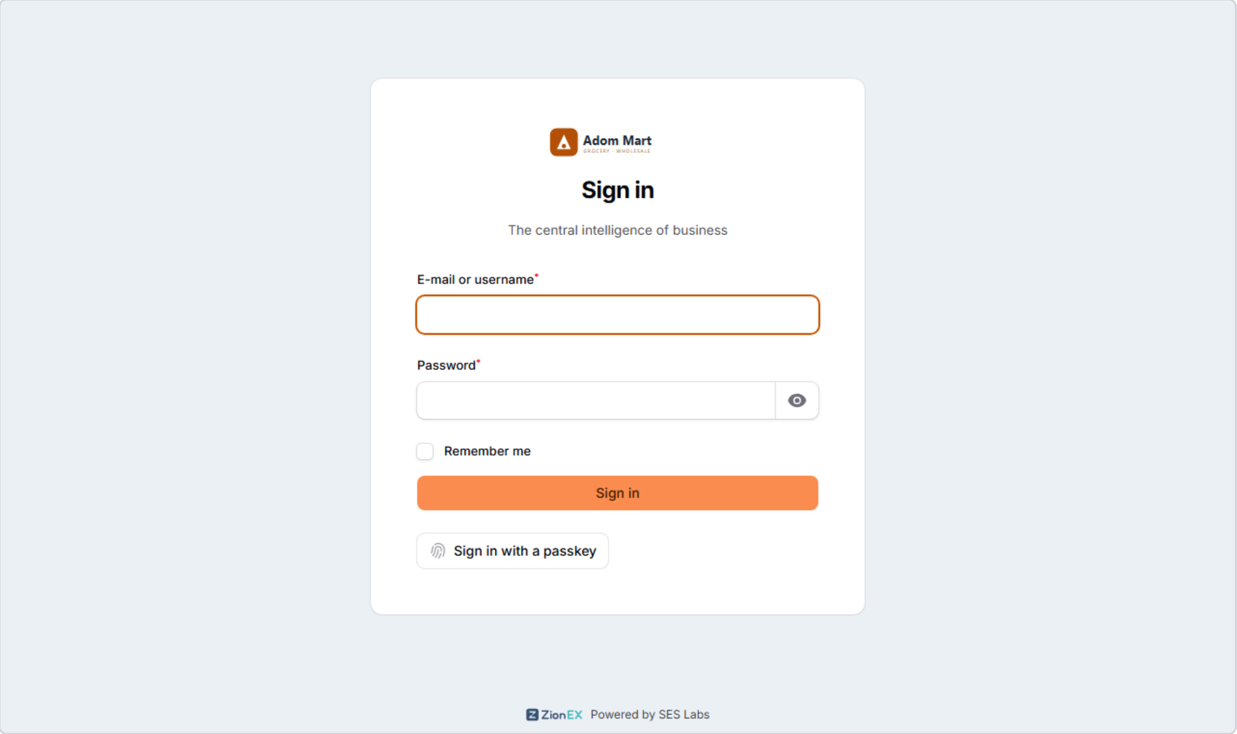
Step 8 (setup B only): let the tills through the Windows firewall:

```
New-NetFirewallRule -DisplayName "SES Labs server" -Direction Inbound -Protocol TCP -LocalPort 80,443 -Action Allow -Profile Private,Domain
```

> **Why:** Windows blocks incoming connections by default. This opens only the web server's two ports, and only on private (shop) networks, not public Wi-Fi. The database, the PHP workers and live updates stay reachable from this PC only.

> **Check:**

1. In a browser on the server, open `https://localhost/up` (setup A) or `https://<server-address>/up` (setup B). It shows a page saying *Application up*, with no certificate warning.
2. Open `…/admin`: the sign-in page appears:

   
   Adom Mart
   Sign in
   The central intelligence of business
   E-mail or username*
   Password*
   ☐ Remember me
   Sign in
   Sign in with a passkey
   ZionEX Powered by SES Labs

3. Setup B: on another PC that trusts the server (4.9), open `https://<server-address>/admin`. The same page appears.

> **Tip:** to restart the server, open **Task Scheduler**, click *Task Scheduler Library*, right-click *SES Labs Web* (or a *PHP*, *Scheduler* or *Live* task), and choose **End**, then **Run**.

4.9 Let tills and other PCs trust this server (servers without an internet name)

On the shop network (and on an Ubuntu server reached by its address, Chapter 5) the server uses **its own certificate**. A certificate is what proves to a browser that it is really talking to your server and that nobody can read along. Each till, tablet, handheld, kitchen screen and office PC must be told **once** to trust it. Without this, browsers show a warning, and the till refuses to enrol: it only uses its security functions on a trusted address.

On a Windows PC (also for the ZionEX Till and ZionEX Suite desktop apps): open PowerShell **as administrator** and type, with your server's address:

```
Invoke-WebRequest http://192.168.1.10/ca.crt -OutFile $env:TEMP\seslabs.crt
Import-Certificate -FilePath $env:TEMP\seslabs.crt -CertStoreLocation Cert:\LocalMachine\Root
```

On an Android tablet or phone:

1. In Chrome, open `http://<server-address>/ca.crt`. It downloads a file.
2. Open **Settings** → **Security (and privacy)** → **More security settings** → **Encryption & credentials** → **Install a certificate** → **CA certificate**, confirm with **Install anyway**, and choose the downloaded file. The wording differs a little between makes.

If the installer said port 80 is in use:

- Another program on the server, such as IIS, already uses port 80, so `http://…/ca.crt` is not available. The server still works normally over https.
- Copy the file `C:\SESLabs\SESLabs-certificate.crt` from the server to each device with a USB stick or a shared folder.
- On Windows, import it with `Import-Certificate -FilePath <the file> -CertStoreLocation Cert:\LocalMachine\Root` (PowerShell as administrator).
- On Android, choose the copied file in the *Install a certificate* step.

> **Check:** open `https://<server-address>/pos/`. It opens with a padlock and no warning.

Why this is safe: the certificate is made by your own server, for your own server. Its secret key never leaves the server and is readable by administrators only.

> **Note:**

- The certificate is valid for the server's address and computer name at the time of installation. If you ever give the server a different address, run the installer again, then repeat this step on each device.
- A server with an internet name uses a public Let's Encrypt certificate instead, which every device already trusts. Skip this section.
- Live updates (instant price and stock changes on tills) are set up by the installer and need nothing more.

4.10 Next steps

- Activate the licence: **Chapter 6**.
- First-time set-up in the back office: **Chapter 7**.
- Set up the tills: **Chapter 9**.
- Set up backups **before** you go live: **Chapter 12**.

5. Installing the server on Ubuntu Linux (setups B, C, D and branch servers)

Ubuntu is the recommended system for a head-office, central or cloud server, and for any shop server with more than four devices. It runs for months without restarts and handles many tills at once.

This chapter assumes **Ubuntu Server 24.04 LTS**, freshly installed, on a cloud server (VPS) or a computer at your premises. You run every command as a user that may use `sudo` (the cloud provider's default user, often `ubuntu` or `root`).

5.0 The quick ways (recommended)

A. The install script — does everything in 5.2–5.9 on a plain Ubuntu server:

1. Do 5.1 (get the server, connect to it). Copy the release ZIP and `install.sh` (both from SyncLytics) to the server, e.g. with WinSCP.
2. Run it. Add your own domain and e-mail if the server has an internet name (setup D); leave both out for a server used only inside your network:

   ```
   sudo bash install.sh --zip ~/ses-labs-<customer>-<build>.zip --domain erp.yourcompany.com --email you@yourcompany.com
   ```

   For a branch server add `--role branch --node-id B01`.
3. Write down the **backup passphrase** it shows, then answer its questions to create your company and the owner.

The server is then reached over **https** only:

- **With a domain:** a free Let's Encrypt certificate is fetched and renewed automatically.
- **Without one:** the server uses its own certificate for its network address and name. Each till, tablet and PC trusts it once, as in **4.9**, from `http://<server-address>/ca.crt`.

The firewall is switched on with only SSH, 80 and 443 open (443 also for HTTP/3).

Running it again with a newer ZIP updates the software and keeps your data, `.env` and keys. During an update the scheduled tasks pause while the files are replaced, and start again at the end. If a large import is running, the update waits for it (up to 15 minutes). **Why:** a task that starts while files are being replaced fails and fills the log with errors. An older installation that used nginx is moved to the new web server (Caddy) automatically.

A shared server for small clients (SyncLytics staff). The standard is **one server per client**. For small businesses that are just starting, several clients can share one server, each as its **own separate copy** of the system on its own internet name:

```
sudo bash install.sh --zip <release>.zip --instance kofi-pharmacy --domain kofi.synclytics.example --email you@example.com
```

- `--instance` is a short name for the client: lowercase letters, digits and dashes.
- Each copy gets its own:
    - folder (`/var/www/seslabs-kofi-pharmacy`) and database (`erp_kofi_pharmacy`);
    - system user, so one copy cannot read another's files, backups or passwords;
    - PHP workers, which stop when idle so a quiet client uses little memory;
    - live-updates port, backups, licence (its own installation code) and update timing.
- Give each copy its own **DNS A record** pointing to the same server (5.1), and its own `--domain`.
- **Update one client:** run the same command with the new ZIP and the same `--instance`. It backs that copy up first and leaves the others alone.
- **How many:** a server with 12 GB of memory (for example Contabo Cloud VPS 6) holds about **3 small clients** (one or two shops each). Move a client to a server of its own when it grows, or when *System → Health → Slow pages* starts filling up: back it up, install it on the new server, restore the backup there (Chapter 12.5), and point its internet name to the new server.
- **Remove a client's copy** (after a final backup and when the contract has ended):

    ```
    sudo systemctl disable --now seslabs-<name>-reverb
    sudo rm /etc/cron.d/seslabs-<name> /etc/caddy/sites/seslabs-<name>.caddy /etc/php/8.4/fpm/pool.d/seslabs-<name>.conf
    sudo systemctl restart php8.4-fpm caddy
    ```

    Then delete its folder and database (`sudo -u postgres dropdb erp_<name>`) only once you are sure the backup is safe elsewhere.

Why separate copies, not one copy holding several companies: one copy shares one licence, one backup and one update for everyone in it. Restoring one client's backup would roll back the others, and a fault in one would reach them all.

B. Docker (if your host prefers containers):

1. Install Docker (`curl -fsSL https://get.docker.com | sudo sh`).
2. In the release folder: `cp installer/docker/.env.docker.example installer/docker/.env` and edit it: `APP_URL` , a long random `DB_PASSWORD` , a `BACKUP_PASSPHRASE` (12+ characters, written down), and e-mail settings (Chapter 16).
3. `sudo docker compose -f installer/docker/docker-compose.yml up -d --build`
4. The first start prints an `APP_KEY` and `ERP_AUDIT_KEY` in `docker compose logs app` : copy both into `installer/docker/.env` straight away — they must never change.
5. Create your company: `sudo docker compose -f installer/docker/docker-compose.yml exec app php artisan erp:install` .
6. Put HTTPS in front (your host's load balancer, or Caddy / Traefik).

Why Docker: the whole server (PHP, web server, scheduler, live updates) is one image that runs the same everywhere; the database and your files live in Docker volumes that survive updates (`up -d --build` with a new release).

The rest of this chapter is the manual way.

5.1 Get the server and connect to it

Cloud server (setup D)

1. Rent a VPS from any provider (for example a provider with a data centre in or near West Africa or Europe; with Contabo, the *Cloud VPS 6* with 100 GB NVMe storage in the *European Union* region, with Auto Backup). Choose **Ubuntu 24.04**, at least 2 vCPU, 4 GB RAM, 80 GB SSD; see the sizing table in Chapter 3. Note the server's **public IP address** and the password or key you are given.
2. Buy or use a **domain name** and create a **DNS A record** such as `erp.yourshop.com` pointing to that IP address (in your domain provider's control panel). It can take up to an hour to work. **Why:** tills, browsers and the HTTPS certificate need a name, not just a number.

Server at your premises (setups B, C, branch servers)

1. Install Ubuntu Server 24.04 from a USB stick (`https://ubuntu.com/download/server`). During installation choose **Install OpenSSH server**.
2. Reserve a fixed IP address for it in your router (see 4.1, step 4).

Connect from your Windows PC

Open PowerShell on your own PC and type (replace the user and address):

```
ssh ubuntu@<server-ip-address>
```

Type `yes` the first time, then the password. You are now typing commands **on the server**.

> **Tip:** to copy files to the server with a mouse, install **WinSCP** (`https://winscp.net`) on your PC and connect with the same address, user and password.

5.2 Update the system and set the time

```
sudo apt update && sudo apt upgrade -y
sudo timedatectl set-timezone Africa/Accra
sudo timedatectl set-ntp true
```

> **Why:** security fixes first; and the correct time is essential for receipts, shifts and sync.

> **Check:** `timedatectl` shows *System clock synchronized: yes*.

5.3 Install PostgreSQL 17

```
sudo apt install -y postgresql-common
sudo /usr/share/postgresql-common/pgdg/apt.postgresql.org.sh -y
sudo apt install -y postgresql-17
```

> **Why:** Ubuntu 24.04 ships an older PostgreSQL; the official PostgreSQL repository provides version 17, which the software is tested with.

Create the database and its user (replace `<db-password>` with a long random password and write it down):

```
sudo -u postgres psql -c "CREATE ROLE erp LOGIN PASSWORD '<db-password>';"
sudo -u postgres psql -c "CREATE DATABASE erp OWNER erp ENCODING 'UTF8';"
```

> **Check:** the commands answer `CREATE ROLE` and `CREATE DATABASE`.

5.4 Install PHP 8.4, Caddy, Composer and Node.js

```
sudo add-apt-repository -y ppa:ondrej/php
sudo apt install -y debian-keyring debian-archive-keyring apt-transport-https curl
curl -fsSL https://dl.cloudsmith.io/public/caddy/stable/gpg.key | sudo gpg --dearmor -o /usr/share/keyrings/caddy-stable-archive-keyring.gpg
curl -fsSL https://dl.cloudsmith.io/public/caddy/stable/debian.deb.txt | sudo tee /etc/apt/sources.list.d/caddy-stable.list
sudo apt update
sudo apt install -y caddy unzip git php8.4-fpm php8.4-cli php8.4-pgsql php8.4-intl php8.4-bcmath php8.4-mbstring php8.4-xml php8.4-curl php8.4-zip php8.4-gd
curl -sS https://getcomposer.org/installer | sudo php -- --install-dir=/usr/local/bin --filename=composer
curl -fsSL https://deb.nodesource.com/setup_22.x | sudo -E bash -
sudo apt install -y nodejs
```

> **Why:**

- **PHP** runs the software, and **PHP-FPM** lets several requests run at the same time.
- **Caddy** is the web server that receives the tills' and browsers' requests. It gets and renews https certificates by itself. It comes from its makers' own signed package source.
- **Composer** and **Node.js** are used during installation.

> **Check:** `php --version` shows 8.4, `caddy version` shows v2, `composer --version` shows 2.x, `node --version` shows v22 or later, and `php -m` lists `pdo_pgsql`, `intl`, `bcmath` and `sodium`.

5.5 Put the software in place

1. Copy your release ZIP to the server (with WinSCP, into your home folder).
2. Unpack it into `/var/www/seslabs`:

```
sudo mkdir -p /var/www/seslabs
sudo unzip ~/ses-labs-*.zip -d /var/www/seslabs
sudo chown -R $USER:www-data /var/www/seslabs
cd /var/www/seslabs/server
```

> **Check:** `ls` shows `artisan`, `app`, `public` and other folders.

5.6 Configure the server

```
composer install --no-dev --optimize-autoloader
cp .env.example .env
nano .env
```

In the editor, change these lines (move with the arrow keys; save with **Ctrl+O**, **Enter**, exit with **Ctrl+X**):

```
APP_ENV=production
APP_DEBUG=false
APP_URL=https://erp.yourshop.com
SESSION_SECURE_COOKIE=true
DB_PASSWORD=<db-password>
ERP_NODE_ROLE=hq
ERP_NODE_ID=HQ
```

- For a server reached by its address on your network, use `APP_URL=https://<server-ip-address>`. It is still https: the server makes its own certificate (5.7).
- For a **branch server**: `ERP_NODE_ROLE=branch` and its id (e.g. `B01`), see Chapter 10.

Then:

```
php artisan key:generate
php artisan erp:audit-key
php artisan migrate --seed --force
php artisan erp:install
npm install && npm run build
php artisan optimize
sudo chown -R www-data:www-data storage bootstrap/cache
sudo chmod 640 .env && sudo chown $USER:www-data .env
```

The reasons for each command are the same as in Chapter 4, section 4.6. The last two lines let the web server write its logs and cache, and stop other users of the machine from reading `.env`.

> **Note:** Skip `php artisan erp:install` on a branch server (Chapter 10).

If you received the source code instead of a release, also build the till app:

```
cd /var/www/seslabs/pos && npm install && npm run build:server && cd ../server
```

5.7 Configure the web server (Caddy)

First let Caddy reach PHP:

```
sudo usermod -aG www-data caddy
```

> **Why:** PHP-FPM listens on a socket that only the `www-data` group may use.

Replace Caddy's settings file (the install script instead writes a short main file that imports one file per copy from `/etc/caddy/sites/`; the manual way keeps everything in one file):

```
sudo nano /etc/caddy/Caddyfile
```

Delete what is in it and paste the text below. This example is for a server on your network with address `192.168.1.20` and name `shopserver`; change both everywhere.

For a server with a domain (setup D):

- write the domain instead of `192.168.1.20, shopserver`;
- remove the whole `http://…` block and the `tls internal` line;
- add `email you@yourshop.com` under `admin off`.

```
{
	admin off
	skip_install_trust
	pki {
		ca local {
			name "SES Labs on shopserver"
			root_cn "SES Labs shopserver Root"
			intermediate_cn "SES Labs shopserver Intermediate"
		}
	}
}

http://192.168.1.20, http://shopserver {
	handle /ca.crt {
		root * /var/lib/caddy/.local/share/caddy/pki/authorities/local
		rewrite * /root.crt
		header Content-Type application/x-x509-ca-cert
		file_server
	}
	handle {
		redir https://{host}{uri} permanent
	}
}

192.168.1.20, shopserver {
	tls internal
	root * /var/www/seslabs/server/public
	encode zstd gzip
	request_body {
		max_size 25MB
	}
	header -Server
	@hidden {
		path_regexp /\.
		not path /.well-known/*
	}
	handle @hidden {
		respond 404
	}
	@live path /app/* /apps/*
	handle @live {
		reverse_proxy 127.0.0.1:8080
	}
	handle /pos/* {
		header /pos/sw.js Cache-Control no-cache
		try_files {path} /pos/index.html
```

```
			file_server
		}
		handle {
			php_fastcgi unix//run/php/php8.4-fpm.sock {
				read_timeout 300s
			}
			file_server
		}
}
```

Check and start it:

```
sudo -u caddy caddy validate --config /etc/caddy/Caddyfile --adapter caddyfile
sudo systemctl restart php8.4-fpm caddy
```

> **Why:**

- *Plain http* only hands out the server's own certificate (so devices can trust it, 4.9) and sends everything else to https.
- *Hidden files* such as `.env` are never served.
- *Live updates* use the same https address.
- *The till app* falls back to its own start page, so it works offline.
- *Everything else* goes to PHP.
- With a domain, Caddy fetches a Let's Encrypt certificate and renews it by itself.
- `admin off` closes Caddy's local control port. After a change to this file, use `systemctl restart caddy` (a *reload* needs that port).

Server on your network: trust the certificate on the server itself, and keep a copy for devices:

```
sudo cp /var/lib/caddy/.local/share/caddy/pki/authorities/local/root.crt /usr/local/share/ca-cer
tificates/seslabs.crt
sudo update-ca-certificates
```

> **Check:** `validate` ends with *Valid configuration*. `curl https://192.168.1.20/up` (or your domain) answers without a certificate error. On another device that trusts the server (4.9), `https://192.168.1.20/admin` shows the sign-in page.

5.8 Firewall

```
sudo ufw allow OpenSSH
sudo ufw allow 80/tcp
sudo ufw allow 443/tcp
sudo ufw allow 443/udp
sudo ufw enable
```

> **Why:** the firewall blocks everything except remote administration (SSH) and the web server's ports. `443/udp` is HTTP/3, which browsers use when they can: it needs fewer round trips and copes better with a slow or unsteady internet line, which matters for a server far from the shop. The database, PHP and live updates are reachable from the server itself only. If SSH on your server uses another port than 22, allow that port instead of `OpenSSH` **before** `ufw enable`, or you lock yourself out.

Why HTTPS always:

- HTTPS encrypts everything between tills and the server: PINs, sales and customer details.
- Browsers only let tills enrol, install as apps and work offline from https addresses (or from `localhost`).

> **Warning:** Never run a server that is reachable from the internet without HTTPS.

5.9 Background jobs

The scheduler runs the routine jobs. Add it to the system's timer (cron):

```
echo "* * * * * www-data cd /var/www/seslabs/server && php artisan schedule:run >> /dev/null 2>&1" | sudo tee /etc/cron.d/seslabs
```

> **Why:** every minute it sends changes to online stores, syncs branch servers, posts to the accounts, sends reminders and, once a day, checks the licence.

Live updates: prices and stock reach tills within seconds, instead of within 60 seconds.

1. In `.env` set (three random values of your own, letters and digits, at least 20 characters each; your address or domain as `REVERB_HOST`):

   ```
   REVERB_APP_ID=<random-1>
   REVERB_APP_KEY=<random-2>
   REVERB_APP_SECRET=<random-3>
   REVERB_HOST=192.168.1.20
   REVERB_PORT=443
   REVERB_SCHEME=https
   REVERB_PUBLISH_HOST=127.0.0.1
   REVERB_PUBLISH_PORT=8080
   REVERB_PUBLISH_SCHEME=http
   ```

   > **Why:** tills connect through the https address. The server itself sends to the live-updates service directly on the same machine.

2. Create the service:

```
sudo nano /etc/systemd/system/seslabs-reverb.service
```

```
[Unit]
Description=SES Labs live updates
After=network.target

[Service]
User=www-data
WorkingDirectory=/var/www/seslabs/server
ExecStart=/usr/bin/php artisan reverb:start --host=127.0.0.1 --port=8080
Restart=always
RestartSec=5

[Install]
WantedBy=multi-user.target
```

3. Start it:

```
cd /var/www/seslabs/server && php artisan optimize
sudo systemctl daemon-reload
sudo systemctl enable --now seslabs-reverb
```

> **Check:** `systemctl status seslabs-reverb` shows *active (running)*.

5.10 Next steps

Continue with the licence (**Chapter 6**), the first-time set-up (**Chapter 7**), devices (**Chapter 9**) and backups (**Chapter 12**).

6. Activating the licence

Every installation needs a **licence key from SyncLytics** (hotline 024 563 6889, info@synclyticsenterprisesystems.com). Until it is activated, only the Licence page works.

6.1 What the licence controls

- How many **branches**, **tills and devices** and **active users** you may have. Logins in the practice company (copies of your staff, for training) and the SyncLytics support account do not count.
- Which **industry modules** you may switch on (Grocery, Pharmacy, Restaurant, Electronics, Fashion, Services…).
- Whether **Finance** (accounts) and **HR & payroll** are included.
- Until when it is valid.

Parts that are not licensed are hidden from the menus.

Reading the licence page (System → Licence):

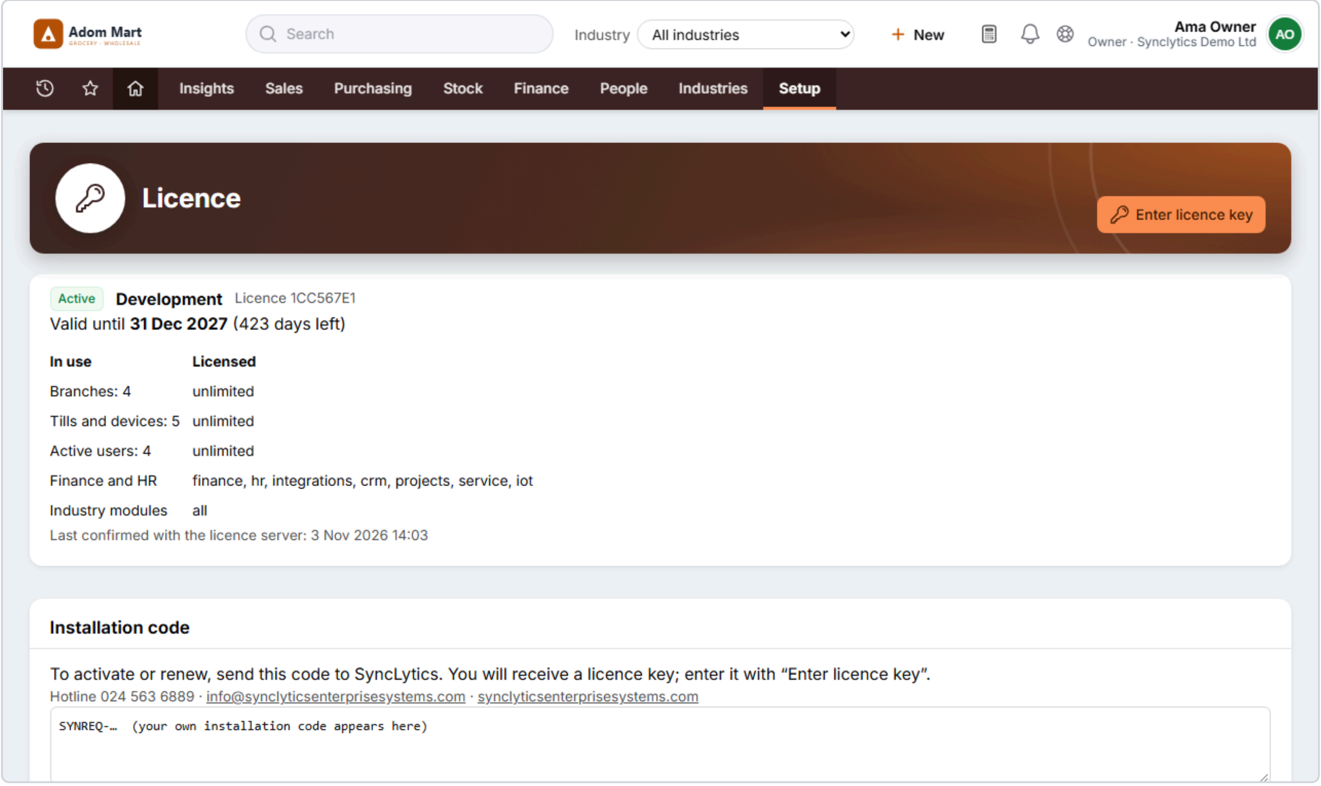
- **In use** beside **Allowed** shows how much of the licence you are using. **Limit reached** means nothing more can be added, so ask for a new key first. Why: a new branch, till or user is refused once its limit is full.
- **Branches** include head office. A company with head office and one shop is already using 2.
- **Features** and **Paid add-ons** list what is included. Anything not listed stays out of the menus.
- **Industry modules** list the kinds of business you may switch on. *Not available yet* means the licence covers it but it has not been built, so choose another module. A supermarket uses **Grocery / Supermarket**.

6.2 Activate

1. Open the back office at `https://<server-address>/admin` (Windows) or `https://<your-domain>/admin` (Ubuntu) and sign in as the **owner** (Chapter 7, steps 1–2 explain the first sign-in and two-factor set-up).
2. You are taken to **System** → **Licence**. It shows the **installation code** (also as a QR code), which starts with `SYNREQ-`.
3. Send that code to SyncLytics: copy and paste it into an e-mail to info@synclyticsenterprisesystems.com, or a WhatsApp message to the hotline 024 563 6889. The Licence page shows these contacts too. Tell them which modules and how many branches, devices and users you need.
4. You receive a **licence key** starting with `SYN1-`. On the Licence page click **Enter licence key**, paste it and confirm.

> **Why:** the key is signed by SyncLytics and checked on your server without the internet. It is tied to this installation, so a copied server or database asks to be activated again.

> **Check:** the Licence page shows **Active**, your company name, the expiry date and the limits:



Licence

Development Licence 1CC567E1

Valid until **31 Dec 2027** (423 days left)

In use	**Licensed**
Branches: 4	unlimited
Tills and devices: 5	unlimited
Active users: 4	unlimited
Finance and HR	finance, hr, integrations, crm, projects, service, iot
Industry modules	all

Last confirmed with the licence server: 3 Nov 2026 14:03

Installation code

To activate or renew, send this code to SyncLytics. You will receive a licence key; enter it with “Enter licence key”.

Hotline 024 563 6889 · info@synclyticsenterprisesystems.com · synclyticsenterprisesystems.com

```
SYNREQ-…  (your own installation code appears here)
```

6.3 Daily check-in

Head office contacts the SyncLytics licence server once a day (03:15, by the scheduler). It collects renewals automatically, so you normally never paste a key again.

- If the server cannot reach the internet for **30 days**, a yellow banner appears; after another **14 days** the software becomes read-only until it checks in.
- **Branch servers** receive the licence from head office; they do not check in themselves.

Hourly health report (support console)

Every hour head office also sends SyncLytics support a short **health report**: the checks on **System → Health** (backups, restore check, disk, memory, server load, scheduler, licence…), tills with problems, the numbers of open problem reports, and whether support access is open. **No sales, customers, prices, stock or staff details are sent.**

> **Why:** support sees a failed backup, a full disk or a silent till on its console and can call you before it becomes a problem. To stop it: **System** → **Settings** → **Security** → **Share system health with SyncLytics support** → off (the daily licence check-in stays).

For SyncLytics (licence server): Licences → **Support console** lists every client, worst first: *Needs attention* (a red check), *Not reporting* (no report for 3 hours: server, internet or scheduler down), *Warning*, *OK*. **Details** shows every check, the tills with problems and the open problem reports; the dashboard's *Needs attention* list includes the same red and silent servers.

For SyncLytics (issuing licences): licences are made with the **SES Licensing** desktop app on the owner's PC, which works without internet. Its own guide is the README in the `SES-Licensing` folder. If the licence server is running, connect the app once so the two stay the same:

1. On the licence server, open **/vendor** → **Set-up** → **Sync devices** → **New device**, name the PC and copy the one-time code. **Why:** only a PC you allow can change licences, and the code works once, for 15 minutes.
2. In the app, open **Sync**, type the server's address and the code, and click **Connect**. **Why:** the app makes its own private key and sends only the public half, so no password travels or sits on the server.
3. To stop a lost PC from syncing, click **Revoke** beside it on the same page.

Customers, plans, add-ons, licences, invoices and payments then stay the same in both places. **Why:** a payment recorded on either side shows on both, so nobody chases a client who has paid.

> **Check:** the device shows *Active* and a recent *Last seen*, and the licences issued in the app appear under **Licences**.

To update the app itself, copy `latest.json` and the installer that `npm run release` made into `storage/app/ses-licensing/` on the licence server. **Why:** connected copies of the app find the new version there and offer it under **Settings** → **Updates**.

6.4 Licence states

State	When	What works
Not activated	New installation	Only the Licence page
Active	Normal	Everything you bought
Attention (yellow banner)	30 days before expiry; no check-in for 30 days; software moved to another server	Everything
Grace	Up to 14 days after expiry	Everything, with a banner
Read-only	After the grace period; licence withdrawn; no check-in for 44 days; moved and not re-activated within 14 days; software modified	You can **view** everything, but nothing can be changed, sold or printed. Tills keep their unsent sales until the licence is renewed. **Nothing is deleted.**

6.5 Renewing, adding branches or modules

Ask SyncLytics for a new key. It reaches head office at the next daily check-in, or paste it on the Licence page straight away.

6.6 Moving to a new server

The licence is tied to the server it was activated on. After moving (Chapter 12), the Licence page shows *Attention: moved*. Send the new installation code to SyncLytics and enter the new key within **14 days**.

Warning: Do not try to copy or alter the software to get around the licence. Modified copies stop working and are reported at check-in; printed documents carry hidden references to the licence they were made with.

7. First-time set-up in the back office

Do these steps once, at head office, in this order. Allow a day for a single shop, longer for a chain (most of the time goes into products and prices).

> **Tip:** Practise first on a **test installation** with made-up products: `php artisan erp:demo-catalog --as=<owner-email>` fills it with sample products and stock. Never run that command on your live system.

Step 1 — Sign in for the first time

1. Open the back office: `https://<server-address>/admin` (Windows server) or `https://<your-domain>/admin` (Ubuntu).
2. Sign in with the owner's **email (or username) and password** (from `erp:install`). Everyone can type either their e-mail address or the username they use on the till.

Step 2 — Set up two-factor sign-in

You are asked to set up an **authenticator app**:

1. Install an authenticator app on your phone (Google Authenticator, Microsoft Authenticator or similar).
2. In the app choose *Add account → Scan a QR code* and scan the code on the screen.
3. Type the 6-digit code the app shows.
4. **Save the recovery codes** shown next: print them or write them down and keep them somewhere safe, **not** on the same phone.

> **Why:** a stolen password alone is then not enough to get in. Everyone from security level L3 (supervisors and above) must use two-factor sign-in.

> **Tip:** the back office can be **Light**, **Dark** or follow your device (**System**): open your profile menu (top right) and choose the sun, moon or screen icon. Each person's choice is kept in their browser.

> **Note:** Later you can add a **passkey** (fingerprint or face) under your profile. Passkeys work only on an HTTPS address or `localhost`.

Step 3 — Activate the licence

See **Chapter 6**.

Step 4 — The setup wizard

The wizard opens by itself the first time.

1. **Deployment** — how the system runs: *Standalone single PC* (setup A/B), *Central on-premise server* (setup C), or *Cloud HQ* (setups D/E). This records your starting layout; you can grow later.
2. **Company** — trading name, country, base currency and time zone. Optionally the **legal entity**: registered name, **TIN** and **VAT number**, needed for VAT invoices (you can add it later under *Company*).
3. **Your brand** (optional) — the company **logo**, a **square icon** and the **brand colour** (leave the colour empty and it is taken from the logo). You can do this later under *Company → Branding* (below).
4. **Branches & modules** — one row per branch:
    - **Codes the system gives:** branches, stores (`PHAOSU-W1`), tills (`T01`), vans (`V01`), departments, price lists and tax groups (`STAFF_PRICE`), kitchen stations, routes, work centres, recipes, styles, asset classes and membership plans all get their code from the name or the next number, and keep it. **Why:** no two can clash, and receipts, files and reports that use them never break. Codes that match something real stay yours to type, with the next one suggested: account numbers, shelf bin labels and table numbers. Unit codes (PC, BOX) are typed as before.
    - The branch's **code** is made from its name (*Pharmacy Osu* → `PHAOSU`). **Why:** it appears in every receipt number, so the system makes sure it is unique and never changes it.
    - **Name**, the number of **tills**, and whether it is the **head office** (exactly one).
    - **Modules** for that branch (Grocery, Pharmacy, Restaurant, Electronics, Fashion, Services, Online Storefront...). You can change these later (Chapter 8).
5. Click **Finish setup**.

> **Why:** each branch automatically gets a main warehouse and the tills you asked for. Receipt numbers look like `HQ-T01A-000123`: branch, till, device series and number.

> **Check:** *Company → Branches* lists your branches with their warehouses and tills.

Your brand: logo, icon and colour

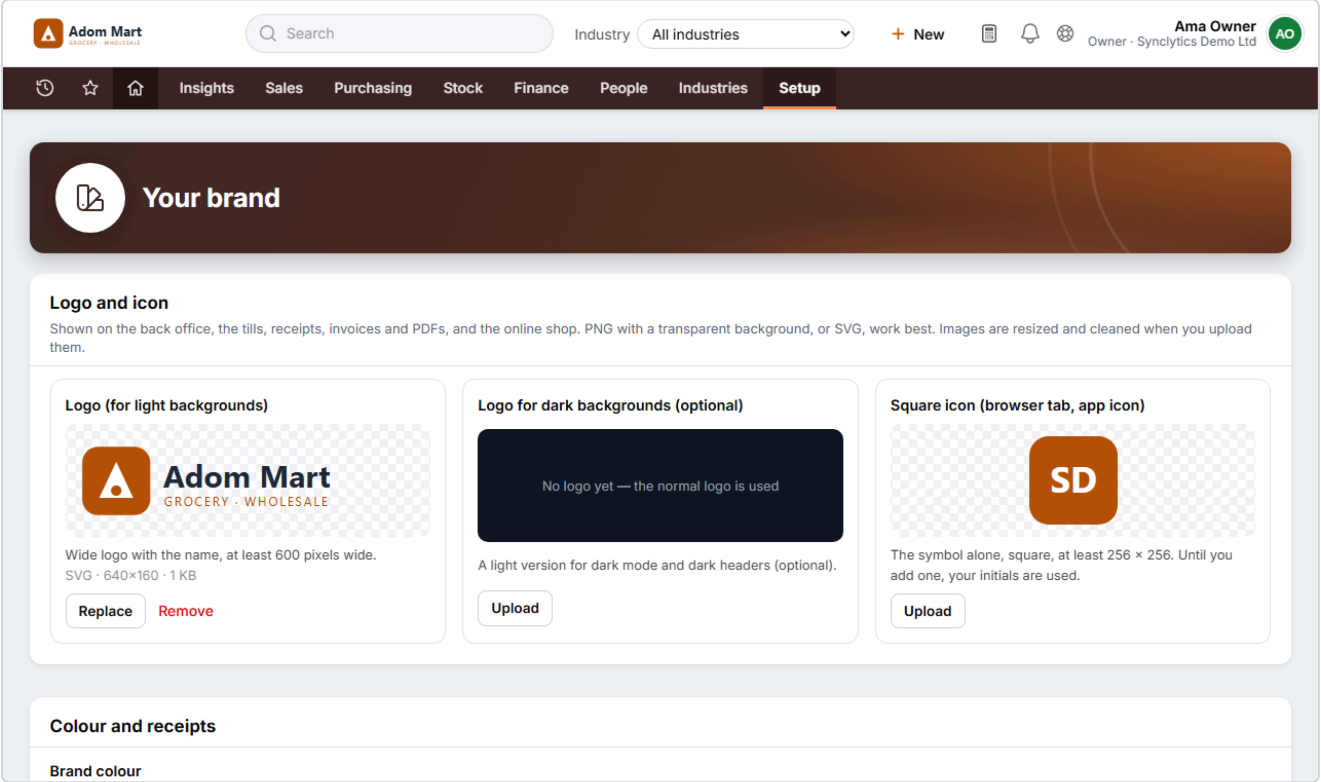
Which name prints: receipts, dosage labels, drawer slips and invoices carry the name of the company the branch trades as — its industry's legal entity (Company → Legal entities and Company → Industries), e.g. *Asante Pharmacy Ltd*, not the group *Asante Holdings*. The receipt top is short: that name, the branch and its phone on one line, the TIN and VAT number. In the back office the same name shows at the top for people who work in one industry (or after choosing one at the top); people who see several industries see the group name.

Company → Branding. Your brand is what staff and customers see: the back office's top bar, the till, printed receipts, invoices, statements, PDF exports and your online shop.

1. **Logo** — your logo with the name, on a transparent background (PNG) or as SVG, at least 600 pixels wide. Choose **Upload**.
2. **Logo for dark backgrounds** (optional) — a light version for dark mode and dark headers. Without it, the normal logo is used.
3. **Square icon** — the symbol alone, at least 256 × 256 pixels. It is the browser-tab icon and the small icon on the till. Until you add one, your **initials on your colour** are used.
4. **Brand colour** — pick it, type it (`#0F766E`), or press **Use ... from your logo**. If the colour is too light to read on white, the page says so: choose a darker shade. **Menu colour** (optional) — the colour of the menu band, the side menu and each page's banner. Leave it empty and it is a dark shade of

your brand colour, so it always goes with your logo; or pick your own dark colour (white writing must read on it). **Automatic** goes back to the shade.

5. **Print the logo on till receipts** — receipt printers print it in black and white (it is converted for them).
6. Press **Save colour and receipts**, and look at the **Preview**: the till accent, a receipt and an invoice.


Adom Mart
GROCERY · WHOLESALE
Search
Industry
All industries
+ New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Adom Mart
GROCERY · WHOLESALE
SD
Replace
Remove
Upload
Upload

Your brand

Logo and icon

Shown on the back office, the tills, receipts, invoices and PDFs, and the online shop. PNG with a transparent background, or SVG, work best. Images are resized and cleaned when you upload them.

Logo (for light backgrounds)

Wide logo with the name, at least 600 pixels wide.

SVG · 640×160 · 1 KB

Logo for dark backgrounds (optional)

No logo yet — the normal logo is used

A light version for dark mode and dark headers (optional).

Square icon (browser tab, app icon)

The symbol alone, square, at least 256 × 256. Until you add one, your initials are used.

Colour and receipts

Brand colour

> **Why:** customers trust a receipt or invoice that carries the shop's own logo, and staff recognise their own system. The images are checked and made smaller when you upload them, and they reach branch servers and backups like the rest of your data.

> **Note:** a small line *"Powered by ZionEX"* stays at the bottom of the back office, receipts and documents. A **white-label licence** removes it (ask SyncLytics).

Step 4b — Industries (if you run more than one kind of business)

Each branch runs one industry (Pharmacy, Grocery, ...), switched on per branch in the wizard or under Company → Modules. Head office can stay company-level. See **Chapter 8.0** for the industry switcher, separate catalogues and industry managers.

Books: by default each industry gets **its own company and books**. Fill in each company's registered name and TIN under **Company** → **Legal entities** before the first sale, so receipts print them. For one set of books instead, switch off **System** → **Settings** → **Finance** → **Separate books for each industry** now, before anything is posted (Chapter 8.0, *Accounts by industry*).

Step 5 — People and access

For each staff member:

1. **People & access** → **Users** → **New user**: name, username, email (for back-office users), and a **Security level (ceiling)** (the highest level of what they can ever be given — see Chapter 18).
2. On the next screen, **Grant access**: choose a **role** (Cashier, Supervisor, Branch Manager, Storekeeper, Stock Controller, Accountant, Pharmacist, Waiter, Stylist / Therapist…) and **where it applies** (a till, a branch, a warehouse or the whole company). A person's own **security level** must be at least the role's (e.g. Stock Controller L3, Warehouse Manager L4); raise the level on the user's page first if the role is refused. **Products** (create and edit, categories, brands, units) need Stock Controller, Warehouse Manager or a manager role — for one branch or the whole company; there is one product list for all branches. **Prices** need a manager role (*Change prices, price lists and tax set-up*), for the whole company.
3. For anyone who sells or clocks in on a till: **Set POS PIN** (4–8 digits; simple PINs like 1234 are refused).
4. **How the person signs in to the back office** (on the user's page):
    - **Company rule** (default): the authenticator app from the level set under System → Settings → Security (L3 unless you change it), password only below it;
    - **Password and the authenticator app** — always, whatever the level;
    - **Password and a code by WhatsApp or e-mail** — a 6-digit code is sent at each sign-in, works once for 10 minutes. It goes to the **Mobile (WhatsApp) number** when your WhatsApp Business provider is set up (16.3), otherwise to the e-mail address. *Send it by e-mail instead* is on the code screen;
    - **Password only** — allowed for staff up to supervisors (L3).

> **Why:** a second step stops someone who learned a password. People without a smartphone app can get a code by WhatsApp or e-mail instead; low-risk staff who only look things up can use the password alone — your choice, person by person, and every change is in the audit log.

> **Why:** people see and do only what their role allows, only where it applies. A cashier with access to *Till 1* cannot sell on *Till 2* or open the back office.

> **Tip:** For temporary cover (e.g. a supervisor on leave), grant access **with an end date and a reason**: it stops by itself.

The SyncLytics support account: the list may show *SyncLytics support (Superadmin)* at level L7. SyncLytics adds it on your server so support can help you; nobody in the company can change it. It always signs in with an authenticator app, it does not count as one of your licensed users, and the owner and general manager get a notification each time it signs in. Everything it does is in the audit log (System → Audit log).

Check who can do what: People & access → **Who can do what** lists the risky combinations first — one person who orders from suppliers and also receives the goods, adds suppliers and pays them, collects customer payments and can write debts off, sells and approves till shortages, adds staff and pays wages, or changes what roles allow — with what each risk means, what to do, and who has it where. Below:

people switched off or not signed in for 60 days who still have access, temporary access with its end date, and everyone's roles. **Download (CSV)** keeps a copy.

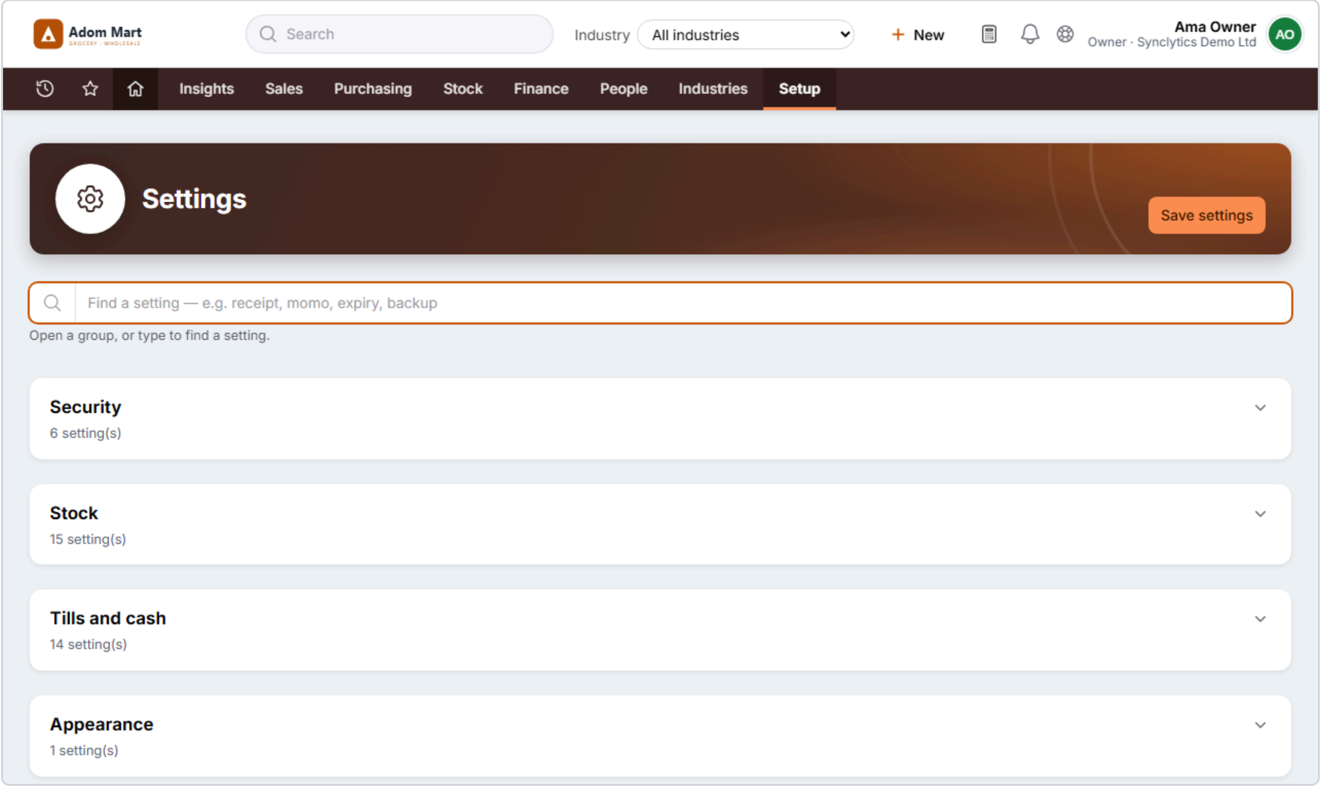
> **Why:** most thefts in a small business need one person to hold two jobs that should check each other. In a small team some overlap cannot be avoided; then look at that person's work in **System → Audit log** now and then. Look at this page every few months and after staff changes.

> **Warning:** Do not share one account between several people. Every sale, refund and change is recorded under the person who signed in.

Finding a page: at the top of the side menu, **Search the menu** — type part of a name (*adjust*, *expiry*, *momo*, *po*) and only the matching pages stay; **Enter** opens the first, **Esc** clears. Press `/` anywhere to jump there. To find a record (a product, a customer, a receipt), use the search at the top (Ctrl+K).

Step 6 — Settings

System → Settings — the page lists the groups (click one to open it), and **Find a setting** at the top shows only the settings with the words you type (e.g. *receipt*, *momo*, *expiry*), whatever group they are in. **Save settings** is at the top and the bottom. Leaving the page (or any form in the back office) with changes not saved asks first, so nothing typed is lost by mistake; the till does the same for a sale on screen, and the handheld for a delivery or count not yet posted. Review each group. The most important:


Adom Mart
GROCERY · WHOLESALE
Search
Industry
All industries
+ New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Save settings

Settings

Find a setting — e.g. receipt, momo, expiry, backup

Open a group, or type to find a setting.

Security
6 setting(s)

Stock
15 setting(s)

Tills and cash
14 setting(s)

Appearance
1 setting(s)

Setting	Why it matters
Require two-factor sign-in from security level	Default L3. Lower it for more security.
Wrong POS PINs before the PIN is locked / POS PIN lockout (minutes)	Stops PIN guessing at the till.
Offline till sign-in lasts (hours)	How long a cashier can sign in on a till without the server (default 72). Shorter is safer, longer helps during long outages.
Stop sales when the system shows no stock	Off by default (tills can sell items the system thinks are out of stock, and you count them later).
Show stock left on the till while selling	On by default (switch off if cashiers should not see stock). On: each item in the sale, and in the till's search list, shows how many are left at that till's store (e.g. *14 TAB left*), counting what is already in the sale. Useful where cashiers should tell customers what is available. Tills pick it up at their next sign-in.
Message at the bottom of receipts	A thank-you line and your returns policy. Each line you type, and each sentence, prints on its own line (*Thank you for shopping with us.* then *Goods sold are not returnable.*). **Why:** short separate lines read easily on narrow receipt paper.
Returns accepted within (days, with receipt)	How long after a sale a return with the receipt is accepted.
Each teller sells from their own vault / How each teller's cash is kept	Off by default. On: each teller has their own vault (cash tray) and a till opens only on theirs (Step 13b). **Why:** nobody can sell in someone else's name.
Lock the till after this many idle minutes	Default 5; 0 = never. **Why:** a till left signed in cannot be used by the next person: it locks with the name on it, and only that person's PIN unlocks it (the sale on screen stays).
Cash control with vaults	Off: whoever closes a till counts the drawer. On: tellers, a Cashier and an Income Controller check the cash at every step (Step 14).

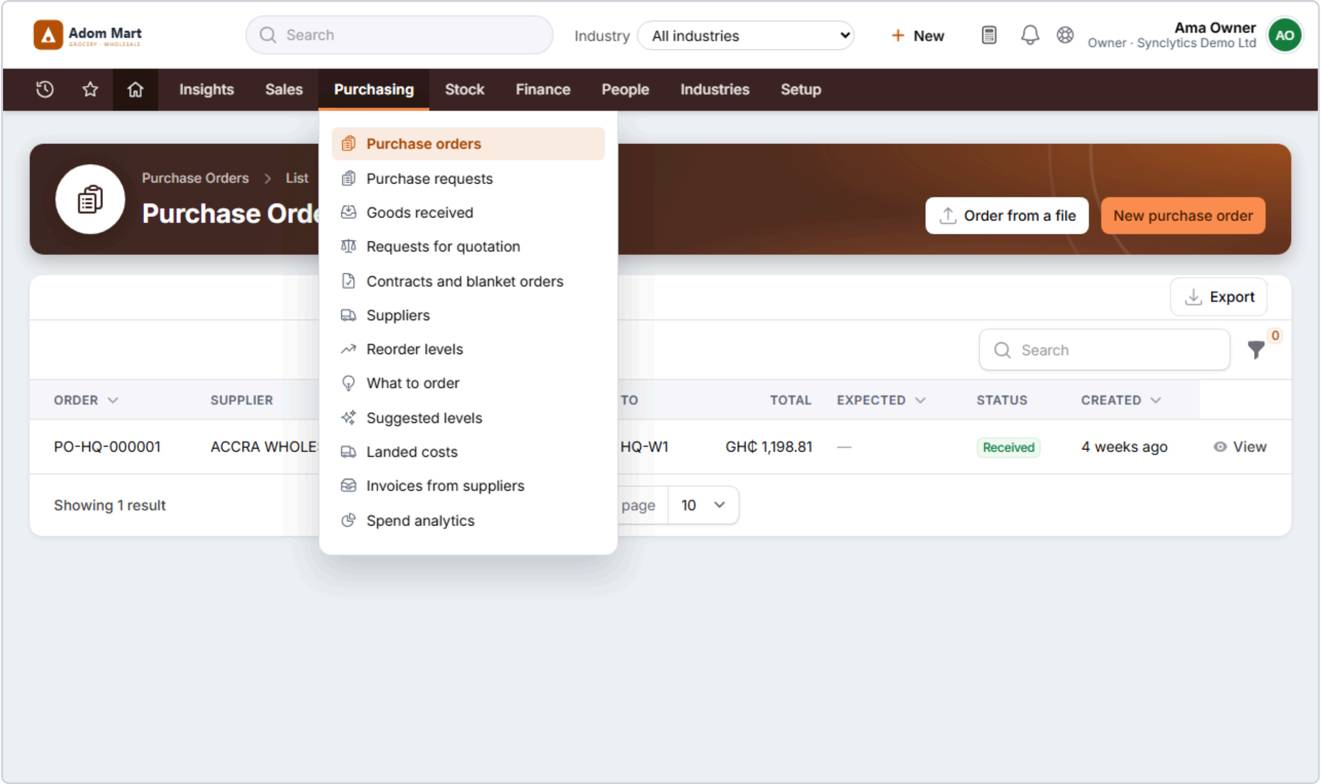
Setting	Why it matters
Automatic shifts / Till trading hours / Straight day from (hours)	Off by default. On: each till session is named by the shift it opens in (Morning, Afternoon… from **Cash → Shifts and vaults**; by default Morning 07:00–15:00 and Afternoon 14:00–23:00, shifts may overlap by at most 1 hour), and one of 10 hours or more is a **Straight day**. A till opens only within the trading hours (default 07:00–23:00): outside them, a manager (Branch Manager or above) types their username, PIN and the reason on the till. Whether this setting is on or off, someone who already closed their own shift on that till today needs a manager to open another one, even if the next shift has a different name. This stops empty shifts from someone signing in again after closing. Another teller taking over the till (a handover) opens their own shift without a manager. The till shows who is signed in on the *Open shift* screen and above every manager's PIN, with a *Sign out* link, so the manager sees whose shift they allow. Closing and counting are always allowed. A branch can choose differently. **Why:** every session is in the right shift for the reports, and nobody sells after closing time or reopens a counted till without a manager knowing, while a normal handover between tellers needs nobody's PIN.
Payment alerts (group)	Off by default. *Each payment*, *each payment and a daily summary* or *the summary only*, for mobile money and/or card; single alerts only from an amount you choose; also by e-mail, SMS or WhatsApp (to each person's e-mail and **Mobile (WhatsApp) number**); the summary time (default 21:00). Who gets them: anyone with *receive payment alerts* — the Owner, General Manager, Finance Manager and Accountant by default. Alerts show on the bell in the back office. **Why:** the owner sees MoMo and card money arrive without asking the till, and the evening summary is checked against the MoMo statement.

Module settings (Grocery, Pharmacy, Restaurant, Fashion, Services) are explained in Chapter 8.

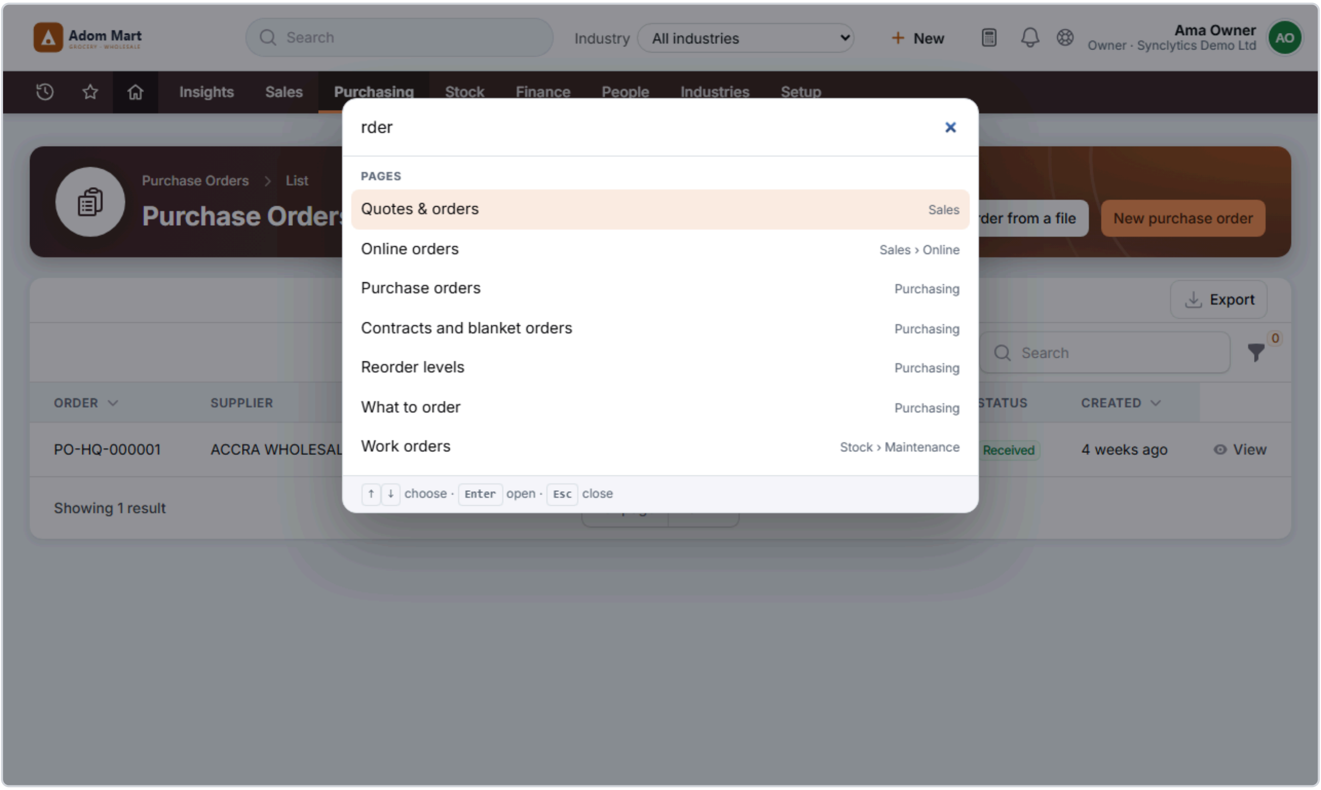
Step 6b — How the back office looks, favourites and profile photos

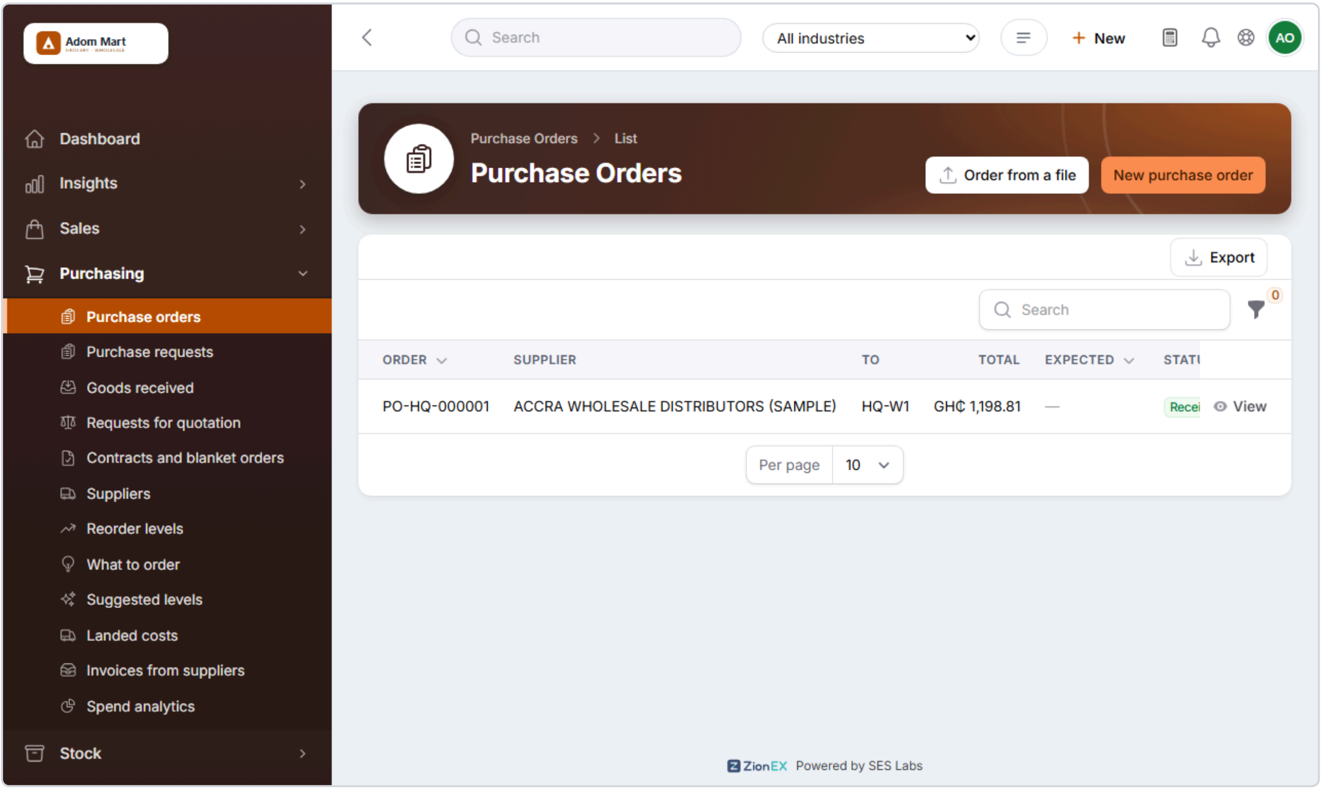
The back office has three looks. **Tabs across the top** is the standard one:

- **Tabs across the top** (standard): like a desktop accounting program. The top row has your logo, a wide **Search** (for records: products, customers, orders…), **+ New** (create any record you are allowed to create, e.g. an order or a product, without looking for its page first) and your name with your role. Under it, a band in your menu colour starts with three icons, then the tabs (Insights, Sales, Purchasing, Stock, Finance, People, your industry, Setup). Point at a tab and its pages open in a panel. The panel always fits inside the window, however narrow.
    - **Long lists:** the column headings stay at the top while you scroll, the first column stays put when a wide list scrolls sideways, and the page title shrinks to one line with its buttons. The rows per page you choose are remembered on that computer for each list. **Why:** you always know which column and which row you are reading.
    - **Clock — recent pages and Find a page:** the pages you opened last; type part of a page's name to find any page. The **/** key opens it from anywhere.
    - **Star — favourites:** open a page you use often, click the star, then *Add this page to favourites*. Your favourites follow you to any computer (up to 20).
    - **House — home:** your home page.
- **New look: side menu:** a menu in your menu colour down the whole left side, with your logo at the top and one row per area (Dashboard, Insights, Sales, Purchasing…); click an area to open its pages. The ‹ button folds it to icons for more room.
- **Classic:** the original layout.


Adom Mart
Search
Industry
All industries
+ New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Purchase orders
Purchase requests
Goods received
Requests for quotation
Contracts and blanket orders
Suppliers
Reorder levels
What to order
Suggested levels
Landed costs
Invoices from suppliers
Spend analytics
Purchase Orders > List
Purchase Orde
Order from a file
New purchase order
Export
Search
0
Showing 1 result
page
10

ORDER	SUPPLIER	TO	TOTAL	EXPECTED	STATUS	CREATED	
PO-HQ-000001	ACCRA WHOLE	HQ-W1	GH₵ 1,198.81	—	Received	4 weeks ago	View


Adom Mart
Search
Industry
All industries
+ New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Purchase Orders > List
Purchase Order
der from a file
New purchase order
Export
Search
0
ORDER
SUPPLIER
STATUS
CREATED
PO-HQ-000001
ACCRA WHOLESAL
Received
4 weeks ago
View
Showing 1 result
rder
×
PAGES
Quotes & orders — Sales
Online orders — Sales › Online
Purchase orders — Purchasing
Contracts and blanket orders — Purchasing
Reorder levels — Purchasing
What to order — Purchasing
Work orders — Stock › Maintenance
↑ ↓ choose · Enter open · Esc close


Adom Mart
Dashboard
Insights
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Purchase orders
Purchase requests
Goods received
Requests for quotation
Contracts and blanket orders
Suppliers
Reorder levels
What to order
Suggested levels
Landed costs
Invoices from suppliers
Spend analytics
Stock
Search
All industries
New
AO
Purchase Orders > List
Purchase Orders
Order from a file
New purchase order
Export
Search
0
Per page 10
ZionEX Powered by SES Labs

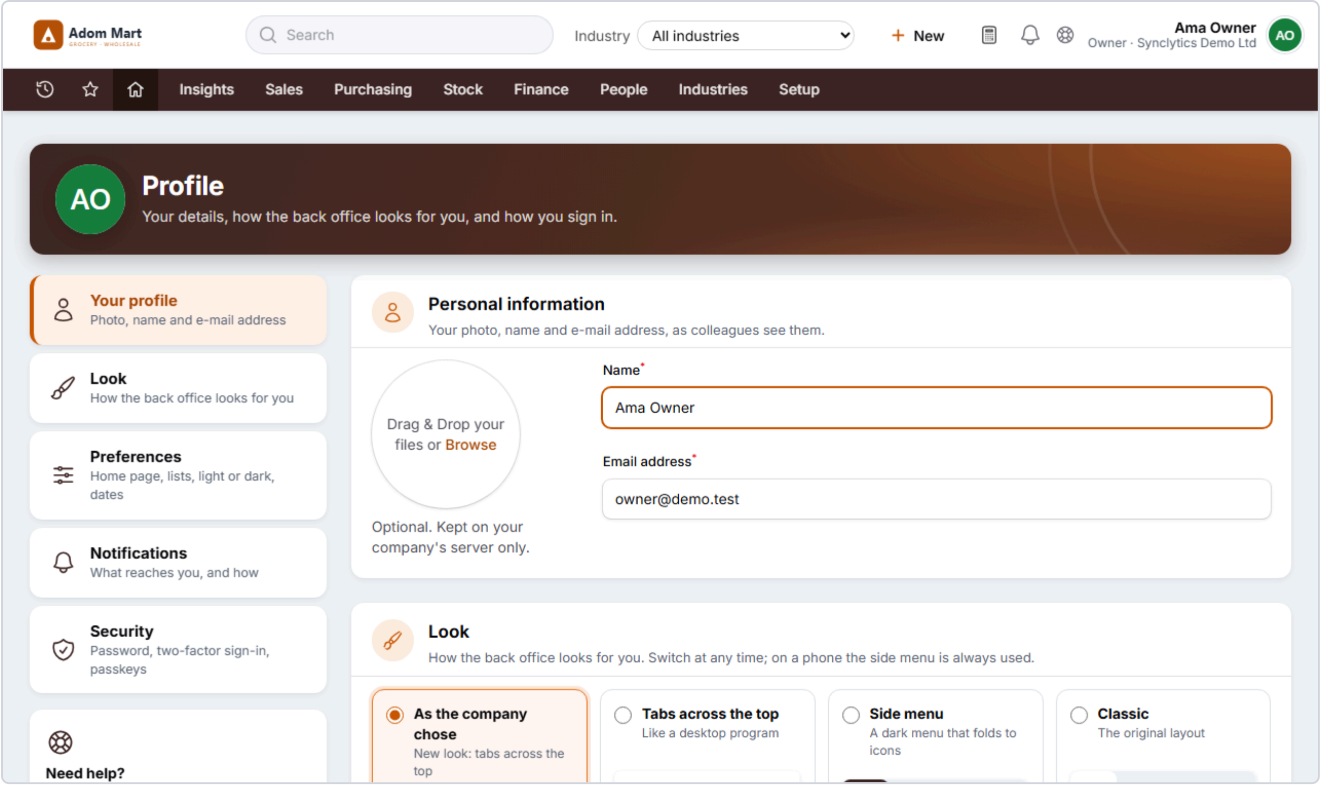
ORDER	SUPPLIER	TO	TOTAL	EXPECTED	STATU	
PO-HQ-000001	ACCRA WHOLESALE DISTRIBUTORS (SAMPLE)	HQ-W1	GH₵ 1,198.81	—	Recei	View

In both new looks each page opens with a banner in your menu colour (the page's icon, its name and its buttons), and sections are white cards, so every page looks the same way. All colours follow your brand (Step 4, *Your brand*).

Where the menu sections are in the tabs: this guide names pages by their menu section, e.g. *System → Settings*. With the tabs, the sections are grouped like this: **Insights** (Insights, Reports) · **Sales** (Sales, Cash, Online) · **Purchasing** · **Stock** (Stock, Catalog, Pricing & tax) · **Finance** · **People** (HR & payroll, People & access) · your **industry** (e.g. Pharmacy) · **Setup** (Company, Messages, System). So *System → Settings* is **Setup → Settings**. Or press **/** and type *settings*.

Why a choice: most people find the tabs faster, and they fit more on the screen. Anyone who prefers another look can keep it.

1. **Each person, for themselves:** top right, click your photo or initials → **Profile → Look** → choose → **Save changes**. *As the company chose* follows the company's choice.
2. **For everyone: System → Settings → Appearance → How the back office looks**. This applies to everyone who has not chosen for themselves.
3. **On a phone** the side menu is always used, because the tabs do not fit.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Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
AO
Your profile
Photo, name and e-mail address
Look
How the back office looks for you
Preferences
Home page, lists, light or dark, dates
Notifications
What reaches you, and how
Security
Password, two-factor sign-in, passkeys
Need help?
Drag & Drop your files or Browse

Profile

Your details, how the back office looks for you, and how you sign in.

Personal information

Your photo, name and e-mail address, as colleagues see them.

Name*: Ama Owner

Email address*: owner@demo.test

Optional. Kept on your company's server only.

Look

How the back office looks for you. Switch at any time; on a phone the side menu is always used.

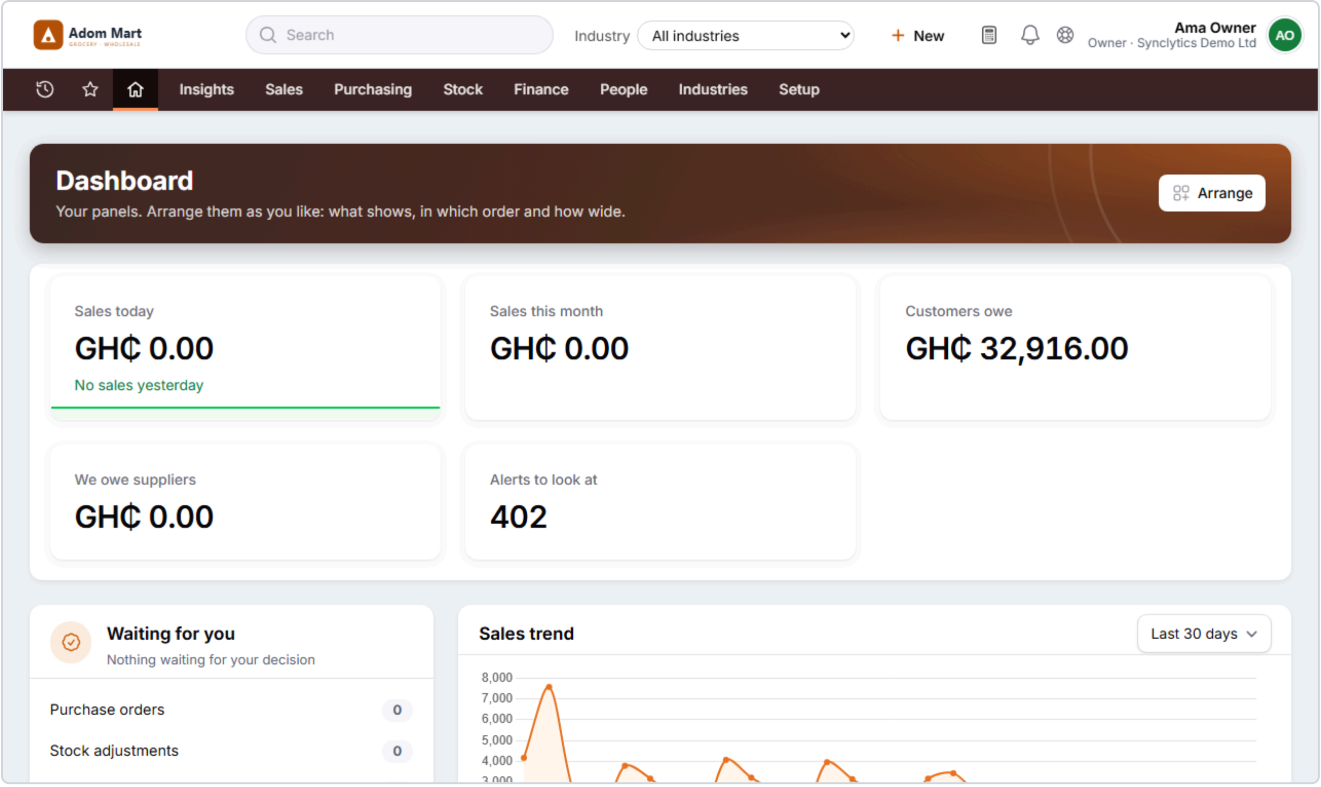
- ● **As the company chose** — New look: tabs across the top
- ○ **Tabs across the top** — Like a desktop program
- ○ **Side menu** — A dark menu that folds to icons
- ○ **Classic** — The original layout

The **Profile** page lists its parts on the left (Your profile, Look, Security) with SyncLytics' support contacts, and shows a card for each on the right: your photo and details, the look (pictures of each, in your colours), your password (with how strong it is), two-factor sign-in and passkeys.

Profile photo: on the same **Profile** page, drop a photo on the circle (or click *Browse*), adjust it with the pencil if you like, and **Save changes**. It shows beside your name in the top bar and in the list of users. **Why:** colleagues see at a glance who did what. The server keeps only a small square copy, with the hidden details of the original removed (where and when it was taken, the camera). It stays on your company's server and is shown only to people signed in to your company. People without a photo show their initials, drawn on your own server (no name is sent to an outside website).

Step 6c — Your dashboard, preferences and notifications

The dashboard (the house icon, or *Dashboard*) is each person's home page: panels with figures, charts and lists — sales today and this month, the sales trend, how customers paid, sales by branch, best sellers, stock to act on, buying, spend by month, what customers owe by how late, cash, pharmacy, people, and **Waiting for you** (orders, stock adjustments, expense claims and leave waiting for your approval) with **Shortcuts** (your favourite pages and quick *new* links).


Adom Mart
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All industries
+ New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Arrange
Last 30 days
8,000
7,000
6,000
5,000
4,000
3,000

Dashboard

Your panels. Arrange them as you like: what shows, in which order and how wide.

Sales today
GH₵ 0.00
No sales yesterday

Sales this month
GH₵ 0.00

Customers owe
GH₵ 32,916.00

We owe suppliers
GH₵ 0.00

Alerts to look at
402

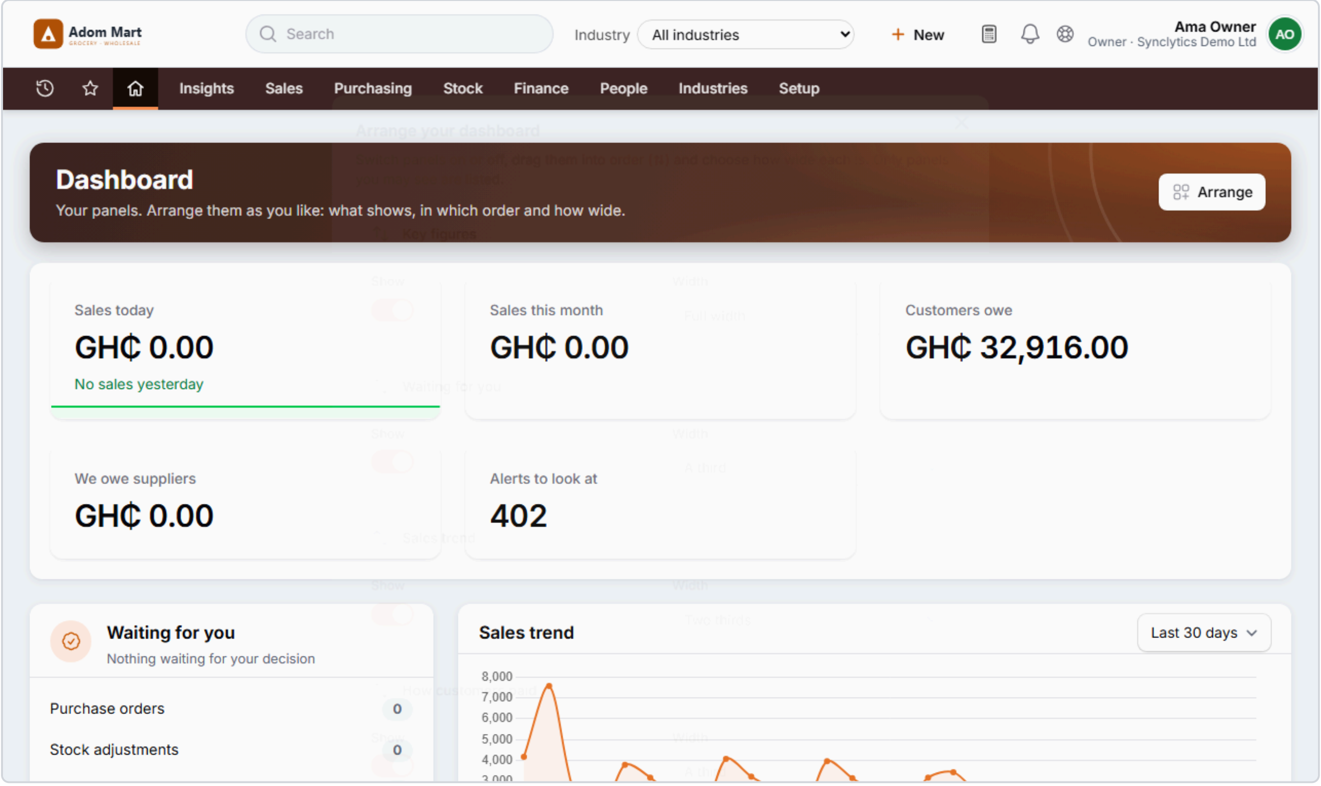
Waiting for you
Nothing waiting for your decision

Purchase orders	0
Stock adjustments	0

Sales trend

> **Why:** each person sees first what they act on. A cashier sees the cash and shortcuts; a buyer sees buying and stock; the owner sees the whole business. Panels show only what the person may see, for their own branches.

1. Each role starts with its own set of panels.
2. **Arrange** (top right of the dashboard): switch panels on or off, drag them into order (⇅) and choose how wide each is (a third, half, two thirds, full width) → **Save my dashboard**.
3. **Back to my role's** brings the role's panels back.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Sales
Purchasing
Stock
Finance
People
Industries
Setup
Arrange
Last 30 days
8,000
7,000
6,000
5,000
4,000
3,000

Dashboard

Your panels. Arrange them as you like: what shows, in which order and how wide.

Sales today: GH₵ 0.00 — No sales yesterday

Sales this month: GH₵ 0.00

Customers owe: GH₵ 32,916.00

We owe suppliers: GH₵ 0.00

Alerts to look at: 402

Waiting for you — Nothing waiting for your decision

Purchase orders: 0

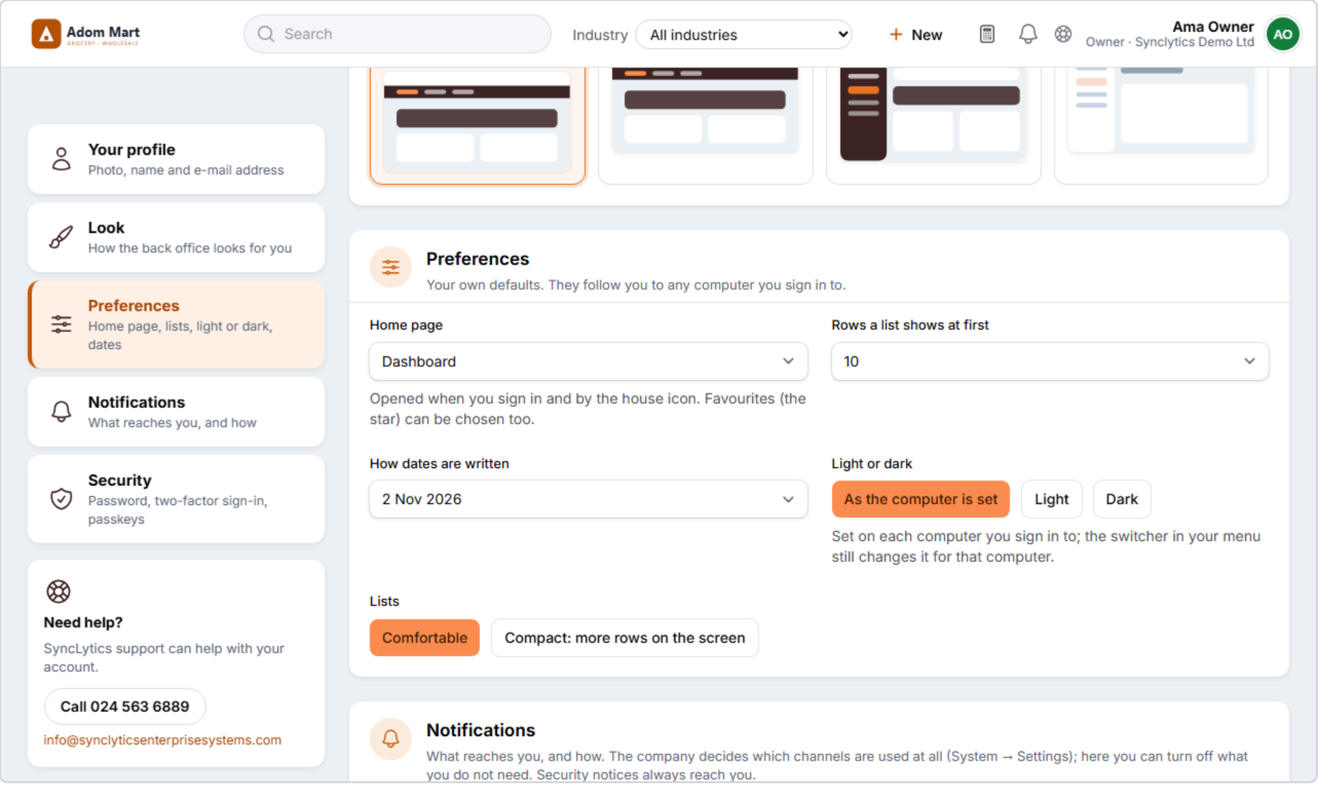
Stock adjustments: 0

Sales trend

Preferences (top right → **Profile** → *Preferences*) — they follow you to any computer:

- **Home page:** the dashboard, *Today*, or one of your favourite pages — opened when you sign in and by the house icon.
- **Rows a list shows at first:** 10, 25, 50 or 100.
- **How dates are written:** *2 Nov 2026*, *02/11/2026* or *2026-11-02*.
- **Light or dark**, and **Lists**: comfortable or compact (more rows on the screen).

Notifications (Profile → *Notifications*): for each kind of notification — waiting for your approval, buying, stock, prices to review, customers and deals (CRM), projects, service desk and maintenance, sensor alarms, cash, pharmacy, payments received, the daily summary, server and devices — tick how it reaches you: the bell in the back office and, where the company uses them, e-mail, SMS or WhatsApp. **Why:** people get what they act on, not everything. The company still decides which channels are used at all (System → Settings → *Payment alerts*, *Daily summary*), and security notices (SyncLytics support signing in) always reach the owner.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AO
Your profile
Photo, name and e-mail address
Look
How the back office looks for you
Preferences
Home page, lists, light or dark, dates
Notifications
What reaches you, and how
Security
Password, two-factor sign-in, passkeys

Need help?

SyncLytics support can help with your account.

Call 024 563 6889

info@synclyticsenterprisesystems.com

Preferences

Your own defaults. They follow you to any computer you sign in to.

Home page: Dashboard

Opened when you sign in and by the house icon. Favourites (the star) can be chosen too.

Rows a list shows at first: 10

How dates are written: 2 Nov 2026

Light or dark: **As the computer is set** · Light · Dark

Set on each computer you sign in to; the switcher in your menu still changes it for that computer.

Lists: **Comfortable** · Compact: more rows on the screen

Notifications

What reaches you, and how. The company decides which channels are used at all (System → Settings); here you can turn off what you do not need. Security notices always reach you.

Step 7 — Tax set-up (with your accountant)

Pricing & tax → **Tax groups** and **Tax rates.** The starter Ghana set-up (VAT 15 %, NHIL 2.5 %, GETFund 2.5 %, all on the price before tax) is marked **Not verified**.

1. Ask a qualified accountant to confirm the rates and the method against current GRA rules.
2. They (or you, on their advice) adjust the rates, then use **Record accountant check** on each group.

> **Warning:** Do not go live until the tax groups show the accountant's check. Wrong tax set-up means wrong receipts and wrong tax returns.

Step 8 — Price lists

Pricing & tax → **Price lists.** You start with **Retail** (default) and **Wholesale**, both including tax. Add lists for special customers if needed. Tick *Cashier can switch to it at the till* only for lists cashiers may use.

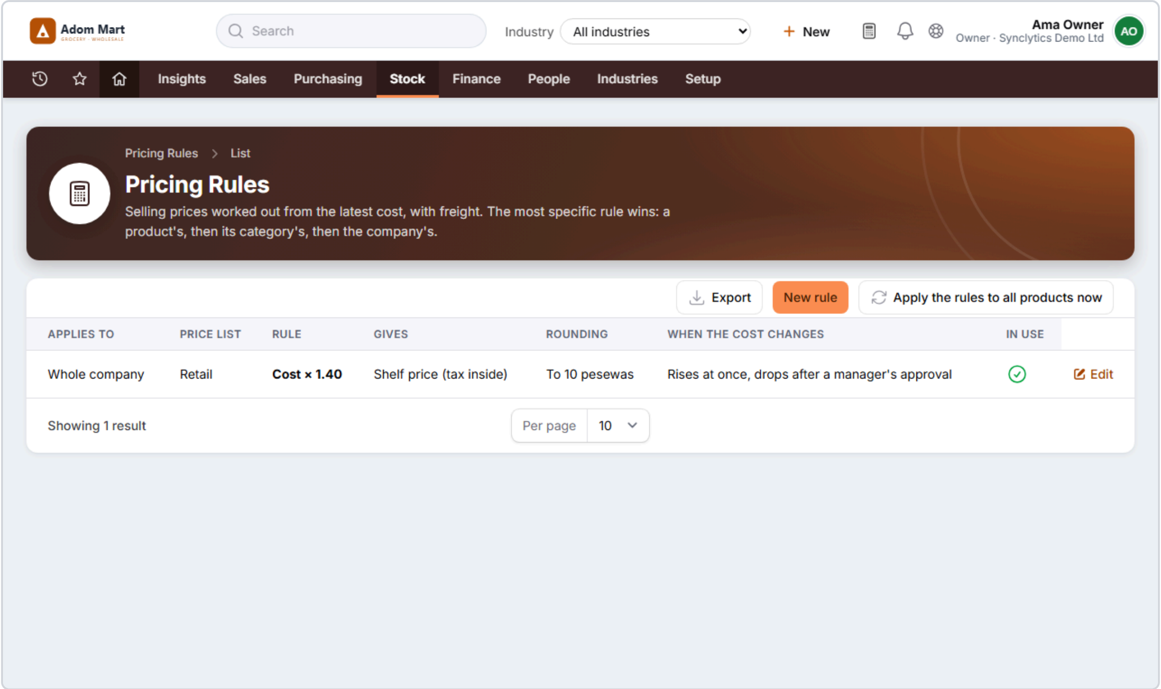
> **Note:** whether a list's prices include tax cannot change once it has prices.

Step 8b — Pricing rules (selling prices worked out for you)

If your selling prices come from what you pay (for example *cost × 1.40*), let the system work them out instead of using a calculator. **Why:** every delivery at a new cost would otherwise mean typing new prices by hand, and a missed one means selling below cost or overcharging.

1. **Pricing & tax** → **Pricing rules** → **New rule.**

2. **Applies to:** *Whole company* for your usual rule. Add more rules later for a **category** (e.g. a lower markup on baby milk) or a single **product**; the most specific rule wins (product, then its category or the one above it, then the company). A rule set to **Manual** keeps a product or category out of automatic pricing.
3. **Price list it sets:** usually *Retail*. A second rule can set *Wholesale* with a lower figure.
4. **How the price is worked out:**
    - **Multiply the cost** — e.g. 1.40 (a cost of 10.00 gives 14.00);
    - **Markup** — a percentage added to the cost (40 % gives the same as × 1.40);
    - **Margin** — the percentage of the selling price you keep (25 % margin on a cost of 10.00 gives 13.33);
    - **Cost + a fixed amount.** The form shows an example as you type.
5. **The result is:** *Shelf price* (VAT and levies inside — the price the customer pays, usual) or *Before tax* (taxes are added on top for taxable items).
6. **Rounding:** to 10 pesewas (usual), 50 pesewas or a whole cedi; *to the nearest* or *always up*. **Why:** prices like 13.37 are awkward at the till and on shelf labels.
7. **When a delivery changes the cost:**
    - *Rises at once, drops for approval* (recommended) — you never sell below the new cost, and a manager decides whether to lower prices;
    - *Always at once*; or *Always for approval*.


Adom Mart
Search
Industry
All industries
+ New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Pricing Rules > List
Export
New rule
Apply the rules to all products now
Showing 1 result
Per page 10

Pricing Rules

Selling prices worked out from the latest cost, with freight. The most specific rule wins: a product's, then its category's, then the company's.

APPLIES TO	PRICE LIST	RULE	GIVES	ROUNDING	WHEN THE COST CHANGES	IN USE	
Whole company	Retail	**Cost × 1.40**	Shelf price (tax inside)	To 10 pesewas	Rises at once, drops after a manager's approval	✓	Edit

8. **Save**, then **Apply the rules to all products now**. **Why:** the rule otherwise starts with the next delivery; this sets every product with a cost straight away.

What "cost" means: the cost of one piece on the **latest posted delivery**, with its share of freight, duty and clearing once a landed cost is posted. Before the first delivery, the average cost of the stock is used. Every unit for sale (piece, strip, carton…) is priced from it.

Prices to review (Pricing & tax, also on the dashboard under *Waiting for you* and in the bell): prices waiting for approval, with the cost, the current price and the new one. **Approve** them (several at once with the boxes) or **Keep current**.

On the product form, the *Prices* part says which rule sets the price and the latest cost, and warns in red when a price is below the cost.

> **Note:** only the standard price of each unit (from quantity 1, no dates) is set by a rule; bulk prices and dated promotions stay as you typed them. With branch servers, prices are set at head office: deliveries posted at a branch are priced within the hour after they reach head office.

> **Tip:** need people to set rules? They need a manager role with *Change prices, price lists and tax set-up* for the whole company.

Step 9 — Products

1. **Catalog** → **Categories** and **Brands**: create the groups you want for reports.
2. **Catalog** → **Units of measure**: check the units (PC, CTN, KG with 3 decimals…). A wrong code can be corrected (**Edit**); two codes for the same thing (PCS and PC) are joined with **Merge into…**, which moves every product over.
3. **Catalog** → **Products** → **New product** for each item:
    - **Name**, category, brand, **tax group**, **base unit** (the unit stock is counted in, usually PC). The **SKU** (product code) is given by the system when you save, so no two products can share one: an industry's products get its code and a number (`PHARMA-000123`), products shared by every industry a plain number (`100123`). It does not change, except when the product is moved to another industry (it then takes that industry's series).
    - **Barcodes** — scan them into the field.
    - **Other units** — e.g. a carton of 24 with its own barcode.
    - **Prices** — one row per list and unit. Add rows with *From qty* for bulk prices or with dates for promotions.
    - Industry details (medicine, serial numbers, sizes and colours, services) appear when the module is on — see Chapter 8.

> **Why:** products are never deleted (old receipts refer to them). Untick *For sale* instead.

> **Tip:** with many products, use **Import products (CSV)** instead of typing them one by one: download the template, fill it in Excel (name, category, barcodes, a price column per list; leave SKU empty for new products) and import it. Chapter 14.2 explains how.

Step 10 — Opening stock

Stock → **Stock on hand** → **Opening stock / adjustment**, choosing *Opening balance*, for each product and warehouse: the quantity you counted and the **cost per unit**.

> **Why:** the cost is used for the value of your stock and the cost of goods sold in the accounts. If you do not have the costs, leave them empty and use a **cost run** afterwards (14.3b), so nothing stays at zero.

> **Tip:** count on a day you are closed, enter the counts, then start selling. For many products use **System → Import data (CSV / Excel) → Opening stock** (warehouse, SKU, quantity, unit cost per row). Medicines and other batch-tracked products: **one row per batch**, with its `batch_no` and `expires_on`; a medicine with three batches on the shelf has three rows. Each batch can be loaded once; once a product has been sold, received or adjusted, further changes go through a stock adjustment.

Step 11 — Customers and suppliers

- **Sales → Customers**: customers who buy on credit or collect loyalty points. Credit limits and payment terms are set at head office. Customers who owe you money get an **opening balance**.
- **Purchasing → Suppliers**: suppliers you buy from, with an **opening balance** if you owe them.
- Both can be imported from CSV files (**Import customers / suppliers (CSV)** on their lists).

Step 12 — Finance and HR (if licensed)

- **Finance → Chart of accounts:** the Ghana SME template is already installed. Your accountant can rename and add accounts.
- **Opening balances** on the go-live date: customers and suppliers (step 11), stock (step 10), and everything else (bank, cash, MoMo, assets, loans, capital) under **Finance → Journals → Opening balances**. The account *Opening balances (to clear)* must end at zero.
- **Withholding tax on supplier payments** (System → Settings → *Withholding tax*): off until you switch it on. **Why:** Ghana's Income Tax Act makes a business keep back part of what it pays suppliers (by default 3 % for goods, 7.5 % for services, 5 % for works, on the amount before VAT, nothing below GHS 2,000) and pay it to GRA by the 15th of the next month. Have your accountant confirm the rates and the threshold there. Then:
    1. On each supplier, set **They supply** to goods, services or works, and enter their **TIN**.
    2. A supplier with a GRA exemption adds it as a document (*Withholding tax exemption certificate*, with its expiry); nothing is kept back while it is in date.
    3. **Pay a supplier:** type the amount to settle; the form shows the tax kept back and what to pay them. The bill is settled in full; the bank goes down by what you paid; the tax waits in *Withholding tax payable*. Press **WHT certificate** on the payment and give it to the supplier.
    4. **Finance → Withholding tax:** last month's list with every supplier's TIN, the base, the rate and the tax, and the total to pay GRA. **Download for GRA (CSV)** for the return. Pay GRA from the bank against the *Withholding tax payable* account.
- **HR & payroll → Employees:** add staff with their pay and link each to their sign-in. Check **Payroll profile** (PAYE and SSNIT rules) with your accountant.

Step 13 — Set up the devices

Register and enrol the tills, handhelds and kitchen screens: **Chapter 9**.

Step 13b — Each teller's own vault (recommended with several tellers)

> **Why:** with one shared drawer, a teller who signs in on a till left open by a colleague can sell under the colleague's name, and nobody can say whose cash is short. With her own vault, each teller only ever sees and opens her own.

1. **Switch it on: System → Settings → Tills and cash → Each teller sells from their own vault**:
    - *a manager recounts each closed vault* (recommended): after the teller closes and counts, a manager counts the tray again without seeing the teller's figure; the manager's count is the one booked.
    - *the teller's own count stands*: no second count.
2. **How the cash is kept** (same page): *own removable cash tray* (default: each teller has a tray they put in the drawer, on any till of the branch), *own till and drawer each*, or *one shared drawer* (the vault is then a record of whose cash is whose).
3. **Give each teller a vault: Cash → Shifts and vaults → Add a vault**: a name (*Vault 1*), the teller who holds it, the till (only for *own till*), and the standard float. One vault per person.
4. **On the till:** at *Open shift* the teller sees only their own vault, confirms they counted its float, and sells. Their name and vault show in their own colour at the top of the till.
5. **Someone away?** A manager lends the absent teller's vault to the person covering, for today only: on the till (*Covering for someone?*, the manager's username and PIN) or in **Cash → Shifts and vaults → Lend for today**. **Why:** the cover is recorded with the manager's name and reason, and ends at midnight.
6. **Closing:** the teller counts blind and is signed out. With *a manager recounts*, the shift shows *Waiting for the manager's count* in **Sales → Shifts**; the manager presses **Count the vault** and counts the tray (one recount if the two counts differ). Only then is the over/short booked.

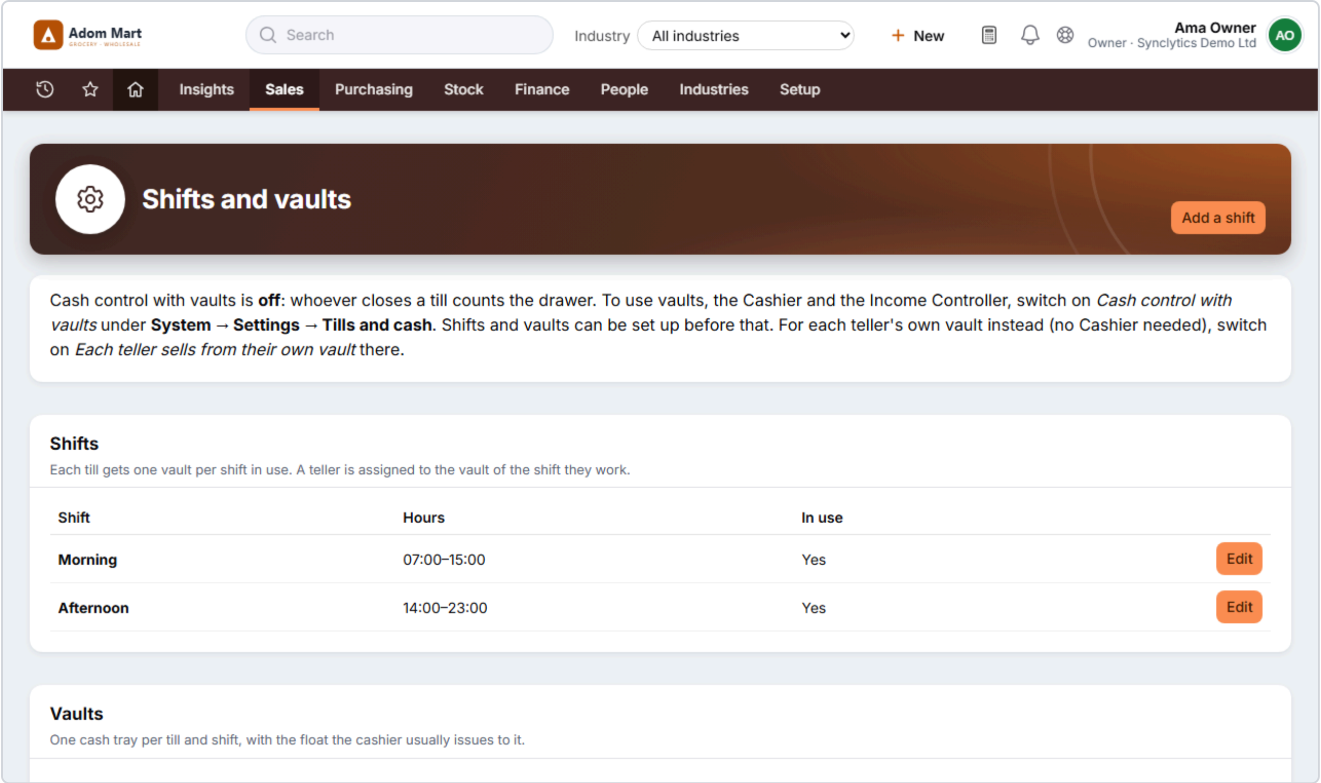
Staff cards (optional, faster sign-in). System → Users → (person) → Staff card prints a card with a QR code. At the till the teller scans it instead of typing her username, then types her PIN. **Why:** no shared or mistyped usernames; a lost card alone cannot sign anyone in (the PIN is still needed); **Card lost** stops it at once.

> **Check:** sign in as one teller: only her vault is offered; open, sell, close; as a manager, **Count the vault** in Sales → Shifts.

Step 14 — Cash control: vaults, Cashier and Income Controller (optional)

A shop with several tellers can check its cash at every step, as banks do.

- **Teller:** sells at a till and counts the tray at the end of the shift.
- **Cashier:** gives each teller a cash tray with a float (a **vault**), receives it back and counts it again.
- **Income Controller** (accounts): receives the day from the Cashier, counts it again, decides any difference and records the bank deposit.



Shifts and vaults

Cash control with vaults is **off**: whoever closes a till counts the drawer. To use vaults, the Cashier and the Income Controller, switch on *Cash control with vaults* under **System → Settings → Tills and cash**. Shifts and vaults can be set up before that. For each teller's own vault instead (no Cashier needed), switch on *Each teller sells from their own vault* there.

Shifts

Each till gets one vault per shift in use. A teller is assigned to the vault of the shift they work.

Shift	Hours	In use	
Morning	07:00–15:00	Yes	Edit
Afternoon	14:00–23:00	Yes	Edit

Vaults

One cash tray per till and shift, with the float the cashier usually issues to it.

Nobody sees the figure they are checking against before they have counted: these are **blind counts**.

> **Why:** a difference is found at the step where it happened and is tied to the person responsible. Cash never moves without a second person checking it.

1. **Switch it on: System → Settings → Tills and cash → Cash control with vaults**.
2. **Shifts: Cash → Shifts and vaults** lists the shifts, *Morning* (07:00–14:00) and *Afternoon* (14:00–21:00). Change the times, or **Add a shift** (e.g. *Evening*). Each till gets one vault per shift, e.g. *T01 · Morning*.
3. **Standard floats:** on the same page, **Edit** each vault and enter the float the Cashier usually puts in it. It is suggested every day, and the Cashier can change it.
4. **People** (**People & access → Users**, Step 5):
    - give each till person the **Teller** role, at their till or branch;
    - give one or more people the **Cashier** role, at their branch;
    - give the **Income Controller** role (security level 4) to someone in accounts, usually company-wide.

    The Cashier and the Income Controller do not sell, so they must not also have the Teller role.
5. **Every day:**
    1. The Cashier opens **Cash → Cash desk** and clicks **Issue float** on each vault, choosing the teller.
    2. The teller signs in at that till, ticks *I have counted the float* and opens the shift.

3. At the end of the shift the teller counts the tray note by note and closes the till, then hands the tray to the Cashier.
4. The Cashier clicks **Receive and count** and counts again. If the two counts differ, they are asked to count once more; the second count stands.
5. At the end of the day the Cashier clicks **Hand over to the income controller**.
6. The Income Controller opens **Cash → Income control**:
    - **Receive and count** the bundle;
    - **Decide** each difference: *cash over/short*, or *charged to the staff member*, recovered from pay through HR & payroll;
    - **Record deposit** with the bank slip number.

> **Check:** on a test day, issue a float, sell, close, receive, hand over and receive again. *Income control* then shows the day's reconciliation (system, teller, cashier, income controller), and *Finance → Trial balance* shows *Cash on hand – tills* and *Cash with cashiers* back at zero.

The full guides for each person are in the training material (Chapter 17.7).

The till drawer (with or without vaults). In **System → Settings → Tills and cash**:

- **Expenses paid from the till: a supervisor approves above** — default 10000 pesewas (GHS 100). **Why:** small shop expenses (water, a taxi) can be paid on the spot, while larger ones need a second person.
- **Cash limit in a till drawer** — off by default; for example 200000 (GHS 2,000). **Why:** above it the till asks for a *cash drop* to the safe and supervisors are told, so little cash sits in the drawer.
- **One shift per person on a till** — on by default. **Why:** each person's sales and cash stay apart. When someone signs in on a till where another person's shift is still open, the till asks them to close it (count the drawer) and open their own; sales into someone else's shift are refused (and flagged in **Sales → Review** when made offline). **Sales → Till shifts** shows *Sold by …* under *Opened by* when more than one person sold in a shift. Switch it off where several people share one till and one drawer (e.g. waiters).
- **Opening the drawer without a sale needs a manager** — on by default. **Why:** an open drawer with no sale is the easiest way to take cash unseen, so a manager types their username and PIN each time. Who counts as a manager is the permission *Approve opening a till drawer without a sale* (Branch Manager, General Manager and the owner unless you change the roles in **System → Roles**); supervisors cannot approve it. If the till is offline, the PIN is checked on the till and anything the person was not allowed to approve appears in **Sales → Review**.

Each pay-out is also recorded under **Expenses → Expense claims** as paid from the till, against an operating-expense account (the categories the till offers). **Check:** on the test day, pay out a small amount for transport, do a cash drop with a supervisor's PIN (a slip prints each time; check it on the receipt printer), and open the drawer once without a sale with a manager's PIN (a supervisor's PIN is refused). Then **Sales → Till drawer** lists all three, and the cash still balances at the close.

Step 15 — Test before going live

Do a full test day with made-up sales:

1. Open a shift on a till, sell with cash, MoMo and card, print receipts, do a return, close the shift (count the cash, and type the MoMo, card, voucher and coupon totals: each should show as balanced).
2. Unplug the till's network cable (or switch off Wi-Fi), sell two items, reconnect, and watch the sales arrive (*Sales → Till sales*).
3. Check *Sales → Review queue*, *Stock → Stock on hand* and, if licensed, *Finance → Reports → Trial balance*.
4. Take a backup and **restore it on another PC** (Chapter 12).

> **Check:** when all four work, remove the test data the easy way — make a fresh installation for go-live, or ask SyncLytics for help — and enter your real opening figures.

8. Installing and setting up modules

Modules add the features of a type of business. They are **already part of the software** — "installing" a module means three things:

1. your **licence** includes it (Chapter 6);
2. it is **switched on for each branch** that needs it;
3. its **set-up checklist** below is done.

A branch runs **one industry module** (Pharmacy, Grocery, Restaurant, ...), plus any company-wide features and add-ons such as Sales & Distribution. A supermarket with a pharmacy counter is set up as two branches at the same address, so each keeps its own stock, staff and figures.

8.0 Industries

When a company runs more than one kind of business, each kind is an **industry**: for example *Pharmacy* with the branches Pharmacy Osu and Pharmacy Tema, and *Grocery* with Grocery Spintex.

> **Why:** each industry has its own staff, products and figures. Separating them keeps a pharmacist's screens free of grocery items, and gives the owners each industry's results on its own as well as the whole company's.

- **An industry appears by itself** when its module is switched on for a branch (8.1). The branch then belongs to it. *Company → Industries* lists the industries. You rarely need to add one by hand: switching on a module for a branch creates it. Its **code** is given automatically from the name and starts its product codes (e.g. GROCER-000001).
- **Head office or a central warehouse** can belong to no industry (*company-level*). Only people with company-wide access see it.
- **The industry switcher** at the top of the back office (*All industries*, *Pharmacy*, *Grocery*, ...) is shown to people who can see more than one industry.
  - With an industry chosen, every screen shows only that industry: branches, stock, prices, purchasing, sales, reports, and its own dashboard. The pharmacy dashboard shows prescriptions and expiring batches.
  - With *All industries*, the dashboard shows each industry's sales side by side.
- **Each industry has its own catalogue.**
  - When you create a product or price list, choose its industry; the industry chosen at the top is filled in for you.
  - That industry's tills sell only its products.
  - Products created before industries are *shared* (sold everywhere) until you give them an industry (Catalog → Products → edit → Industry).
- **Every product belongs to an industry.** The Industry field is required, and shows only the industries you work in. Why: an item left in "All industries" shows on every shop's tills.
- **The same item in two shops** (Lactogen at the pharmacy and at the supermarket): each industry has its own product, with its own name, price and stock, and both keep the barcode on the pack. A till, a handheld and a stock file read the barcode in their own shop's industry. Within one industry a

barcode is still on one product only. **Why:** each shop prices and names its goods its own way, while the scanner still knows which product is meant. At head office (no industry) a barcode on two shops' products is refused: give the SKU instead.
- **Shared items** are for things every shop really sells, such as carrier bags. Only someone with company-wide access can choose *Shared by all industries*. To stop them altogether, use Settings → Stock → *Products shared by every industry*.
- **Tidying shared items:** Catalog → Shared products lists every product in no industry, with where it has stock or sales.
  - Tick products and press **Move to suggested industry**. Products with no suggestion stay where they are.
  - Or press **Move all to one industry**. This suits a company that traded in one industry before adding another. Stock and prices do not change.
- **The switcher** shows only to people who cover several industries or the whole company. Everyone else is always in their own industry and does not see the others' products.
- **Staff:**
  - give a person a role **at their branch** (People & access → Users → Grant access → Applies to: *Branch*). A pharmacist at Pharmacy Osu then sees that pharmacy only;
  - an **industry manager** gets the role with *Applies to: Industry*, and sees every branch of that industry, and nothing of the others;
  - **company-wide access** shows everything, so it is limited to managers and above. Settings → Security can limit it to the owner and SyncLytics support;
  - **People & access** → **Company-wide access** lists anyone who has it but should not, because the system cannot know where they work. Press **Give a branch** for each person. Until then they keep seeing everything, so do this soon after updating.

Only what the business runs is offered. Roles (such as *Waiter* or *Tailor*), settings sections, message templates and device types that belong to an industry appear only where that industry is switched on. A pharmacy, for example, is never offered a kitchen display.

Menus follow the choice. With an industry chosen, only its own module menus show (for example *Pharmacy* but not *Restaurant*). With *All industries*, every installed industry's menus show. Grocery's menu has **Markdowns** (near-expiry stickers and expired write-offs at the grocery stores) and **Scale items** (the weighed products to load into the scales).

> **Check:** choose *Pharmacy* at the top. *Stock → Stock on hand* shows only the pharmacy branches, and *Catalog → Products* shows only pharmacy (and shared) products. Choose *All industries* to go back.

Stock between branches and industries

- **Sister branches** (Pharmacy Osu and Pharmacy Tema), or head office and a branch: use a normal transfer (*Stock → Transfers*), or let the branch that needs stock ask for it (*Stock → Stock requests → Ask for stock*). The branch asked clicks **Send** (entering 0 for anything it cannot spare), or **Reject** with a reason. Sending creates the transfer; the asking branch receives it as usual.

- **Between two industries** (a grocery branch asking a pharmacy for bottled water) only a stock request works, and it needs two approvals before anything moves:

1. a manager of the asking branch, other than the person who asked;
2. a manager of the supplying branch.

The supplier then clicks **Send** and, for each item, chooses **its own product** that matches (the two catalogues are separate). The goods leave at the supplier's cost and arrive as the asking branch's product, with the same batch numbers and expiry dates.

> **Why:** each industry keeps its own stock and figures. Two approvals mean neither side's stock leaves or arrives without a manager knowing, and moving at cost means neither side makes a profit on the other.

Who: storekeepers, supervisors, pharmacists and managers can ask (*Ask another branch for stock*). Branch, warehouse and general managers can approve (*Approve stock requests between industries*). Sending needs the transfer permission at the supplying branch.

Accounts by industry

Two ways, chosen per installation under **System** → **Settings** → **Finance** → **Separate books for each industry**:

Setting	What it means	Suits
On (the default)	Each industry is its own company with its own books, TIN and VAT return. See *Separate books* below.	Groups whose industries are registered separately, or will be.
Off	**One set of books** (one TIN, one VAT return); each industry is a profit centre.	A single registered company running several kinds of shop.

Change it only at the start of a financial year, because switching moves the branches' future postings to other books. Companies already running when this option arrived were left on *Off*.

With either choice, everything a branch posts is labelled with its industry:

- **Finance** → **Reports** → **Profit by industry** shows each industry's income, costs and profit side by side. A *Company* column holds head office and shared costs.
- The **Industry** filter on the other reports (trial balance, profit and loss, balance sheet, account ledger, budget) shows one industry, or *Company* for head office lines. It starts on the industry chosen at the top of the screen.
- **Stock moved between industries** leaves one industry's stock and joins the other's at cost, so neither makes a profit on it.

> **Why:** the owners see what each line of business earns, without the cost and paperwork of separate companies.

Sharing head office costs (optional). By default head office costs (rent, admin salaries, the accountant) stay in the *Company* column.

1. To share them, switch on **System** → **Settings** → **Finance** → **Share head office costs between industries by sales**.

2. At month end open **Profit by industry** with the *To* date in that month, and click **Share head office costs**. The window shows each industry's share of that month's sales.
3. Click **Post the journal**.

One journal moves each cost from *Company* to the industries in proportion to their sales, to the pesewa; the company's total profit does not change. A month can be shared only once. To redo it, reverse that journal (*Finance → Journals*) and share again.

> **Note:** an industry's balance sheet on its own does not balance, because bank accounts, money owed and stock in transit sit partly at company level. Use *All industries* for the company's balance sheet.

Separate books (the default)

When an industry is first switched on, the system creates its company under **Company → Legal entities**, named after it (for example *Demo Ltd – Pharmacy*), and a **head office company** if there is none.

1. **Company → Legal entities:** open each company and enter its **registered name, registration number, TIN, VAT number and address**.
    - Its branches' receipts and invoices print that name and TIN.
    - Their sales go to GRA E-VAT under that TIN.
2. **Head office company:** head office lines, and branches without an industry, are booked there. Only one company can be head office at a time; ticking *Head office company* on another moves it.
3. **Two industries under one registered company:** choose the same company for both under **Company → Industries → Books**. To give an industry no company of its own, choose the head office company.
4. Changing an industry's company is a **start-of-year** step. Balances from before (stock, money owed) stay in the old books; ask your accountant to move them with a journal on the first day.

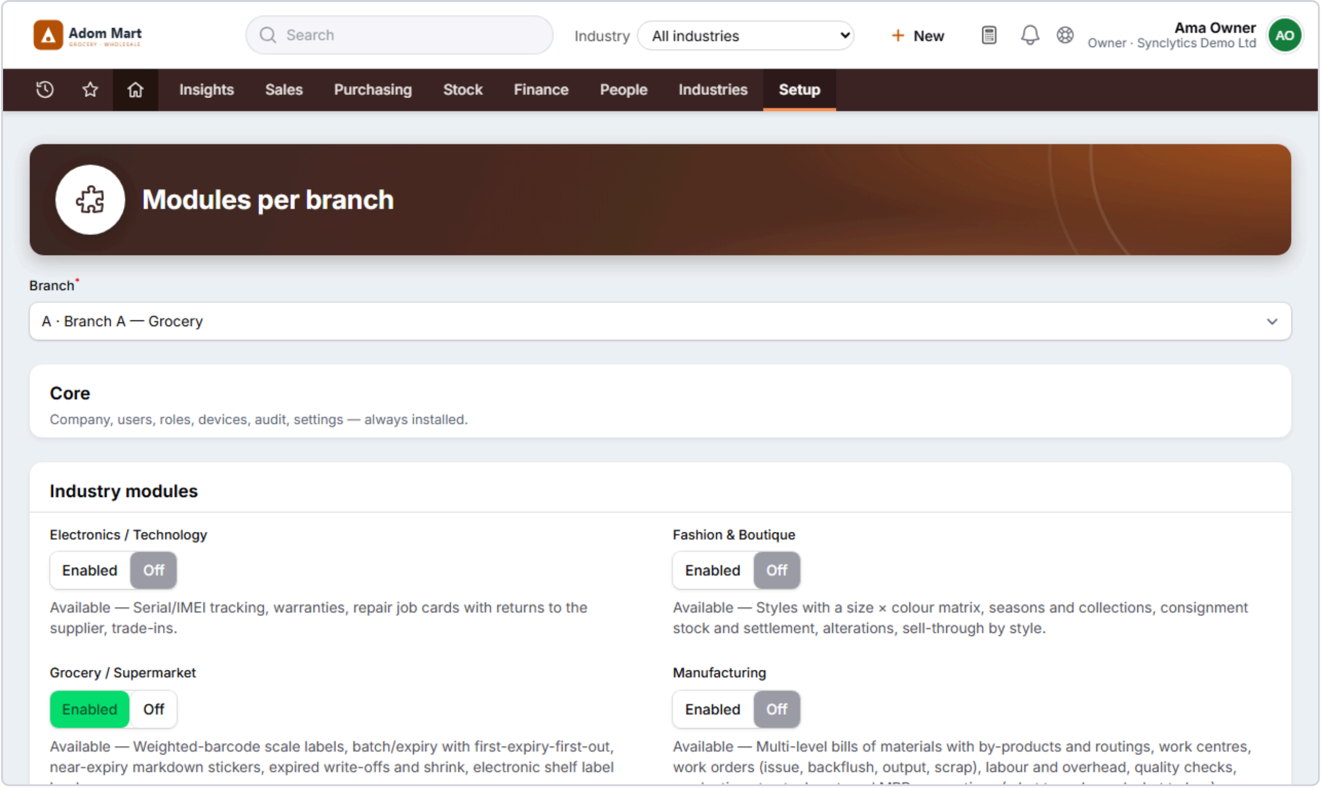
What changes:

- **Every entry balances inside each company.** When one company pays for another, for example head office paying the pharmacy's electricity, the system records what the pharmacy now owes head office on **Due from / to group companies**, in both sets of books. Stock sent between the companies at cost is recorded the same way.
- **Finance → Reports → Books:** choose a company for its own trial balance, profit and loss, balance sheet and ledger. *Whole group (consolidated)* adds them together, and what they owe each other cancels out.
- **VAT return:** each company files its own. Choose it under *Books*.
- **Settling up:** when one company pays the other, enter a journal with each company's bank line (*Company* on lines without a branch). The amount owed clears by itself.
- **Ledger check** (*Finance → Reports → Ledger check*) also proves that each entry balances in every company's books, and that both sides agree on what they owe each other.

> **Why:** a separately registered company must keep its own books and file its own returns. Recording what they owe each other automatically means no entry is ever missed or one-sided.

8.1 Switching a module on or off

1. **System → Modules.**
2. Choose the **Branch**, then for each module choose **Enabled** (or **Off**; a planned module offers **Reserved**).
3. Type the **Reason for change** and click **Save modules**.



Modules per branch

Branch*: A · Branch A — Grocery

Core

Company, users, roles, devices, audit, settings — always installed.

Industry modules

Electronics / Technology: Off

Available — Serial/IMEI tracking, warranties, repair job cards with returns to the supplier, trade-ins.

Fashion & Boutique: Off

Available — Styles with a size × colour matrix, seasons and collections, consignment stock and settlement, alterations, sell-through by style.

Grocery / Supermarket: Enabled

Available — Weighted-barcode scale labels, batch/expiry with first-expiry-first-out, near-expiry markdown stickers, expired write-offs and shrink, electronic shelf label

Manufacturing: Off

Available — Multi-level bills of materials with by-products and routings, work centres, work orders (issue, backflush, output, scrap), labour and overhead, quality checks,

> **Why:** modules are per branch so that, for example, only the pharmacy branch shows medicine screens. Every change is recorded in the audit log.

Status shown	Meaning
Available	Released: can be switched on now.
Planned · Phase N	Not built yet. You may **reserve** it for a branch; it switches on by itself when that phase is released. Nothing is shown as working before it exists.

Only modules you have bought are shown. The set-up wizard, this screen and the industry choices list only the modules in your licence, and the industry switcher and dashboards list only industries switched on at a branch.

> **Why:** staff never see, or try to set up, a business type the company has not bought. To add one, ask SyncLytics for a new licence key (Chapter 6); the module then appears here, ready to switch on. If a licence renewal drops a module, its screens are hidden but its records are kept, and they come back if it is added again.

> **Note:** switching a module **off** hides it and stops new work. It **never deletes** data; switching it on again brings everything back.

Who can do it: the owner and the IT Admin (permission *Manage modules*).

> **Check:** after switching a module on, the module's menu appears in the back office (you may need to refresh the page), and its buttons appear on that branch's tills after their next catalog refresh (within 60 seconds, or press *Refresh* on the till).

8.2 Available modules and their checklists

Online Storefront (built-in online shop)

1. Switch it on for the branch that will fulfil online orders.
2. **Online** → **Channels** → **New channel**, type *Built-in online shop*: choose the fulfilling branch, the stock warehouse, the price list and the shop's web address (`/shop/<address>`).
3. Set **safety stock** on products that sell fast in the shop too.
4. Click **Activate**.

> **Why:** safety stock is held back from the shop so walk-in and offline sales do not oversell online.

> **Check:** open `https://<your-domain>/shop/<address>` — your products appear with prices and availability.

Grocery / Supermarket

1. Products with use-by dates: tick **Batches and expiry dates** (works in any shop, but essential for grocery).
2. **Weighed items:** give each its **scale code (PLU)** and sell it in **KG** (3 decimals). Set the label scale to print EAN-13 labels: `21` · item code (5 digits) · weight in grams (5 digits) · check digit.
3. **System** → **Settings** → **Grocery:** check the scale label prefixes. The markdown rules (default 30 % off at 7 days left, 50 % at 3 days) are under **System** → **Settings** → **Stock**.
4. Electronic shelf labels (optional): enter their gateway address in System → Settings → Grocery.
5. **Grocery** → **Scale items** lists every weighed product with its scale code, unit and retail price. **Export** it to load the scales, and again after price changes.
6. **Grocery** → **Markdowns** (daily): apply the suggested markdowns, print the stickers, and write off what has expired.

> **Why:** the till reads the weight and item from the scale label, and sells the batch expiring first, so old stock does not go to waste.

> **Check:** print a test label on the scale and scan it at a till: the item appears with its weight and price.

Pharmacy

1. On each medicine (Catalog → Products → **Medicine**): **schedule** (OTC, Prescription only, Controlled), generic name, strength, form, FDA registration number, **active ingredients**. Switch on **Batches and expiry dates** (required for medicines).

2. **Pharmacy → Drug interactions → Add the starter list:** about 55 well-known interactions between common medicines (warfarin with painkillers and some antibiotics, methotrexate with co-trimoxazole, sildenafil with nitrates, statins with some antibiotics, antacids and iron with ciprofloxacin or doxycycline, the pill with rifampicin…). Pairs already in your table are left alone. The superintendent pharmacist then reads the list, changes or deletes rows, and adds others. The top of the page says how many medicines have **neither a generic name nor active ingredients**: warnings cannot work for those, so fill them in on the products. **Why:** the till warns only about pairs in this table, and it recognises a medicine by its generic name or active ingredients. The starter list is a beginning, not a complete reference.

3. **System → Settings → Pharmacy:** the Pharmacy Council premises licence number and the superintendent pharmacist, printed on the controlled-drug register.

4. Give pharmacists the **Pharmacist** role (and technicians **Pharmacy Technician**) at the branch.

5. **Dosage:** on each medicine, **Usual dosage (directions)** — e.g. *1 TABLET 3 TIMES DAILY AFTER MEALS FOR 5 DAYS* (or the `dosage` column of the products file). At the till the directions appear under the medicine on the sale and can be changed for that patient. They print on the receipt and on a **dosage label** for each medicine: the pharmacy's name and phone, the patient (from *Details*), the medicine and quantity, the directions in large letters, the date, the receipt number and *Keep out of reach of children*. The desktop till prints the labels by itself on the receipt printer (each one cut); in a browser, **Print dosage labels** on the receipt screen. **Why:** patients take medicines correctly when the directions are on the pack, not only told at the counter.

6. **System → Settings → Pharmacy → Selling prescription-only / controlled medicines:** by default a pharmacist approves every such sale with the prescription. If tellers sell them in your pharmacy, choose *Prescription details only*, *Patient only* or *Sell freely* (a branch can differ: Company → Branches → the branch → *Pharmacy rules for this branch*), and how they are checked afterwards (a pharmacist signs them off on Pharmacy → Sales to sign off, or a daily summary). A quantity limit, major warnings and *Controlled drugs always need a pharmacist* keep the riskiest sales with a pharmacist.

7. **Refill reminders** (medicines taken every day — blood pressure, diabetes, asthma…):
    - On each such medicine, **Used a day (refill reminders)**: how much one patient takes a day in the base unit — *1* for one tablet a day, *2* for twice a day (or the `daily_use` column of the products file).
    - Sell it to a **named customer** (choose the customer on the till) who agreed to **WhatsApp or SMS** messages (the customer's consent ticks). The quantity sold tells how long it lasts: 30 tablets at 1 a day last 30 days.
    - **System → Settings → Refill reminders:** switch on; *Days before it runs out* (3) and *Send at* (09:00). A few days before the medicine runs out the customer gets one message listing every medicine due, with the branch's phone number. Buying again replaces the old reminder.

  - **Pharmacy → Refill reminders** lists what is waiting and what was sent; **Send now** or **Cancel** (patient changed medicine, asked not to be reminded). *No phone or consent* shows customers to ask next time.
  - It needs an SMS or WhatsApp provider (Chapter 16). The wording is under Messaging → Templates (*Refill reminder*); WhatsApp needs the approved template `refill_reminder`.

> **Why:** patients on long-term medicine who run out often stop taking it, or buy elsewhere. A reminder before it runs out helps them stay on treatment and brings them back to your pharmacy. Only customers who agreed are texted, as the Data Protection Act requires.

8. **NHIS at the till** (NHIS-accredited pharmacies):

   - On each medicine on the NHIS Medicines List, **NHIS price (per base unit)**: what the NHIS pays for one tablet, capsule or bottle (or the `nhis_price` column of the products file, Chapter 14). Empty means the NHIS does not cover it.
   - The patient's **NHIS membership no.** and **NHIS card expires** on their customer record (Patient details).
   - **System → Settings → Pharmacy → Take NHIS at the till:** switch on (a branch can differ).
   - At the till, choose the patient and add the medicines. In **Payment**, **NHIS pays ...** shows what the NHIS covers: the NHIS price of each listed medicine, never more than the line. Press it, and the patient pays the rest as usual. Without a valid card (missing or expired) the button does not appear. Keep the NHIS prescription form and the patient's signature as the NHIA requires.
   - **Pharmacy → NHIS claims → Prepare a claim** (e.g. monthly): the branch, and *Sales up to and including* (the last day of the month). It gathers every NHIS part not claimed yet; returns come off. **Download the list (CSV)** and send it to the NHIA with your claim forms, then **Mark as sent** with the NHIA's reference.
   - When the NHIA pays: **Record what the NHIA paid** (amount, date, bank or cheque). Whatever it did not pay is written off as *NHIS claims not paid*. A claim not sent yet can be **withdrawn**; its sales go back for the next claim.
   - Who: the owner, managers, the accountant and the superintendent pharmacist (*NHIS claims* permission).

> **Why:** the NHIS part of a sale is money the NHIA owes you, not cash in the drawer. The books show it as *Due from NHIS* until the NHIA pays, so you can see how much you are owed and how much was cut at vetting.

9. **Near-expiry markdowns: Pharmacy → Near-expiry markdowns** (daily or weekly) lists batches close to expiry with a suggested discount (rules under System → Settings → Stock). **Apply**, print the stickers, stick them on the packs; the till sells the sticker at its price from that batch. Expired stock is written off from the same page. Pharmacists can do this; for staff whose role was set up before this release, the owner adds *Mark down near-expiry batches* to their role (People & access → Roles). **Why:** a medicine sold at a discount before it expires is better than a medicine thrown away after.

> **Why:** the Pharmacy Council expects a pharmacist to approve prescription-only and controlled medicines, and the prescription is recorded; controlled drugs go into the Controlled Drug Register. Where a pharmacy works otherwise, the system still records who sold what so a pharmacist can check it.

> **Check:** sell a POM medicine on a till: with the default setting the till asks for the patient, the prescription and the pharmacist's PIN. With NHIS on, sell a listed medicine to a patient with a valid card: **NHIS pays ...** appears in Payment.

Restaurant / Café / Bar

1. **Restaurant** → **Kitchen stations:** e.g. `KITCHEN`, `BAR`. Tick *Print tickets* for a station that has a printer instead of a screen.
2. **Restaurant** → **Floor plan:** each table with its number, area, seats and position.
3. **Restaurant** → **Modifiers:** groups of choices (e.g. *Spice: Mild / Hot*). A choice can be charged as a product.
4. On each **dish** (Catalog → Products → **Restaurant**): the station, the choices offered, and the **recipe** (ingredients per portion). Switch **Track stock** off for the dish itself; the ingredients are stocked products.
5. **System** → **Settings** → **Restaurant:** service charge rate and its product (create a *Service charge* product with *Track stock* off, and copy the SKU the system gives it into this setting), tips on/off, booking reminders and the SMS gateway address (only with your own SMS provider's details).
6. Give staff the **Waiter / Bartender** or **Kitchen Staff** role.
7. Set up the **kitchen screens** (Chapter 9).

> **Why:** bills live on the server so every till, waiter tablet and kitchen screen shares the same tables. For a restaurant with unreliable internet, use a **branch server** (Chapter 10) so this keeps working during outages.

> **Check:** open a table on a till, send a dish, and see it on the kitchen screen.

Electronics / Technology

1. On serial-numbered products (Catalog → Products → **Serial numbers and warranty**): tick **Serial / IMEI numbers**, enter the **warranty** in months.
2. Stock already on hand: **Electronics** → **Serial numbers** → **Register numbers for stock on hand**.
3. Create a **labour** product (no stock) for repair charges.
4. Give technicians the **Repair Technician** role.

> **Why:** every unit is then traced from receipt to sale, warranties are checked by number, and stolen or wrong units cannot be sold.

> **Check:** add a serial-numbered product at a till: it asks for the number and refuses one that is not in stock there.

Fashion & Boutique

1. **Fashion** → **Styles** → **New style:** code, name, tax, season, collection, sizes, colours and price. Then **Create sizes and colours** to make one product per size and colour, each with SKU and barcode.
2. **Consignment** (goods owned by a supplier): on the product, choose the consignor and their share.
3. **Alterations:** create an *Alteration* product with *Track stock* off, and copy the SKU the system gives it into System → Settings → Fashion.
4. Give tailors the **Tailor** role.

> **Check:** pick any variant of a style at the till: the size × colour grid opens with the stock at that till.

Sales & Distribution (vans, routes, reps)

1. **Distribution** → **Vans** → **New van:** branch, name (the van code `V01`, `V02`... is given automatically), registration, rep and the warehouse it is loaded from. The van is created as a warehouse **with its own till**.
2. Register the rep's phone as a **POS till** device on that van's till and enrol it (Chapter 9.4/9.5). It works **fully offline**.
3. Give the rep the **Van Sales Rep** role at the branch and a PIN; in **HR & payroll** → **Employees** set their **Commission %** if they earn commission.
4. **Routes:** Distribution → Routes → New route — the van, rep, weekdays and customers in visiting order.
5. **Targets:** Distribution → Rep targets — monthly sales target per rep.
6. **Returnable containers** (crates, bottles, gas cylinders): mark the container products as *returnable containers* (their price is the deposit) and choose the container on each product sold in them (Product → Returnable containers).
7. Daily: **Load van** in the morning; in the evening **Start end-of-day settlement**, count, submit, and a supervisor approves (Chapter 14 and `docs/distribution.md`).

> **Why:** a van as a warehouse with its own till means every sale, receipt, shift and cash count works exactly as in a shop — offline too — and the daily settlement catches shortages the same day.

> **Check:** load a van, sell from the phone, then settle: the van's stock in *Stock on hand* matches what is left on it.

Manufacturing (factories, bakeries, workshops, food processing)

1. **Switch it on** for the branch that produces (System → Modules → Manufacturing). *Why:* the screens and work orders only work where the module is on.
2. **Work centres:** Manufacturing → Work centres → New — each machine or bench (e.g. `MIX` Mixer, `OVEN` Oven), its **cost per hour** (wages of the people there plus running costs such as power) and the hours a day it runs. *Why:* every minute worked there is charged to the product, so its cost is real; the hours show how full each machine is.
3. **Bills of materials:** Manufacturing → Bills of materials → New — the product made, how many one batch makes, the **materials** per batch (with any expected loss %), **by-products** (e.g. bran from

milling), and the **operations** in order with their work centre and minutes. Then **Put in use**. *Why:* a BOM in use is locked so open orders and history stay true; to change it, start a **New revision**.
    - A material that has its own BOM is a **sub-assembly** (e.g. dough for bread): it is made on its own work order.
    - **Backflush** (default) means the material or time is charged automatically when output is reported; switch it off for items you issue by hand or time you record.
    - Tick **Quality check before stock** if output must be inspected first.
4. **Roles:** *Production Manager* (sets up, runs orders, checks quality), *Machine Operator* (runs orders), *Quality Inspector* (passes or fails output). *Why:* the person who makes the goods is not the one who passes them.
5. **Daily:** Manufacturing → **Production planning (MRP)** shows what to make and what to buy; **Raise work order**, then on the work order **Release** → **Report output** → **(Quality check)** → **Finish**. Print the **Job card** for the shop floor.

> **Why:** production at actual cost (the owner's choice) means each product enters stock at what its materials, labour and machines really cost, and the accounts show work in progress until it is finished.

> **Check:** make a small order of a product, then look at *Stock on hand*: the materials went down, the product went up, and the work order's *Actual cost* section shows the cost per unit.

Services & Packages (salons, spas, gyms, clinics, consultancies)

1. **Services:** on each service (Catalog → Products → **Services**) enter the **duration** in minutes and switch **Track stock** off. Optionally a **commission** rate of its own.
2. **Staff:** in **HR & payroll** → **Employees**, switch on **Takes appointments** for each person who does services, and set their **Commission %**. Give them the **Stylist / Therapist** role (or another role with *appointments*).
3. **Prepaid packs** (e.g. *5 massages for GHS 800*): a product with *Track stock* off and, under **Services** → **Prepaid pack**, the services it covers and the number of sessions, and how many days it is valid.
4. **Memberships and subscriptions:** create a product to invoice them as (e.g. *Gym membership*, *Track stock* off), then **Services** → **Membership plans** → **New plan**: price per period, how often it is billed (monthly, every 3 or 6 months, yearly) and any sessions included each period.
5. **System** → **Settings** → **Services:** the calendar's start and end times and time steps, appointment reminders (hours before), and the SMS gateway address (leave empty to use the restaurant one; only with your own SMS provider).
6. **Software licence keys** (optional): on the product tick **Hands out a licence key** (*Track stock* off), then **Services** → **Licence keys** → **Load keys** for each branch that sells it.

> **Why:** the duration fills the appointment book; the person who did each service earns their commission, which payroll pays each month; packs and memberships let customers prepay and use their sessions at the till.

> **Check:** book an appointment in **Services** → **Staff calendar**, check it out at a till with **Appointments (F10)**, and see the commission under **Services** → **Commissions**.

> **Note:** memberships are invoiced by **head office** each day (01:30) for every period that has started; the invoices go on the customer's account (Sales → Invoices) and are paid like any invoice.

8.3 Company-wide features (licence options)

Feature	What it adds	Set-up
Finance	Chart of accounts, automatic postings from sales, purchases, stock and payroll; supplier bills; bank reconciliation; month-end; reports	Chapter 7, step 12; your accountant reviews the chart and opening balances
HR & payroll	Employees, clock-in at tills, leave, loans, monthly payroll with PAYE and SSNIT, payslips	Chapter 7, step 12; payroll profile checked with your accountant
CRM (paid add-on)	Leads, deals on a pipeline, follow-ups with reminders, campaigns by SMS, WhatsApp and e-mail	Section 8.3a below
Projects (paid add-on)	Projects with tasks and milestones, timesheets, costs from stock, bills, expenses and time, invoicing, profit per project	Section 8.3b below
Service desk and maintenance (paid add-on)	Tickets from WhatsApp, e-mail, the web and staff with deadlines; equipment register, servicing plans and work orders	Section 8.3c below
Sensors (paid add-on)	Fridge, cold-room, freezer and room temperature and humidity, power and doors; alarms by WhatsApp / SMS; temperature log	Section 8.3d below

8.3a CRM: leads, deals, follow-ups and campaigns (paid add-on)

The **CRM** menu (under the *Sales* tab) helps you win and keep customers: nobody who asked about buying is forgotten, every sale you are working on is visible with its value, and customers who agreed hear about your offers.

Availability: a paid add-on. The menu appears once your licence includes **CRM** (ask SyncLytics; a new licence key is entered under *System → Licence*). **Why:** it is priced separately, so shops that do not need it do not pay for it.

Who can use it (People → Roles): *See leads, deals…* to look; *Work leads and deals…* to add and change them; *Send campaigns…* for campaigns; *Set up the sales pipeline stages* for the stages. Managers, branch managers, customer service and marketing roles get them when CRM arrives; give them to others as needed. People given one branch see and work only that branch's leads and deals. **Why:** a branch's prospects stay with the people who serve them.

Step 1 — Check the pipeline stages

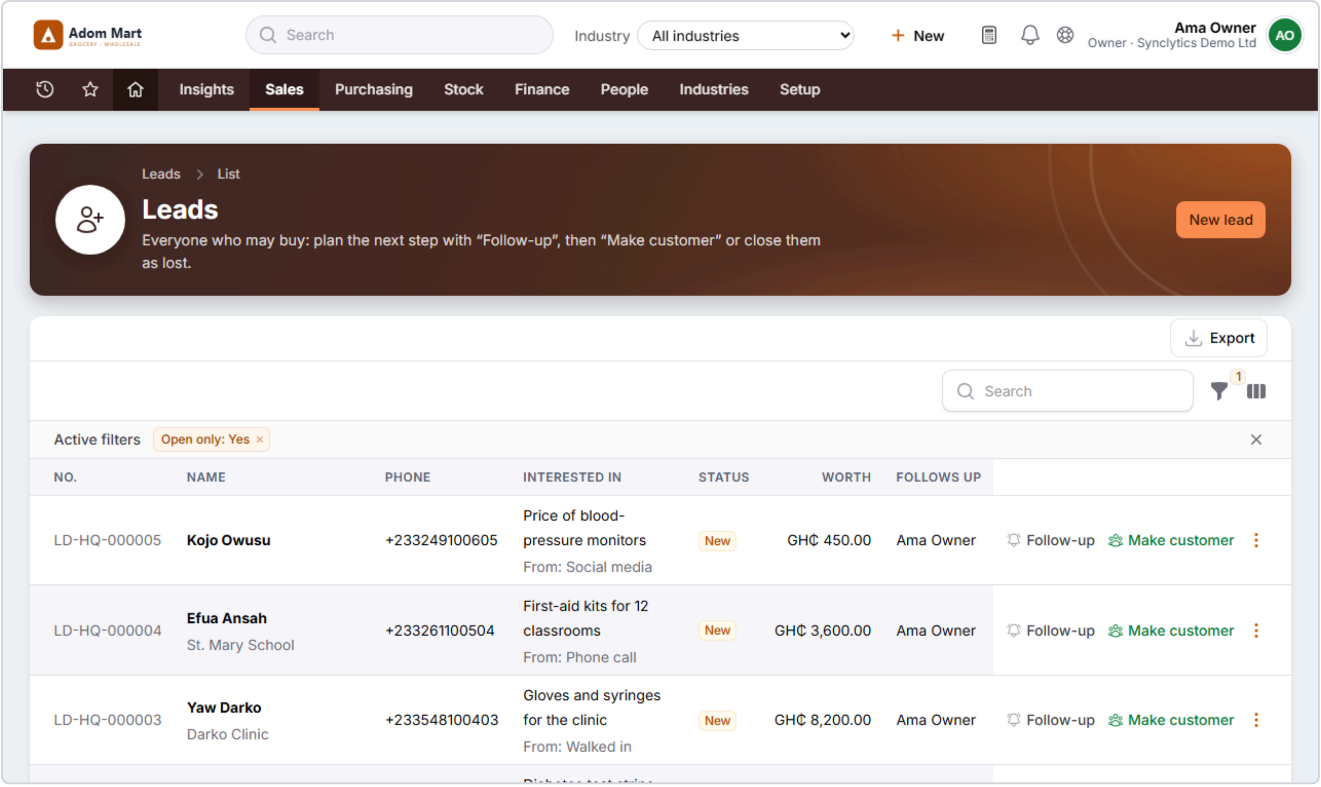
CRM → Pipeline stages. You start with *New → Contacted → Quotation sent → Negotiation*, then *Won* and *Lost*. Each stage in progress has a **chance of winning** (10 %, 20 %, 50 %, 75 %); rename stages, change the chances or add your own (e.g. *Sample given*), and set the order with *Order*. Keep one *Won* and one *Lost* stage. **Why:** the chances give the **weighted** value of the pipeline — a realistic figure of what is likely to come in.

Step 2 — Add leads

CRM → Leads → New lead for everyone who asks about buying: a walk-in, a call, a WhatsApp message, a referral. Enter the name, the business (if any), a phone number or e-mail, where they came from, what they want and what it may be worth, and who follows it up. Tick the **marketing consent** they gave (SMS, WhatsApp, e-mail). **Why:** a lead written down is followed up; one in someone's head is often lost. Consent is needed before campaigns.

On each lead:

- **Follow-up** plans the next call or visit (with a date and time and who does it) or notes what happened. A call noted moves a new lead to *Contacted*.
- **Make customer** turns the lead into a customer — with their consents and follow-ups — and can open a deal at once. If a customer already has the same phone number, the lead is linked to them instead of making a duplicate.
- **Lost** (in the ⋮ menu of the row, with *Edit*) closes the lead with the reason (too far, bought elsewhere...). **Why:** the reasons show what to improve.


Adom Mart
GROCERY · WHOLESALE
Search
Industry
All industries
+ New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Leads > List
New lead
Export
Search
1
Active filters
Open only: Yes ×
Follow-up
Make customer
⋮

Leads

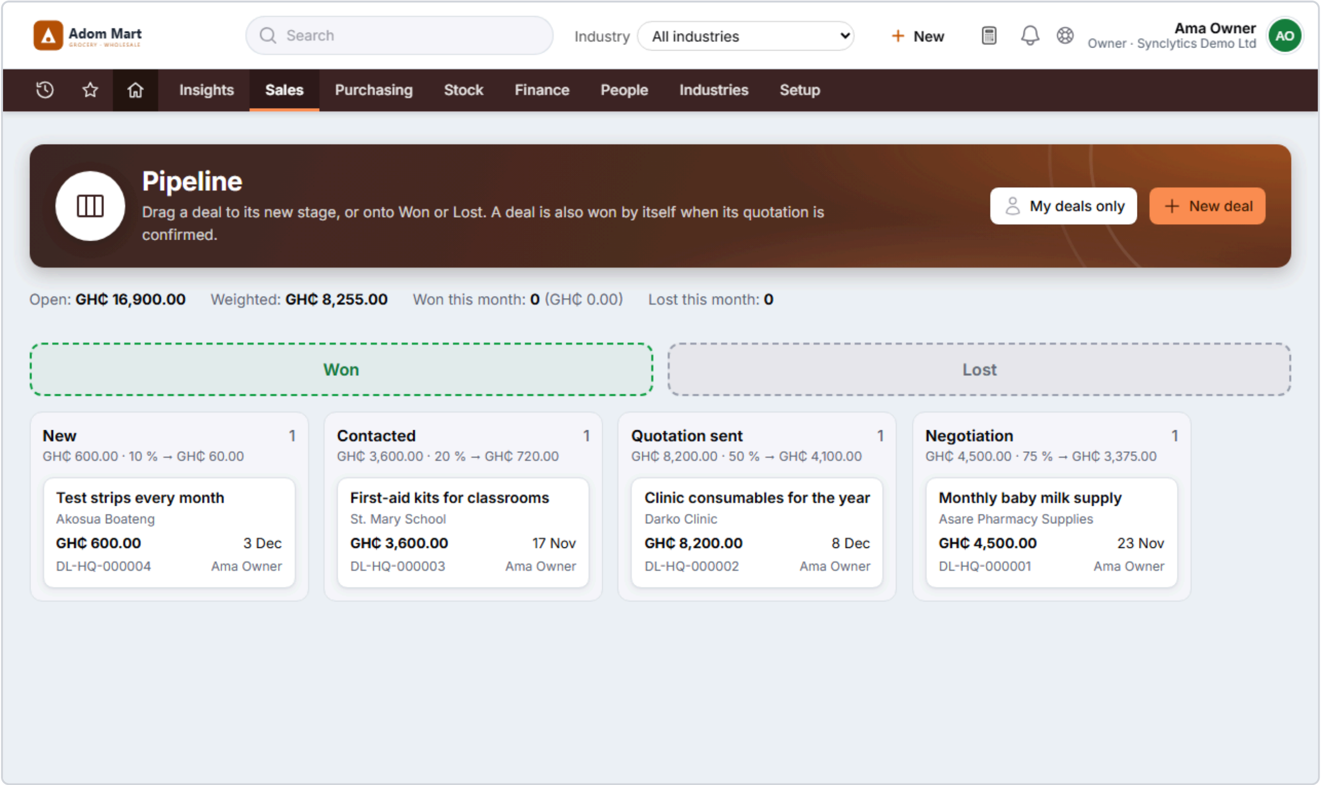
Everyone who may buy: plan the next step with “Follow-up”, then “Make customer” or close them as lost.

NO.	NAME	PHONE	INTERESTED IN	STATUS	WORTH	FOLLOWS UP
LD-HQ-000005	**Kojo Owusu**	+233249100605	Price of blood-pressure monitors From: Social media	New	GH₵ 450.00	Ama Owner
LD-HQ-000004	**Efua Ansah** St. Mary School	+233261100504	First-aid kits for 12 classrooms From: Phone call	New	GH₵ 3,600.00	Ama Owner
LD-HQ-000003	**Yaw Darko** Darko Clinic	+233548100403	Gloves and syringes for the clinic From: Walked in	New	GH₵ 8,200.00	Ama Owner

Step 3 — Work deals on the pipeline

A **deal** is a sale you are working on. Add one from **CRM → Deals → New deal**, from a lead (*Make customer*), or from a customer's page (**New deal**). Give it a title, its value and the date you expect to close it.

CRM → Pipeline shows the open deals as cards, in a column per stage, with each column's total and weighted value. **Drag a card** to its next stage, onto **Won**, or onto **Lost** (you are asked why). *My deals only* shows yours. On a phone, open the deal and change its stage. In the list of deals, *Won*, *Lost* and *Reopen* are in the ⋮ menu of each row.



Pipeline

Drag a deal to its new stage, or onto Won or Lost. A deal is also won by itself when its quotation is confirmed.

Open: **GH₵ 16,900.00** Weighted: **GH₵ 8,255.00** Won this month: **0** (GH₵ 0.00) Lost this month: **0**

New 1 GH₵ 600.00 · 10 % → GH₵ 60.00	Contacted 1 GH₵ 3,600.00 · 20 % → GH₵ 720.00	Quotation sent 1 GH₵ 8,200.00 · 50 % → GH₵ 4,100.00	Negotiation 1 GH₵ 4,500.00 · 75 % → GH₵ 3,375.00
Test strips every month Akosua Boateng **GH₵ 600.00** 3 Dec DL-HQ-000004 Ama Owner	**First-aid kits for classrooms** St. Mary School **GH₵ 3,600.00** 17 Nov DL-HQ-000003 Ama Owner	**Clinic consumables for the year** Darko Clinic **GH₵ 8,200.00** 8 Dec DL-HQ-000002 Ama Owner	**Monthly baby milk supply** Asare Pharmacy Supplies **GH₵ 4,500.00** 23 Nov DL-HQ-000001 Ama Owner

Make quotation on a deal starts its quotation (Sales → Quotes & orders) for the customer, with the deal number as the customer's reference, and moves the deal to *Quotation sent*. Add the items and send it as usual. **When the quotation is confirmed as an order, the deal is won by itself**, at the order's value. **Why:** nobody has to remember to update the CRM after the real work is done.

Step 4 — Follow-ups

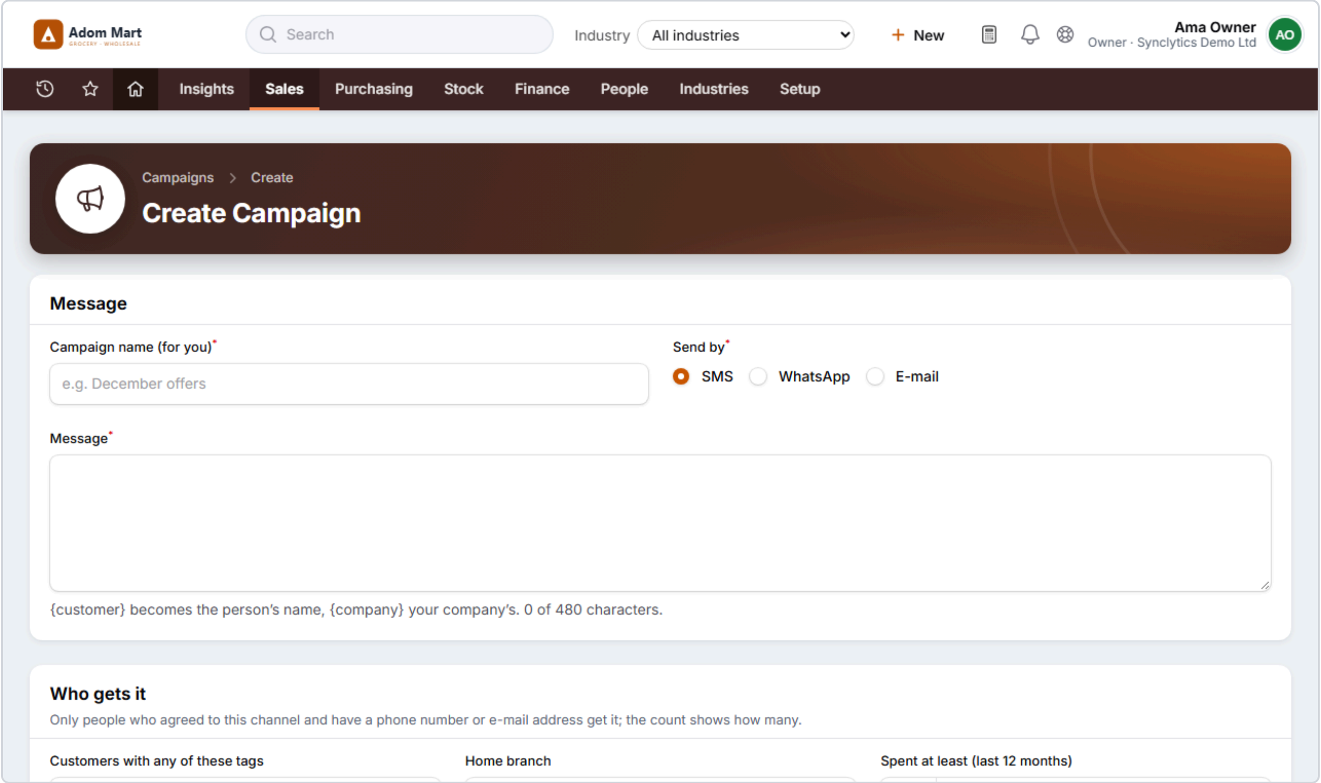
CRM → Follow-ups lists what to do and when — your own by default, overdue ones in red — with **Done** (and what came of it) and **Another time**. The person a follow-up is given to is told in the bell, and **reminded when it is due**. The menu shows how many are due today. Plan follow-ups from a lead, a deal or a customer's page (**Follow-up**).

Step 5 — Campaigns

CRM → Campaigns → New campaign: one message to many people.

1. **Name** it (for you) and choose **SMS, WhatsApp or e-mail**. Write the message; `{customer}` becomes each person's name and `{company}` your company's name. Keep SMS under 480 characters (3 SMS).
   - **WhatsApp** only delivers business messages that use a **template approved by Meta**: create the template in your WhatsApp Business account first, then type its name; the text must match it.

    - **E-mails** get a link at the end to stop them. The link asks the person to confirm, so e-mail scanners that open links do not unsubscribe anyone. Their choice is kept on their customer record.
2. **Who gets it**: customers with certain **tags** (e.g. *VIP*), a **home branch**, who **bought in the last … days**, who have **not been back for … days** (to win them back), who **spent at least …** in the last 12 months; and, if you like, open leads. The count shows how many it reaches: **only people who agreed to that channel and have a number or address** get it. **Why:** sending to people who did not agree is against data-protection law and annoys customers.
3. **When**: leave empty to send when you press **Send now**, or pick a date and time (**Schedule**). Avoid nights.
4. Save, check the campaign in the list, then **Send now** (or **Schedule**). The confirmation says how many people it goes to; it cannot be called back.



Create Campaign

Message

Campaign name (for you)*

e.g. December offers

Send by*

● SMS ○ WhatsApp ○ E-mail

Message*

{customer} becomes the person's name, {company} your company's. 0 of 480 characters.

Who gets it

Only people who agreed to this channel and have a phone number or e-mail address get it; the count shows how many.

Customers with any of these tags

Home branch

Spent at least (last 12 months)

The list shows how many it was sent to, how many were left out (no consent, no number) and how many arrived. Every message is also in **Setup** → **Message log**. A campaign is sent with the SMS, WhatsApp or e-mail set-up you already use for receipts and reminders (Chapter 16). **Copy** reuses a campaign.

Dashboards: the **Customers and deals** panel shows the open pipeline and its weighted value, deals won this month, new leads and your follow-ups due today (add it with *Arrange*).

> **Note:** with branch servers, CRM is used at head office (or on the cloud server), like the customers' accounts.

8.3b Projects and job costing (paid add-on)

The **Projects** menu (under the *Sales* tab) runs jobs from start to invoice: a shop fit-out, an installation, a consultancy assignment, or an internal job such as a new store. Each project **collects its costs by itself** and shows what it earns.

Availability: a paid add-on. The menu appears once your licence includes **Projects** (ask SyncLytics; a new licence key is entered under *System* → *Licence*).

Who can use it (People → Roles): *See projects…* to look; *Run projects…* to create them, plan tasks and milestones, issue stock and set hourly rates; *Log your own time on projects*; *Approve other people's timesheets*; *Invoice projects*. Managers get them when Projects arrives; supervisors, technicians and service staff can log time; accountants can invoice. Give them to others as needed.

Step 1 — Hourly rates

Projects → **Hourly rates**: for each person who logs time, what an hour **costs the company** (their pay and overheads divided by the hours they work, e.g. GHS 3,000 a month ÷ 176 hours ≈ GHS 17) and, if they have their own price, what an hour is **billed at**. **Why:** approved time is valued at the cost rate, so each project shows its true cost; only people who run projects see these rates.

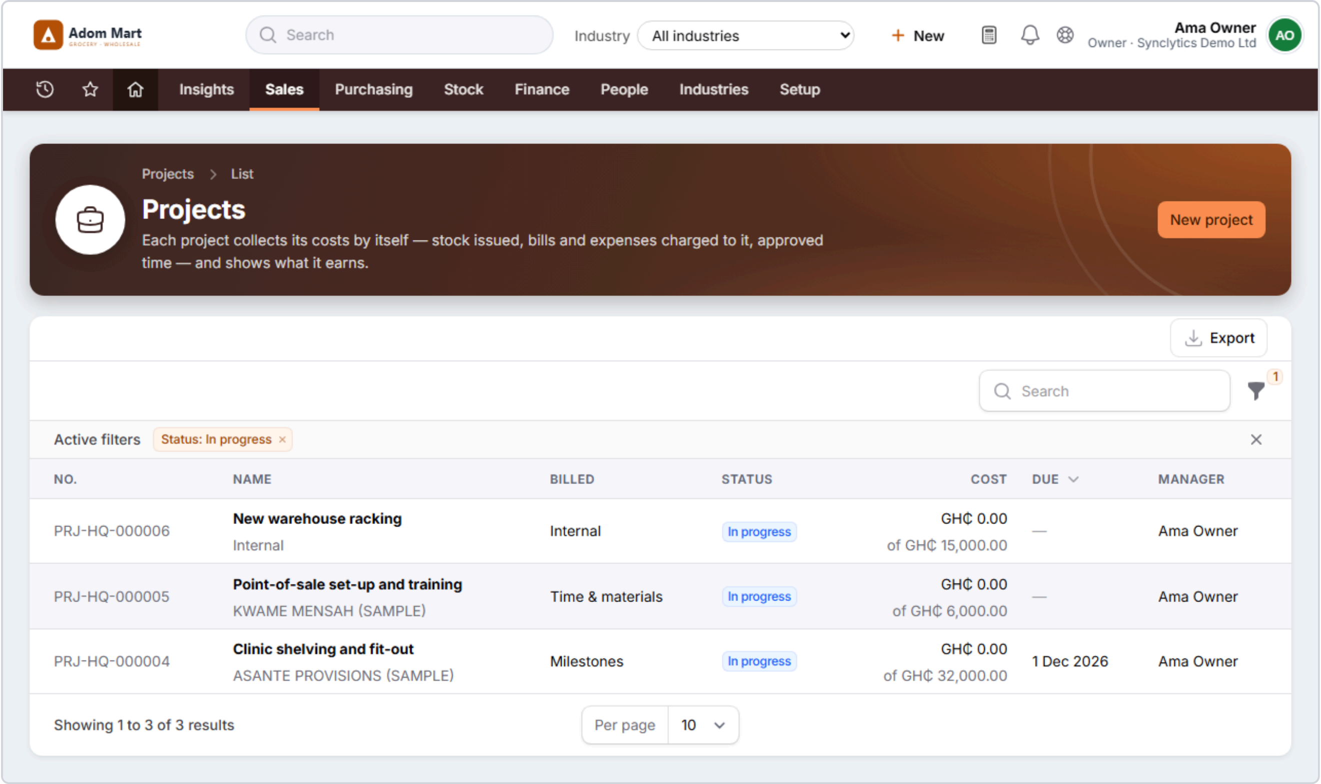
Step 2 — A project

Projects → **Projects** → **New project**: name, customer, branch, manager, dates, and the **budget** (what it should cost). Choose **how it is billed**:

- **Fixed price, by milestones** — give the **agreed price**, then add milestones on the project (e.g. *Deposit 30 %*, *Delivery 50 %*, *Handover 20 %*). They cannot add up to more than the agreed price.
- **Time and materials** — the **hourly rate billed** (for people without their own) and the **markup on materials** (e.g. 10 %: items costing 100 are billed at 110).
- **Internal** — not billed (your own work, e.g. a new branch).

For a billed project, choose the **service on invoices** — a product that is a service, not a stock item (e.g. *Professional services*, created under Catalog → Products with *Track stock* off). **Why:** invoices need an item to show time, milestones and bought-in costs, with its tax.

On the project, add **tasks** (who does them, by when, hours planned).



Projects

Each project collects its costs by itself — stock issued, bills and expenses charged to it, approved time — and shows what it earns.

Active filters: Status: In progress ×

NO.	NAME	BILLED	STATUS	COST	DUE	MANAGER
PRJ-HQ-000006	**New warehouse racking** Internal	Internal	In progress	GH₵ 0.00 of GH₵ 15,000.00	—	Ama Owner
PRJ-HQ-000005	**Point-of-sale set-up and training** KWAME MENSAH (SAMPLE)	Time & materials	In progress	GH₵ 0.00 of GH₵ 6,000.00	—	Ama Owner
PRJ-HQ-000004	**Clinic shelving and fit-out** ASANTE PROVISIONS (SAMPLE)	Milestones	In progress	GH₵ 0.00 of GH₵ 32,000.00	1 Dec 2026	Ama Owner

Showing 1 to 3 of 3 results

Per page: 10

Step 3 — Costs arrive by themselves

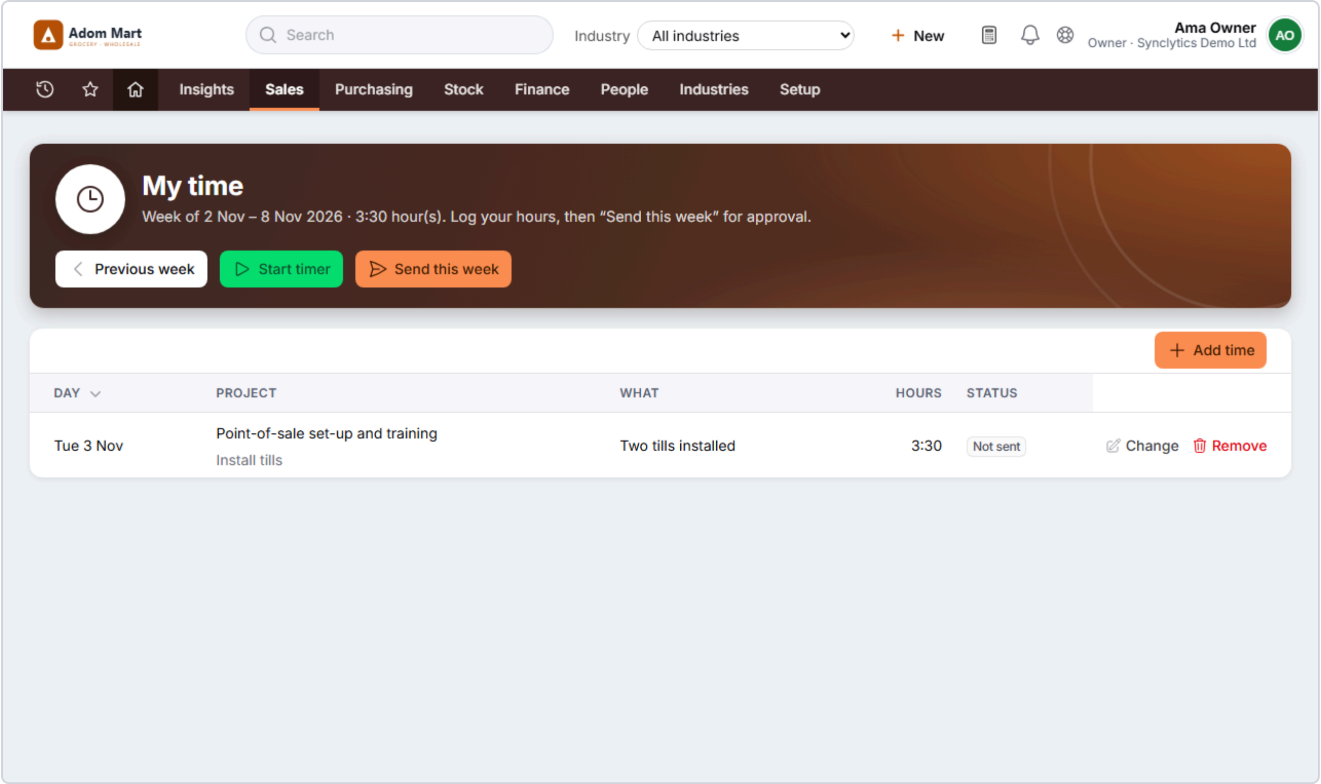
- **Stock**: **Issue stock** on the project takes items out of a store at their cost; **Return stock** brings unused items back at what the project was charged. **Why:** the store's stock and the project's cost stay right together.
- **Bought in**: on a supplier bill (*Other charges*) or a purchase order's service line, choose **For project**.
- **Expenses**: on an expense claim, choose **For project**; it counts when the claim is paid.
- **Time**: approved timesheets (Step 4), at each person's cost rate.

The **Costs** list on the project shows each one; the top of the project shows the cost against the budget, what was invoiced and the profit so far.

Step 4 — Timesheets

Everyone who logs time uses **Projects** → **My time**: **Start timer** when they begin (and **Stop timer** after), or **Add time** for a day (project, task, hours, what they did). At the end of the week, **Send this week**.

Managers open **Projects** → **Time to approve** (also under *Waiting for you* on the dashboard): **Approve** (one or many at once) or **Send back** with what to correct. Nobody approves their own time. Approved hours become project costs and, for time-and-materials projects, billable time. **Why:** only checked hours are costed and billed.


Adom Mart
GROCERY · WHOLESALE
Search
Industry
All industries
New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Previous week
Start timer
Send this week
Add time
Change
Remove

My time

Week of 2 Nov – 8 Nov 2026 · 3:30 hour(s). Log your hours, then "Send this week" for approval.

DAY	PROJECT	WHAT	HOURS	STATUS
Tue 3 Nov	Point-of-sale set-up and training Install tills	Two tills installed	3:30	Not sent

Step 5 — Invoicing

- **Milestones**: on the project, **Invoice** next to a milestone when it is reached.
- **Time and materials**: **Invoice time and materials** puts everything not yet invoiced on one invoice — hours at the billing rate, materials at cost plus markup, bought-in costs and expenses. To invoice only some, tick them in the **Costs** list and use **Invoice the selected**.

Invoices go to the customer's account like any other (Sales → Invoices) and show the project number as the customer's reference.

Step 6 — Finish

Finish (or **Cancel project**) when the work is done: open tasks close, and nothing more can be charged to it. **Reopen** if needed.

In the books (Finance): every journal line carries its project, so reports can show profit per project. For a billed project, stock, bought-in costs and expenses wait in **Project work in progress** (an asset) until they are invoiced — each milestone takes the costs so far, a time-and-materials invoice the costs it bills — or the project is finished; then they move to **Project costs**. Invoices go to **Project and service income**. An internal project's costs go straight to project costs. Staff time is a management figure only: salaries are already in the books through payroll. **Why:** costs are matched to the income they earn, as accountants expect.

Dashboards: the **Projects** panel shows projects in progress (and those over budget), time and materials not yet invoiced, your hours this week and timesheets to approve.

8.3c Service desk and maintenance (paid add-on)

Two menus: **Service desk** (under *Sales*) for requests and complaints, and **Maintenance** (under *Stock*) for the equipment you look after.

Availability: a paid add-on. Both appear once your licence includes **Service desk and maintenance** (ask SyncLytics; a new licence key is entered under *System → Licence*).

Who can use it (People → Roles): *Ask for help* (most staff: their own requests); *See service desk tickets*; *Handle tickets* (customer service, supervisors, technicians); *Run the service desk* (managers: pass on anyone's tickets, deadlines, late tickets); *See equipment*; *Do maintenance work* (technicians); *Run maintenance* (managers: equipment register, plans, invoicing customers' jobs).

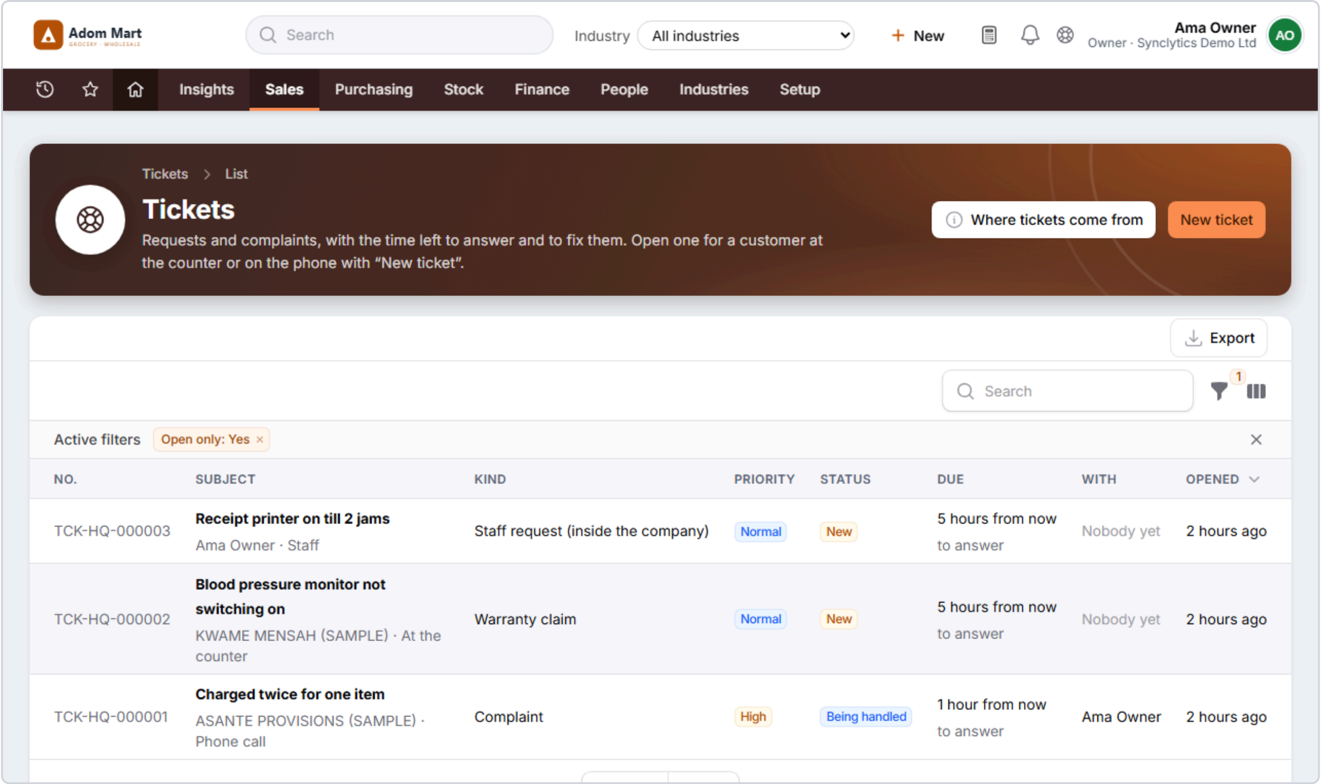
Service desk — Step 1: deadlines

Service desk → **Deadlines**: how fast a ticket is **answered** and **fixed**, by priority — you start with Urgent 1 h / 4 h, High 4 h / 1 day, Normal 8 h / 3 days, Low 1 day / 7 days, counted around the clock. Change them to suit. **Why:** everyone knows what is late; the person handling a ticket is warned when 75 % of the time is gone, managers are told when it is missed. While you wait for the customer's answer the clock stops.

Step 2: where tickets come from

Service desk → **Tickets** → **Where tickets come from** shows the exact links for your company.

- **At the counter or on the phone**: **New ticket** — what kind (complaint, return, warranty, question, request), the customer (or their name and number), the receipt number for a return or warranty (the warranty is checked on the sale), the equipment if it is about one.
- **WhatsApp**: messages to your WhatsApp Business number arrive under *Online → WhatsApp orders*; for a problem rather than an order, use **Make a ticket**. Their next messages are added to the ticket by themselves.
- **Web form**: switch on *Customers can report a problem on a web page* (System → Settings → *Service desk and maintenance*) and put the link on your website, receipts or a QR code. A hidden field stops robots.
- **E-mail**: type a long secret key in *Key for e-mails to tickets* (same settings), then have your e-mail provider (Mailgun, SendGrid, Postmark…) forward incoming mail to the address shown. A customer's reply that keeps the ticket number in the subject joins the ticket. **Why:** the key keeps strangers from creating tickets; keep the address private.
- **Staff**: anyone with *Ask for help* sees **Ask for help** (and *My requests*) for their own problems — a printer, a stock problem — and is told when it is resolved.


Adom Mart
Search
Industry
All industries
New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Tickets > List
Where tickets come from
New ticket
Export
Search
Active filters
Open only: Yes ×

Tickets

Requests and complaints, with the time left to answer and to fix them. Open one for a customer at the counter or on the phone with "New ticket".

NO.	SUBJECT	KIND	PRIORITY	STATUS	DUE	WITH	OPENED
TCK-HQ-000003	**Receipt printer on till 2 jams** Ama Owner · Staff	Staff request (inside the company)	Normal	New	5 hours from now to answer	Nobody yet	2 hours ago
TCK-HQ-000002	**Blood pressure monitor not switching on** KWAME MENSAH (SAMPLE) · At the counter	Warranty claim	Normal	New	5 hours from now to answer	Nobody yet	2 hours ago
TCK-HQ-000001	**Charged twice for one item** ASANTE PROVISIONS (SAMPLE) · Phone call	Complaint	High	Being handled	1 hour from now to answer	Ama Owner	2 hours ago

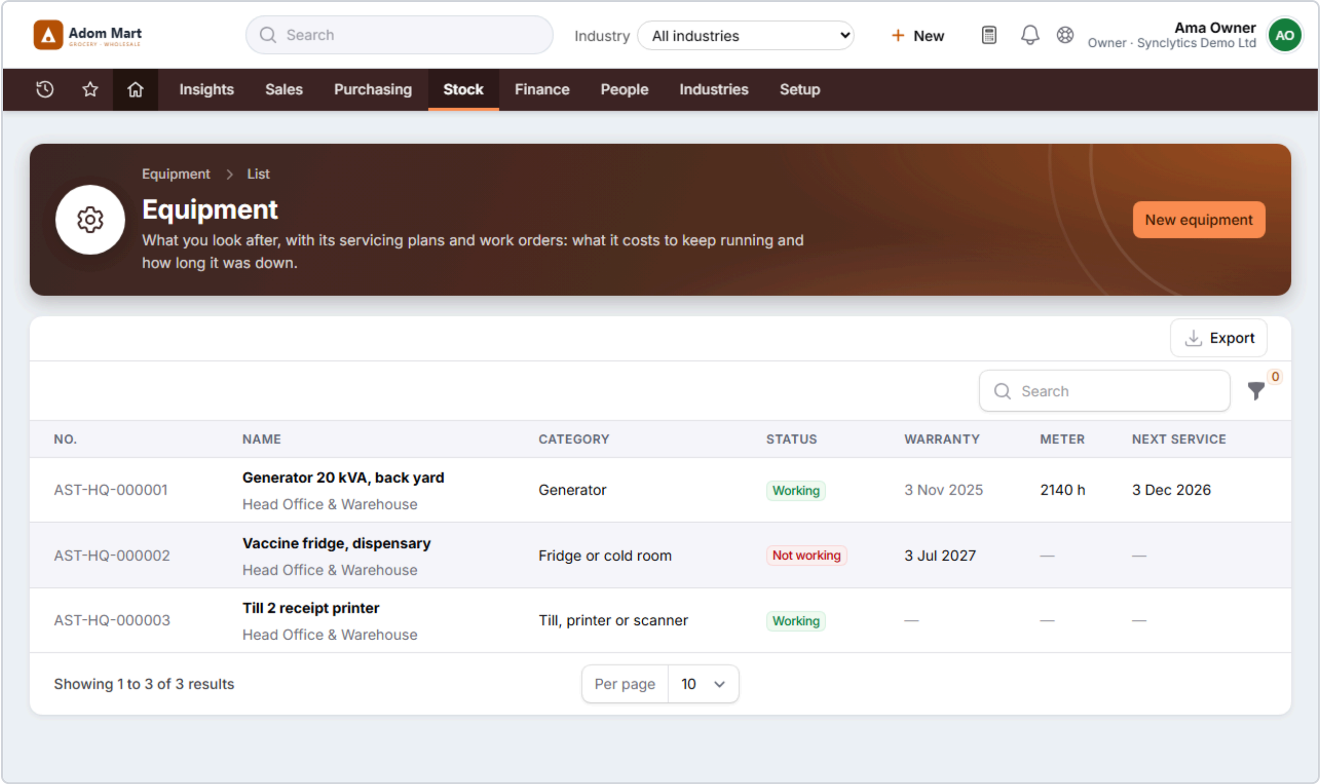
Step 3: working a ticket

Open the ticket: **Reply** by WhatsApp, SMS or e-mail (tick *Wait for their answer* to stop the clock), add a **Note** (staff only), **Pass on** to someone else or change the priority, open a maintenance **Work order** for the equipment it is about, and **Resolve** it — optionally telling the customer, with a link to rate the service from 1 to 5. **Why:** the whole conversation is on one page, so anyone can pick it up. A resolved ticket closes by itself after 7 days (*Close resolved tickets after*); a reply from the customer before then reopens it.

> **Note:** a free WhatsApp reply reaches the customer only within 24 hours of their last message (WhatsApp's rule); after that, answer by SMS or e-mail.

Maintenance — Step 1: the equipment register

Maintenance → Equipment → New equipment: your own equipment (generator, fridge, air conditioner, van, tills, computers) with its branch and place, make, model, serial number, supplier, warranty date, the fixed asset it is in the books as, and its **meter** (running hours or km) if it has one. Add **customers' equipment** you sold or service: the customer, the warranty and any **service contract** date. **Why:** you see each machine's history, cost and downtime.



Equipment > List

Equipment

What you look after, with its servicing plans and work orders: what it costs to keep running and how long it was down.

NO.	NAME	CATEGORY	STATUS	WARRANTY	METER	NEXT SERVICE
AST-HQ-000001	**Generator 20 kVA, back yard** Head Office & Warehouse	Generator	Working	3 Nov 2025	2140 h	3 Dec 2026
AST-HQ-000002	**Vaccine fridge, dispensary** Head Office & Warehouse	Fridge or cold room	Not working	3 Jul 2027	—	—
AST-HQ-000003	**Till 2 receipt printer** Head Office & Warehouse	Till, printer or scanner	Working	—	—	—

Showing 1 to 3 of 3 results

Step 2: servicing plans

On the equipment, **New plan**: what is done (e.g. *Oil and filter change*), **every so many days** and/or **every so many hours / km**, when it was last done, who does it, how many days early to open the work order, and a **checklist**. **Why:** a work order opens by itself when servicing is due — by date or when **New meter reading** passes the mark.

Step 3: work orders

Maintenance → **Work orders**: planned servicing (from the plans), breakdowns (**Report a breakdown** on the equipment, from a ticket, or from a fridge sensor alarm once sensors are connected) and **customer visits**. On each: **Start**, **Log time**, **Use parts** (out of a store, at cost; **Parts back** for unused ones), **Waiting for parts**, and **Done** with what was done, the checklist ticked and the meter reading. A breakdown marks the equipment *Not working* and **Done** puts it back to work, counting the downtime. An outside technician's bill is charged to the work order with *For work order* on the supplier bill's charge.

Customers' jobs: under warranty or contract they cost you; otherwise **Invoice the customer** — parts at the customer's price list, labour at the amount you type, as a service product.

Technician time is valued at the person's hourly cost (Projects → Hourly rates, if you have Projects) or at *Technician time costs per hour* (System → Settings → *Service desk and maintenance*).

In the books: parts for your own equipment go to **Repairs and maintenance**; for a customer's under warranty or contract to **Warranty and after-sales costs**; for a chargeable job to **cost of sales** (it is invoiced).

Dashboards: the **Service desk and maintenance** panel shows open and late tickets, the customers' rating, equipment not working and open work orders.

8.3d Sensors: fridges, cold rooms, freezers, rooms, power and doors (paid add-on)

The **Sensors** menu (under *Stock*) watches temperature and humidity around the clock, tells the people on duty the moment something is wrong, and keeps the temperature log inspectors ask for (FDA Ghana, Good Pharmacy Practice).

Availability: a paid add-on. It appears once your licence includes **Sensors** (ask SyncLytics; a new licence key is entered under *System → Licence*).

Who can use it (People → Roles): *See sensors…* (anyone who should see the log); *Respond to sensor alarms and sign fridge checks* (pharmacists, technicians, storekeepers, supervisors — the people on duty); *Set up sensors…* (managers: they are also told when an alarm goes unanswered). Give each person their **phone number** on their user record so alarms reach them by WhatsApp or SMS.

Step 1 — Choose the sensors

Any sensor that can send its readings over the internet works. Three ways:

1. **Straight to ZionEX (recommended)** — Wi-Fi or LoRa devices that can post to a web address: for example Shelly (H&T for rooms; a Shelly with a wired temperature probe for fridges, the probe inside and the device outside so Wi-Fi reaches it), ESP32 / Tasmota devices, or LoRaWAN sensors through The Things Network. No subscription.
2. **A model SyncLytics recommends and sets up** — ask us; we test it and configure it for you.
3. **A sensor cloud** — if you already have **SensorPush** (with its gateway) or **Tuya / Smart Life** sensors, ZionEX reads them from their cloud every 5 minutes (Step 3).

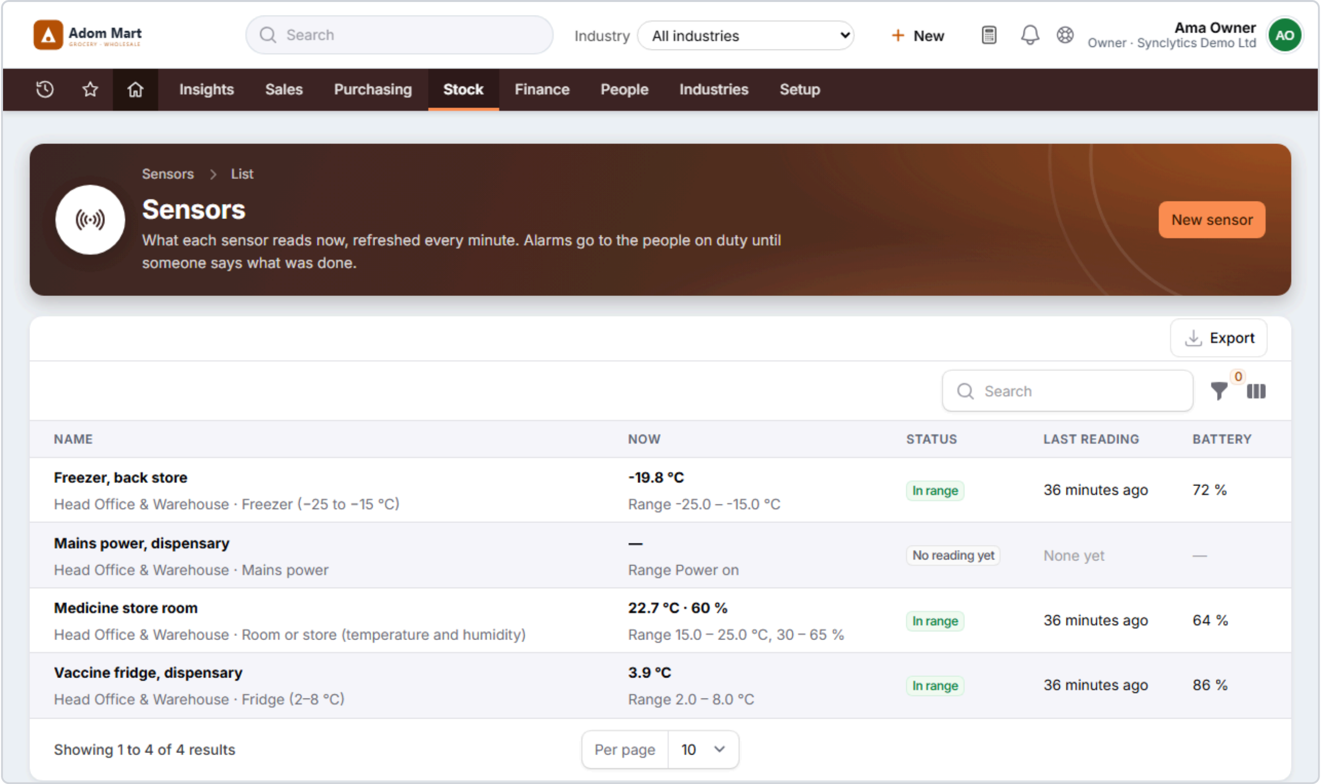
> **Why:** a probe sensor gives the temperature of the fridge's air and contents, independent of the fridge's own thermostat; a sensor that reports on its own needs nobody to remember to look.

Step 2 — Add each sensor

Sensors → Sensors → New sensor: a name (e.g. *Vaccine fridge, dispensary*), what it watches (fridge, cold room, freezer, room, mains power, door), the branch, and the **equipment** it is on (Maintenance → Equipment) so an alarm opens a breakdown work order. The range fills in by itself — fridges and cold rooms 2–8 °C, freezers −25 to −15 °C, rooms 15–25 °C and 30–65 % humidity — change it to your products' storage conditions.

- **Alarm after … minutes out of range** (10 by default): opening the door to take out a vial is not an alarm.
- **Alarm when no reading for … minutes** (30): a silent sensor (flat battery, Wi-Fi down, power cut) could hide a warm fridge.

Save, then **Sensor address** shows the secret address the device sends to, with examples (JSON for most devices; a link with `?temp=…&hum=…` for devices that can only call a web address, such as Shelly's). Enter it in the device's settings. The sensor shows **In range** once its first reading arrives. **New address** replaces it if a device is replaced or the address was shared.



Sensors

What each sensor reads now, refreshed every minute. Alarms go to the people on duty until someone says what was done.

NAME	NOW	STATUS	LAST READING	BATTERY
Freezer, back store Head Office & Warehouse · Freezer (−25 to −15 °C)	**-19.8 °C** Range -25.0 – -15.0 °C	In range	36 minutes ago	72 %
Mains power, dispensary Head Office & Warehouse · Mains power	— Range Power on	No reading yet	None yet	—
Medicine store room Head Office & Warehouse · Room or store (temperature and humidity)	**22.7 °C · 60 %** Range 15.0 – 25.0 °C, 30 – 65 %	In range	36 minutes ago	64 %
Vaccine fridge, dispensary Head Office & Warehouse · Fridge (2–8 °C)	**3.9 °C** Range 2.0 – 8.0 °C	In range	36 minutes ago	86 %

Showing 1 to 4 of 4 results

Per page 10

Step 3 — Sensor clouds (only for SensorPush or Tuya sensors)

Sensors → Sensor clouds → Add a sensor cloud: for SensorPush the account's e-mail and password; for Tuya the Access ID and Secret of a cloud project (iot.tuya.com) linked to the Smart Life app account, and its data centre. Then on each sensor choose *SensorPush cloud* or *Tuya cloud*, the account, and the sensor's id in it. **Read now** checks it works; a problem shows on the line.

Step 4 — When an alarm goes off

The people on duty for the branch are told at once — the bell, and **WhatsApp or SMS** (System → Messaging must be set up; WhatsApp needs the template *sensor_alarm* approved by Meta, otherwise SMS is used) — and again **every 15 minutes** until someone acknowledges it. After **30 minutes** the managers are told that nobody has. A fridge, cold room or freezer with its equipment chosen also gets an **urgent breakdown work order** (Service desk and maintenance add-on).

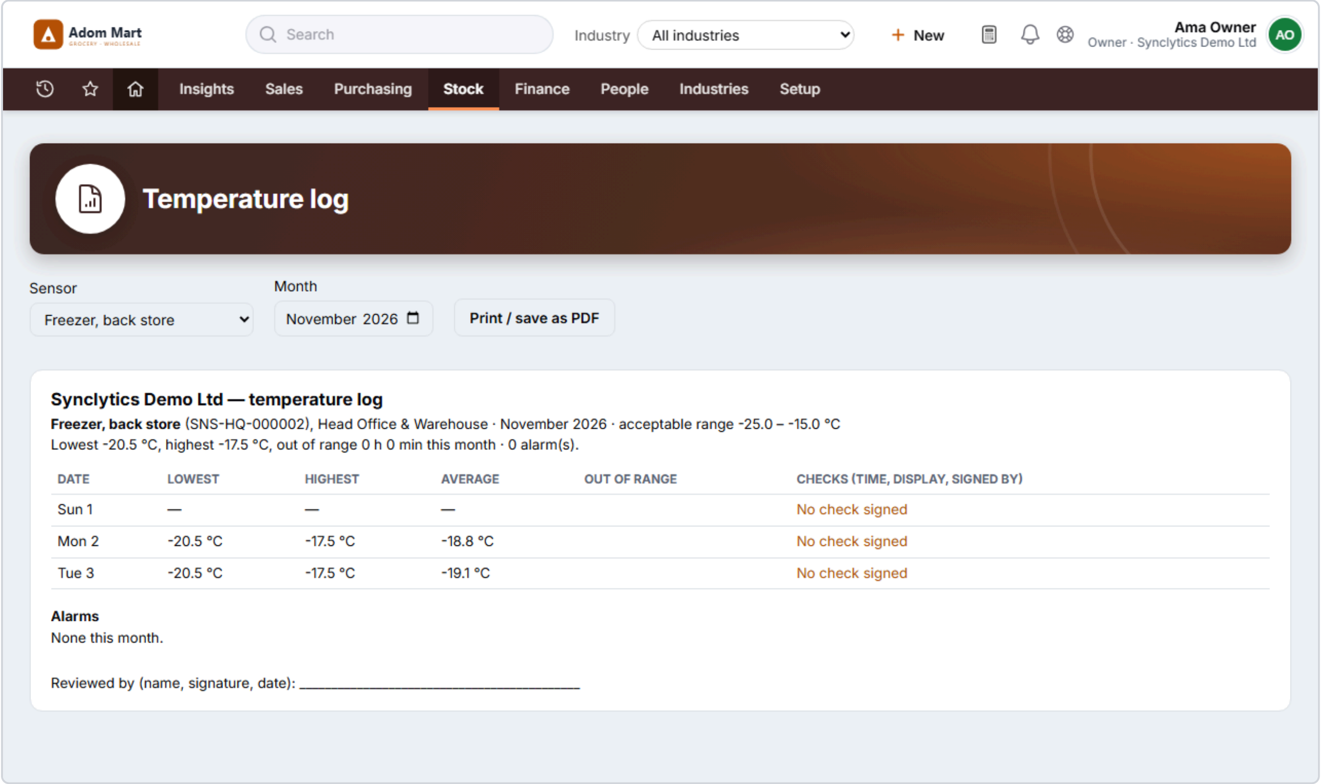
Sensors → Alarms → Acknowledge: check the stock (how warm, for how long?) and say what was done — stock fine, moved to another fridge, quarantined or discarded (name the product, batch and quantity), fault fixed, false alarm. The alarm closes when it is acknowledged and the reading is back in range. **Why:** inspectors look for what was done about every excursion, and by whom.

Each person chooses under **Profile → Notifications → Sensor alarms** whether alarms reach them by the bell, WhatsApp or SMS.

Step 5 — Daily checks and the temperature log

On a fridge's page, **Record a check**: what the fridge's own display shows and whether all is well, signed with your name. Most pharmacies check morning and afternoon. The dashboard's **Sensors** panel shows how many fridges are checked today.

Sensors → **Temperature log**: choose the sensor and the month — each day's lowest, highest and average, minutes out of range, the signed checks, and every alarm with what was done. **Print / save as PDF** for the inspector's file, and sign the *Reviewed by* line. Every reading is kept.


Adom Mart
GROCERY · WHOLESALE
Search
Industry
All industries
+ New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Temperature log

Sensor: Freezer, back store

Month: November 2026

Print / save as PDF

Synclytics Demo Ltd — temperature log

Freezer, back store (SNS-HQ-000002), Head Office & Warehouse · November 2026 · acceptable range -25.0 – -15.0 °C

Lowest -20.5 °C, highest -17.5 °C, out of range 0 h 0 min this month · 0 alarm(s).

DATE	LOWEST	HIGHEST	AVERAGE	OUT OF RANGE	CHECKS (TIME, DISPLAY, SIGNED BY)
Sun 1	—	—	—		No check signed
Mon 2	-20.5 °C	-17.5 °C	-18.8 °C		No check signed
Tue 3	-20.5 °C	-17.5 °C	-19.1 °C		No check signed

Alarms

None this month.

Reviewed by (name, signature, date): ____

8.4 Planned modules

These appear as **Planned** and can be reserved: **Manufacturing** (bills of materials, production at actual cost), **Logistics & Delivery**. This guide gets a checklist for each as it is released.

9. Setting up tills and other devices

9.1 How devices are set up

Every till, handheld and kitchen screen is set up the same way:

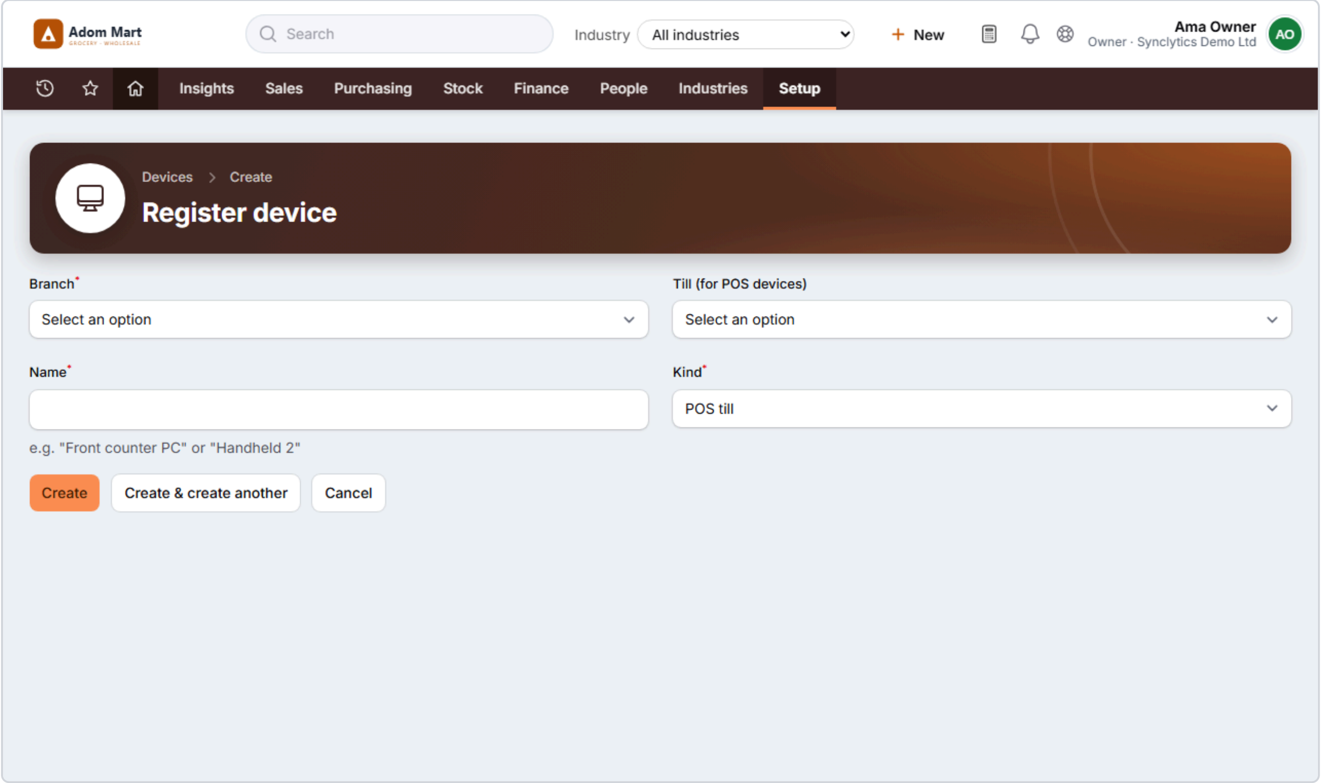
1. **Register** it in the back office: **System** → **Devices** → **New device**. You get a **one-time enrolment code** (valid 24 hours by default).
2. On the device, open the till app address in Chrome or Edge and **enter the code**.
3. People **sign in** on it (cashiers with username and PIN; kitchen screens need no sign-in).

> **Why:** only devices you registered can sell or see your data. Each device creates its own security key when it is enrolled, so a copied browser cannot pretend to be your till. A lost device is switched off with one click (**Revoke**).

Register a device

System → **Devices** → **New device**:

Field	What to enter
Branch	Where the device is used
Till (for POS devices)	The till it sells on (e.g. *Till 1*). Each selling device needs a till.
Name	Something you will recognise: *Front counter PC*, *Waiter tablet 2*, *Handheld 1*
Kind	**POS till** (tills and waiter tablets) or **Handheld scanner** (warehouse). **Kitchen display** (kitchen or bar screen) is offered only at a branch running the Restaurant module.


Adom Mart
Search
Industry
All industries
New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Devices > Create

Register device

Branch*: Select an option

Till (for POS devices): Select an option

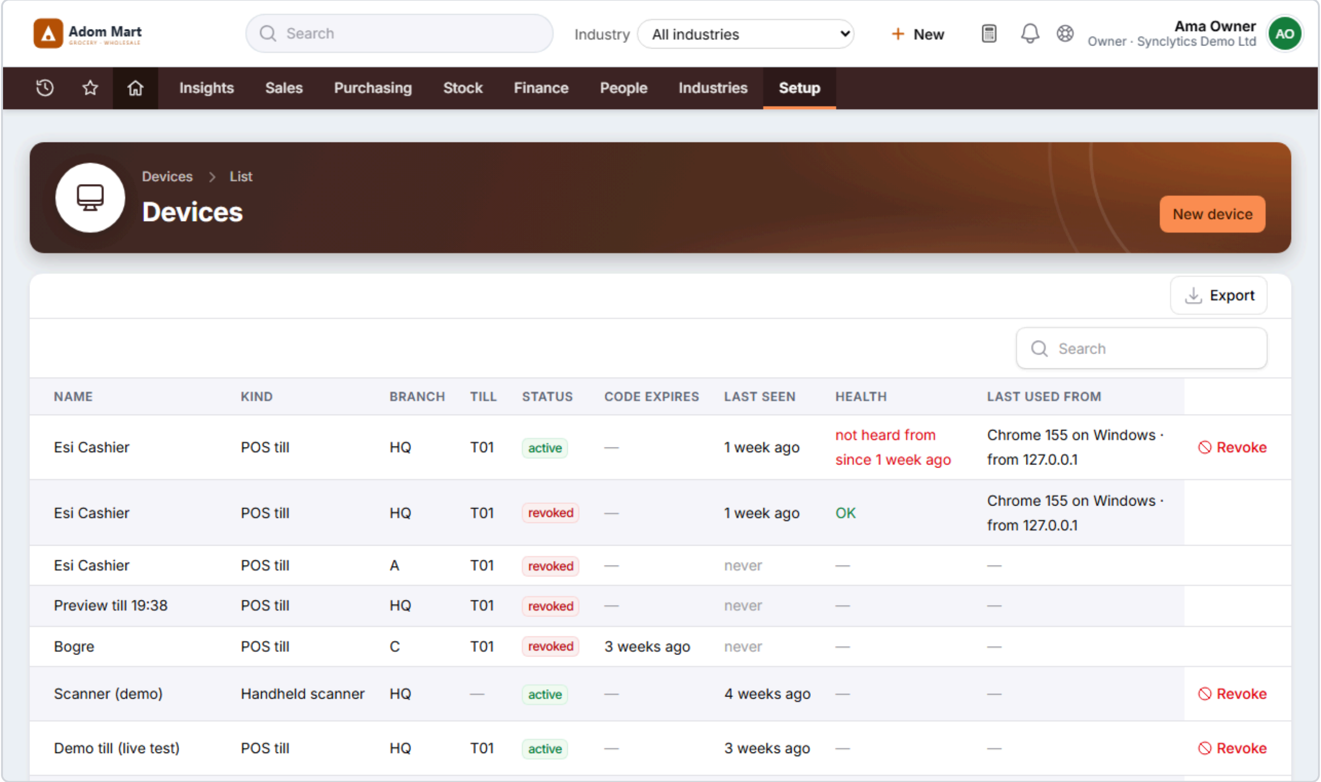
Name*:

e.g. "Front counter PC" or "Handheld 2"

Kind*: POS till

Create
Create & create another
Cancel

Click **Create**. The code is shown **once**: write it down or keep the page open while you set up the device. If it expires, use **New code** on the device's row.


Adom Mart
Search
Industry
All industries
New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Devices > List
New device
Export
Search

Devices

NAME	KIND	BRANCH	TILL	STATUS	CODE EXPIRES	LAST SEEN	HEALTH	LAST USED FROM	
Esi Cashier	POS till	HQ	T01	active	—	1 week ago	not heard from since 1 week ago	Chrome 155 on Windows · from 127.0.0.1	⃠ Revoke
Esi Cashier	POS till	HQ	T01	revoked	—	1 week ago	OK	Chrome 155 on Windows · from 127.0.0.1	
Esi Cashier	POS till	A	T01	revoked	—	never	—	—	
Preview till 19:38	POS till	HQ	T01	revoked	—	never	—	—	
Bogre	POS till	C	T01	revoked	3 weeks ago	never	—	—	
Scanner (demo)	Handheld scanner	HQ	—	active	—	4 weeks ago	—	—	⃠ Revoke
Demo till (live test)	POS till	HQ	T01	active	—	3 weeks ago	—	—	⃠ Revoke

9.2 The till app address

Setup	Address to open on the device
A (on the server PC itself)	`https://localhost/pos/`
B (Windows server on the shop network)	`https://<server-address>/pos/` (e.g. `https://192.168.1.10/pos/`). Trust the server first (4.9).
B/C (Ubuntu server on the network, no domain)	`https://<server-address>/pos/` . Trust the server first (4.9).
D (cloud, with HTTPS)	`https://<your-domain>/pos/`
E (branch server)	the **branch server's** address, as in B

9.3 Trust the server's address

Browsers allow a web app to use its **security functions** (the till's own signing key, the offline PIN check), to **work offline** and to **install as an app** only when it comes from a secure address: `https://…` or `localhost` . **This is required**: without it the till cannot even enrol.

- **A server without an internet name** (Windows or Ubuntu): it is reached over `https` with its own certificate. Each device trusts it once, as in **4.9**. Nothing else is needed.
- **Setup D and servers with an internet name:** nothing to do. The certificate is a public one.
- **Only for a server you set up by hand without https** (not recommended): each till, tablet, handheld and kitchen screen must be told once that the plain `http://` address is trusted:

1. In Chrome, open `chrome://flags/#unsafely-treat-insecure-origin-as-secure` (in Edge: `edge://flags/#unsafely-treat-insecure-origin-as-secure`).
2. In the box, type the server address **without** `/pos/` , e.g. `http://192.168.1.10` .
3. Set the option to **Enabled** and click **Relaunch**.

> **Why:** the till proves it is your device by signing with a key it creates and keeps itself, and checks PINs offline; browsers only offer these functions to trusted addresses.

> **Check:** the till app shows *Set up this device* and enrolling with the code works.

> **Note:** This is safe on your own shop network. Never use plain `http://` over the internet; use HTTPS (Chapter 5).

9.4 Till PC (Windows)

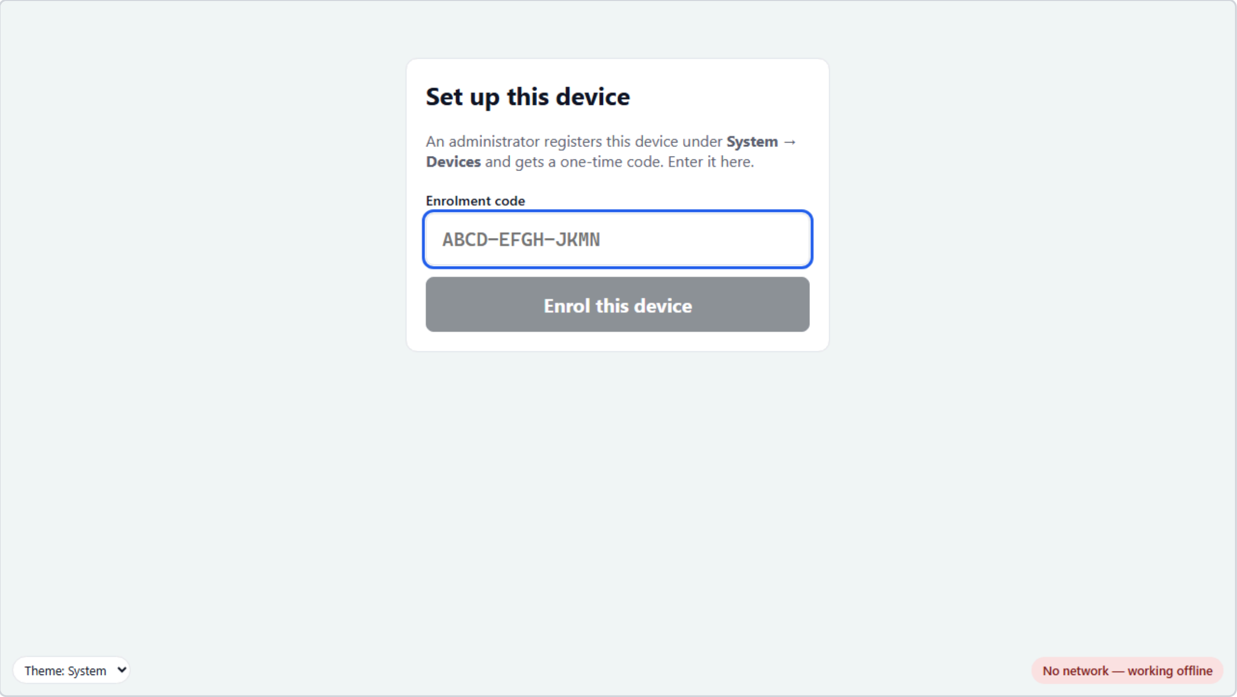
1. Install **Google Chrome** or use **Microsoft Edge**. Set the PC's clock to update automatically (as in 4.1, step 2).

2. Do 9.3 if your setup needs it.

3. Open the till address (9.2).

4. **Install it as an app:** in Chrome, menu ⋮ → **Cast, save and share** → **Install page as app...**; in Edge, menu **...** → **Apps** → **Install this site as an app**. A *POS* icon appears on the desktop and Start menu; use it from now on. **Why:** the app opens full-screen without the address bar and starts even when the server is off.

5. Enter the **enrolment code** and click **Enrol this device**:



Set up this device

An administrator registers this device under **System** → **Devices** and gets a one-time code. Enter it here.

Enrolment code: ABCD-EFGH-JKMN

6. A cashier signs in with **username and PIN**, counts the drawer and **opens the shift**.

> **Check:** sell an item and print the receipt. Then pull out the network cable (or switch off Wi-Fi): the top shows *working offline*, and you can still sell. Plug it back in: the yellow banner disappears as the sale is sent.

9.5 Tablet (Android)

1. Update Chrome from the Play Store. Do 9.3 if needed (Chrome on Android has the same flags page).
2. Open the till address, then Chrome menu ⋮ → **Add to Home screen** → **Install**.
3. Enrol and sign in as on a PC.

> **Tip:** waiter tablets in a restaurant are registered as **POS till** devices on a till of their branch.

Van reps: register the rep's Android phone as a **POS till** on the **van's till** (each van has one, created with the van). Allow the app to use the phone's **location** when asked, so visits record where they happened. On the phone the till shows **Route (F3)**, **Take order** and **Empties returned**. Install it as an app (step 2) so it opens without signal.

9.6 Choosing light or dark

Every till, handheld and kitchen screen has a **Theme** switch at the bottom left: **System** (follows the device's own setting), **Light** or **Dark**. The choice is remembered on that device. Dark is easier on the eyes in a dim bar or kitchen and saves battery on phones.

9.7 Receipt printer and cash drawer

1. Install the printer's **Windows (or Android) driver** from the manufacturer, and print a Windows test page.
2. In the printer's **Printing preferences**, choose paper **80 mm** (or 72 mm) receipt roll. For a cash drawer connected to the printer, switch on the driver's **cash drawer / open drawer before printing** option.
3. Make it the **default printer** on that till PC.
4. The first time you click **Print receipt**, in the browser's print window choose the printer, set **Margins: None**, **Scale: 100**, and untick **Headers and footers**. The browser remembers it.
5. **Print without the print window** (recommended): edit the desktop shortcut of the till app (right-click → **Properties**), and at the end of **Target** add a space and `--kiosk-printing`. Start the till from that shortcut.

> **Why:** `--kiosk-printing` sends receipts straight to the default printer, which is faster and avoids mistakes at busy times.

> **Check:** a sale prints a full-width receipt and the drawer opens.

The desktop till app (silent printing, cash drawer, scale)

Name: the desktop till is **ZionEX Till**. Until version 0.29 Windows listed it as *SES Labs Till*; installing 0.30 or later removes that copy by itself and keeps the server address, the printer, drawer and scale settings and any unsent sales. **Why:** one app on each till PC, under the product's name.

For a Windows till PC with a receipt printer, a cash drawer or a weighing scale, install **ZionEX Till** (`ZionEX Till_<version>_x64-setup.exe`, from SyncLytics) instead of using the browser:

1. Install the receipt printer's Windows driver (or note the IP address of a network receipt printer).
2. Trust the server on this PC (4.9). Then run the ZionEX Till setup. On first start it asks for the **server address**: exactly what you type in a browser to open the back office, without `/admin`. For a cloud server that is its name, e.g. `https://erp.clientdomain.com`, **never its IP address** (the security certificate is for the name, so the IP is refused). For a server in the shop it is the address used in 4.9, e.g. `https://192.168.1.10`.
3. **Receipt printer:** choose it from the list (or type `tcp://<printer-ip>:9100` for a network printer); tick **Open the cash drawer** if the drawer is plugged into the printer; **Print a test receipt**.
4. **Scale:** choose its COM port and speed (usually 9600), then **Read the scale** with something on it.
5. **Save and open the till.** The app first checks that the server answers. If it does not, it says so and stays on this page: correct the address and save again. (**Use this address anyway** keeps an address

that cannot be reached yet, for example when the server is not switched on.) Then enrol and sign in as with any till (9.1).

The app also tells the server **which computer it is** (its name, network card and address on the shop network), so the audit log and **System → Devices** (*Last used from*) show which till did what (12.1). **Open the back office** on its set-up page opens the back office in the app, so changes made there are recorded with the computer's details too.

Receipts then **print by themselves** the moment the sale is done, without a print window, the drawer opens after cash sales, and items sold by weight show a **Weigh** button. **Printer and scale set-up** on the receipt screen opens these settings again.

Till settings (desktop till 0.28 or later): the ⚙ **Settings** button at the top of the till. Anyone can look and press the test buttons; **Save** needs a manager's username and PIN (permission *Change a till's printer, drawer, scale and screen settings*; general manager, branch manager, IT Admin), and the change is recorded in the audit log against that till.

- **Receipt printer:** the receipt printer is picked for you (e.g. *EPSON TM-T20III Receipt*). Printers that only save files (*Microsoft Print to PDF*, *XPS*, *OneNote*, fax) cannot be chosen. **Why:** a receipt sent to them makes Windows open a "Save as" window after every sale. **Print a test receipt** prints a ruler across the paper: check it fits and the paper is cut.
- **Paper** 80 mm (48 characters a line) or 58 mm (32); **copies** of each receipt (2 when the shop keeps one); **company logo** on or off.
- **Dosage labels** on their own label printer, or on the receipt printer.
- **Cash drawer:** open after cash sales; **Open the drawer (test)** needs a manager's PIN, like opening it without a sale.
- **Weighing scale:** port and speed, and **Read the scale**.
- **Barcode scanner:** scan into the box to see exactly what the scanner sends; a **beep** after each scan (high: found, low: not found).
- **Screen:** light, dark or like the computer; text size normal, larger or largest. Saving needs the server (so the change is approved and recorded). Printer, drawer and scale are kept on the till computer itself.

Starting the till (0.28 or later): a till already set up opens straight on the till: the window stays hidden until the till is drawn, so there is no white window and no set-up page in between. The set-up page appears only on first start, from *Till set-up*, or when the server never answered. To print only when the cashier presses **Print receipt**: **System → Settings → Tills and cash → Desktop till prints each receipt by itself** (on by default).

Slow network: every till (desktop or browser) waits at most a few seconds for the server to accept a sale. If it has not answered by then, the sale is kept on the till, the receipt shows and prints at once (marked *Recorded on this till while offline*), and the sale is sent from the till's queue moments later; the server never records it twice. No other call to the server waits longer than 20 seconds. **Why:** a customer should never wait for the network; the receipt number and the sale are the same either way.

> **Why:** the app sends the receipt to the printer in the printer's own language (ESC/POS), which is faster and more reliable than a browser print window, and can talk to the drawer and scale, which a browser cannot.

> **Note:** the app only opens your ZionEX server — no other website — so the printer and drawer cannot be reached from anywhere else.

When the till starts: a till already set up shows only *Opening the till…* for a moment, then the till (version 0.26 or later; older versions showed a white window and the set-up page first). It opens the server straight away if it answers. If the server has worked on this PC before but cannot be reached now (internet down), the till still opens and sells offline. If the saved address has **never** worked, the set-up page comes back with a message, so a wrong address can always be corrected.

Updating the app: install the new `ZionEX Till_<version>_x64-setup.exe` over the old one (also over *SES Labs Till*, the earlier name). The server address, printer, drawer and scale settings are kept.

Starting the desktop till again from scratch

Uninstalling the app keeps its settings (the server address, printer and scale), so a reinstall comes back with the same settings. To clear them:

1. Close ZionEX Till (also from the system tray, if it shows there).
2. Open **PowerShell** (Start → type *PowerShell*) and run these two lines:

   ```
   Remove-Item "$env:APPDATA\app.seslabs.till" -Recurse -Force -ErrorAction SilentlyContinue
   Remove-Item "$env:LOCALAPPDATA\app.seslabs.till" -Recurse -Force -ErrorAction SilentlyContinue
   ```

3. Open ZionEX Till again (or reinstall it). It starts at the set-up page with empty fields.

> **Why:** the first folder holds the till's settings; the second holds what the till app saved for working offline. Clearing both makes the PC behave like a new till. Then delete the old entry under **System → Devices** in the back office and enrol the till again (9.1).

> **Note:** sales not yet sent to the server are in the second folder. Clear it only when the till shows nothing waiting to send, or when the till never worked.

9.8a The ZionEX Suite desktop app (office computers)

Office staff can use the back office in **ZionEX Suite**, an app installed on the computer, instead of a browser tab. It is free with every licence. **Why:** the back office gets its own icon and window (no address bar or tabs to close by mistake), the server address is remembered, exports go straight into *Downloads*, and the audit log records which computer each change was made from.

1. Trust the server on this computer first if it is on the shop network (9.3). A cloud server with a name needs nothing.
2. Run `ZionEX Suite_<version>_x64-setup.exe` (from SyncLytics). It installs for every user of the computer.
3. On first start it asks for the **server address**: what you type in a browser to open the back office, without `/admin` — for a cloud server its name, e.g. `https://erp.clientdomain.com`, **never its IP**

address. Click **Save and open**, then sign in with your usual username and password (and your second step, as in the browser).

Using it:

- Everything works as in the browser: the same pages, menus, look and rights. New features appear in it by themselves.
- **Export** files and documents are saved in your **Downloads** folder and open by themselves; a file of the same name is never overwritten (*report (2).xlsx*).
- **Print** opens the normal print window.
- Links to other sites (help pages, WhatsApp, a supplier's website) open in your normal browser; the app only ever shows your own server.
- **F5** reloads the page; **Ctrl + / Ctrl −** zoom; **Ctrl+Shift+S** goes back to the set-up page to change the server address.
- If the server cannot be reached when the app starts, it says so with **Try again** instead of an error page.

Updating: install the new `ZionEX Suite_<version>_x64-setup.exe` over the old one; the server address is kept. Version 1.0.2 replaces *ZionEX Back Office* (the earlier name) by itself, so there is one app on the computer.

9.8 Barcode scanner

Tablets and phones without a scanner use the camera: the 📷 button beside the till's search box (Chrome on Android). Hold the barcode steady in the frame; the item is added as if scanned.

1. Plug it in (USB) or pair it (Bluetooth).
2. Open Notepad and scan a product: the number appears and the cursor moves to a **new line**.
3. If it does not move to a new line, scan the **"Add Enter / CR suffix"** setting barcode from the scanner's manual.

> **Why:** the till adds the product when the scanner "presses Enter" after the code.

9.9 Handheld (warehouse)

1. Register a device of kind **Handheld scanner** for the branch.
2. On the phone: Chrome, do 9.3 if needed, open the till address (9.2), enrol with the code.
3. Storekeepers sign in with **username and PIN**. They need a warehouse role (Storekeeper, Warehouse Manager…), not till access.

Working offline: the handheld keeps working when the Wi-Fi or the server drops, like a till. Receiving deliveries, moving stock between bins, receiving transfers and counting are saved on the handheld and sent by themselves when the server is back (*Working offline* and *N action(s) waiting to be sent* show at the top). Scans are recognised from the handheld's own copy of the products and shelf labels, refreshed every 30 minutes while connected. The person can even sign in offline, as long as they signed in on that handheld while it was connected before. Each action is done once on the server, even if it is sent twice.

What the server refuses (e.g. a count already closed in the back office) is listed on the menu under *Refused by the server* with the reason: **Send again** after fixing the cause, or **Discard** what was done again correctly.

What needs the connection: the list of what is in a bin (Put away), and products or orders created since the handheld last connected. **Why:** warehouse Wi-Fi is often weakest at the back of the store, where the work is.

4. **Install it as an app:** in Chrome, menu (⋮) → **Add to Home screen** (or **Install app**). It then opens full screen from its own icon. To keep staff in it, use Android's **screen pinning** (Settings → Security → App pinning) or the brand's kiosk launcher (e.g. Zebra Enterprise Home Screen).

Which handhelds work

Any Android handheld with **Android 10 or later** and **Google Chrome**, and any phone. The system does not need a special app or driver from the maker: the built-in scanner only has to **type the code like a keyboard and finish with Enter** (Tab also works). Phones and tablets without a scanner use the **camera**: the 📷 button beside each scan box (Chrome on Android).

What to look for when buying:

	Recommended
Android	11 or later, with Google Play (so Chrome stays up to date)
Memory	3 GB or more
Scanner	a **2D imager** (reads shop barcodes, QR codes and codes on screens)
Wi-Fi	dual band (2.4 and 5 GHz)
Battery	a full shift; a spare or swappable battery for long days
Build	drop-rated (1.2 m or more) for warehouses

Well-known models that fit: **Sunmi** L2s / L2H, **Urovo** DT40 / DT50, **Newland** MT90 / N7 (good value); **Honeywell** EDA52 / CT30 XP, **Zebra** TC22 / TC27 (mid-range); **Zebra** TC58 / MC33, **Honeywell** CT45 / CT60, **Datalogic** Memor 11 (rugged, for large warehouses). A Bluetooth or USB ring / pocket scanner paired with an ordinary phone or tablet also works.

Setting up the scanner, by brand

On each handheld, open the maker's scanner settings once and set: **output as keyboard (keystrokes)** and **Enter after the code**. Then test in the handheld's scan box: the item should be found and the device beeps.

Brand	Where	What to set
Zebra	**DataWedge** app → Profile0 (or a new profile for Chrome)	*Keystroke output*: on. *Basic data formatting → Send ENTER key*: on. *Intent output*: off
Honeywell	Settings → Honeywell Settings → Scanning → Internal scanner → Default profile → Data Processing	*Wedge*: on, method *Keyboard*. *Suffix*: `\n` (Enter)

Brand	Where	What to set
Sunmi	Settings → Scanner (or the *Scanner* app)	*Output mode*: simulate keyboard / keyboard input. *Add suffix*: Enter
Urovo	Settings → Scanner settings	*Output mode*: keyboard. *Send Enter key* (or suffix Enter): on
Newland	Scan Settings (*Quick Setting*)	*Output mode*: simulate keystroke. *Suffix*: CR / Enter
Datalogic	Settings → Scanner settings → Wedge	*Keyboard wedge*: on. *Formatting → Label suffix*: Enter
Bluetooth / USB scanner	The scanner's own setting barcodes (in its manual)	*HID keyboard* mode, and the *Add Enter / CR suffix* barcode

If a scan does nothing: tap the scan box and scan again; if the code appears but nothing happens, the Enter suffix is missing. The handheld also catches a scan when the cursor is not in the box, so a code is not lost after tapping a button.

Why keyboard output: it works the same on every brand and model, with nothing to install or update from the maker; changing brand later needs no change to the system.

9.10 Kitchen screen (Restaurant)

1. Register a device of kind **Kitchen display** for the branch.
2. On the screen's PC or tablet: Chrome, do 9.3 if needed, open the till address, enrol with the code.
3. Choose the **station** it shows (e.g. *KITCHEN* or *BAR*). No one signs in.
4. Press **F11** (PC) for full screen.

> **Check:** send a dish from a till: it appears on the screen within a few seconds.

9.11 Counting stock with a handheld

During a stock count (Chapter 14.4) storekeepers scan the bin, then each product, and type the quantity. Counts from several handhelds go into the same count.

9.12 Label scale (Grocery)

Set the scale to print **EAN-13** labels starting with `21` (see Chapter 8, Grocery). The scale's own software holds each item's **PLU** (5-digit code) and price per kg; keep its PLU codes the same as the products' **scale codes**.

9.13 When a device is lost, stolen or replaced

- **Lost or stolen: System** → **Devices** → **Revoke** at once. Its access and every sign-in on it end immediately.

- **Replacing a till that still has unsent sales:** on the old till press **Export unsent sales** (in the yellow banner). In the back office: **System** → **Sync Center** → **Import till file**. Then register and enrol the new device.
- **Automatic copy (desktop till 0.27 or later):** while any sales are waiting, the desktop till saves the same file by itself every 15 minutes in **Documents** → **ZionEX Till** → **Unsent sales** on that PC (*SES Labs Till* on tills older than 0.30), keeping the newest 48 (12 hours). **Why:** if the till PC breaks or its data is wiped before the sales reach the server, the newest file there is loaded with *Import till file*, even if nobody pressed the button. Loading a file twice is safe: sales already received are skipped.
- **A shift left open overnight:** the till shows *This shift was opened on … and not closed*, and **Sales** → **Shifts** marks it *left open since …* in red. Close it on the till (count the cash) and open a new shift. **Why:** otherwise today's sales and cash run into yesterday's shift, and neither day balances on its own.
- **Re-installing the same device:** register it as a new device (it gets the next receipt series letter, so receipt numbers never repeat).

9.14 Device health and remote fixes

Every till and handheld tells the server how it is doing every few minutes: its version, sales waiting to be sent, the connection to the server, free storage and the printer. **System** → **Devices** shows it in the **Health** column — *OK*, or what is wrong in plain words (hover for the details) — and IT is told once a day per problem: a till not heard from during trading hours, sales stuck for more than 30 minutes, items refused by the server, storage nearly full, an old till version.

With the **Remote access** add-on, someone allowed to control devices can **Send a fix** to a till: reload it, send waiting sales now, refresh the catalogue, clear its cache (catalogue and pictures — never unsent sales) or collect its logs. The till does it within about five minutes and the result shows under **Fixes**. **Why:** most till problems are solved by one of these, without anyone travelling to the shop.

SyncLytics support never needs the add-on: the owner opens **System** → **Support access** for a few hours or until a problem is closed (see Chapter 13).

Seeing and controlling the screen: once the remote support relay is set up (Chapter 20) and the agent is installed on a device (**System** → **Devices** → **Install remote support**), the device shows **View screen** and **Take control**. Who may connect, and whether the person at the device must accept, follow the same rules as above; every connection is in the audit log and recorded on the relay.

10. Branch servers and several sites (setup E)

A **branch server** is a second copy of the server at a branch. The branch's tills, back office, kitchen screens and handhelds use it over the local network, so the branch keeps working fully when the internet to head office is down. It exchanges changes with head office every minute when the link is up.

> **Why:** in setup D, tills sell offline during an outage, but kitchen screens, shared restaurant bills, handhelds and stock look-ups need the server. A branch server puts a server inside the branch.

10.1 What flows where

Information	Direction	Who decides
Company, branches, tills, modules, settings, licence	HQ → branches	Head office
Users, roles, access, PINs	HQ → branches	Head office
Products, units, barcodes, prices, tax, promotions	HQ → branches	Head office
Sales, shifts, devices, review items	Branches → HQ	The branch
Stock movements, transfers, counts, purchase orders	Both ways	Where it happened
Customers, balances, bookings, passes, memberships	Both ways	Field by field / where it happened

> **Note:** on a branch server, master data (products, prices, users) is **read-only**: change it at head office and it arrives within a minute. Nothing is ever overwritten — sales and stock movements are only added.

10.2 Install the branch server

1. Head office must already be installed and activated (setup C or D).
2. At **head office**: **System** → **Sync Center** → **Add branch server**. Enter the branch server's **id** (e.g. `B01`, unique per server) and a name. Note the **command** and the **secret** that are shown **once**. **Why:** the secret proves to head office that the branch server is yours; every exchange is signed with it.
3. Install the branch server on its PC or small server with **Chapter 4** (Windows) or **Chapter 5** (Ubuntu), with these differences:
    - In `.env`: `ERP_NODE_ROLE=branch`, `ERP_NODE_ID=B01` (the id from step 2), and `APP_URL` = the branch server's own network address.
    - **Do not run** `php artisan erp:install` (the company comes from head office).
    - `php artisan migrate --seed --force` on a branch server only installs the permissions.
4. Join head office (on the branch server):

```
php artisan erp:node:join https://<hq-address> <HQ node id> <company id>
```

Paste the secret when asked. The exact command, with the head office id and company id filled in, is the one shown in step 2. **Why:** this downloads everything the branch needs: company, people, products, prices and stock.

5. Make sure the scheduler runs on the branch server (Chapter 4.8 or 5.9). It runs the sync every minute.

6. Register and enrol **that branch's devices on the branch server's own back office** (`http://<branch-server>/admin` → System → Devices), and point the tills at the branch server's address (Chapter 9).

> **Check:**

- Branch server back office → **System** → **Sync Center**: head office shows **OK** and a recent *Last sync*.
- Sell on a branch till; within a minute the sale appears at head office under **Sales** → **Till sales**.
- Change a price at head office; within a minute it appears on the branch till.

> **Note:** two-factor sign-in secrets and passkeys are not copied to branch servers (each server has its own encryption key). Managers who use the branch server's back office set up two-factor sign-in there once.

10.3 The Sync Center (System → Sync Center)

What you see / do	Use it when
Status per server: last sync, waiting, failed, last error	Daily glance
Sync now	You want changes across immediately
Choose what syncs per group: Automatic, Manual (preview first), Off	Normally leave everything on *Automatic*
Review incoming	A group is on *Manual*
Export / Import sync file (USB)	The link is down for days: carry the changes on a USB stick
Full resync from this peer	A branch seems out of step; safe to repeat
Import till file	A till was replaced before it could send its sales

10.4 Keeping servers in step

- **Upgrade head office first**, then branch servers. Changes a branch server cannot apply yet wait as *failed — update this node* and apply by themselves after the upgrade.
- Server clocks must agree within 5 minutes (automatic time on every server).
- To stop a branch server (closed branch, replaced PC): at head office **Sync Center** → **(row menu)** → **Disable this peer**. Nothing is deleted. Keep a final backup of the branch server.

11. Online stores, delivery apps and SMS

These connect your system to other services. Each one is used **only with your own account and keys** at that service. ZionEX never creates accounts for you.

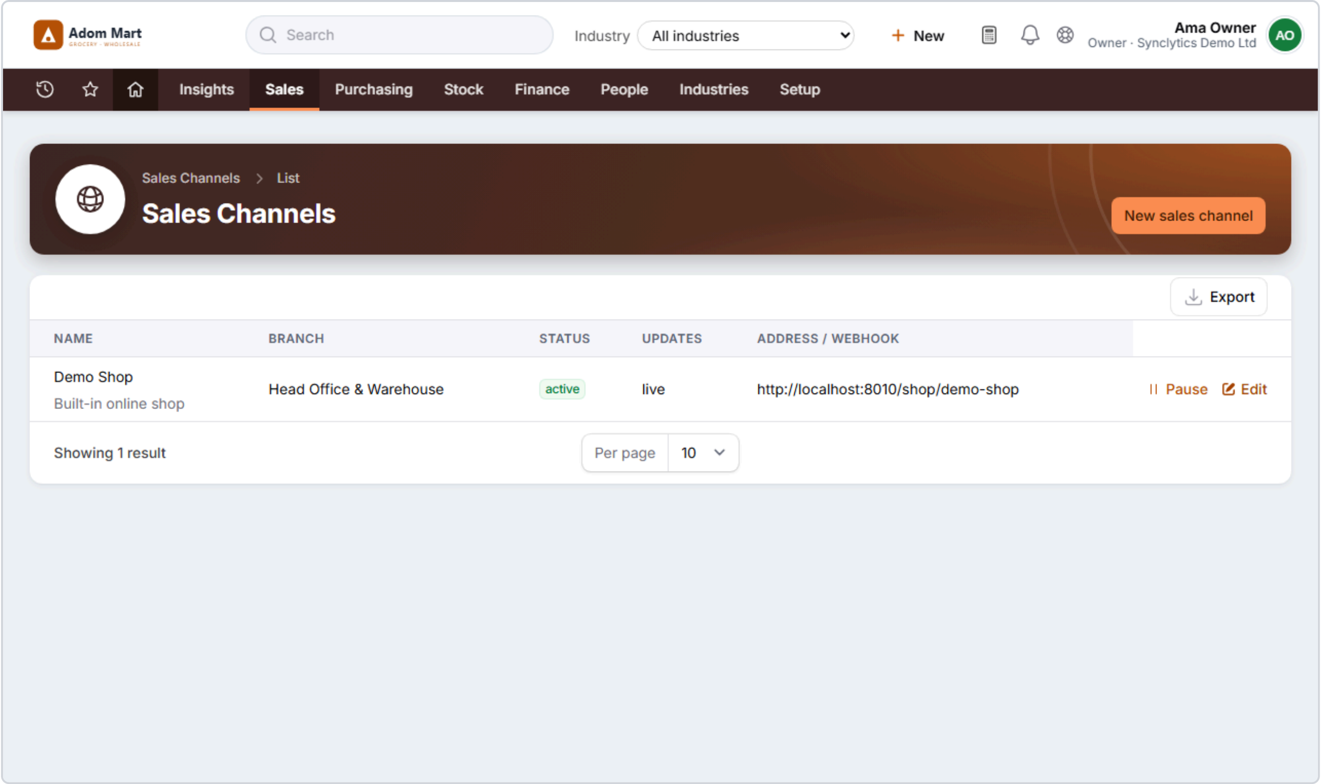
> **Warning:** treat every key and secret you paste here like a password. They are stored encrypted and never shown again.

11.1 Requirements

- A server reachable from the internet with **HTTPS** (setup D, or head office in setup E). Online stores send orders to your server; they cannot reach a PC inside a shop.
- The **scheduler** running (it sends changes every minute and checks the stores every 15 minutes).

11.2 Online stores (Online → Channels)

Type	Set-up at the store	In ZionEX
Built-in online shop	none	Online Storefront module on (Chapter 8); channel type *Built-in online shop*; choose the address
WooCommerce	WooCommerce → Settings → Advanced → REST API: create a key with **Read/Write**. Webhooks: *Order created* and *Order updated* to the address ZionEX shows, with your webhook secret	Channel type *WooCommerce*: store address, key, secret, webhook secret
Shopify	Create a custom app with Admin API access to products, inventory and orders; note the access token and your location id. Webhook *Order creation* to the address ZionEX shows	Channel type *Shopify*: shop address, token, location id, webhook secret
Other shops (generic)	Must accept JSON product updates and send signed order webhooks	Channel type *Generic REST*


Adom Mart
GROCERY · WHOLESALE
Search
New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Sales Channels
List
Sales Channels
New sales channel
Export
Pause
Edit

Industry: All industries

NAME	BRANCH	STATUS	UPDATES	ADDRESS / WEBHOOK
Demo Shop Built-in online shop	Head Office & Warehouse	active	live	http://localhost:8010/shop/demo-shop

Showing 1 result

Per page: 10

For every channel:

1. **New channel**: type, name, the **fulfilling branch**, **stock warehouse** and **price list**.
2. Enter the secrets (they stay hidden; leave blank to keep the current one).
3. Copy the **webhook address** shown in the channel list into the store's webhook settings.
4. Set **safety stock** on products where needed.
5. Click **Activate**. Every product is sent once; afterwards only changes are sent.

> **Why:** the store is offered *on hand − reserved − safety stock*, so the shop and the till do not sell the same last item.

> **Check:** change a price; within a minute it is on the store. Place a test order on the store; it appears under **Online** → **Online orders** with its stock reserved.

> **Tip: Compare with store** lists differences between the store and ZionEX and fixes them; it also runs automatically every 15 minutes.

11.3 Delivery apps (Restaurant)

Bolt Food, Glovo and Chowdeck orders can arrive by webhook (with the keys from **your** merchant account) or be entered by hand at the till. Set them up under **Online** → **Channels** with the type *Bolt Food (delivery app)*, *Glovo* or *Chowdeck*, then copy the webhook address into the app's merchant portal.

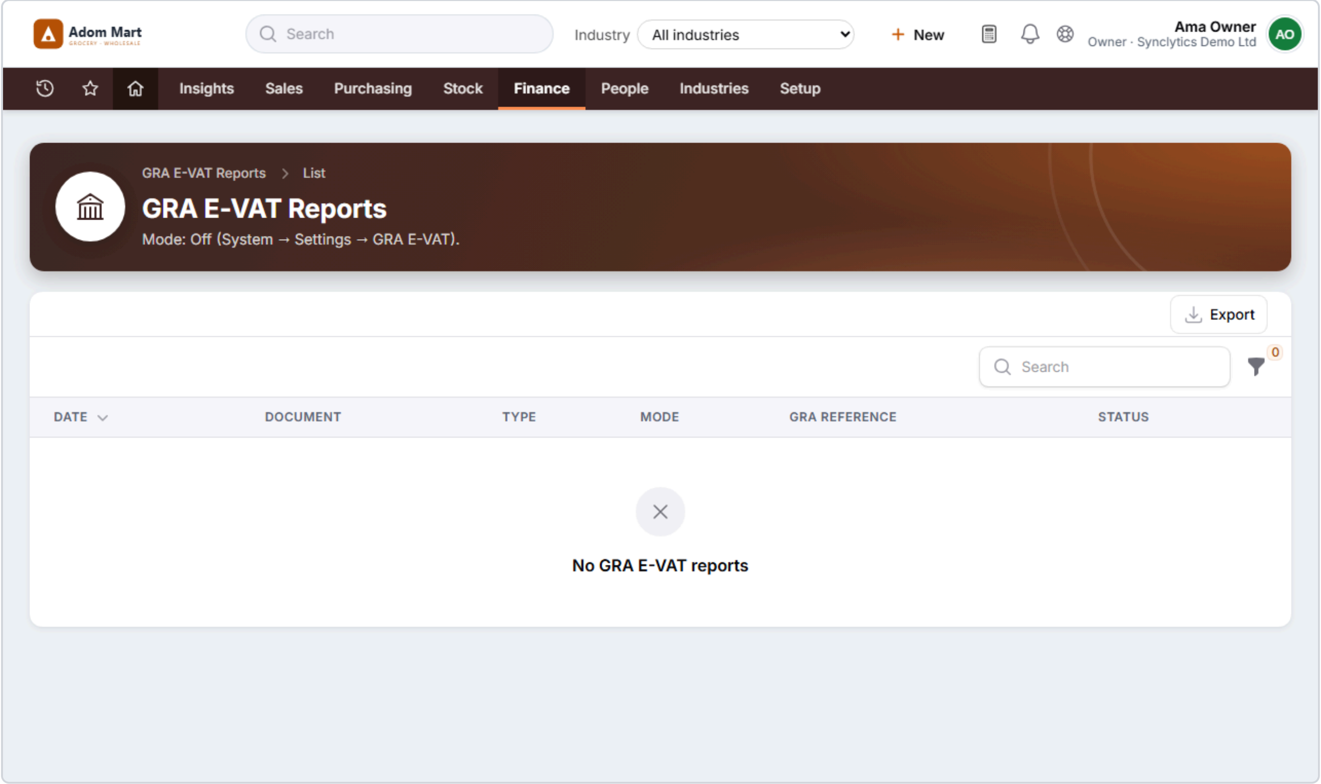
11.4 SMS, WhatsApp and e-mail

Booking and appointment messages, digital receipts, invoice e-mails and scheduled reports go through **your own** SMS, WhatsApp and e-mail accounts. Setting them up is explained in **Chapter 16**.

> **Note:** if you set an *SMS gateway address* under Settings → Restaurant or Services in an earlier version, it keeps working for those messages until you choose an SMS provider in Chapter 16.

11.5 GRA E-VAT (electronic VAT invoicing)

The system talks to GRA's **VSDC** (Virtual Sales Data Controller) the way GRA's *E-VAT API version 8.2* describes. Once GRA has onboarded your business for E-VAT (apply through GRA's E-VAT team; they first give you **staging** access to test), GRA gives you three things: the VSDC address, your **company reference** (your TIN with a suffix, e.g. `C00XXXXXXXX-001`) and your **security key**.


Adom Mart
GROCERY · WHOLESALE
Search
Industry
All industries
New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
GRA E-VAT Reports > List
GRA E-VAT Reports
Export
Search
0

Mode: Off (System → Settings → GRA E-VAT).

DATE	DOCUMENT	TYPE	MODE	GRA REFERENCE	STATUS

No GRA E-VAT reports

1. Put them in the server's `.env` (head office), then `php artisan optimize`:

    ```
    GRA_EVAT_URL=https://vsdcstaging.vat-gh.com/vsdc
    GRA_EVAT_COMPANY_REFERENCE=C00XXXXXXXX-001
    GRA_EVAT_KEY=<the security key from GRA>
    GRA_EVAT_TIN=C00XXXXXXXX
    ```

    Start with GRA's **staging** address; change to the production address GRA gives you once the tests below pass. **Separate books** (Chapter 8): each company with its own TIN has its own GRA account — add them on one line: `GRA_EVAT_ACCOUNTS={"C00YYYYYYYY": {"reference": "C00YYYYYYYY-001",` `"key": "…"}}`. **Check:** System → Health shows **GRA E-VAT: GRA's VSDC answers** once a reporting mode is chosen (step 2).

2. **System** → **Settings** → **GRA E-VAT** → **Report to GRA**, choose one:

Mode	What happens	Receipts
Each sale and invoice as it happens	Every till sale and invoice is sent at once. If GRA or the internet is down, it is sent automatically later.	Carry GRA's code and QR code when GRA answers within a few seconds
Till sales in a daily batch	Till sales are sent together after the batch time (e.g. 23:30); invoices still go one by one	No GRA code
Invoices and credit notes only	Only invoices to business customers are reported	—

3. **Finance** → **GRA E-VAT** lists every report: *Accepted by GRA* (with GRA's reference), *Waiting to send*, or *Refused by GRA* (with GRA's reason — fix it, e.g. the customer's TIN, then **Send again**).

> **Why:** GRA requires VAT-registered businesses to report invoices electronically. The mode lets you start with invoices only, and move to real-time when your tills are reliably online.

On the receipt: with each sale reported as it happens, the receipt prints GRA's signature block — **SDC ID and receipt number, SDC time, internal data, signature, MRC** — and GRA's QR code, which customers can scan to check the receipt with GRA. A sale GRA has not answered within a few seconds prints *GRA E-VAT: being reported* and is sent automatically.

Returns and credit notes are reported to GRA as a *partial refund* of the original receipt or invoice. **Levies** go per item: NHIL, GETFund, COVID-19 (none from 1 January 2026), CST and tourism, as GRA requires.

> **Note:** test with GRA's staging VSDC first: make a few sales, a return and an invoice, and compare them in Finance → GRA E-VAT with what GRA's portal shows, before switching to the production address. `docs/evat.md` describes the fields sent.

11.6 Mobile money that confirms itself (payment requests)

Without this, the cashier types the transaction ID from the customer's MoMo message, and a wrong or made-up ID is only found when the wallet is checked at closing. With a **payment request**, the till sends the amount to the customer's phone; the customer approves with their PIN; the sale goes through only once the payment provider confirms the money arrived, and the payment shows as **verified** with the provider's transaction ID.

What you need: a merchant account with a payment provider that collects from **MTN MoMo, Telecel Cash and AT Money**:

Provider	Where	What to copy into the system
Paystack	dashboard.paystack.com → Settings → API Keys & Webhooks	The **secret key** (`sk_live_…`). Never the public key.

Provider	Where	What to copy into the system
Hubtel	Hubtel business account → Developers / API keys; Receive Money must be enabled for your account	**API ID**, **API key** and the **POS Sales ID** (merchant account number). Ask Hubtel to allow your server's IP address for the status check.
Any other provider (Flutterwave, ExpressPay, Korba, Theteller, MTN's own MoMo API, your bank…)	A developer sets up a small *gateway* service that follows the system's custom gateway contract (`docs/momo-custom-gateway.md`, about a day's work)	Choose **Other provider (custom gateway)**: the gateway's address (https only) and the shared secret agreed with the developer.

The provider pays the money into your bank account or wallet on its own schedule and charges its fee (about 2 %). **Why a provider:** it is the only way for the system to know, by itself, that a payment really arrived.

Set it up (owner or finance manager):

1. **System** → **Mobile money** → **Payment provider:** choose the provider, paste the keys, tick **Send payment requests through this provider**, save. The keys are stored encrypted on this server and never shown again.
2. Copy the **callback address** shown in the same window into the provider's dashboard (Paystack: *Webhook URL*; Hubtel: the callback is sent with each request, nothing to do). **Why:** the provider then tells the system the moment a customer pays, even after the till stopped waiting.
3. **System** → **Settings** → **Mobile money** → **How the till takes mobile money:**
   - *Payment request; typing the ID still allowed* — to start with;
   - *Payment request only* — once staff are used to it. Typing an ID is then possible only when the till is offline (the sale is still recorded and checked at closing).
4. Try it with a small amount on your own phone before go-live. With Paystack you can first use the test key (`sk_test_…`).

At the till: Pay (F12) → **Mobile money** → choose the network → type the customer's number → **Send payment request**. The customer gets a prompt on their phone and enters their PIN. The till shows *Waiting for the customer…* and adds the payment by itself when it is paid (some Telecel and AT payments send the customer a code first: type it in and **Confirm code**). If the customer does not approve within 3 minutes, send the request again or take another payment.

System → **Mobile money** lists every request: paid, not paid, timed out. **Paid, not on a sale** shows money that arrived after the till stopped waiting — ring up the sale with it, or refund the customer from the provider's dashboard.

Branch servers: set the provider up on each server the tills sign in to (the keys are not copied between servers).

11.7 Integrations: every connection on one page (paid add-on)

Setup → **Integrations** shows a card for each connection to the outside world — e-mail, SMS, WhatsApp, mobile money, GRA E-VAT, online shops, other systems (API), branch servers, backups off the server, the AI assistant and accounting exports. Each card says whether it is **set up**, **when it last worked** and **how many failed this week**, with the way to its set-up page and its test.

> **Why:** a connection that stops (an SMS account out of credit, a GRA certificate expired, a backup place full) is found the same day, from one page, instead of when a customer complains.

Availability: a paid add-on. It appears once your licence includes **Integrations** (ask SyncLytics; a new licence key is entered under *System → Licence*). Without it, every connection still works and is set up on its own page as described above.

12. Daily running, backups, updates and moving

12.1 Every day

Who	What	Where
Cashiers	Open the shift with the float; close it with the cash count	Till
Supervisor	Look at the **Review queue** and resolve items	Sales → Review queue
Manager	Check no till shows a red banner (*refused*) or a long yellow queue	Tills; System → Devices (*Last seen*)
IT / owner	Check **System → Health** is all green (backup made last night, scheduler running)	System → Health
Head office	Check branch servers synced recently (setup E)	System → Sync Center
Owner or branch manager	**Close the day** before leaving	Cash → Close the day

Till shifts (Sales → Till shifts): each shift's *Expected cash* is only what should be in the drawer (float, cash sales, pay-ins, less pay-outs and drops); *Sales total* is everything the shift sold (less returns), with cash, each MoMo network, card, vouchers, NHIS and on-account on hover, and the *returns refunded* by how the money went back. A drawer or MoMo phone that is over by exactly a refund usually means the refund was keyed but never paid out: check that return under Sales → Sales. The till shows the same list when the shift closes. **Why:** cash is only part of the day; the total tells you what the shift really sold.

Shift report (the printer button on a closed shift): one A4 page with the branch's name, address and phone under the company. At the top, the **grand total sold** (every payment method, less returns refunded), the number of sales, the average sale, returns, and the cash expected, counted and over / short. Then who opened and closed it and when, the sales by payment method (sold, refunded, net and the grand total row), the cash drawer, the MoMo / card / voucher checks, the drawer's pay-outs and drops and coupons (left out when there were none), and lines for the teller and the manager to sign. **Why:** the manager sees at a glance what the shift sold and whether the money is right, and files a signed copy.

Finding sales and payments (the filter button on each list): **Sales** → **Till sales** by **teller**, **total** (from / up to), **how it was paid** (cash, mobile money, card… — a sale paid two ways shows under both) and the days it was sold; click the *Teller*, *Total* or *Paid by* heading to sort (*Paid by* sorts by the method that paid most of each sale). The same filters are on **Till shifts** (teller, days, over / short), **Mobile money** payments (cashier, amount, days), **Customer payments** (method, who received it, amount, days), **Invoices**, **Online orders** and **Quotes & orders** (total, days, who entered it) and **Till drawer** (who). **Export** downloads exactly the filtered list. **Why:** to answer questions like "what did Ama take by mobile money last week?" in a few clicks, and to give the accountant just that list.

Close the day (Cash → *Close the day*; owner, general manager and branch manager): one screen per branch shows whether the day is wrapped up — ✓ every till closed and counted, the cash counted and any shortage by cashier, cash handed over and banked (when you use cash control), every sale received from the tills, medicine sales signed off (pharmacies), approvals waiting, and last night's backup. Enter the cash **taken to the bank** with the **deposit slip number** and the **bank account**, and what was **kept in the**

safe, then **Close the day**. The deposit goes into the books as money moved from the tills' cash to that bank account (with cash control, each handover's deposit is recorded on the Cash desk instead). Once the bank statement is imported (Finance → Bank reconciliation), **Recent days** shows for each deposit *on the statement ✓*, *on the statement — match it*, or *NOT on the bank statement — check with the bank*; a deposit missing for more than two days also shows as a check on the next close. A red ✗ (e.g. a till still open) needs a sentence saying why before the day can be closed; amber ! items are reminders. Closing again later (after the till was closed) updates the record. **Recent days** lists the last two weeks: who closed, cash counted, over / short, banked, slip, safe. The daily summary says, per branch, who closed the day or *not yet*. **Why:** one look tells the owner the day is finished and the money is where it should be; a missing deposit or an open till is noticed the same evening, not at the month-end.

Every month: verify the audit log (**System** → **Audit log** → **Verify chain**), check stock (`php artisan erp:stock-check`), take the USB backup drive home if you use one, and every three months practise a full restore on a spare PC (12.3).

You are told when something is wrong. Every hour the server checks itself; when a check turns red — a backup failed or is more than 26 hours old, the weekly restore check failed, the backup passphrase is missing, the disk or memory is nearly full, the server is overloaded, the licence is read-only — everyone who looks after the system settings (owner, IT Admin) gets a notification on the bell and an e-mail, once a day per problem, until it is fixed. **System** → **Health** also shows the server's **load** and **memory** (on Linux servers). **Why:** a backup that quietly stopped is only found when it is needed; now it is found the next morning.

The daily summary (System → Settings → *Daily summary*, off until you switch it on): each evening at the time you choose (default 20:00), the owner and general manager (anyone with *receive the daily summary*) get the day in one message — sales by branch against the same day last week, cash / mobile money / card, cash over or short at the till closes, low, expired and expiring stock, today's losses and unusual ones, and what is waiting for approval. On the bell, by e-mail (on by default) and, if WhatsApp Business is set up, a short WhatsApp. **Why:** the owner sees the whole business in one minute, wherever they are.

Today — the owner's phone view (top of the menu; anyone with *receive the daily summary*): the day on one screen that reads well on a phone and updates every minute — sales against the same day last week, cash / mobile money / card, till closes and cash over or short, what needs attention (server health, refused till work, approvals, expiring stock; tap to open), and per branch the sales, tills still open, who closed the day and whether the deposit is on the bank statement. **Yesterday** is one tap away. On the phone, open the back office, go to **Today**, then the browser menu → **Add to Home screen**: it opens like an app. **Why:** the owner sees the business from anywhere in ten seconds, without the full back office.

The SyncLytics support account can be switched off by the company: **System** → **Settings** → **Security** → **The SyncLytics support account may sign in**. Switch it back on when you want support to look at something.

Who did what, and from which computer

System → **Audit log** lists every change and every sign-in: who, when, what changed, and where from.

Column	What it shows
IP address	The address the server saw. For a cloud server it is the shop's internet router, the same for every PC in that shop. For a server in the shop it is the PC's own address.
Computer	The name you gave the computer (see below), the browser and system (e.g. *Edge 141 on Windows*), and for the **desktop till app** (version 0.25 or later) also the **computer name**, its **MAC address** (network card) and its **address on the shop network**.

Open a row for the details; click the IP address to see roughly where it is (town and internet provider, on ipinfo.io). The **Computer** filter shows everything one computer did; the **Sign-ins** filter shows back-office sign-ins, sign-outs and failed attempts, and till sign-ins.

Name your computers: System → Computers. Every computer is listed the first time someone signs in to the back office from it, with who used it last, when and from which address. Click **Name it** and type a name you will recognise, such as *Front desk PC* or *Manager's laptop*. From then on the audit log shows that name. **What it did** opens its audit log. A computer you do not recognise means someone signed in from a computer you did not expect: change that person's password.

Lists, one line per row: every list keeps each row on one line, so it stays easy to read. Where a value has more to it — the payment methods behind a shift's sales total, who else sold in a shift, a long name or note — it is underlined with dots: hover over it (or tap it on a tablet) for the details. The record's own page and its prints still show everything. **Why:** totals and names line up and can be compared at a glance; the details are one hover away.

Renamed accounts: records keep the name a person had when they did something, so renaming an account never rewrites history. Where the account has been renamed since, a small ⇄ icon shows beside the name; hover over it for today's name. **People & access → Users → the person → Name history** lists every change of name and username, when and by whom. Give every person their own account; do not rename one person's account for someone else, or their work will look like the new person's.

Why browsers show less: for your security, a web browser never tells a website the computer's name or MAC address. So the system gives each browser its own lasting ID instead (kept like a cookie). Clearing the browser's data or using another browser makes it a new, unnamed computer in the list. The desktop till app does report the computer's name and MAC address, also when the back office is opened in it (on its set-up page: **Open the back office**). These are what the computer says about itself: good for telling computers apart, not proof that nobody pretended.

12.2 Backups (automatic, encrypted)

> **Why:** a theft, fire, failed disk or ransomware attack can destroy the server. A backup kept **somewhere else** is the only way to get your sales, stock and accounts back.

The server makes a backup **every night at 02:00** by itself. Each backup holds everything needed to start again: the **database**, the `.env` **settings** (with the keys that unlock stored secrets and seal the audit log) and the **uploaded files**. It is **encrypted** with your backup passphrase, so a lost USB drive or a leaked cloud folder reveals nothing. Every Sunday the newest backup is **checked**: decrypted, read, and — where the database user may create a scratch database — fully restored into it and compared, without touching your live data.

The backup passphrase

The installers create it and show it once. Otherwise, in the `server` folder run `php artisan erp:backup:key`.

> **Warning:** write the passphrase on paper and keep it **away from the server** (with the owner, or in a safe). Without it no backup can be restored.

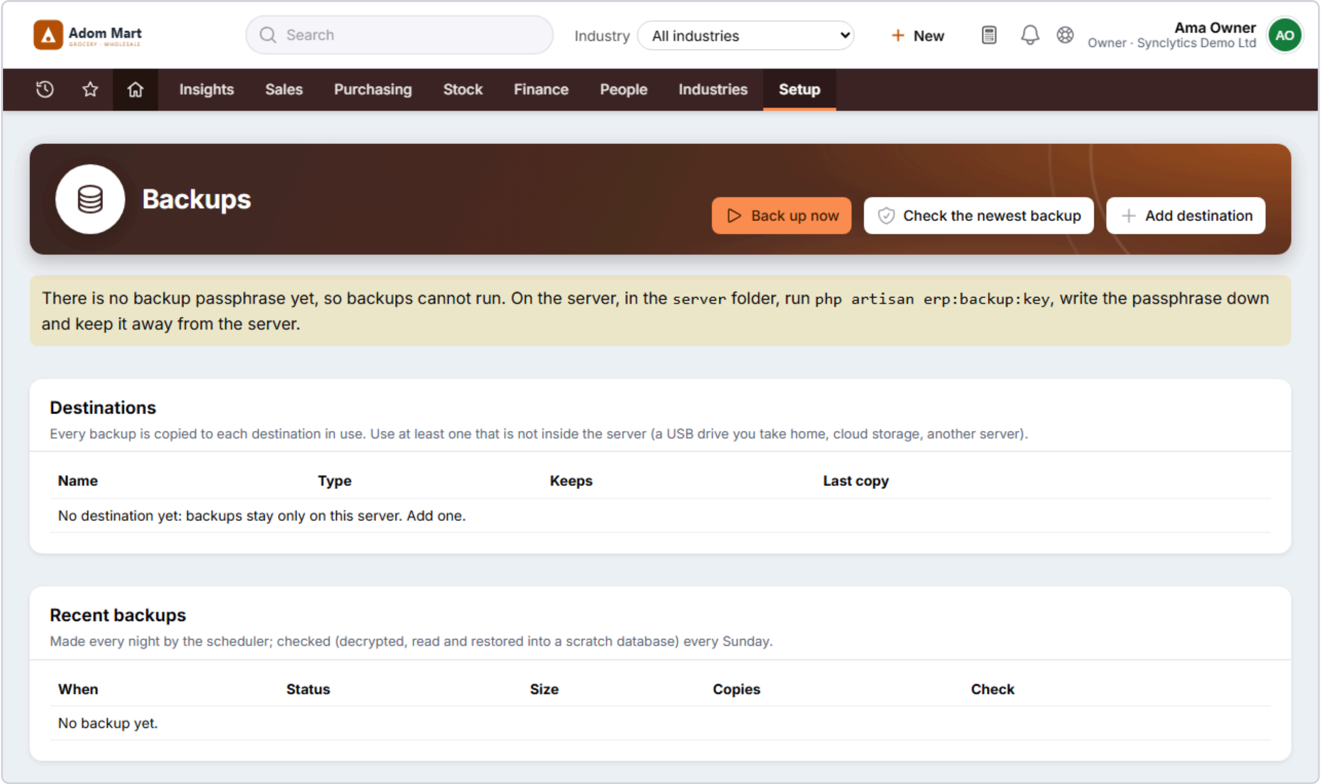
Where backups go

The server keeps the last 7 itself. Add at least one place **outside the server** under **System → Backups → Add destination** (a test file is written and removed before it is saved):

Destination	What to fill in	Good for
Folder — USB drive, second disk, network share	The folder, e.g. `E:\Backups` (Windows) or `/mnt/usb/backups`	Shops: rotate two USB drives, one always at home
S3-compatible storage — Backblaze B2, Wasabi, AWS	Endpoint, region, bucket, access key and secret (from your storage account; give the key access to that bucket only)	Cloud copies at low cost
Google Drive	OAuth client ID and secret, a refresh token, and the Drive folder ID (from a Google Cloud project in **your** Google account — your IT person sets this up once)	Businesses already on Google Workspace
Another server (SFTP)	Address, port, user, password or private key, folder	Head office or a hosting account

Which are included: a **folder** (a USB drive you take home, a second disk, a network share) is always included. **S3-compatible storage, Google Drive and another server (SFTP)** are the **off-site backup** add-on on the licence (System → Licence → *Paid add-ons*); without it, those types show *(paid add-on)* and cannot be saved. Copies a company set up before 7 November 2026 keep working without the add-on. **Why:** every client can always keep a copy away from the server on a drive; copies to the cloud are a service SyncLytics supports and charges for.

Keep the newest … says how many backups each destination keeps; older ones there are deleted automatically.


Adom Mart
GROCERY · WHOLESALE
Search
Industry
All industries
+ New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Backups
Back up now
Check the newest backup
+ Add destination

There is no backup passphrase yet, so backups cannot run. On the server, in the `server` folder, run `php artisan erp:backup:key`, write the passphrase down and keep it away from the server.

Destinations

Every backup is copied to each destination in use. Use at least one that is not inside the server (a USB drive you take home, cloud storage, another server).

Name	Type	Keeps	Last copy
No destination yet: backups stay only on this server. Add one.			

Recent backups

Made every night by the scheduler; checked (decrypted, read and restored into a scratch database) every Sunday.

When	Status	Size	Copies	Check
No backup yet.				

> **Check: Back up now**, then the backup shows ✓ for each destination, and **Check the newest backup** reports *Readable* or *Restore drill passed*. **System** → **Health** then shows *Backups OK*.

> **Note:** on Windows set `BACKUP_PG_BIN=C:\SESLabs\pgsql\bin` (the installer does this). For the full restore drill the database user needs permission to create a scratch database: the installers give it; on an existing server an administrator runs `ALTER ROLE erp CREATEDB;` in PostgreSQL.

12.3 Restoring from a backup

Use this after a disaster, or to practise on a spare PC.

1. Install the server software on the new machine with the installer or script (Chapter 4 or 5) — it creates an empty company you will replace.
2. Copy the newest `.synbak` backup file to it (from the USB drive, cloud storage or **System** → **Backups** → **Download** on the old server if it still works).
3. Put your **backup passphrase** in the new server's `.env` as `BACKUP_PASSPHRASE=…`, then in the `server` folder:

   ```
   php artisan erp:backup:restore C:\path\ses-labs-hq-20261029-020000.synbak
   ```

   It checks the backup, asks for confirmation, and replaces the data.
4. The backup's settings are saved next to `.env` as `.env.from-backup`. Copy `APP_KEY` and `ERP_AUDIT_KEY` from it into `.env` (they must match the backup), then `php artisan optimize`.
5. Restart the server (Task Scheduler on Windows, or reboot).

> **Check:** `php artisan erp:audit-verify` reports the chain intact, `php artisan erp:stock-check` passes, and yesterday's sales are in **Sales** → **Till sales**. If this is a new machine, re-activate the licence (Chapter 6.6).

> **Note:** tills keep any sales made since the backup in their queue and send them when the server is back, so they are not lost.

12.4 Updating the software

SyncLytics sends you a new release. Update outside trading hours.

The simple way (recommended):

- **Windows:** run the new `SESLabs-server-setup-<version>.exe` on the server and choose the same folder.
- **Ubuntu:** `sudo bash install.sh --zip <new release>.zip` with the same options as the first time.

Either one **takes a backup first** with the version still installed. If that backup fails, it stops and changes nothing. Then it replaces the software, keeps your `.env`, uploads, backups and keys, updates the database, prepares the caches, applies the speed settings and restarts the services. Tills update by themselves (step 6 below).

Why the backup first: if anything goes wrong during the update, that backup puts the business back exactly as it was.

By hand (only if SyncLytics support asks you to):

1. **Take a backup** (run the backup task now) and check it.

2. Stop the server: Windows — Task Scheduler → *End* both SES Labs tasks; Ubuntu — `sudo systemctl stop caddy`.

3. Extract the new ZIP **over** the existing folder (`C:\SESLabs` or `/var/www/seslabs`). Your `.env` and `storage` are kept.

4. In the `server` folder:

   ```
   composer install --no-dev --optimize-autoloader
   php artisan migrate --seed --force
   php artisan optimize
   php artisan filament:optimize
   ```

   Why the last two: they prepare the settings, routes, screens and icons once, so pages do not rebuild them on every request.

   (Ubuntu: then `sudo chown -R www-data:www-data storage bootstrap/cache`.) If you install from source code, also rebuild the till: in `pos`, `npm install` and `npm run build:server`.

5. Start the server again (Windows — *Run* both tasks; Ubuntu — `sudo systemctl start caddy` and `sudo systemctl restart php8.4-fpm`).

6. **Tills** download the new version by themselves (they look every 30 minutes) and switch to it at the **sign-in screen**: the next time someone signs out or the cashier changes, the till reloads once and shows the sign-in screen again. **Why:** nobody is in the middle of a sale there, so nothing is interrupted. Tills from before 29 October 2026 need one last full close and reopen of the till app (twice) to get this.

Updating to 0.34.2 or later from an older version: use the simple way above. It switches on PHP's code cache (OPcache) and other speed settings, which older Windows installations did not have, and pages then open several times faster. *System → Health* shows *PHP code cache: On* afterwards. The first update to 0.34.2 also adds database indexes; on a large database this takes a few minutes, so let it finish.

Optional on older installations: in `server\.env` set `LOG_LEVEL=warning` (the log no longer records every detail) and `CACHE_STORE=file` (one less database round-trip for each till request), then run `php artisan optimize` again. New installations have both already.

Why `migrate --seed`: it adds new tables and new permissions safely; it never removes your data or changes your own settings.

With branch servers: update **head office first**, then each branch server.

12.5 Moving to a new server

1. Install the new server (Chapter 4 or 5) without creating keys, tables or the company, as in 12.3 step 1.
2. Take a fresh backup on the old server **after closing trading** and stop the old server.
3. Restore it on the new server (12.3).
4. Point the devices to the new address: if the address changed, open the new till address on each device and enrol it again (System → Devices → New code), after its queue has been sent.
5. Re-activate the licence within 14 days (Chapter 6.6).

12.6 Trying changes on the build PC (preview)

To look at a new version before it goes to a client, on the PC with the source code:

```
node pos/scripts/preview.mjs
```

It builds the till into the server, brings the local database up to date, and starts the server: the back office at **http://127.0.0.1:8010/admin** and the till at **http://127.0.0.1:8010/pos/** — one address, nothing else to set up. In the Claude app, the same is the **ERP preview** entry in the browser pane. Stop it with **Ctrl+C**. **Why:** one command shows exactly what the client will get, without the separate development server on port 5173.

13. Troubleshooting

Find the symptom, then try the fixes in order.

Installation

Symptom	Cause and fix
`php` is not recognised	PHP is not on the path. Repeat 4.3 step 6, then **close and reopen** PowerShell.
`could not find driver` or `pdo_pgsql` missing	The extension is not switched on. Check `php.ini` (4.3 step 5) and that `extension_dir = "ext"`. Run `php -m` to confirm.
PHP does not start: *VCRUNTIME140.dll missing*	Install the Visual C++ Redistributable (4.3 step 1).
`SQLSTATE[08006] … password authentication failed for user "erp"`	`DB_PASSWORD` in `.env` differs from the one in 4.2/5.3. Fix `.env`, then `php artisan optimize`.
`SQLSTATE[08006] … Connection refused`	PostgreSQL is not running. Windows: Services → *postgresql-x64-17* → Start. Ubuntu: `sudo systemctl start postgresql`.
`composer install` fails with a PHP extension message	Switch on the extension it names in `php.ini` (Windows) or install `php8.4-<name>` (Ubuntu).
`ERP_AUDIT_KEY is missing`	Run `php artisan erp:audit-key` (only on a new installation — never replace an existing key).
The page shows *500 Server Error*	Look at the end of `server/storage/logs/laravel.log` for the reason. After changing `.env`, always run `php artisan optimize`.
Ubuntu: *Permission denied* in the log	`sudo chown -R www-data:www-data storage bootstrap/cache`
Uploading a CSV file does nothing, or *failed to upload*	PHP cannot write its temporary upload file. Windows: check `upload_tmp_dir` in `php.ini` points to an existing folder (4.3 step 5) and restart the server task.
`/pos/` says the till app is not installed	Build it: in `pos`, `npm install` then `npm run build:server` (4.7).

Back office

Symptom	Cause and fix
Lost the authenticator phone, or deleted the app	Use a **recovery code**: type it where the 6-digit code is asked for. If those are lost too, someone who manages users (an owner or the IT Admin) opens **Users**, picks the person and presses **Reset two-step sign-in**, after checking by phone or in person that it is really them. **Why:** whoever can reset it can take over the account. Nobody can reset themselves, nor anyone above their own level.

Symptom	Cause and fix
The owner is locked out (no app, no recovery code)	Nobody in the company is above the owner, so SyncLytics support does it on the server after checking it is really the owner: `sudo -u www-data php artisan erp:mfa-reset --email=owner@example.com` (in the `server` folder). The owner then signs in with username and password and scans a new code. **Why on the server:** only the people who run the server can type it, and it is kept in the audit log. Afterwards, save the new recovery codes away from the phone.
A menu is missing	The module is off for your branches (Chapter 8), your licence does not include it (Chapter 6), or your role does not allow it (ask the owner). Type part of the page's name in **Search the menu** (top of the side menu, or press `/`) to check.
You can't open this page	The page says **why** (for example *it needs the permission "View users and their access"*) and **who can give access**. Press **Ask for access**: they get a notification with the page and your message, and decide. A link you cannot use should not show at all; if you find one, tell SyncLytics support.
This is managed at head office	You are on a branch server; change master data at head office (Chapter 10).
Everything is read-only, with a red banner	Licence expired, withdrawn or not checked in for 44 days (Chapter 6.4). Renew; nothing is lost.
Passkey button does nothing	Passkeys need HTTPS or `localhost`. Use the authenticator app.
Could not reach the server pops up	Parts of the back office refresh themselves (the bell, the dashboard, Today). One refresh missed on a shaky line is ignored; the message shows only when the connection stays down, once, and *Connected again* follows when it is back. When it appears right after you clicked something, what you did may not have gone through: check the connection and do it again. If it happens often, check the shop's internet or Wi-Fi. *You were signed out* means the back office was left unused for a while: press F5 and sign in. *This did not go through* says what failed: a detail left empty, a value already used, someone else changing the same thing, or a fault in the program. Correct it and try again, or send SyncLytics support the **reference** in the message, which finds the full error in the server log. The message stays until you close it. Personal details never appear in it. **Why:** the message names the cause, so you know whether to wait, sign in or call.
Open a shift on this till before selling, and signing in again goes straight back to selling	The till's shift was closed (by mistake, on another till or in the back office). Update the server (release of 2026-10-28 or later), then sign out and in on that till: *Open shift* appears; type the cash in the drawer as the float (a manager allows it with username and PIN when the till asks). **Why:** the till keeps its own note of the open shift so it can sell offline; it now drops it when the server says the shift is closed.
Receipts print several seconds after the sale, while online	Update the server (release of 2026-10-28 or later): before it, every sale made every till download the whole product list again, which kept the server busy and held up the next sale. Updating with the installer also gives the server more workers (from its memory, 8 to 32; it was 5). **Check:** under System → Health, *Slow pages* should no longer list `/api/v1/pos/catalog` many times a day. **Why:** a sale waits for a free worker; with the product list sent only when products or prices change, workers are free for sales.

Symptom	Cause and fix
Pages are slow	Open **System** → **Health**. *PHP code cache* must say *On* (if not, update with the simple way, Chapter 12.4). *Slow pages* lists what was slow, how often and how many database reads it made: send it to SyncLytics support. Run `php artisan erp:load-test --force` outside trading hours for a full speed check (it leaves nothing behind). For a page-by-page check, `php artisan erp:page-timings` (in the `server` folder) opens every page as the owner and lists the slowest with their database reads; it changes nothing. Send its output to SyncLytics support. Its times are a little slower than in the browser (the code cache is usually off for commands), so a page with few, quick reads that shows about one second is normal there; *Slow pages* under System → Health shows what people actually waited.

Tills

Symptom	Cause and fix
Till shows *working offline* but the server is up	Network or Wi-Fi between till and server. Open `https://<server-address>/up` on the till's browser. On Windows servers, check the firewall rule (4.8, step 8).
The browser warns *Your connection is not private*	The device does not trust the server's certificate yet (4.9), or the server's address changed since installation (run the installer again, then 4.9 on each device).
The till does not work at all when the server is off	Offline was never enabled on that device: setups B/C/E need 9.3; then open the till once while the server is on.
Enrolling a till fails on a network address (`http://192.168…`)	The browser does not trust the address yet: do 9.3 on that device.
Cashier cannot sign in offline	They must have signed in on **this** till while it was online within the offline period (System → Settings → Offline till sign-in lasts).
Your sign-in on this till has ended (payment screen)	The cashier's sign-in lasts 12 hours from the last time the till was used, so a till left idle (e.g. overnight) asks again. Enter your own username and PIN in the box; the sale completes and nothing in the cart is lost. **Why:** a till nobody is using should not stay signed in for anyone who walks up to it. Older tills showed *No offline sign-in for you on this till* instead; update the till (close and reopen it).
PIN locked	Too many wrong PINs. A manager resets it: People & access → Users → the person → **Set POS PIN**.
Red banner: items refused	A supervisor opens **Sales** → **Review queue** (*sync failed*) to see why, fixes it, then presses **Retry refused** on the till.
The price … changed. Refresh the till	The price changed at head office while the item was in the cart. Remove the line and add it again.
Open a shift on this till before selling	Open the shift (count the float).
The till clock is ahead of the server	Set the till's clock to automatic (Windows Settings → Time).
Receipt is tiny or cut	Printer paper size 80 mm, margins *None*, scale 100 %, headers off (9.7).
Scanner types the code but nothing happens	The scanner does not send Enter (9.8).

Symptom	Cause and fix
Desktop till: *Command … not allowed by ACL* when printing, or the drawer does not open	The till app is older than the server expects. Install the latest till app (9.7) over the old one; settings are kept. **Why:** the app decides which of its commands the till page may use, so this is fixed on the PC, not on the server.
Desktop till: *Could not reach …* on the set-up page, or a blank window	The server address is wrong. Use the server's name exactly as in the browser (`https://erp.clientdomain.com`), not its IP address. Correct it and click **Save** (9.7). Old app versions open the saved address with no way back: clear the settings as in 9.7, *Starting the desktop till again from scratch*.
Till did not pick up a new version	Close the till app completely and open it again (twice if needed).

Branch servers and online stores

Symptom	Cause and fix
Sync Center: *Sync request time … off*	Server clocks differ by more than 5 minutes; set automatic time on both.
Changes *failed — update this node*	Update the branch server to the same version as head office (12.4).
A branch seems out of step	Sync Center → **Full resync from this peer**, then `php artisan erp:stock-check`.
Online store not updating	Is the scheduler running (4.8 / 5.9)? Is the channel **Active**? Look at the channel's failed updates and **Retry failed updates**.
No SMS arrive	Check the SMS gateway address in Settings and that the phone number is on the booking. Test your provider separately.

Products imported twice

If a product file was imported more than once with **Products — create new** before version 0.35 (when one import did both), each product may exist two or three times. On the server, in the `server` folder:

```
php artisan erp:merge-duplicate-products
```

This only **shows** what it would do: for each product, the copy created first is kept, and the later copies (same name, industry and unit, from a later import) are listed. Then:

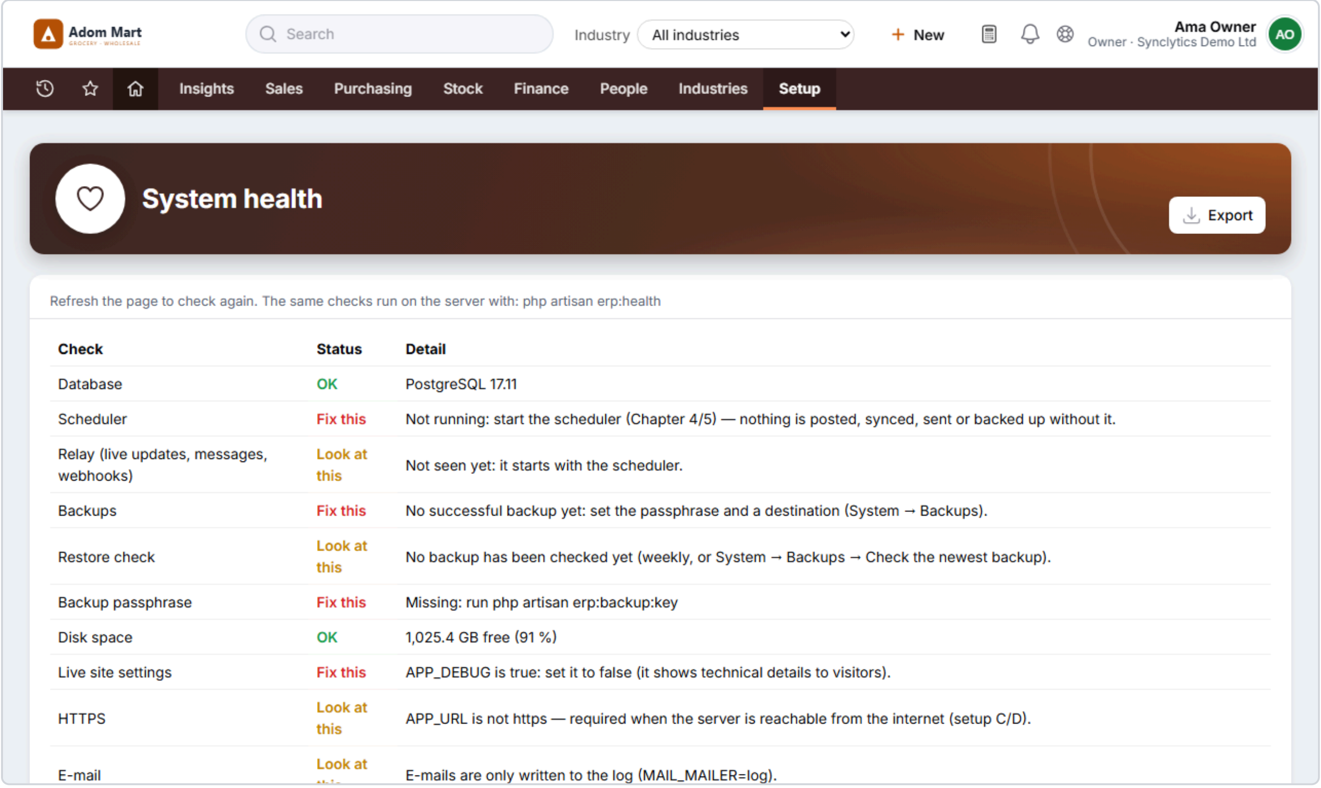
```
php artisan erp:merge-duplicate-products --apply
```

Barcodes on the later copies move to the kept product; a later copy that was never used is removed; one that has stock, sales or orders is switched off instead (it keeps its history). Every step is in the audit log. Two different products with the same name in one file stay two products.

Why on the server: removing products is permanent, so it is a deliberate step by the person looking after the server, after the automatic backup the update takes.

Health checks

System → Health shows the state of the server at a glance: database, scheduler, live updates, last backup and its check, disk space, live-site settings, HTTPS, e-mail, licence, PHP's code cache, slow pages, and anything stuck (messages, webhooks, GRA E-VAT, sync). Each line says what to do. Below the checks, **Slow pages** lists every page and till call that took over a second in the last 7 days, worst first.



System health

Refresh the page to check again. The same checks run on the server with: php artisan erp:health

Check	Status	Detail
Database	OK	PostgreSQL 17.11
Scheduler	Fix this	Not running: start the scheduler (Chapter 4/5) — nothing is posted, synced, sent or backed up without it.
Relay (live updates, messages, webhooks)	Look at this	Not seen yet: it starts with the scheduler.
Backups	Fix this	No successful backup yet: set the passphrase and a destination (System → Backups).
Restore check	Look at this	No backup has been checked yet (weekly, or System → Backups → Check the newest backup).
Backup passphrase	Fix this	Missing: run php artisan erp:backup:key
Disk space	OK	1,025.4 GB free (91 %)
Live site settings	Fix this	APP_DEBUG is true: set it to false (it shows technical details to visitors).
HTTPS	Look at this	APP_URL is not https — required when the server is reachable from the internet (setup C/D).
E-mail	Look at	E-mails are only written to the log (MAIL_MAILER=log).

The same checks run with `php artisan erp:health` (useful for monitoring tools: it fails when something needs fixing).

Individual checks (run in the `server` folder)

Command	What it checks
`php artisan about`	PHP, database and settings in use
`php artisan erp:audit-verify`	The audit log is complete and untouched
`php artisan erp:stock-check`	Stock levels match the stock ledger
`php artisan erp:gl-check`	The accounts agree with sales, stock, customers and suppliers (Finance)
`php artisan license:status`	Licence state, limits and last check-in
`php artisan erp:load-test --force`	Speed: posts test sales and times the heavy reports, searches and the till catalogue on your real data, then undoes everything. Lines over a second are marked *SLOW*

Website copiers and robots

The system is a private business system. Tools that try to download a whole website (HTTrack, Cyotek WebCopy, Offline Explorer, wget and similar) are refused with *Copying this site is not allowed* — by the web server and again by the application — and anyone not signed in who asks for more than 120 pages a minute is told to wait a minute. Search engines and web archives are asked not to index or keep any page.

What a copier could get anyway: only the sign-in page and the till's screen code, which is compressed and does nothing without your server, an enrolled till and a valid licence. The system's code, the database, passwords and keys never leave the server. **Why:** copying the pages cannot copy the system, and the blocking keeps such tools from loading the server.

If a legitimate tool of yours is refused (for example a monitoring service), give it a normal browser name, or ask SyncLytics support.

Getting help from SyncLytics

Hotline (support)	024 563 6889
Office	055 668 6975 · 050 693 8589
E-mail	info@synclyticsenterprisesystems.com
Website	synclyticsenterprisesystems.com
Address	Puff Cl, Race Course St, Tabora, Accra (GhanaPost GPS GC-027-0884)

The quickest way is **Report a problem** in the back office (top right) or on the till: it attaches the technical details for you, and a picture of the screen (the till takes it by itself; in the back office press **Win+Shift+S**, then **Ctrl+V** in the box). Whoever may give support access can tick *Let SyncLytics support connect* there.

Support access (free). System → Support access → Give support access lets SyncLytics support reach your tills and computers to help — *see only* or *see and control*, every branch or one, for a few hours or until a problem report is closed. Everyone signed in sees a banner while it is open, and you can withdraw it at any time. **Why:** support can fix things without travelling, and only when you say so. The company's own staff connecting to devices (training, IT) is the paid *Remote access* add-on.

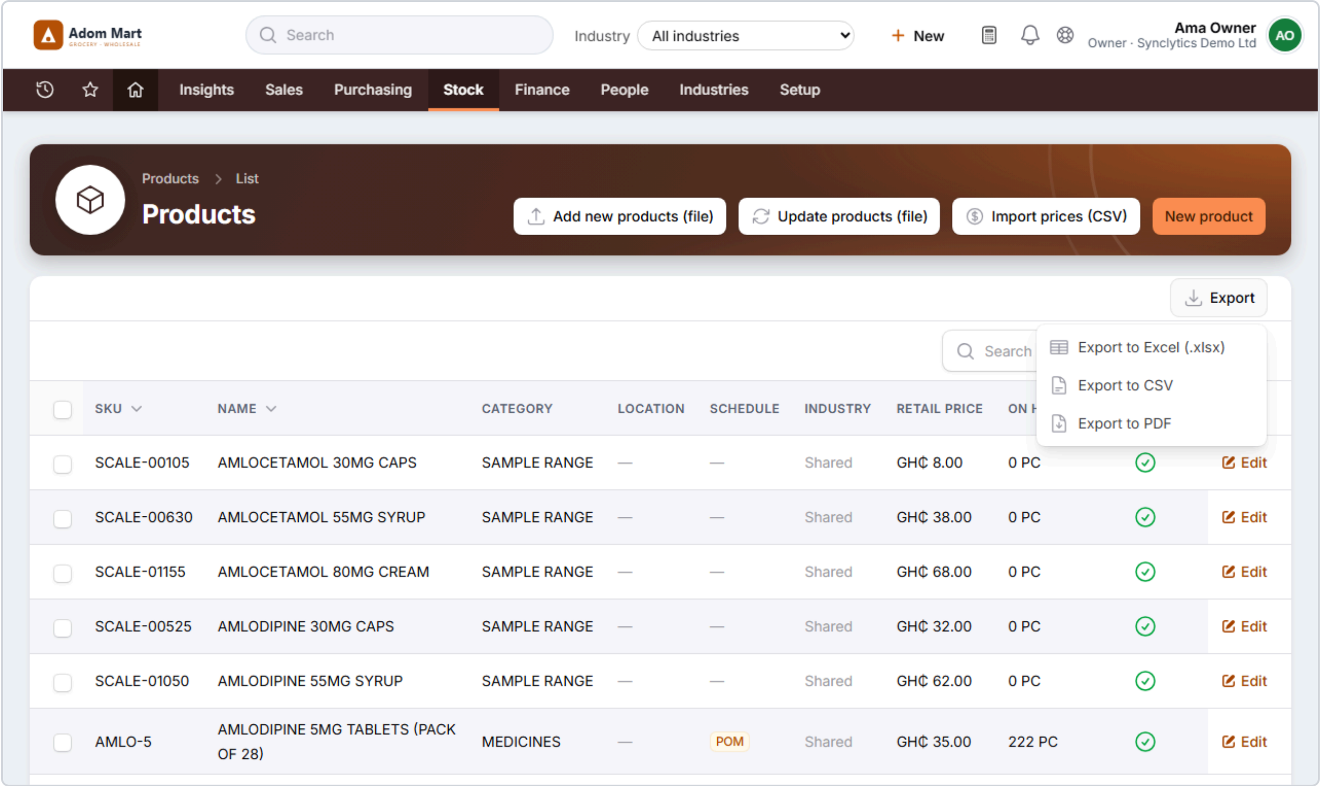
Remote connections and trading hours. System → Settings → Remote access: by default a connection is unattended outside trading hours (07:00–21:00 unless changed) and must be accepted by the person at the device during them; a branch can choose differently (Company → Branches).

Otherwise send: what you were doing, the exact message (a screenshot helps), the time it happened, and the last lines of `server/storage/logs/laravel.log`. **Never** send your `.env` file, passwords, PINs or recovery codes.

14. Exporting and importing data, stock adjustments and stock takes

14.1 Exporting (downloading) anything you can see

Every list in the back office (products, customers, invoices, stock on hand, sales, audit log…) and every report page (financial reports, aged debts, shrink, commissions, timesheets, the staff calendar…) has an **Export** button at the top right:


Adom Mart
Search
Industry
All industries
+ New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Products > List
Products
Add new products (file)
Update products (file)
Import prices (CSV)
New product
Export
Search
Export to Excel (.xlsx)
Export to CSV
Export to PDF

☐	SKU	NAME	CATEGORY	LOCATION	SCHEDULE	INDUSTRY	RETAIL PRICE	ON H		
☐	SCALE-00105	AMLOCETAMOL 30MG CAPS	SAMPLE RANGE	—	—	Shared	GH₵ 8.00	0 PC	✓	Edit
☐	SCALE-00630	AMLOCETAMOL 55MG SYRUP	SAMPLE RANGE	—	—	Shared	GH₵ 38.00	0 PC	✓	Edit
☐	SCALE-01155	AMLOCETAMOL 80MG CREAM	SAMPLE RANGE	—	—	Shared	GH₵ 68.00	0 PC	✓	Edit
☐	SCALE-00525	AMLODIPINE 30MG CAPS	SAMPLE RANGE	—	—	Shared	GH₵ 32.00	0 PC	✓	Edit
☐	SCALE-01050	AMLODIPINE 55MG SYRUP	SAMPLE RANGE	—	—	Shared	GH₵ 62.00	0 PC	✓	Edit
☐	AMLO-5	AMLODIPINE 5MG TABLETS (PACK OF 28)	MEDICINES	—	POM	Shared	GH₵ 35.00	222 PC	✓	Edit

Choice	What you get	Use it for
Export to Excel (.xlsx)	An Excel workbook (numbers stay numbers; opens in Excel, Google Sheets, LibreOffice, Numbers)	Working with the figures: sorting, totals, sending to your accountant
Export to CSV	A plain `.csv` file	Loading into another program
Export to PDF	An A4 PDF with your company name, the title, the filters in use, who printed it, the date and page numbers	Printing, filing, e-mailing

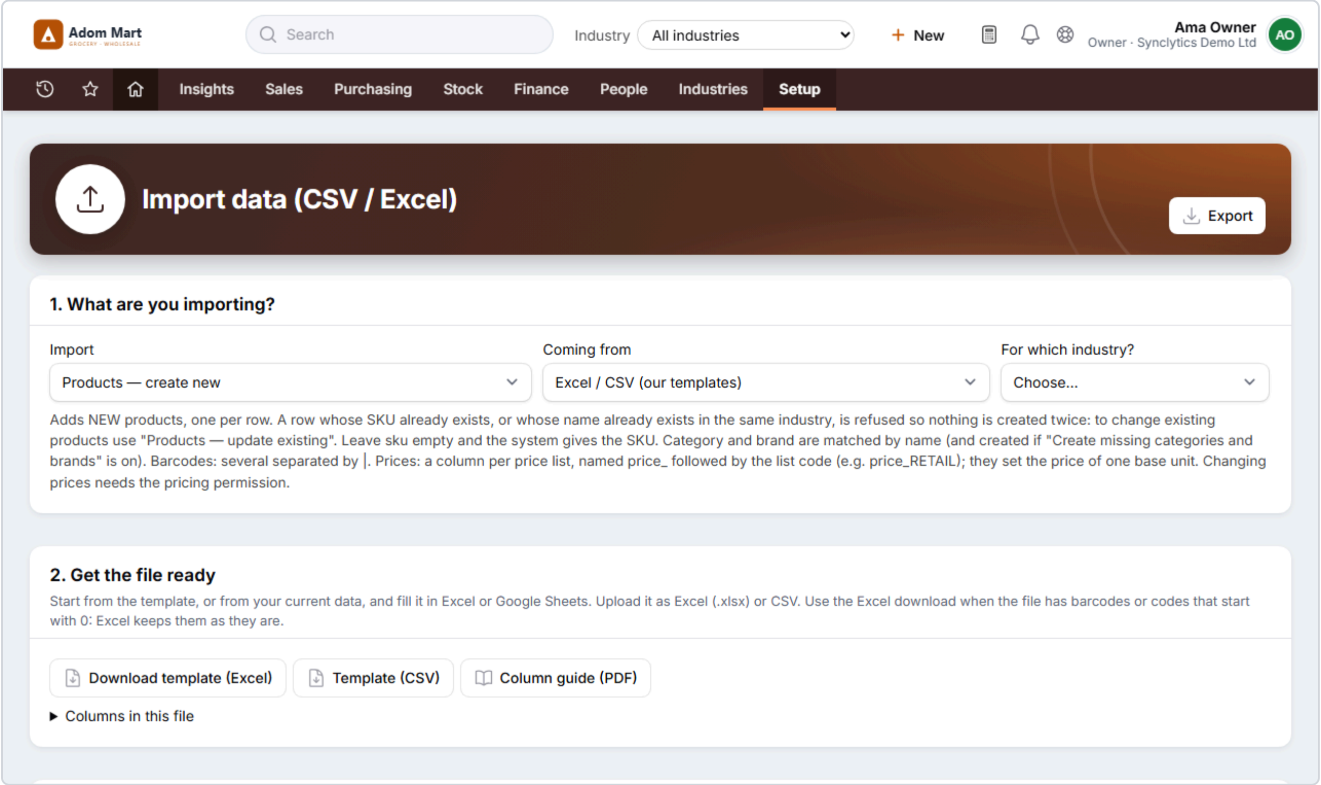
- The file contains **what is on screen**: the same columns, search, filters and sort — but **all pages**, not only the page you are looking at. Filter first (e.g. one branch, one month), then export.
- Amounts appear as on screen (with the currency); dates in your local time.
- A PDF holds up to 5,000 rows per table. For more, use CSV.

> **Why:** every export is recorded in the audit log (who, what, how many rows), because exported files leave the system. When the licence is read-only, exports are blocked like printing.

> **Note:** A CSV opens in Excel by double-clicking it. Accents and the cedi sign show correctly. A cell starting with `=`, `+`, `-` or `@` that is not a number gets a leading apostrophe so that Excel never runs it as a formula — this protects you from "CSV injection".

14.2 Importing data from CSV files

System → Import data (CSV / Excel) — or the **Import … (CSV)** buttons on Products, Customers, Suppliers, Stock adjustments and stock counts.


Adom Mart
Search
Industry
All industries
New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Import data (CSV / Excel)
Export

1. What are you importing?

Import: Products — create new

Coming from: Excel / CSV (our templates)

For which industry?: Choose...

Adds NEW products, one per row. A row whose SKU already exists, or whose name already exists in the same industry, is refused so nothing is created twice: to change existing products use "Products — update existing". Leave sku empty and the system gives the SKU. Category and brand are matched by name (and created if "Create missing categories and brands" is on). Barcodes: several separated by |. Prices: a column per price list, named price_ followed by the list code (e.g. price_RETAIL); they set the price of one base unit. Changing prices needs the pricing permission.

2. Get the file ready

Start from the template, or from your current data, and fill it in Excel or Google Sheets. Upload it as Excel (.xlsx) or CSV. Use the Excel download when the file has barcodes or codes that start with 0: Excel keeps them as they are.

Download template (Excel)
Template (CSV)
Column guide (PDF)
▶ Columns in this file

What you can import

Import	What a row does	Matched by
Products (create and update)	Creates a product (leave SKU empty: the system gives it one), or updates an existing one by its SKU. Prices per list in columns `price_RETAIL`, `price_WHOLESALE` …	SKU
Prices (price list update)	Sets one price (list, unit, from-quantity, optional dates for a promotion)	SKU + list + unit + quantity + dates
Customers	Creates or updates a customer, optionally with credit limit and terms	Code, else phone number
Suppliers	Creates or updates a supplier	Code

Import	What a row does	Matched by
Opening stock	The go-live quantity and cost of a product in a warehouse (one row per batch, with its expiry, for batch-tracked products). Works as a count sheet: list every product, fill in what was counted, and rows left empty (or 0) are skipped	Warehouse + SKU, barcode or exact name (+ batch)
Stock adjustment lines	Increase, decrease or "set to" lines in a stock adjustment	SKU or barcode
Stock count results	Counted quantities in an open stock count	SKU or barcode (+ bin)
Chart of accounts (migration)	Adds an account from the old books (QuickBooks types accepted)	Code, else name
Opening trial balance (migration)	One line of the opening journal on the go-live date	Account code or name
Customer / supplier balances owed (migration)	What a customer owed you / you owed a supplier on the go-live date	Code, phone or name
Loyalty points / gift cards (migration)	Points and unspent cards brought over from the old system	Customer; card number

Reorganising the catalogue: Catalog → Products: tick products → **Change category, brand or tax**. Catalog → Categories (or Brands): **Move products** moves all of them to another one — the way to merge two duplicates — and **Delete** removes one that nothing uses. Every product moves, however many there are, and each product's history records the move. **Why:** history stays intact; only empty lists can go. **Missing category, brand or unit?** On the product form, click **+** next to the list to add it without leaving the page (also suppliers on purchase orders and bills, customers on orders, departments on staff). **Changing a product's unit (e.g. Pack or Strip to Tablet):** open the product and click **Change the unit**, choose the new unit and one of:

- **Convert by pack size** — say how many of the new unit are in one of the current unit (10 tablets in a strip). Stock in every store and batch is multiplied by it, the cost per tablet is the strip cost divided by it (the stock value does not change), the tablet price starts at the strip price divided by it (change it if you like), and reorder levels follow. The strip stays as a selling unit (× 10) with its price and barcode, so you can still sell whole strips.
- **The label was wrong** — the numbers stay (30 Pieces become 30 Packs); prices, barcodes and stock stay.

Many products at once: tick them on the product list → **Change the unit** (same pack size for all), or **System** → **Import data** → **Unit changes** with one row per product (`sku`, `new_unit`, `per`, `mode`) when pack sizes differ. A product on an open document (a draft order, a transfer on the way, an open count) is refused until that is finished. **Why:** the stock history is never rewritten: the change is recorded as stock leaving in the old unit and coming back in the new one, so every figure still adds up, and past receipts and documents keep showing what they showed.

Imported twice by mistake? The system remembers each file: importing exactly the same file again is stopped with when and by whom it was imported. Tick **Import it again** only if you really mean to add it a second time. After an import the page empties, ready for the next file.

How far it has got: a long file shows a card at the bottom right — *Checking products.csv — 150 of 2,000* and about how long is left. Keep the page open until it finishes. **Why:** you can see it is working, so nobody clicks again or closes the page half-way.

Medicines in the products file: `drug_schedule` says what kind of medicine a product is: `otc` (over the counter), `pom` (prescription only: a pharmacist approves it at the till) or `controlled` (pharmacist approval, and it goes in the controlled-drug register). Common words also work (POM, prescription, CD, controlled); `none` means not a medicine. With it: `generic_name`, `strength`, `dosage_form`, `fda_reg_no`, `active_ingredients` (separated by `|`, for allergy and interaction warnings), `daily_use` (refill reminders) and `nhis_price` (what the NHIS pays for one base unit; `none` = not covered). *Download current data* includes them, so you can fill the column in Excel and upload it with **Products — update existing**. **Why:** the till can only ask for a pharmacist, and the register can only list controlled drugs, when the system knows which products they are.

The migration imports are explained in Chapter 17.2. **Coming from** (QuickBooks, another till system, or our templates) matches the old system's column names for you.

Step by step

1. **Choose what you are importing.** The page explains what a row does and lists every column (**Columns in this file**), with an example. Products have **two separate imports**:
    - **Products — create new** adds new products only. A row whose SKU already exists, or whose name already exists in the same industry, is refused, so importing the same file twice cannot create duplicates. Tick **Allow names that already exist** only for genuinely different products with the same name.
    - **Products — update existing** changes existing products only, found by their SKU: start from **Download current data**, change what you need, and import it. Empty cells and missing columns stay as they are. It never creates a product, so importing it twice is harmless.

    Why two: with one import, a row without an SKU always made a new product, so re-importing a file to "update" it created a second copy of everything. For **products**, a company with several industries also chooses **For which industry?** (for example *Pharmacy*, or *Shared by all industries*). New products go into that industry's catalogue, and **Download current data** gives that industry's products only. Check and Import stay refused until you choose. An `industry` column in the file can name a different industry for single rows.

    > **Why:** each industry's tills and screens show only its own products. A product list imported to the wrong industry would show at the wrong shops.

    Products already in the wrong industry: Catalog → Products, tick them (the tick at the top selects the page; *Select all* selects every product in the list), then **Move to industry**. Stock and prices stay; SKUs given by the system take the new industry's series (e.g. `100123` becomes `PHARMA-000124`), while SKUs from your old system stay as they are.

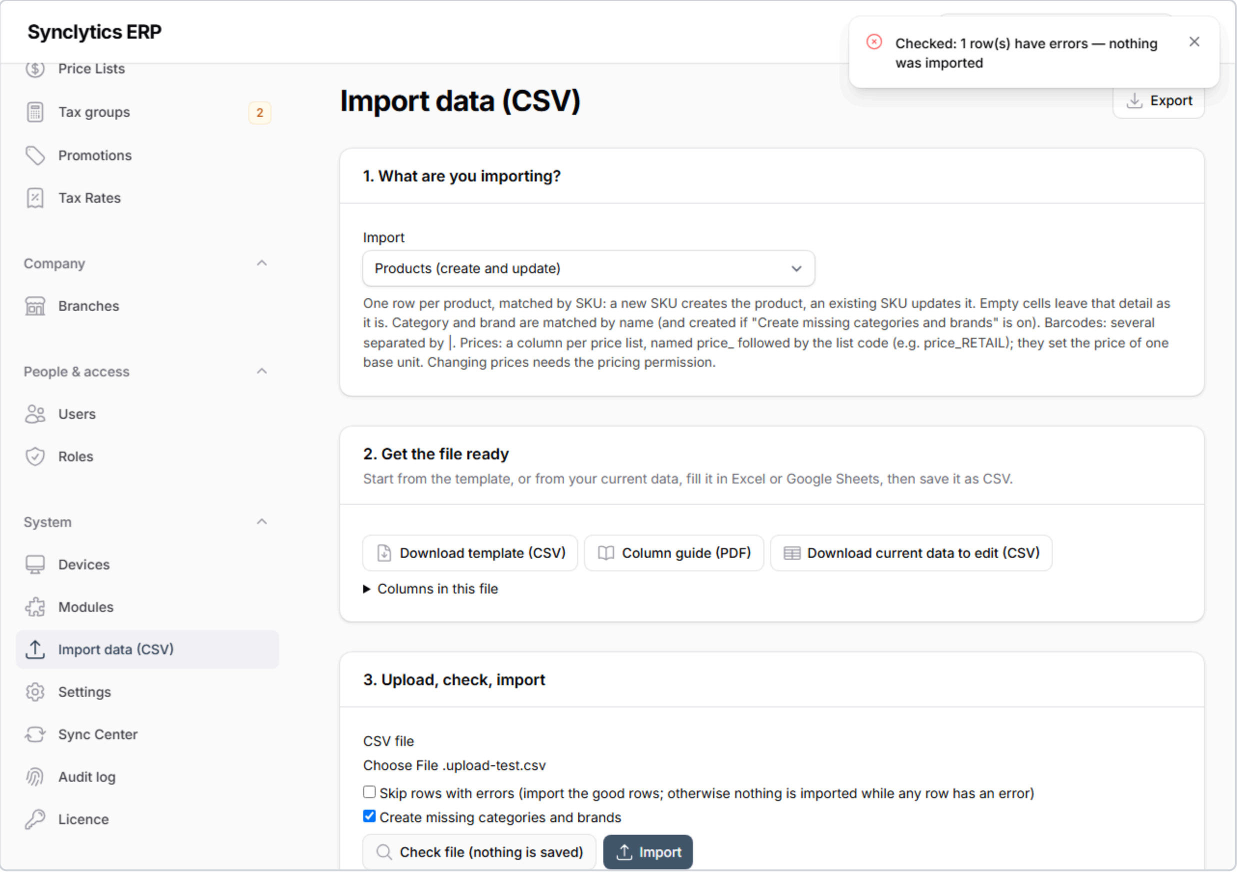
2. **Get the file ready:**

    - **Download template (Excel)** — the headings and one example row (also as **CSV**); or
    - **Download current data to edit (Excel)** — your existing products, prices, customers or suppliers in exactly the import layout (also as **CSV**). Change what you need in Excel and import it back;
    - **Column guide (PDF)** — what to put in each column, to print.

3. Fill it in Excel or Google Sheets and save it, as Excel (.xlsx) or as CSV (Excel: *File → Save As → CSV UTF-8 (Comma delimited)*).

   Barcodes: before typing or pasting barcodes in Excel, select the barcode column and choose *Format Cells → Text*. **Why:** otherwise Excel treats a barcode as a number. It drops the zeros in front (`0012345678905` becomes `12345678905`) and shortens long ones (`6.00107E+12`), so the scanner would no longer find the product. The Excel downloads already have codes as text. On import, zeros Excel dropped are put back when the barcode's check digit proves them, and a shortened barcode is refused with a message.

4. **Choose the file**, then **Check file (nothing is saved)**. Every row goes through exactly the same checks as the screens — then everything is undone. You see each row's result and a list of errors.



Synclytics ERP
Price Lists
Tax groups
2
Promotions
Tax Rates
Company
Branches
People & access
Users
Roles
System
Devices
Modules
Import data (CSV)
Settings
Sync Center
Audit log
Licence
Export

Checked: 1 row(s) have errors — nothing was imported

Import data (CSV)

1. What are you importing?

Import: Products (create and update)

One row per product, matched by SKU: a new SKU creates the product, an existing SKU updates it. Empty cells leave that detail as it is. Category and brand are matched by name (and created if "Create missing categories and brands" is on). Barcodes: several separated by |. Prices: a column per price list, named price_ followed by the list code (e.g. price_RETAIL); they set the price of one base unit. Changing prices needs the pricing permission.

2. Get the file ready

Start from the template, or from your current data, fill it in Excel or Google Sheets, then save it as CSV.

Download template (CSV)
Column guide (PDF)
Download current data to edit (CSV)
▶ Columns in this file

3. Upload, check, import

CSV file: Choose File .upload-test.csv

☐ Skip rows with errors (import the good rows; otherwise nothing is imported while any row has an error)

☑ Create missing categories and brands

Check file (nothing is saved)
Import

5. Fix the errors (**Download error report (CSV)** gives the failed rows with the reason in the first column), and check again.

6. **Import.** By default a file is imported **all or nothing**: if any row has an error, nothing is imported. Tick **Skip rows with errors** to import the good rows and get the others in the error report.

Why the check first: a mistake in a file can change hundreds of products at once. The check shows you exactly what would happen, and what would be refused, before anything changes.

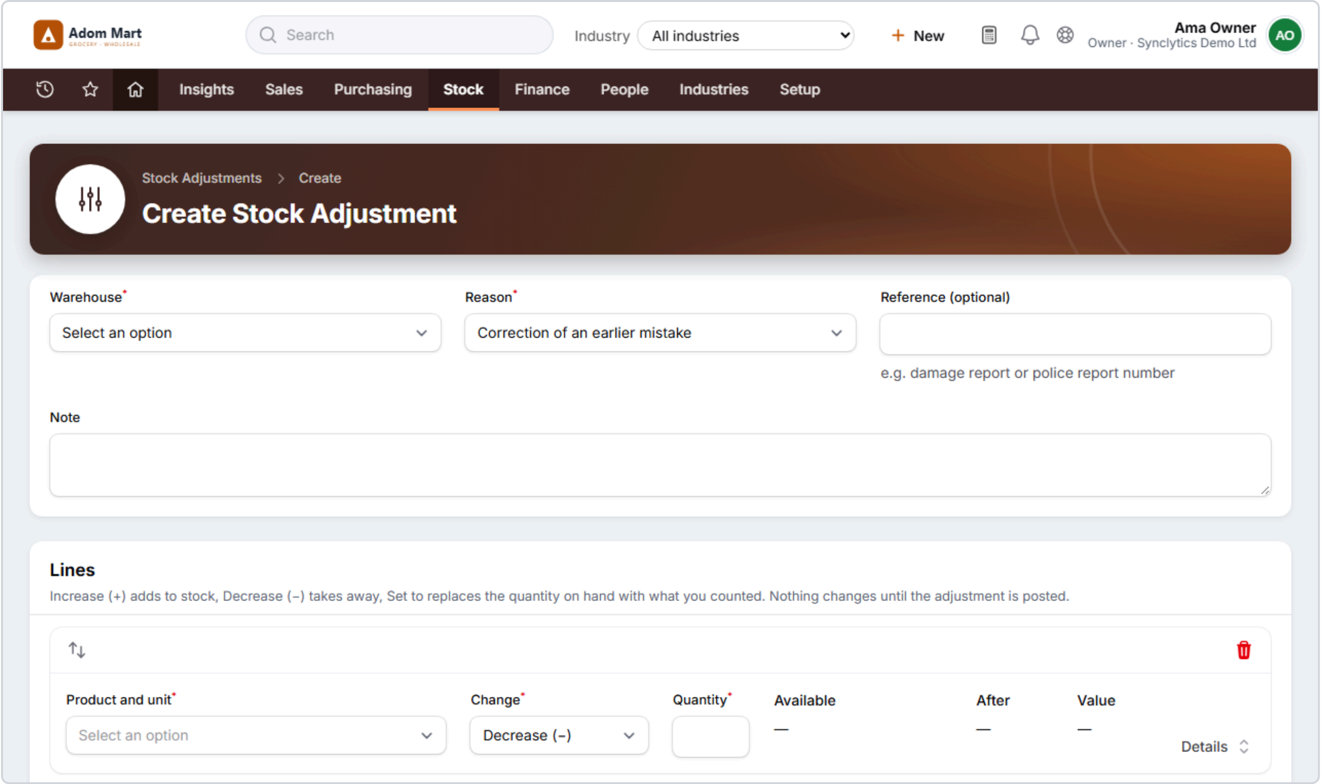
Rules worth knowing

- **Empty cells leave a detail unchanged** when updating. To change only prices, a file with just `sku` and `price_RETAIL` is enough.
- **Headings** are matched loosely: `Unit cost`, `unit_cost` and `UNIT COST*` are the same column. Columns the import does not know are listed and ignored.
- **Units** (`base_unit`, `unit`) can be the code or the name: `TAB`, `Tablet`, `Strips` and `Cont.` all work.
- **Yes/no columns** accept yes, no, y, n, true, false, 1, 0.
- **Numbers** may have thousands separators (`1,250.50`). **Dates**: `2026-10-31` or `31/10/2026`.
- **Location** (products file): where the product is kept, e.g. `Shelf B2`. It shows on the product, the stock sheet and the till's search list.
- **Barcodes and serial numbers**: several in one cell, separated by `|`. In CSV downloads a code starting with 0 appears as `="0123…"`, which is how Excel is told to keep the zeros; leave it like that.
- **Large files** (more than 500 rows) run **in the background** on the server. The page shows *Waiting to start*, then the progress (e.g. *1,200 of 2,467 · About 3 min left*). You can keep working or close the page. When it is done, a message appears on the page if it is open, and the bell has a notification with **Open**. After a **check**, press **Import this file** on the result (or **Open** in *Import history*): it imports the copy already on the server, with the same options, so there is no need to upload it again. If some rows have errors, **Import, skipping the rows with errors** imports the rest. **Why:** a web page may only wait a few minutes for the server, so a big file is never cut off halfway. Only one large file runs at a time, and the same file cannot be sent twice while it waits. Smaller files are done while you wait, as before. The tills and the rest of the back office keep working meanwhile: the file goes in 50 rows at a time. *All or nothing* still holds: the whole file is checked first and imported only if no row has an error. (Opening balances and stock adjustments go in as one piece, since their rows make one document.)
- Each row is checked with **your permissions**: changing prices needs the pricing permission, credit limits need the credit permission.
- Every import is listed under **Import history** on the same page and recorded in the audit log under your name.

> **Warning:** Only import customers who agreed to be on your records, and never use real customer data in a test system.

14.3 Stock adjustments

Stock → Stock adjustments. A stock adjustment changes stock outside sales and purchases — breakage, theft, expiry, samples, items found, or correcting a mistake. One adjustment covers one warehouse and has as many lines as you need.



Create Stock Adjustment

Warehouse*: Select an option

Reason*: Correction of an earlier mistake

Reference (optional):

e.g. damage report or police report number

Note:

Lines

Increase (+) adds to stock, Decrease (−) takes away, Set to replaces the quantity on hand with what you counted. Nothing changes until the adjustment is posted.

Product and unit*	Change*	Quantity*	Available	After	Value	
Select an option	Decrease (−)		—	—	—	Details

1. **New stock adjustment:** the warehouse, the **reason** (damaged, expired, theft / missing, found / surplus, own use / sample, correction...) and an optional **reference** (e.g. a damage report number).
2. **Lines** — for each product, one of:
   - **Increase (+)** — adds to stock (give the unit cost if you know it; otherwise the average cost is used);
   - **Decrease (−)** — takes away from stock;
   - **Set to counted quantity** — the new quantity on hand; the system works out the difference when you post. Batch-tracked products need the batch number; serial-numbered products need one serial number per unit (the box appears only for those products). Each line is numbered and fits on one row: the product, the change, the quantity, what the store holds now (**Available**), what it will hold (**After**, in red if it would go below zero) and the line's **Value** at average cost. The total value shows under the lines, with a note when a manager must approve it (above GHS 500 by default). The batch number, expiry date, unit cost, a different reason, a note and a **photo of the goods** are behind the **Details** drop-down at the end of the line; it opens by itself for batch-tracked products (all medicines). A warning shows when the same product (and batch) is on two lines. **Why:** you see the effect of each line before saving, and most lines need only the product and quantity.
3. Save. The adjustment is a **Draft**: you can still change it, or **Add lines from CSV**.
4. **Post** — the changes go into the stock ledger.

Approval: an adjustment worth more than the limit in **System** → **Settings** → **Stock** (default GHS 500 at average cost) must be **submitted for approval** and **approved and posted by a manager other than the person who prepared it**.

> **Why:** stock adjustments are where stock losses are hidden. A second person checking large adjustments, a reason on every line and a permanent record of who did what are the standard controls.

The adjustment page shows each line's stock **before**, the **change** and **after**, and the value. **Export → Adjustment voucher (PDF)** prints it with a signature line for the approver.

> **Note:** a posted adjustment cannot be changed or deleted. To correct it, post another adjustment. The quick **Opening stock / adjustment** button on *Stock on hand* still exists for a single product.

Photos of damaged or expired goods: under **Details** on a line, **Photo of the goods** (a phone photo, up to 5 MB). The adjustment shows *Photos: Line 1 · Line 3* to open them; only people who may see that store's stock can. **System → Settings → Stock → Photos on stock adjustment lines**: *Optional* (default), *Required for damaged, expired, spoiled and missing stock* (the adjustment cannot be submitted or posted without them), or *Not used*. **Why:** a photo settles questions about a write-off months later, and makes "it was damaged" harder to say without proof.

14.3b Cost runs: products loaded without a cost

Stock → Cost runs gives products a cost worked out from their selling price, when the real cost was never entered. It is free with every licence.

> **Why:** stock with no cost is worth nothing in the books, and each sale of it shows its whole price as profit. Where the real costs cannot be found, an estimate from the selling price (accountants call it the *retail method*, allowed by IAS 2) is far closer to the truth than zero. Ask your accountant to agree the factor and keep the signed run with your stock records.

Before you start:

- Enter any real costs you do have (Chapter 7, Step 10): a real cost is always better than an estimate.
- Do a stock count if you can: the value is only as good as the quantities. Products below zero are left out until they are counted.
- Choose a quiet time for posting: it can take a few minutes on a large catalogue. Tills keep selling; their sales go in right after.

Step by step:

1. Click **New cost run** (or tick products under **Catalog → Products** and choose **Work out cost from selling price**, to run on those only).
2. **How to work out the cost:**
    - *Selling price ÷ a factor*: at 1.4, an item selling at GH₵ 14.00 costs GH₵ 10.00. Use this when you add 40% to your cost.
    - *Selling price less a profit percentage*: at 40%, the same item costs GH₵ 8.40. Use this when 40% of the selling price is your profit.
    - **VAT and levies:** *Use the price the customer pays* if that is the price you multiply your cost by. *Remove VAT and levies first* if you add your profit before tax.

    - Add **a different factor** for a category or brand that earns more or less, e.g. cosmetics at 1.6.
3. **Which products:** leave empty for all, or choose products, categories, brands, suppliers or stores. Products that already have a cost are left as they are unless you switch on *Also replace costs products already have*. **Why:** a real cost from a delivery is worth more than an estimate.
4. **The books:** the *Date in the books* is usually your go-live date, in an open month. *Add the cost of past sales* is explained below. Your accountant may choose another account.
5. Click **Create**. Nothing has changed yet: this page is the **preview**. It shows what is added to the stock value, every product with its old and new cost, and each **skipped** product with the reason. **Export** gives the whole run in Excel for checking. Correct what you can (a missing price, say), then **Work it out again**.
6. **Send for approval.** Someone else with the right to approve opens it (it shows on their dashboard) and clicks **Approve and post**. **Why:** the run changes the balance sheet, so a second person checks it. A company with one manager can turn this off in **System** → **Settings** → **Stock**.

What it changes: the cost of each product in each store, and the stock value in the books. Quantities, batches and bins stay as they are. **Selling prices never change.**

In the books:

- Stock that had no cost goes against **Opening balances (to clear)**, which your accountant then clears to capital.
- A replaced cost goes to **Stock losses and count differences**.
- **Past sales:** sales made with no cost stay as they are, but one labelled entry per branch adds their estimated cost to cost of sales, so your profit is realistic. Sales are only counted once, however many runs follow. Reports per sale still show the old sales at no cost; the accounts carry the correction.
- **Delivered onto no-cost stock:** when a delivery arrives on stock that had no cost, its cost becomes the cost of all of it, but the books only hold the delivered goods. The run brings the books up to that cost, without changing the cost (*Books brought up to its cost* in the preview).

Skipped products: no selling price on the chosen list, stock below zero, batches that do not add up to the stock on hand, consignment goods (their cost is what you pay the consignor), and products with serial numbers.

Undo: Undo this run on a posted run puts the earlier costs back with an equal and opposite run, dated today. Products that had a delivery since keep their new cost, and sales made since keep the cost they were sold at. Costs cannot be undone by editing: a posted run stays on record.

Is the factor right? Stock → **Estimated costs** lists every product with an estimated cost, and what its first real delivery cost, with the real factor (price ÷ real cost). **Why:** after a few weeks of deliveries you can see whether 1.4 was close, and run the rest again with a better factor if it was not.

14.3c Direct purchases: a supplier's invoice with no purchase order

Goods often arrive with an invoice and no purchase order, such as a wholesaler's delivery paid by MoMo at the door. **Purchasing** → **Direct purchases** (also **Enter a supplier invoice** on *Stock adjustments*) records it all on one form. It is free with every licence.

Why not a stock adjustment? An adjustment only changes the stock. It would not show what you bought from whom, the VAT, what you paid or what you still owe, and the value would land in *Stock losses*. This form makes the goods received, the supplier bill and the payment for you, so everything agrees.

Step by step:

1. Click **Enter a supplier invoice**. If AI is switched on, click **Scan the invoice** and add a photo of **every page**. It fills the form; check every line against the paper. Handwriting on the invoice is ignored.
2. **The invoice:** supplier, invoice number, date, store. Switch on *Prices on the invoice include VAT and levies* only if they do: most supplier invoices show prices before VAT. Type the **invoice total as printed**, so a mistyped line shows up.
3. Add the **photo or PDF of every page**, so the paper is kept with the record.
4. **How is it paid?**
    - *Paid on delivery*: how, from which account, and the MoMo or transfer reference.
    - *On credit*: it shows in what you owe the supplier. Leave *Due on* empty to use their payment terms.
    - *Consignment*: see below.
5. **Products**, one line per invoice line: the product and the unit you bought it in, quantity, unit price as printed, batch and expiry date, and the supplier's code. Tax follows the product; change it on a line if the invoice differs.
    - A new product? Click **+** beside the product, and give it the unit it sells in and the pack size, e.g. sold in tablets, 60 in a box.
    - **New selling price:** type it to change the price, and choose the unit it is for (box or tablet). The line shows the margin and how many times the cost it is. Leave it empty to keep today's price.
6. Click **Create**. Nothing has changed yet. The page shows the totals and anything to check: a total that differs from the invoice, goods that expire soon, or a thin margin.
7. **Post**, or **Send for approval** when it is above the limit. A manager (not the person who entered it) then posts it, and it shows on their dashboard. **Why:** receiving goods with no order is an easy way to hide missing stock or fake a purchase, so larger ones get a second look. Set the limit in **System → Settings → Purchasing** (default GHS 500).

The same invoice number from the same supplier is refused, so an invoice cannot be entered or paid twice. Expired batches are refused.

After posting: Print labels gives shelf labels for the products. **The invoice** shows the pages. The goods received note, bill and payment are all listed on the document, and appear under Purchasing and Payables as usual.

Mistakes: a posted purchase cannot be edited. Goods sent back go through **Purchasing → Returns to suppliers**, and a wrong selling price is changed on the product.

Consignment (pay when sold): the goods stay the supplier's. They come in at no cost, and each sale owes the supplier the cost on the invoice. **Purchasing → Consignment to pay** shows what is owed; **Settle** raises a bill to approve and pay as usual. A product is either your own or consigned, never both, because one average cost cannot cover goods you paid for and goods you did not. If you hold stock of your own, the form refuses it: sell that first, or enter this as a purchase.

Stock with no invoice at all (a gift, or goods from the owner's other shop): use a stock adjustment with the reason **Received with a cost, no supplier invoice**, and enter the cost on each line. The value waits in *Stock received, to be cleared*. **Why:** your accountant then decides what it was, such as capital or a gift, so it does not show up as profit.

14.3d Price list: export prices

Pricing & tax → **Price list** shows every product in every unit, with the price before tax, the VAT and levies, and what the customer pays.

1. Choose the **price list** and narrow it with the filters: category, brand, supplier, product, active or inactive, in stock only, or priced only. For a few products, tick them on **Catalog** → **Products** and choose **Export prices**.
2. Use the **columns** button to add *In stock*, *Cost* and *Margin*, or hide what you do not need. Cost and margin appear only for people allowed to set prices.
3. Click **Export**: Excel or CSV for everything, or PDF with your logo for a customer. **Hide cost and margin before sending a list outside the business:** the file holds the columns on screen.

> **Why:** one list can be filtered for the need in hand, such as a wholesale customer's brands, without keeping a spreadsheet that goes out of date.

A PDF holds up to 800 rows. A longer list says so: filter it, for example by category, or use Excel or CSV. The same applies to the **Stock ledger**, which can be filtered by date, product, type, warehouse, category, brand and person. **Why:** a PDF is laid out row by row in the server's memory, and a very long one would stop it.

Batches and expiry dates for many products: on **Catalog** → **Products**, tick products and choose **Switch on batches and expiry**, or click **Switch on batches for all**. Products already on are left as they are. Stock on hand goes into a batch called OPENING with no expiry date, so it gets no expiry warnings and sells after everything dated, until you give it a date with the expiry sheet below. Every new delivery then needs a batch and expiry date. It cannot be switched off once batches exist. **Why:** expiry warnings and earliest-expiry-first selling only work once products carry batches.

Expiry sheet: dating stock with no batch. Stock that had no batch when batches were switched on sits in a batch called OPENING with no expiry date. It gets no expiry warnings, and it **sells last**, after everything with a date. To put that right for many products at once:

1. Go to **Stock** → **Batches & expiry** and click **Expiry sheet**, or **System** → **Import data** → **Expiry sheet**.
2. Click **Download your current data**. The file lists everything still in OPENING, store by store, with how much of each.
3. On the shelf, fill in **expires_on** from the pack. Leave a row empty to come back to it later, so you can do a shelf at a time.
    - **One date on the shelf:** the date is all you need.
    - **Several batches:** copy the row for each one, and fill in **batch_no** and **qty** in the smallest unit (tablets, not packs). A row with no qty takes all that is left.

4. Upload the file and click **Check** first. **Why:** it tries every row and undoes it, so you see any mistakes before anything changes.
5. Import it, then download the list again: it now shows only what is left.

Quantities and the stock value never change, so there is nothing to approve. **A date with no batch number applies in every store**, because one batch has one date; use a batch number if two stores hold different dates. If you do not know a date, use the earliest you can see on that shelf, so you are warned too early and not too late.

14.3e Returns: approval, quarantine and the back office

Manager's approval. System → Settings → Returns → Manager's approval of returns has three choices:

- **Require** (the default): every return needs a manager's username and PIN at the till.
- **Allow:** a manager can approve there and then. If none does, the cashier carries on and it goes to the review queue for a manager.
- **Off:** only returns with no receipt, or past the return period, need a PIN.

To let small returns through, set **Only returns above this value need approval**. **Why:** a return is money leaving with goods coming back, so it is one of the easiest ways to take cash. Managers need their PIN set and the right to approve returns.

Every return has a reason from a list (wrong item, damaged, expired, changed mind, doctor changed the treatment, cashier's mistake, other), so the reasons can be totalled.

Quarantine. Returned goods go back on the shelf, unless they are damaged or expired. Those go to the branch's **quarantine** store, made the first time it is needed. Its stock still has a value, but no till can sell it. Dispose of it or send it back to the supplier from there, or transfer it back if it is fit to sell.

To treat every returned medicine this way, switch on **Send returned medicines to quarantine** in **System → Settings → Returns**. **Why:** once a medicine has left the pharmacy, nobody can vouch for how it was stored. Whether to resell it is your pharmacist's decision, so ask them. Until it is switched on, check a returned medicine before taking it back, and choose **Damaged** or **Expired** as the reason if it should not be sold.

Returns from the back office. For a customer who comes to the office, or a mistake found later:

1. Open the sale under **Sales → Sales** and click **Return goods**.
2. Enter what comes back, tick **Damaged** where it applies, and choose the reason and the store.
3. Choose **Refund as**: store credit, the customer's account, or paid from a cash, bank or MoMo account with its reference.
4. It waits under **Sales → Back-office returns** and on a manager's dashboard. Someone other than the person who entered it clicks **Approve and refund**. Only then does stock move and the refund go out.

Why a second person always: the money leaves with no till or drawer to count it against. The return period does not apply here, because a manager is deciding. Cash paid out of an office account does not go through a till shift, so it never affects a drawer count.

Return slip: the till prints lines for the customer and manager to sign on the return receipt. In the back office, **Return slip** gives a PDF to sign and file.

Sales → **Returns** lists every return from tills and the office, by person, reason, product, date and approval, with who approved it and where the goods went. The line at the top names anyone who did more than half of the last 30 days' returns, once there are five or more. **Why:** a pattern like that is worth a look, though it may have a good explanation, such as one person always working the return counter.

14.3f Fee services and health checks

Services you charge for, such as a blood pressure check, are sold at the till like a product, with no stock.

Set one up: Pharmacy → **Health checks** → **Add a fee service**: the name, the fee, the tax, and the reading taken (blood pressure, blood sugar, weight or temperature), or none for a plain fee. Or use **Catalog** → **Products**: switch off stock, and set **Reading taken** under *Health check*.

At the till: find it by name. If it has a reading, the till asks for it: top and bottom numbers and pulse for blood pressure. **Add without a reading** charges the fee and keeps no reading. Choose the customer (F4) if they are a patient of yours, so it goes on their history. The reading prints on the receipt.

The result: the till warns when a reading is **high**, **low** or **refer now**. These follow common screening limits, such as 140/90 for high blood pressure and 180/120 to refer now. **They are a prompt to refer the person to a clinic, not a diagnosis.** Ask your pharmacist to agree them.

Pharmacy → **Health checks** lists every reading by date, patient, kind and result, for people allowed to see patients. Use **Record a reading** for one taken away from a sale. Filter by patient to see their history. A reading is never changed: if one was wrong, take it again. **Why:** these are health records, so they keep what was measured, when and by whom.

14.3a Barcode and shelf labels

Catalog → **Products**: tick the products → **Print labels** (or **Print labels** on one product, or on a delivery under **Purchasing** → **Goods received** for what came in).

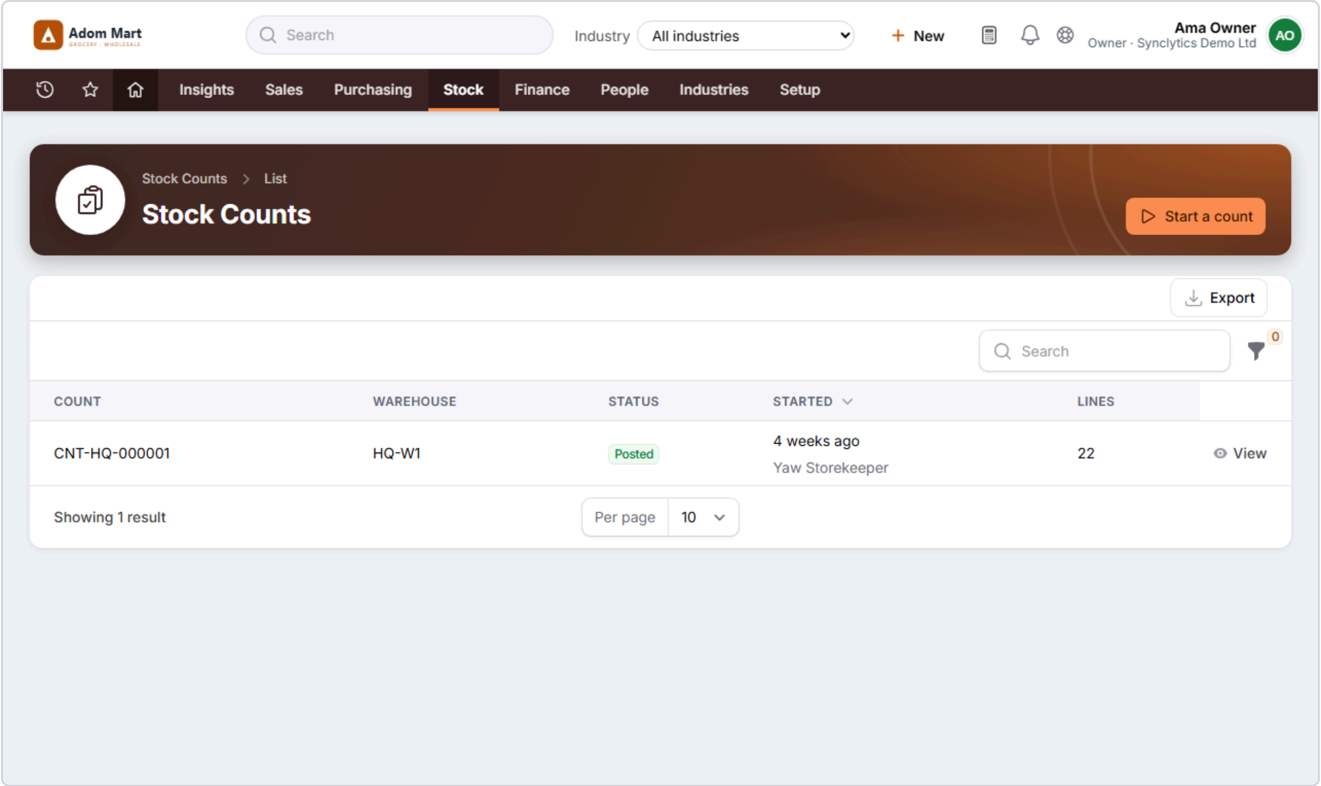
1. **Label:** *Shelf label* (name and price large, barcode small) or *Barcode sticker* (barcode large, name and price small).
2. **Print on:** an A4 label sheet — 24 (3 × 8, e.g. Avery L7159), 21 (3 × 7, L7160) or 65 small (5 × 13, L7651) — on any office printer, or a **label printer** (50 × 25, 38 × 25 or 58 × 40 mm; Zebra, Xprinter and similar, installed as a printer on the computer).
3. **Prices from** the price list, **labels per product**, and on a used A4 sheet **leave empty at the start** the labels already peeled off.
4. **Give products without a barcode an in-store barcode first** (people who may edit products): an EAN-13 starting with *20* (System → Settings → Stock → *In-store barcodes start with*), added to the product so the tills scan it.
5. **Labels ready** → **Open the labels**: a page with the labels; **Print**. In the print window set **Margins: none** and **Scale: 100%**, and choose the label printer when using one.

Tip: print one sheet on plain paper first and hold it against a label sheet to check the alignment.

Why: products with a clear price and a barcode that scans sell faster and with fewer mistakes at the till; items that came without a barcode no longer need typing.

14.4 Stock takes (counting stock)

There are two ways to bring the system in line with what is on the shelves.


Adom Mart
Search
Industry
All industries
+ New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Stock Counts > List
Stock Counts
Start a count
Export
Search
0
View
Per page
10

COUNT	WAREHOUSE	STATUS	STARTED	LINES
CNT-HQ-000001	HQ-W1	Posted	4 weeks ago Yaw Storekeeper	22

Showing 1 result

A. A stock count (recommended for full or regular counts)

1. **Stock** → **Stock counts** → **Start a count** for a warehouse (or one bin). The system freezes what it expects at that moment. Leave **blind** on so counters do not see the expected figure.
2. Open the count and use **Count sheet**:
    - **PDF** to count on paper (SKU, barcode, product, unit, bin, and an empty *Counted* column; signature lines for the counter and the checker);
    - **CSV** to fill in Excel or on a tablet.
3. Enter the counts: scan with a handheld (Chapter 9), type them on the count page, or fill the CSV's `counted_qty` column and use **Import counted quantities (CSV)**. Rows left empty are **not counted** (not set to zero).
4. **Submit**, then a manager **approves and posts**. The differences become stock movements, with their value.

B. Update stock from a stock sheet (quick corrections)

1. **Stock** → **Stock on hand** → **Stock sheet (before a count)**: choose the warehouse and category, whether to show the system quantities (leave them out for a blind count), and PDF or CSV. This is the **current stock standing**.
2. Or, to update stock with a file: **Stock** → **Stock adjustments** → **Update stock from a CSV file**, choose the warehouse and reason, and **Stock sheet — current stock standing (CSV)**. Its `qty` column holds the system quantity; type the **new** quantity where it differs.
    - **Batches and expiry dates:** a product tracked by batch (most medicines) has **one row per batch**, with its `batch_no`, `expires_on` and quantity. Type the batch's new quantity; **add a row** (copy one and change `batch_no`, `expires_on`, `qty`) for a batch found that the system does not know; change `expires_on` to **correct a wrongly typed expiry date**. A new batch needs its expiry date.
    - The `location` (where the product is kept) and `bins` columns are for information: change a location on the product (or the products file), and move stock between bins on the Bins screen.
3. Import it. Rows whose quantity did not change are skipped; the others become **Set to** lines in a new draft adjustment (for a batch: an increase or decrease of the difference, worked out when you import). A corrected expiry date applies at once and is in the audit log.
4. Open the adjustment, check it, and post it (with approval above the limit).

Why two ways: a stock count freezes the expected figures and compares them (best for full counts while trading continues); the stock sheet route is quicker for a few corrections.

> **Tip:** count when the shop is closed, or count one area (bin or category) at a time. Sales during a count are still recorded normally.

C. Cycle counting (a few products every day)

Stock → **My day** lists, for each store, the products to count today and **Start today's count** turns them into a blind stock count (way A). How many a day: **System** → **Settings** → **Stock** → **Cycle count: products per store per day** (default 20; 0 = off).

- The system ranks products by the value that left the store in the last 90 days: the top 80% (A) are counted every 30 days, the next 15% (B) every 60, the rest (C) every 120. Products never counted come first.
- A product with a difference in two of its last three counts is **often wrong** and is counted twice as often.

> **Why:** in a busy pharmacy a full stock-take means closing; a few products a day catches mistakes (and theft) within weeks, and the products that matter most are checked most.

My day also shows the store's low stock, expired and expiring batches, negative stock, products never counted, adjustments waiting for a manager, and transfers and orders on the way — each opens the page where it is dealt with.

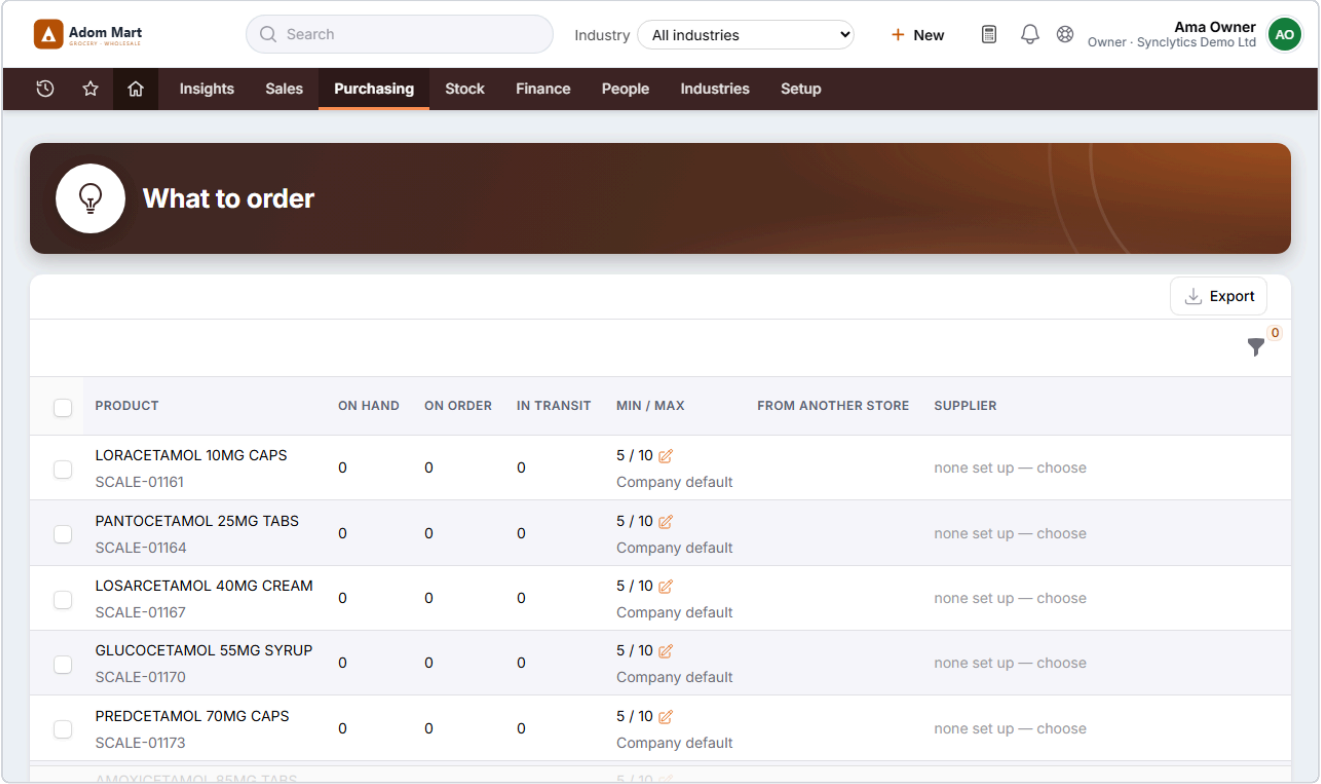
D. Losses, slow stock and checks

- **Stock → Shrink:** stock lost (damaged, expired, spoiled, stolen, count shortages) at cost for a period and store — by reason, by product, the **last six months** month by month, **by category**, **recorded by** whom, and products **lost again and again** (three times or more).
- **Stock → Slow & dead stock:** what has not sold in a store for 60, 90 or 180 days (or, if never sold, since it came in), with its value at cost. For each product a suggestion: **Move** it to the store where it sells (a draft transfer in one click, to check and dispatch), ask the supplier to take it back where they do (Expiry & returns), or mark it down.
- **Stock → My day → Checks:** losses in the last 30 days **larger than usual** for that store (three times the usual loss line and at least GHS 100), products **written off three times or more**, and **possible duplicate products** (same name). These point at what deserves a second look; they are not a verdict.

> **Why:** most stock losses are small and repeated; seeing them by month, by product and by person shows where to look, and stock that does not sell ties up money that could buy what does.

14.5 Minimum stock and reordering

Every product that keeps stock has a **minimum** in each store. When stock (plus what is already on order or on its way) falls to it, **Purchasing → What to order** suggests topping it up to the **maximum** and turns the suggestions into draft purchase orders, one per supplier.



☐	PRODUCT	ON HAND	ON ORDER	IN TRANSIT	MIN / MAX	FROM ANOTHER STORE	SUPPLIER
☐	LORACETAMOL 10MG CAPS SCALE-01161	0	0	0	5 / 10 Company default		none set up — choose
☐	PANTOCETAMOL 25MG TABS SCALE-01164	0	0	0	5 / 10 Company default		none set up — choose
☐	LOSARCETAMOL 40MG CREAM SCALE-01167	0	0	0	5 / 10 Company default		none set up — choose
☐	GLUCOCETAMOL 55MG SYRUP SCALE-01170	0	0	0	5 / 10 Company default		none set up — choose
☐	PREDCETAMOL 70MG CAPS SCALE-01173	0	0	0	5 / 10 Company default		none set up — choose

> **Why:** running out of a medicine loses the sale and the patient; a minimum on every product means nothing is forgotten, without anyone having to type 2,000 figures.

- **The default** (System → Settings → Stock → *Default minimum stock*, 5): 5 of the unit the product is **bought** in — for example 5 packs of 100 tablets — and orders up to twice that (10 packs). New and existing products all get it. 0 switches it off.
- **Your own figure: Purchasing → Reorder levels → Own figure for a product**: choose the store (or every store), the product, the unit and the minimum and maximum. It replaces the default in that store. **Use the default** on the row goes back. Many at once: the products file (Catalog → Products → Update products) has `min_qty` and `max_qty` columns, in the buying unit; type `default` to go back.
- **Smart levels from sales** (Settings → Stock → *Smart reorder levels*): every Monday the system works out each product's minimum and maximum per store from the last 12 weeks of sales, the supplier's delivery time and a safety margin. With **Suggest** (the default) they wait on **Purchasing → Suggested levels**: tick the ones you agree with and click **Accept**. With **Update automatically** they are applied by themselves — but a figure you typed is never changed.
- **Slow movers** (setting, off by default): products that have not sold in a store for 180 days get no default minimum there, so dead stock is not reordered.
- **Low stock:** the dashboard shows how many products are at or below their minimum (click it to open What to order); buyers get a notification each morning; and with *Show stock left on the till* on, the till marks such items **Low** in amber.

Changing a line before ordering: on *What to order*, every value you can change has a **pencil** beside it — click the value. **Supplier** (the product's own suppliers come first, with their last price; the unit follows the supplier), **Order** (unit and quantity) and **Cost** (when the supplier has no price yet) change that line until the orders are made; **Min / max** (if you may set reorder levels) saves the product's levels for that store at once. The product then says *changed by you*; **Undo my changes**, in any of those small windows, puts the line back. **Why:** you change exactly the value you are looking at, without scrolling to a button at the end of the row. **Move before buying:** when another store of the business has more of an item than its own maximum, What to order shows it in **From another store** (e.g. *36 PC from Branch A*) and orders only the rest (the **Order** column then says *nothing — moved instead* when the move covers it all). Tick those lines and **Ask other stores to send**: each store with stock to spare gets one stock request, which it sends as an ordinary transfer. Stores of a different industry are not suggested (their requests need two approvals). Switch it off under System → Settings → Stock → *Suggest moves between stores before buying*. **Why:** stock sitting above one store's maximum is money already spent; moving it costs nothing. Long lists show 25 lines a page (choose 50 or 100 at the bottom); lines you ticked stay ticked when you move to another page, and **Create draft purchase orders** includes them all. Filter by **Supplier** to order one supplier at a time (or *No supplier set up* to find what still needs one). Tick the lines and **Create draft purchase orders**.

> **Why:** the suggestion is a starting point. A cheaper supplier this week, a bigger pack or a slow month are things only the buyer knows.

> **Tip:** set the supplier's *delivery lead time* (Purchasing → Suppliers) and their unit and cost on the supplier's products list — smart levels and what-to-order use them.

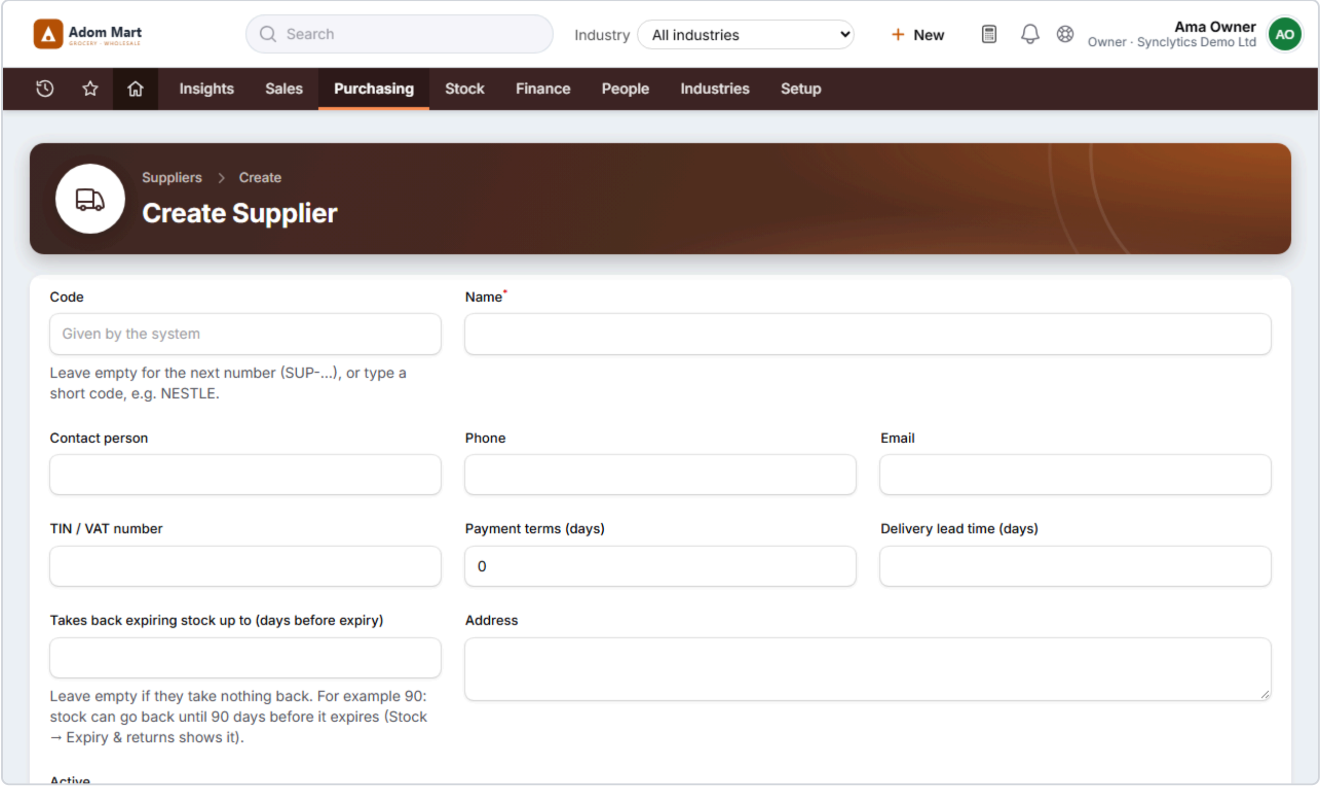
Buying in packs, selling in units

Medicines are bought by the box (or carton) and sold by the tablet or strip. Stock is always kept in the **base unit** (e.g. TAB); each pack says how many base units it holds.

1. **The product** (Catalog → Products): under **Barcodes and units**, add each pack: *Strip* × 10, *Box* × 100, *Carton* × 1200, with its barcode. Mark each **Sold**, **Bought** or both (a carton only *Bought*; a tablet only *Sold*). **Usually bought in:** *Box*. **Why:** purchase orders then offer the box first, and never the tablet.
2. **The supplier** (Purchasing → Suppliers → Products supplied): their **pack** (one supplier may sell a box of 100, another a box of 50: add each supplier's own pack), their code, the **cost per pack**, the **minimum order** and **in multiples of** (e.g. 5 boxes at a time; 0 = any). **Why:** the order says exactly what the supplier sells, and the cost is per pack, as on their invoice.
3. **Purchase orders** offer only units that are bought, the usual one first, each with what it holds: *PARA500 · Paracetamol 500 mg — BOX × 100 (= 100 TAB) · usual buying unit*. Under the quantity: *Supplier sells at least 5, in multiples of 5*. An order below the minimum or off the multiple is refused, with the next right quantity (e.g. *in multiples of 5 BOX (e.g. 10)*).
4. **What to order** suggests whole packs, at least the minimum, rounded up to the supplier's multiple (e.g. 300 tablets needed, boxes of 24, multiples of 5 → 15 boxes).
5. **Receiving** is in the ordered pack (*Received now: 15 BOX*); stock goes up by 1 500 TAB. For medicines with batches, **Scan the pack code**: the square GS1 code on the box fills in the **batch number and expiry** (check them against the box). **Why:** no typing, and no expiry dates typed wrongly.

14.5a Suppliers: approval, documents, bank checks and scorecards

> **Why:** buying only from suppliers you have checked keeps fake medicines and unlicensed sellers out, and a second person checking bank details stops the commonest supplier fraud (a message asking you to pay into a "new account").



Create Supplier

Code: Given by the system

Leave empty for the next number (SUP-...), or type a short code, e.g. NESTLE.

Name*

Contact person

Phone

Email

TIN / VAT number

Payment terms (days): 0

Delivery lead time (days)

Takes back expiring stock up to (days before expiry)

Leave empty if they take nothing back. For example 90: stock can go back until 90 days before it expires (Stock → Expiry & returns shows it).

Address

Active

Supplier controls (System → Settings → Purchasing → *Supplier controls: approval, documents and bank checks*) are **off** until you switch them on, so a one-person business can keep ordering straight away. Switch them on once two or more people buy and pay. Suppliers you already had stay approved.

With the controls on:

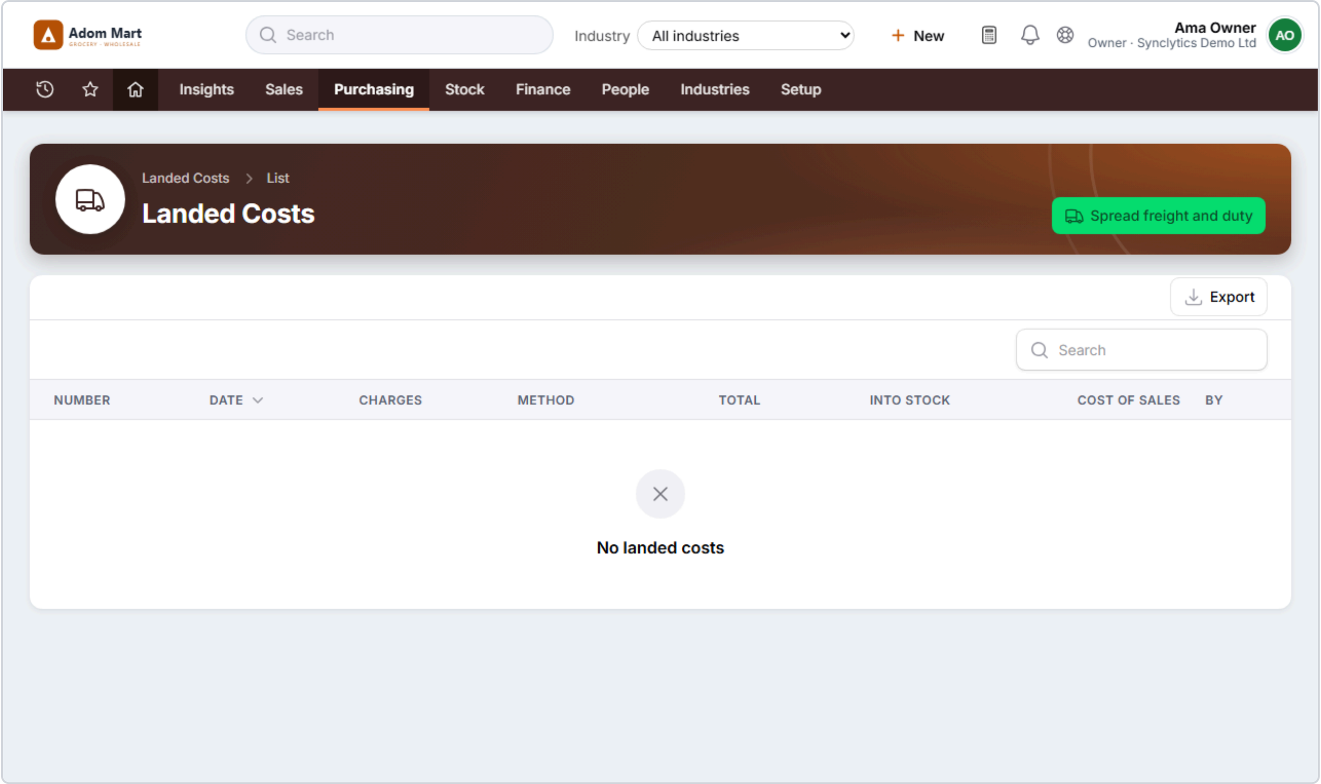
1. **Adding a supplier** (Purchasing → Suppliers → New): fill in the TIN, what they supply (goods, services or works — it sets withholding tax later) and how they are paid. The supplier shows *Waiting for approval*.
2. **Documents** (open the supplier → *Documents* → *Add a document*): a scan or photo and its expiry date. Which ones every supplier needs is set under *Documents every supplier must have on file* (default: TIN certificate and business registration; a pharmacy adds `fda_licence` and `pharmacy_council`). Someone else presses *Checked against the original*.
3. **Approve**: someone with *Approve suppliers* (General Manager, Finance Manager) — never the person who added it — presses **Approve** on the supplier list. Only then can purchase orders go to it.
4. **Expiry:** the buyers get a notice on the bell before a document expires (30 days by default) and when it has expired. A supplier whose required document has expired gets no orders until a new one is added.
5. **Bank details:** a new or changed account or MoMo number shows *Not yet* checked. Someone else calls the supplier **on a number you already had** (never one from the message announcing the change), then presses **Verify bank details**. Until then no transfer or MoMo payment can be recorded to that supplier.

Block a supplier (with the reason buyers will see) to stop new orders, whatever the setting; **Unblock** when settled.

Scorecard (on the supplier list: the *Score* column and the *Scorecard* button): worked out from the last 12 months, nobody types a rating — on time (40 %), fill rate (30 %), goods sent back for faults (20 %; expiry returns are not the supplier's fault and are left out), and price changes on bills (10 %), with the average lead time. **A** from 85, **B** from 70, **C** from 50, else **D**. **Why:** it shows which supplier to order from first, and gives you facts when you negotiate.

14.5b Landed costs: freight, duty and clearing on what you buy

> **Why:** a carton bought for GHS 240 that cost GHS 48 to bring to the shop really cost GHS 288. Without spreading the freight, your stock looks cheaper than it is and your margins look better than they are.

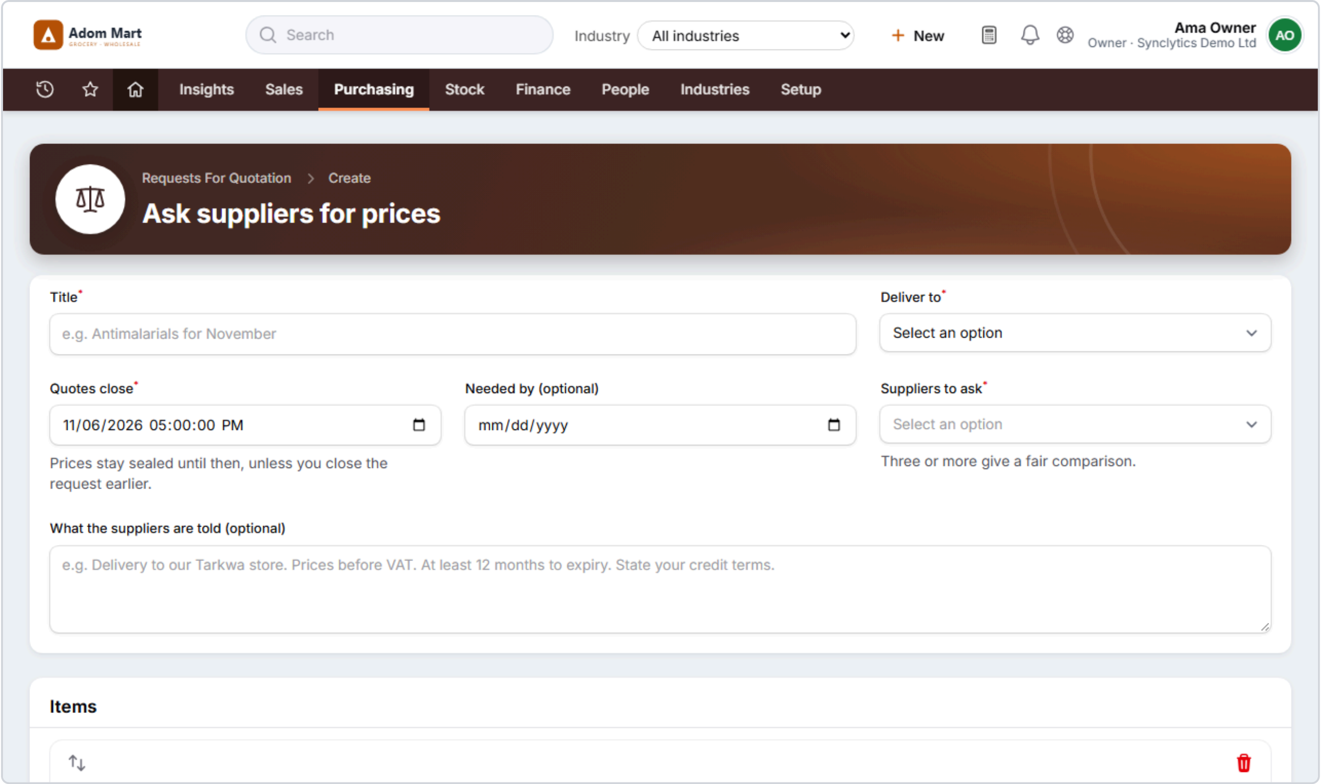

Adom Mart
Search
Industry
All industries
New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Landed Costs > List
Landed Costs
Spread freight and duty
Export
Search
NUMBER
DATE
CHARGES
METHOD
TOTAL
INTO STOCK
COST OF SALES
BY
No landed costs

1. **Enter the charge as a supplier bill** (the transport company, the clearing agent or customs): an expense line to the account **Freight, duty and clearing to spread** (1360). It waits there.
2. **Purchasing → Landed costs → Spread freight and duty:** choose the deliveries (goods received) it belongs to, how to spread it — **by value** (usual: a dearer item carries more) or **by quantity** (bulky goods of similar size) — and the charges (kind, description, amount before VAT). **Post.**
3. What it does, to the pesewa: each delivery line gets its share. The part for goods **still in the warehouse** raises their cost (the average cost per piece goes up); the part for goods **already sold** goes to cost of sales, since it is too late to put it in stock. The books move the total out of *to spread*, which comes back to zero.
4. **Per product** on a landed cost shows each product's share and its cost per unit before and after.
5. **Finance → Reports → Ledger check:** *Stock at cost* includes the part that went into stock, so it still agrees with the stock account. That is why the figure can be higher than the deliveries alone add up to.
6. **With branch servers**, the landed cost reaches them through sync like any other document, and the cost per piece rises there too, so sales there go out at the right cost once it has synced.

A posted landed cost is never edited. A mistake is corrected with another one with a **negative** amount.

14.5c Requests for quotation: asking several suppliers for prices

> **Why:** asking three suppliers for the same list, and comparing their answers fairly, usually saves more than any discount you could ask one supplier for. Quotes are **sealed**: nobody in the company sees a price before the closing time. That way no supplier can be told what to beat.



Ask suppliers for prices

Title*: e.g. Antimalarials for November

Deliver to*: Select an option

Quotes close*: 11/06/2026 05:00:00 PM

Prices stay sealed until then, unless you close the request earlier.

Needed by (optional): mm/dd/yyyy

Suppliers to ask*: Select an option

Three or more give a fair comparison.

What the suppliers are told (optional): e.g. Delivery to our Tarkwa store. Prices before VAT. At least 12 months to expiry. State your credit terms.

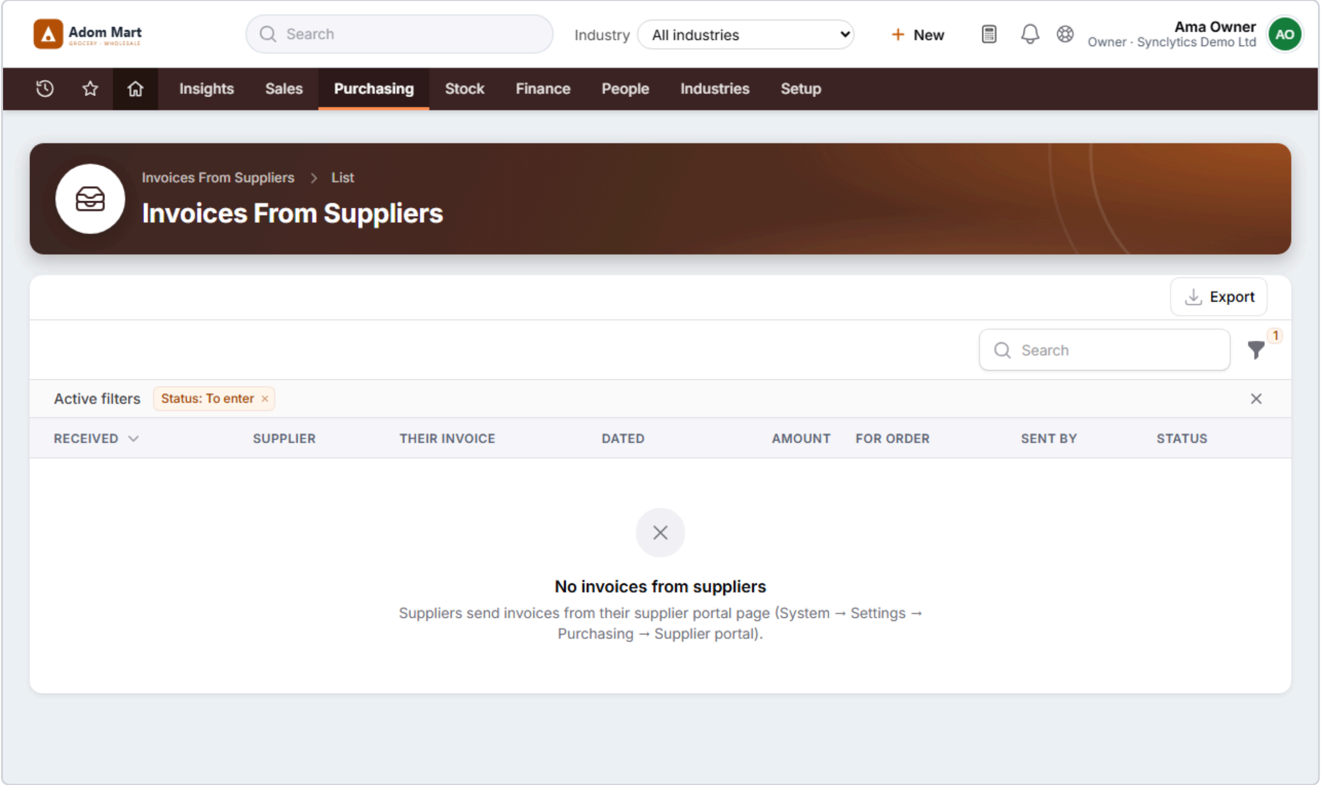
Items

1. **Purchasing → Requests for quotation → Ask for quotes.** Give a title, the store to deliver to, the closing time, the suppliers to ask (three or more is fairest), what you want to tell them (delivery place, minimum expiry, payment terms) and the items with quantities. You can also open a **purchase request** and press **Ask for quotes**. The request then shows *Asking suppliers for prices*.
2. **Send to a supplier** (on the request): by e-mail, by WhatsApp from this computer, or by WhatsApp sent by the system. Each supplier gets **their own link**, with no password. *Supplier links* shows every link, so you can send them another way. *Seen* tells you when a supplier opened it.
3. On their link the supplier enters, for each item, a **price before VAT** and how much they can supply. They also give their delivery time, credit days and how long the prices hold. They can change their offer until the closing time, or decline. If a supplier answers on paper or by phone, enter it with **Enter a quote**. It shows *Entered by* your name.
4. At the closing time the prices open. You can open them earlier with **Close and compare**. Suppliers can then no longer change anything, and the early opening is recorded. The **Comparison** shows each item's price from each supplier, with ★ for the cheapest. It also gives a **score out of 100** for each supplier:
    - price: 50 points;
    - delivery time: 20 points;
    - credit days: 10 points;
    - past performance: 20 points, from the supplier's scorecard (a new supplier gets half).
5. **Award:** choose who supplies each item. It may be one supplier for everything, or different suppliers for different items. Choose *Do not buy now* to leave an item out. If an item does not go to the cheapest price, or a supplier's prices have expired, **write the reason**: it stays on the record. Each supplier chosen gets a **draft purchase order** at their prices, expected after their delivery time. The order then goes for approval as usual.

Cancel gives a purchase request back to the buyers. Suppliers who did not win see "not awarded this time" on their link.

14.5d The supplier portal: suppliers confirm orders, send invoices and see payments

> **Why:** it saves phone calls. The supplier tells you the delivery date before the day and warns the store when goods are on the way. Invoices arrive as files instead of paper that gets lost, and suppliers stop calling to ask whether they have been paid.


Adom Mart
Search
Industry
All industries
New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Invoices From Suppliers > List
Invoices From Suppliers
Export
Search
Active filters
Status: To enter
RECEIVED
SUPPLIER
THEIR INVOICE
DATED
AMOUNT
FOR ORDER
SENT BY
STATUS

No invoices from suppliers

Suppliers send invoices from their supplier portal page (System → Settings → Purchasing → Supplier portal).

The portal is **off** until you switch it on: **System** → **Settings** → **Purchasing** → **Supplier portal**.

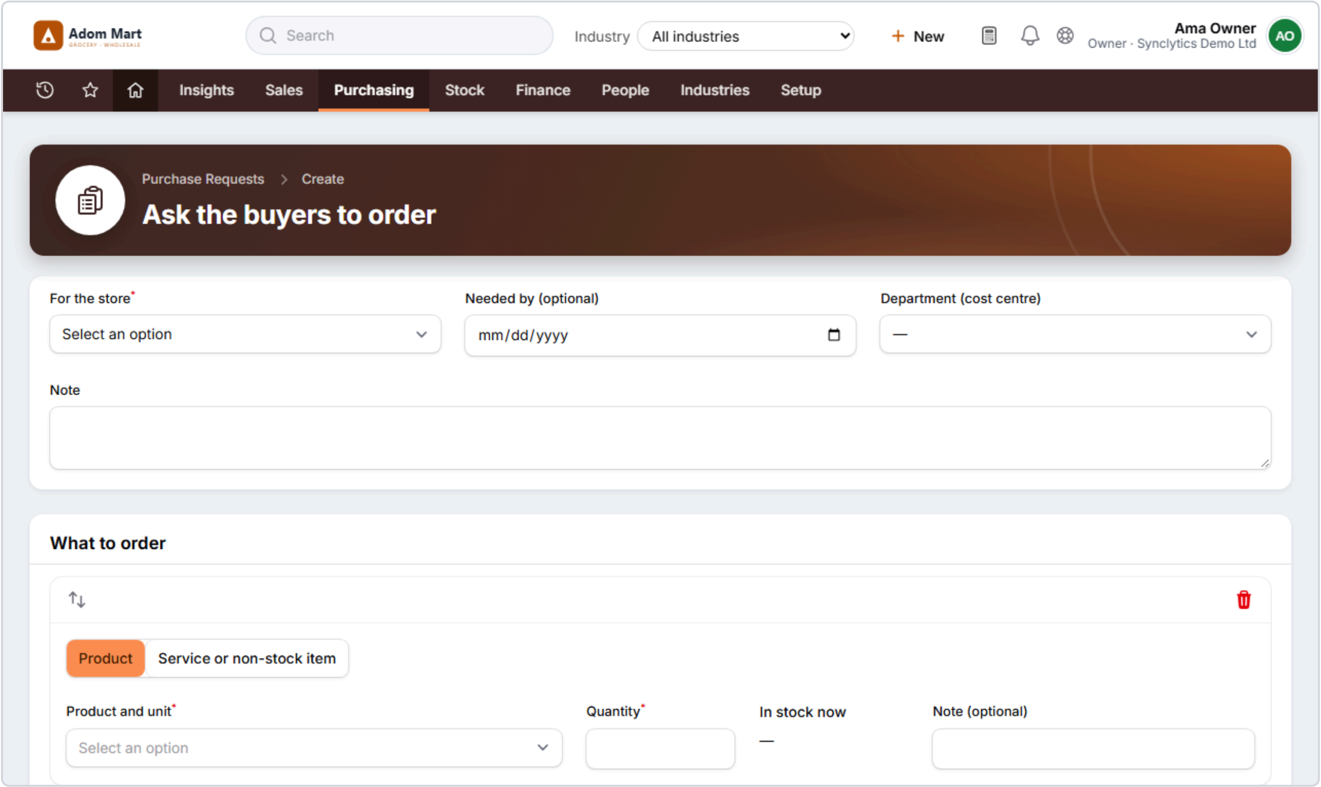
1. **Purchasing** → **Suppliers** → **Portal link** on the supplier. Send it by e-mail or WhatsApp, or choose *just show me the link*. The link is the supplier's own page, with no password. Send it **only to the supplier's own e-mail or number**. Tick *Make a new link* if it reached the wrong person: the old link stops working at once. A blocked supplier's link stops working too.
2. **What the supplier sees and does on their page:**
    - **Requests for quotation:** your open requests to them, with a link to quote.
    - **Orders:** your approved orders from the last six months, each with a PDF.
        - **Confirm the order** with the date they will deliver. The buyers get a notice, and the order shows *Supplier confirmed*.
        - **Goods on the way:** their delivery note or waybill number. The store gets a notice to expect the goods, and the order shows *On the way*.
    - **Their account with you:** what you owe them, bill by bill, and payments from the last 12 months. Each payment shows the tax kept back and a link to its **withholding tax certificate**.

   - **Send an invoice:** the invoice number, date, total with VAT, the order it is for, and the PDF or photo. The same invoice number cannot be sent twice.
3. **Purchasing → Invoices from suppliers** (for accounts) lists what came in. **Open the invoice**, enter it under **Finance → Supplier bills** as usual (matched to the goods received), then press **Mark entered** and choose the bill. If something is wrong, press **Turn back** with the reason, for example "prices differ from the order". The supplier sees the reason on their page.

Everything a supplier does on the portal is recorded under their name: for example, *Kofi (Kofi Wholesale, supplier portal)*.

14.5e Buying services and non-stock items, with departments

> **Why:** a business also buys things it does not sell: repairs, servicing, cleaning, stationery and equipment. Ordering them the same way as stock means someone approves the spending first, and the bill is checked against what was really delivered. The **department** (cost centre) shows who spent the money.


Adom Mart
Search
Industry
All industries
+ New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Purchase Requests > Create

Ask the buyers to order

For the store*: Select an option

Needed by (optional): mm/dd/yyyy

Department (cost centre): —

Note

What to order

Product
Service or non-stock item

Product and unit*: Select an option

Quantity*

In stock now: —

Note (optional)

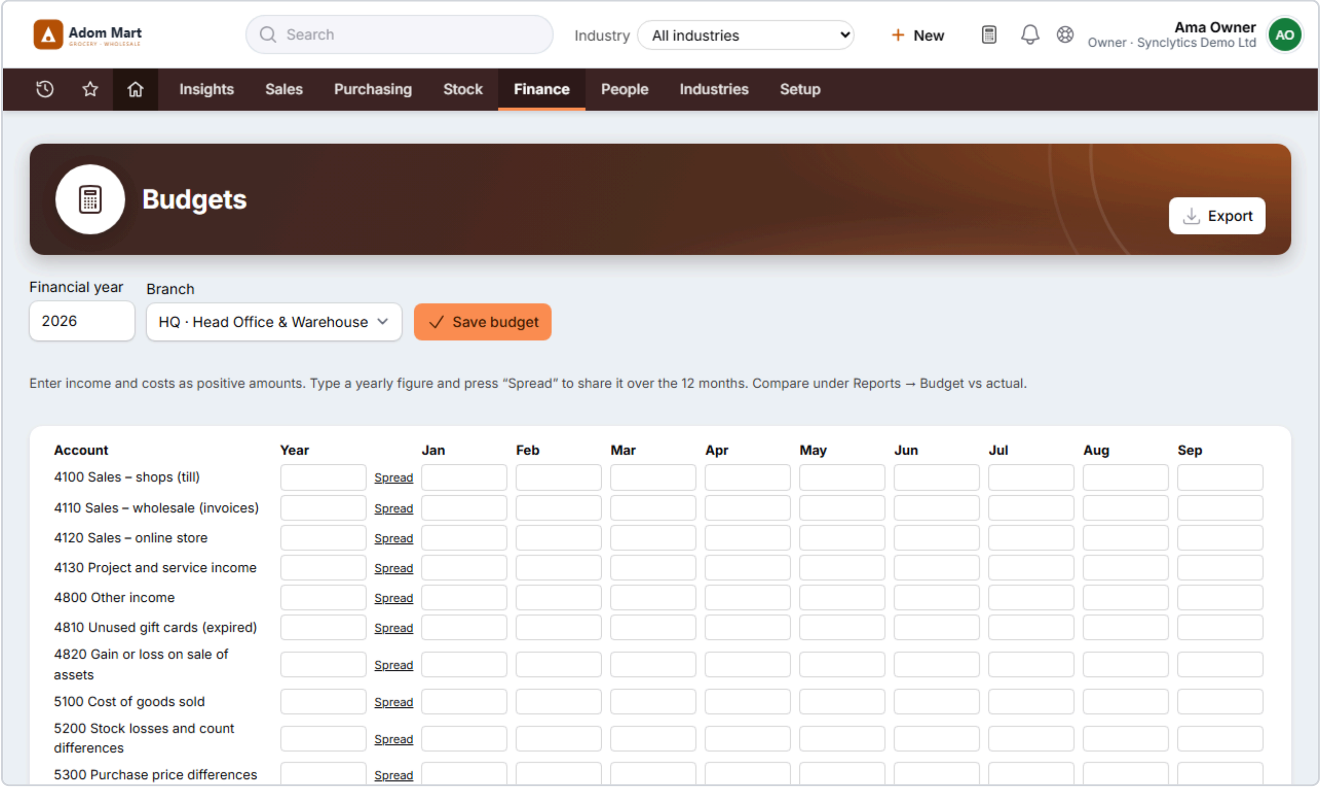
1. On a **purchase request** or **purchase order** line, choose **Service or non-stock item** instead of *Product*. Then describe it (e.g. *Servicing of the generator*), choose the **kind of spending** (the expense account, e.g. repairs), the unit (JOB, HOUR, MONTH...) and, if you want, the **department**. On a request, give an **estimated price**. It is used for the budget check, and the buyer later puts in the real price.
2. The order is submitted and approved as usual.
3. When the work is done or the items are handed over, press **Confirm delivered** on the order. Nothing goes into stock, and goods received and the handheld do not show these lines.

4. **Finance → Supplier bills → Services and non-stock items on orders:** bill what was confirmed. You cannot bill more than was confirmed, and the price is checked against the order like stock. The cost goes to the order line's account and department in the books. *Other charges* can also carry a department.

Departments are the ones under HR → Departments.

14.5f Budget check before buying

> **Why:** the budget you set under Finance → Budgets is only useful if people see it *before* they spend.



Financial year: 2026

Branch: HQ · Head Office & Warehouse

Enter income and costs as positive amounts. Type a yearly figure and press “Spread” to share it over the 12 months. Compare under Reports → Budget vs actual.

Account	Year		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep
4100 Sales – shops (till)		Spread									
4110 Sales – wholesale (invoices)		Spread									
4120 Sales – online store		Spread									
4130 Project and service income		Spread									
4800 Other income		Spread									
4810 Unused gift cards (expired)		Spread									
4820 Gain or loss on sale of assets		Spread									
5100 Cost of goods sold		Spread									
5200 Stock losses and count differences		Spread									
5300 Purchase price differences		Spread									

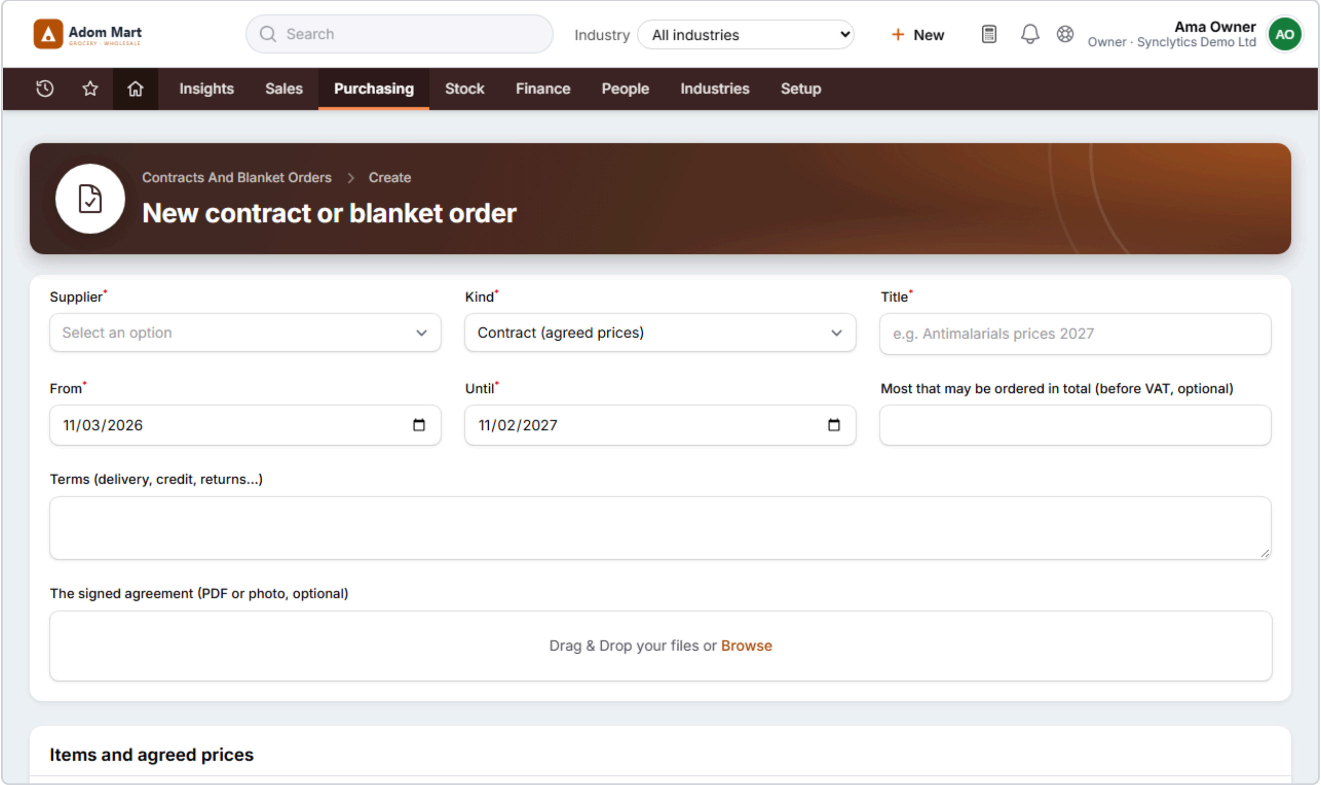
System → Settings → Purchasing → Budget check on service and non-stock buying. It is **off** until you choose:

- **Warn and record:** a request shows a **Budget** box: the year's budget for the branch and kind of spending, what is already spent, what is already ordered and not yet billed, this request, and what is left. An order that goes over is marked **Over budget** when it is submitted, with the figures, and this is recorded.
- **Over budget needs a second approval:** the same, and an order over budget is approved only by someone with **Approve orders over budget** (General Manager, Finance Manager).

Only service and non-stock lines are checked; stock is not spending until it is sold. A kind of spending with no budget counts as over, so set budgets first.

14.5g Contracts and blanket orders

> **Why:** prices agreed for the year are only worth having if every order uses them. A blanket order also stops buyers ordering more than was agreed.


Adom Mart
Search
Industry
All industries
+ New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Contracts And Blanket Orders > Create

New contract or blanket order

Supplier* Select an option

Kind* Contract (agreed prices)

Title* e.g. Antimalarials prices 2027

From* 11/03/2026

Until* 11/02/2027

Most that may be ordered in total (before VAT, optional)

Terms (delivery, credit, returns...)

The signed agreement (PDF or photo, optional)

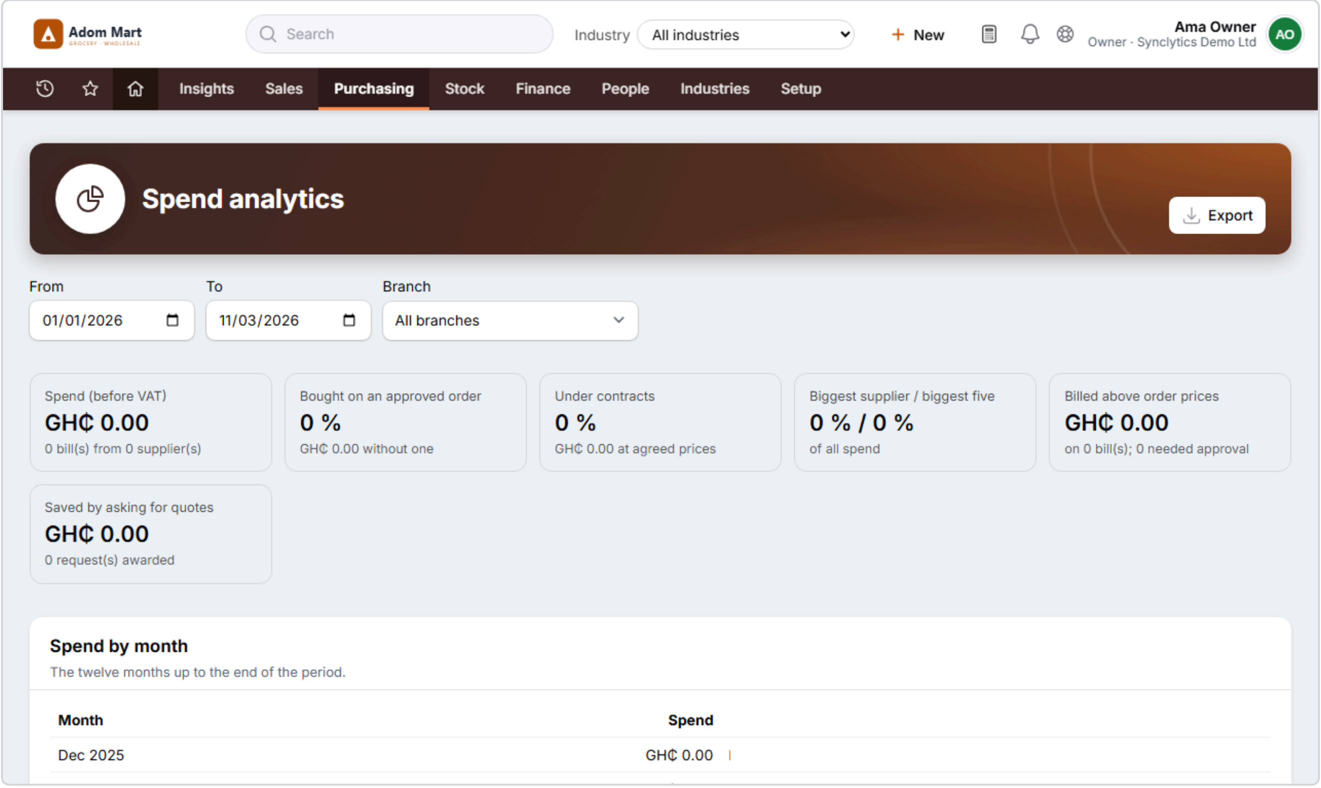
Drag & Drop your files or Browse

Items and agreed prices

1. **Purchasing → Contracts and blanket orders → New agreement:**
    - the supplier;
    - **Contract** (agreed prices) or **Blanket order** (agreed prices plus how much may be ordered);
    - the dates;
    - optionally, the most that may be ordered in total;
    - the terms, and a scan of the signed agreement;
    - each item (a product, or a service) with its **agreed price before VAT**. On a blanket order, add the most of each item that may be ordered.
2. **Put in force:** someone with *Approve purchase orders*, other than the person who made it.
3. From then until its end date:
    - an order to that supplier with no price typed **takes the agreed price**;
    - a higher price is **refused** (a lower one is fine);
    - each order counts against the limits, and an order that would pass them is refused with what is left.
4. **Order against it** (on the agreement): enter the quantities now. A draft purchase order is made at the agreed prices, and you submit it as usual. The agreement shows **Ordered so far** and what is **Left** of each item.
5. The buyers get a notice on the bell before it ends (30 days by default: the same setting as supplier documents) and when it has ended. **Close** it when it is finished.

14.5h Spend analytics

> **Why:** knowing where the money goes is the first step to spending less. It shows which suppliers to negotiate with, which categories cost most, and whether people buy without approval.



Industry: All industries

Spend analytics

From: 01/01/2026

To: 11/03/2026

Branch: All branches

Spend (before VAT)
GH₵ 0.00
0 bill(s) from 0 supplier(s)

Bought on an approved order
0 %
GH₵ 0.00 without one

Under contracts
0 %
GH₵ 0.00 at agreed prices

Biggest supplier / biggest five
0 % / 0 %
of all spend

Billed above order prices
GH₵ 0.00
on 0 bill(s); 0 needed approval

Saved by asking for quotes
GH₵ 0.00
0 request(s) awarded

Spend by month

The twelve months up to the end of the period.

Month	Spend
Dec 2025	GH₵ 0.00

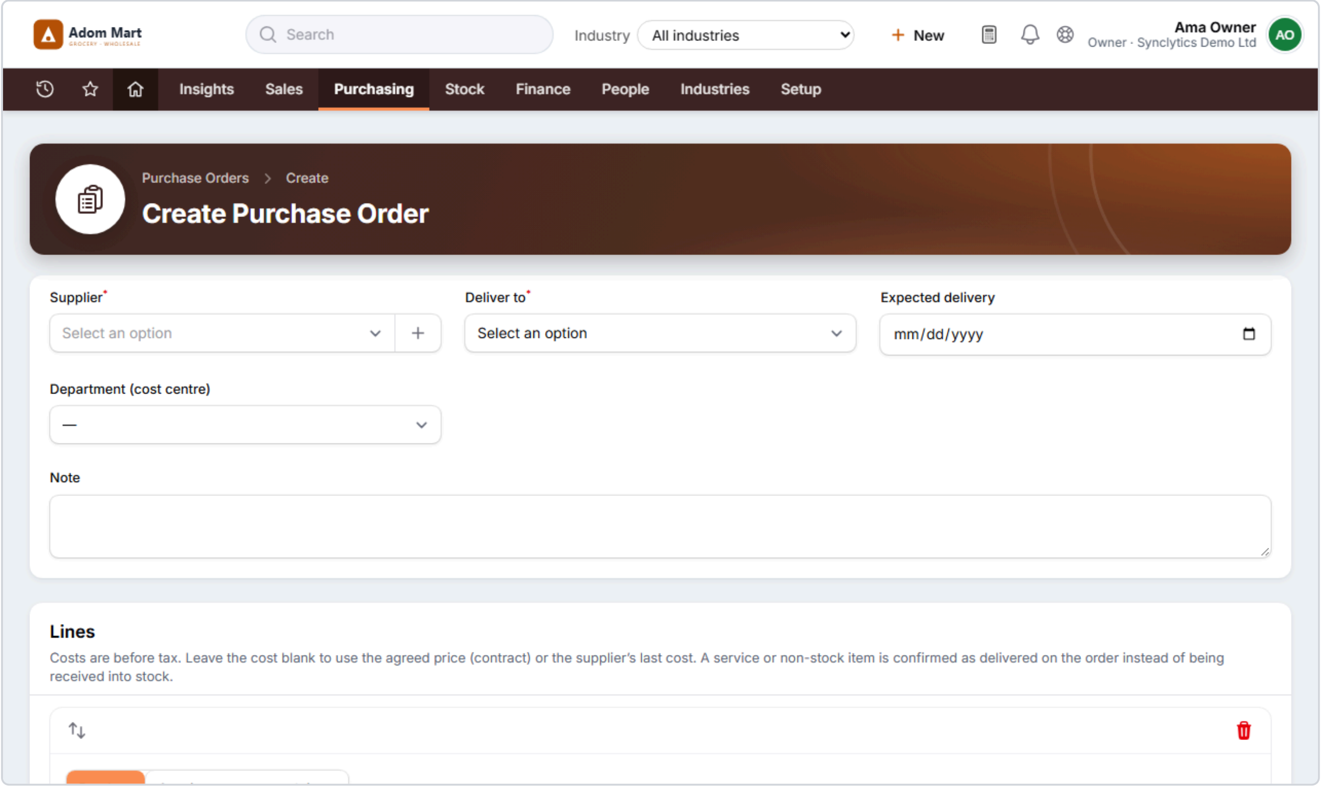
Purchasing → Spend analytics (for people who see supplier bills or approve orders). Choose the dates and, if you want, a branch. It counts posted supplier bills before VAT, by bill date:

- **Spend**, the number of bills and the number of suppliers.
- **Bought on an approved order**, in %. Spend without an order skipped approval. Rent and utilities usually have none. Goods received without an order are worth asking about.
- **Under contracts**, in %: the spend at agreed prices (14.5g).
- **Biggest supplier / biggest five**, as a share of spend. One supplier taking most of it is a risk if they let you down; many tiny ones waste effort.
- **Billed above order prices**, and how many bills needed approval for it.
- **Saved by asking for quotes**: for each item awarded on a request for quotation (14.5c), the average of the other prices quoted minus the price chosen. Also how many quotes each request got, and how many suppliers answered.
- Tables **by month** (the last twelve), **by supplier**, **by category** (the product's category for goods, the account for services and charges), **how it was bought**, and **services and charges by department**.

Export gives every table as Excel, CSV or PDF.

14.6 Purchase orders from a file, ordering again and sending to the supplier

From a file: Purchasing → Purchase orders → Order from a file.


Adom Mart
Search
Industry
All industries
+ New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Purchase Orders
Create
Create Purchase Order
Supplier*
Select an option
Deliver to*
Select an option
Expected delivery
mm/dd/yyyy
Department (cost centre)
—
Note
Lines

Costs are before tax. Leave the cost blank to use the agreed price (contract) or the supplier's last cost. A service or non-stock item is confirmed as delivered on the order instead of being received into stock.

1. Choose **An Excel or CSV list** or **A supplier's pro-forma invoice** (a photo or PDF, read by the AI — it must be switched on under System → Settings → AI).
2. Choose the supplier (for a pro-forma you may leave it empty: the system finds it from the paper) and the store to deliver to, then the file.
3. The system makes a **draft** purchase order and opens it. Check every line and cost, change what is needed, and **Submit for approval** as usual.

The list needs a heading row with a product column — **SKU** (or the supplier's code), **Barcode** or **Product** name — and **Qty**; **Unit** and **Cost** are optional (without a cost, the supplier's last price is used). Lines that match no product, have no quantity or have no cost at all are **left out and listed** on screen, never guessed.

> **Why:** suppliers send their price list or pro-forma; typing 80 lines again wastes an hour and invites mistakes.

Ordering again: on any purchase order that is not a draft, **Order again** makes a new draft with the same supplier, store and lines, at the supplier's latest prices.

Sending to the supplier: once an order is **approved**, **Send to supplier** sends it:

- **E-mail** — with the order attached as a PDF (needs the e-mail settings, 16.1);

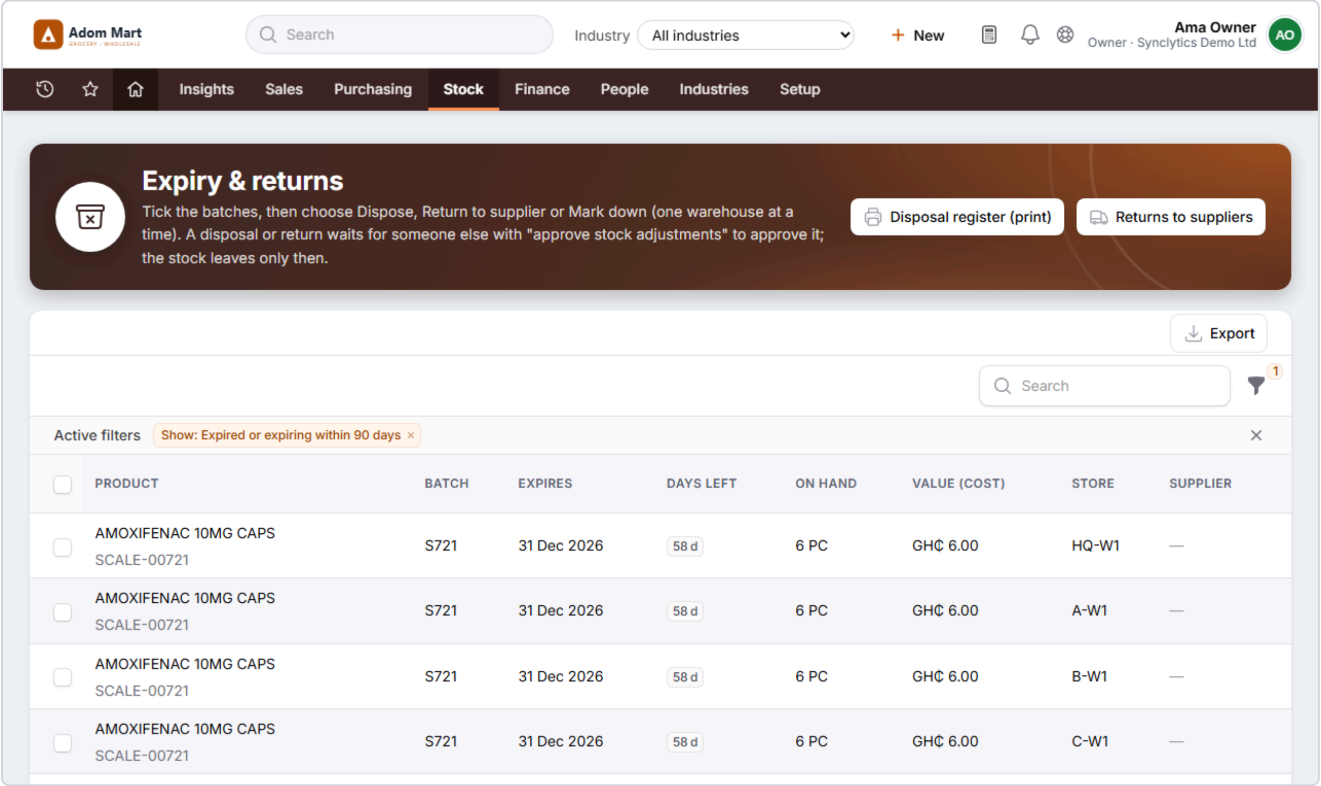
- **WhatsApp from this computer** — WhatsApp opens with the message and a link to download the order; you press send. Works without any WhatsApp provider;
- **WhatsApp (sent by the system)** — only when a WhatsApp Business provider is set up (16.3).

The order then shows **Sent to supplier** with when, how and to whom; **Send again** sends it once more. The download link works for 30 days, only for that order. The wording is under **Messages** → **Templates** → **Purchase order sent to a supplier**.

> **Why:** the supplier gets a clean, numbered order straight away, and everyone can see that it went and how.

14.7 Expired and near-expiry stock: disposals and returns to suppliers

Stock → **Expiry & returns** lists every batch in stock that has expired or expires soon, earliest first. For each batch it shows the days left (red once expired, amber under 30 days), the quantity, its value at cost, the store, the supplier it came from and whether that supplier still takes it back. Controlled drugs are marked.


Adom Mart
GROCERY · WHOLESALE
Search
Industry
All industries
+ New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Disposal register (print)
Returns to suppliers
Export
Search
1
Active filters
Show: Expired or expiring within 90 days ×

Expiry & returns

Tick the batches, then choose Dispose, Return to supplier or Mark down (one warehouse at a time). A disposal or return waits for someone else with "approve stock adjustments" to approve it; the stock leaves only then.

☐	PRODUCT	BATCH	EXPIRES	DAYS LEFT	ON HAND	VALUE (COST)	STORE	SUPPLIER
☐	AMOXIFENAC 10MG CAPS SCALE-00721	S721	31 Dec 2026	58 d	6 PC	GH₵ 6.00	HQ-W1	—
☐	AMOXIFENAC 10MG CAPS SCALE-00721	S721	31 Dec 2026	58 d	6 PC	GH₵ 6.00	A-W1	—
☐	AMOXIFENAC 10MG CAPS SCALE-00721	S721	31 Dec 2026	58 d	6 PC	GH₵ 6.00	B-W1	—
☐	AMOXIFENAC 10MG CAPS SCALE-00721	S721	31 Dec 2026	58 d	6 PC	GH₵ 6.00	C-W1	—

> **Why:** expired medicines must leave the shelf and be destroyed (or returned) with a record. Doing it batch by batch from a list stops anything being missed, and the record satisfies an FDA inspection.

1. Use **Show** to choose what to list: *Expired*, *Expiring within 30 days*, *within the warning period* (default 90 days, **System** → **Settings** → **Stock**) or every batch. Filter by store, supplier, **controlled drugs only** or **only what the supplier still takes back**. Search by product, SKU or batch.
2. **Tick** the batches (or tick the box at the top for all of them). Tick one store at a time.
3. Choose what to do:

    - **Dispose** — for stock that will be destroyed. Give the reason (expired or damaged), how it is disposed of (incineration, destroyed under FDA supervision, an approved waste company...), the date and the **witness** (name and role). If your company asks for it, attach the **disposal certificate** (photo or PDF).
    - **Return to supplier** — choose the supplier (filled in when the system knows it), what the supplier gives (**credit note**, **replacement stock**, **cash / MoMo / bank refund** or **nothing**) and the reason.
    - **Mark down** — for stock that can still be sold at a lower price (needs the markdown permission). Print the stickers from **Stock → Batches & expiry**.
4. The disposal or return is **prepared** and waits for approval. Keep the packs aside, marked "not for sale". Everyone allowed to approve is told.
5. **Someone else** with *approve stock adjustments* opens it and clicks **Approve and post** (disposal, in **Stock → Stock adjustments**) or **Approve and send** (return, in **Stock → Returns to suppliers**). Only then does the stock leave the system. Nobody can approve what they prepared, whatever the value.

Why two people, always: expired stock is an easy place to hide theft ("it expired" when it was sold privately). A second person checking every disposal and return, whatever its value, is the standard control for pharmacies.

Evidence (System → Settings → Stock → Evidence kept when stock is disposed of):

Setting	The person preparing must give
Reason only	the reason
Method and witness (default)	the reason, the method and a witness
Method, witness and certificate	the above and an uploaded certificate

Controlled drugs always need a witness, whatever the setting.

Printing: each disposal has **Disposal record (print)** with signature lines for the pharmacist, the witness and the approver; each return has a **Return note (print)** for the supplier's representative to sign. **Disposal register (print)** on *Expiry & returns* lists every approved disposal for a store and period — keep it in the premises file.

After a return is sent: when the supplier settles, open the return and click **Record the supplier's settlement**: their credit note number, the delivery note of the replacement stock (receive the replacement as a normal delivery) or, for a refund, the amount and where the money went (refunds need the *pay suppliers* permission). A refund smaller than the value at cost books the difference as a stock loss.

What happens in the accounts (automatic): a disposal is a stock loss at average cost. A return moves the stock to *Goods returned to suppliers (awaiting credit)*; a credit note or replacement reduces what you owe the supplier; *nothing* is a stock loss; a refund clears it when the money is recorded.

Each supplier's return window: on the supplier (**Purchasing → Suppliers**), **Takes back expiring stock up to (days before expiry)** — for example 90 if they accept stock until 90 days before it expires. Leave it empty if they take nothing back. *Expiry & returns* then shows "can still go back" while there is time.

Correcting an expiry date, adding or moving a batch (**Stock → Batches & expiry**, choose the store):

- **Correct expiry** on a batch (in *All batches on hand* or *Expired*): type the right date and why. The stock stays in the batch; the change and the reason are in the audit log. Use it for a date typed wrongly at receiving, or one that is missing.
- **Add or move a batch** (top right):
    - *Stock of a batch the system does not have* — found on the shelf, or never entered: the product, batch number, expiry and quantity (and the cost if you know it);
    - *Stock recorded under the wrong batch number*: choose the batch it is wrongly under, then type the right batch number, its expiry and the quantity to move.

  Either way a **draft stock adjustment** opens: check it and post it as usual (a manager approves it above the limit).

Why a wrong batch number is moved, not renamed: the batch number is how every server and every delivery recognises the batch. Moving the stock keeps the history true and lets the next delivery of that batch join it.

Many at once: the stock sheet import (14.2) also corrects dates and adds batches.

Stop selling before expiry: System → Settings → Stock → Stop selling a batch this many days before it expires (default 0 = sell until the expiry date). With 30, the till takes the next batch, or refuses the sale if only stock with less than 30 days is left. Expired stock is never sold.

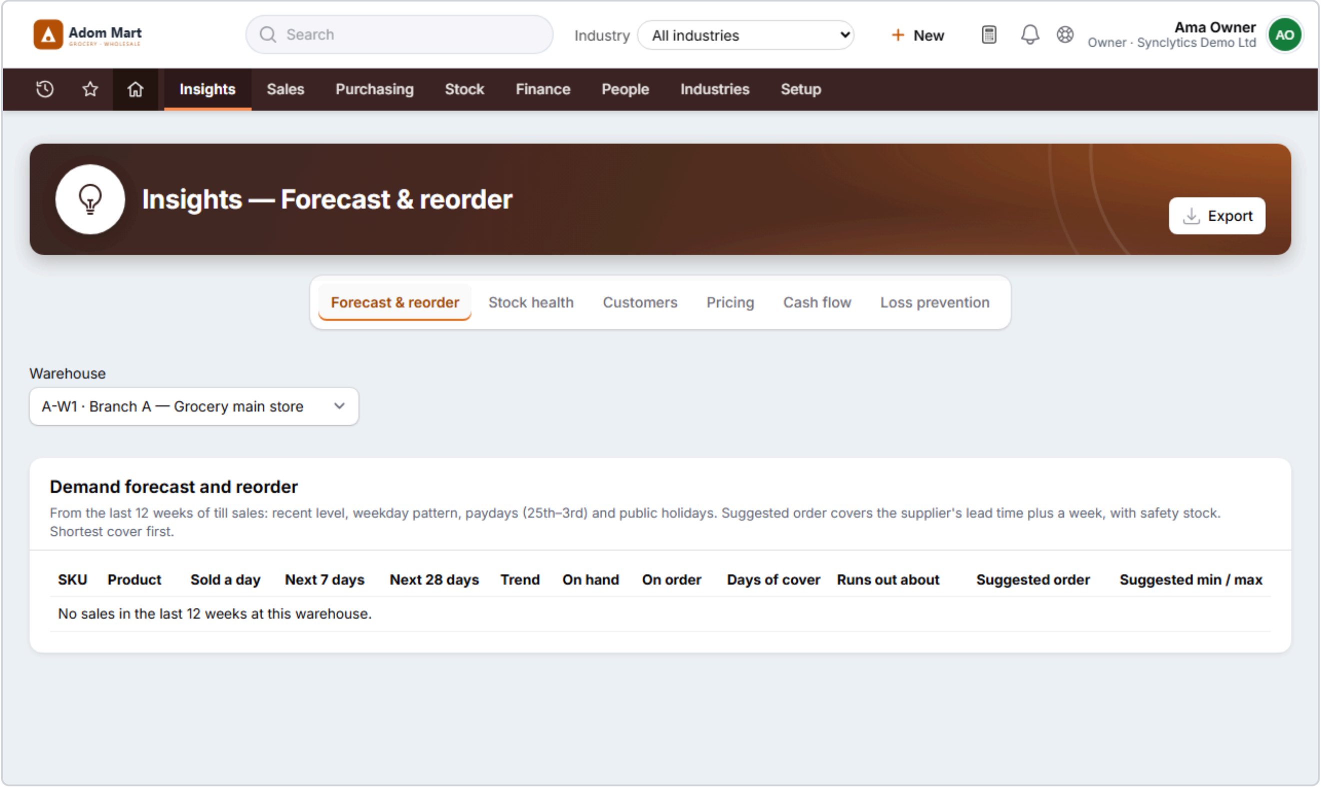
The dashboard (Pharmacy) shows how many batches have expired and how many expire within the warning period; click either to open *Expiry & returns*.

15. Insights, alerts and the AI assistant

15.1 What works straight away (no AI needed)

The **Insights** menu works on every installation, offline too. It is ordinary arithmetic on your own sales and stock — nothing is sent anywhere and nothing needs installing.

Page	What it shows	Who sees it
Insights → Forecast & reorder	Expected sales of every product for the next 7 and 28 days, days of stock left, the day it runs out, and a suggested order quantity and min/max	Managers (*See insights*)
Insights → Stock health	Dead stock (no sale for 90 days), overstock, batches that will expire before they sell, and transfers between branches	Managers
Insights → Customers	Customer segments (champions, loyal, at risk…), regulars who have stopped coming, products bought together	Company-wide managers
Insights → Pricing	Products with a low margin or sold below cost, markdown suggestions, how much each promotion lifted sales	Company-wide managers
Insights → Cash flow	Money expected in and out, week by week, for 8 weeks	Company-wide managers
Insights → Loss prevention	For each cashier: refunds, refunds without a receipt, discounts, drawer over/short, kitchen voids, and a risk score	Owner, general, finance and branch managers
Insights → Alerts	A to-do list built every night from all of the above	Managers


Adom Mart
Search
+ New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Insights — Forecast & reorder
Export
Forecast & reorder
Stock health
Customers
Pricing
Cash flow
Loss prevention

Industry: All industries

Warehouse: A-W1 · Branch A — Grocery main store

Demand forecast and reorder

From the last 12 weeks of till sales: recent level, weekday pattern, paydays (25th–3rd) and public holidays. Suggested order covers the supplier's lead time plus a week, with safety stock. Shortest cover first.

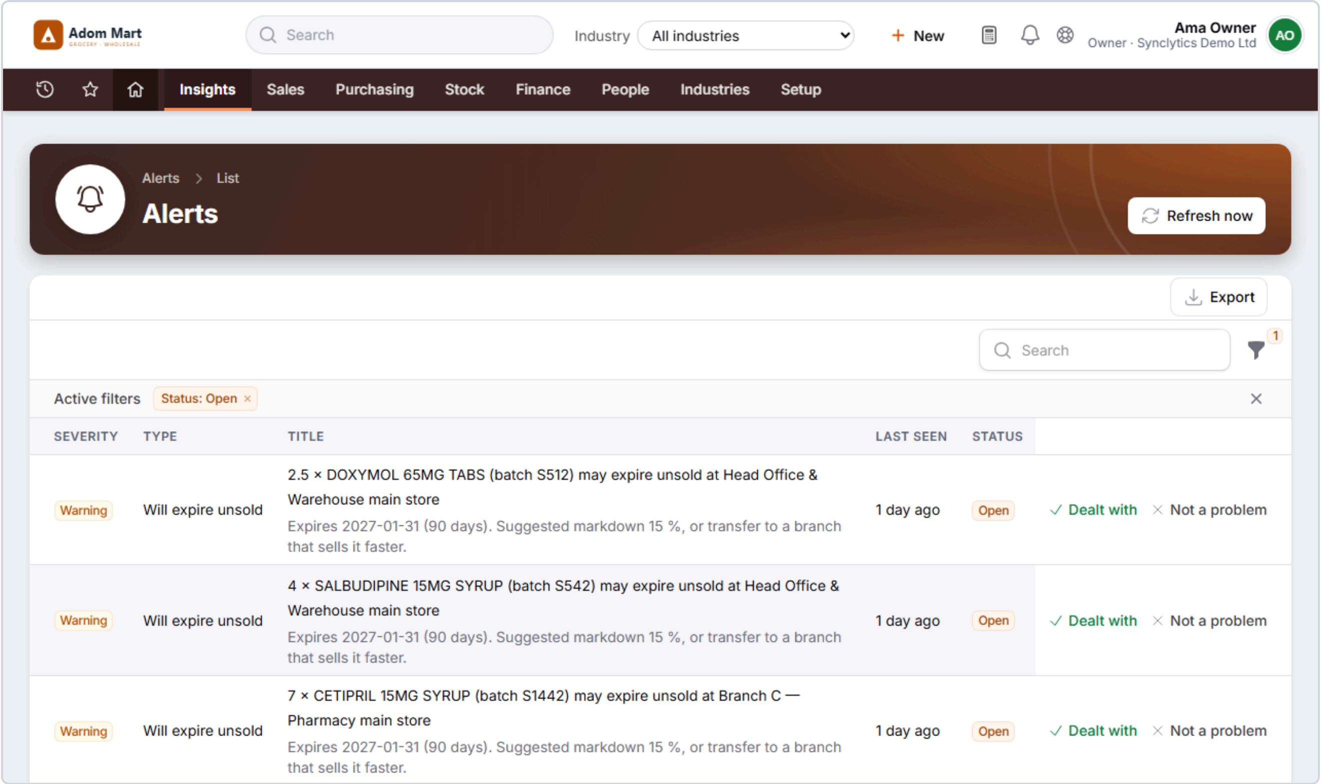
SKU	Product	Sold a day	Next 7 days	Next 28 days	Trend	On hand	On order	Days of cover	Runs out about	Suggested order	Suggested min / max
No sales in the last 12 weeks at this warehouse.											

How the forecast works (in plain words): it looks at the last 12 weeks of till sales, gives the most recent weeks the most weight, and adjusts for the day of the week, for **paydays** (the 25th to the 3rd, when shops in Ghana are busier) and for **public holidays** (fixed holidays, Easter and Farmers' Day; Eid dates change each year and show up through your own history). The suggested order covers the supplier's **lead time** (set it on the supplier) plus a week, with a little safety stock.

> **Why:** you see trouble — a best-seller running out, yoghurt that will expire, a till with too many refunds — before it costs money.

> **Note:** The risk score is a reason to look (receipts, CCTV), never proof. Cashiers with fewer than 20 receipts in the period are listed but not scored.

Alerts



SEVERITY	TYPE	TITLE	LAST SEEN	STATUS	
Warning	Will expire unsold	2.5 × DOXYMOL 65MG TABS (batch S512) may expire unsold at Head Office & Warehouse main store Expires 2027-01-31 (90 days). Suggested markdown 15 %, or transfer to a branch that sells it faster.	1 day ago	Open	✓ Dealt with × Not a problem
Warning	Will expire unsold	4 × SALBUDIPINE 15MG SYRUP (batch S542) may expire unsold at Head Office & Warehouse main store Expires 2027-01-31 (90 days). Suggested markdown 15 %, or transfer to a branch that sells it faster.	1 day ago	Open	✓ Dealt with × Not a problem
Warning	Will expire unsold	7 × CETIPRIL 15MG SYRUP (batch S1442) may expire unsold at Branch C — Pharmacy main store Expires 2027-01-31 (90 days). Suggested markdown 15 %, or transfer to a branch that sells it faster.	1 day ago	Open	✓ Dealt with × Not a problem

1. Every night at 04:30 the scheduler refreshes the alerts (it must be running — Chapter 4/5). To refresh now: **Insights → Alerts → Refresh now**.
2. For each alert choose **Dealt with** (say what you did) or **Not a problem** (it stays quiet for 30 days).
3. An alert closes by itself when its cause goes away (e.g. the stock arrived).

15.2 The AI assistant — off unless you switch it on

The owner chose **no cloud AI for now**, so the AI assistant is **off**. Three features need it:

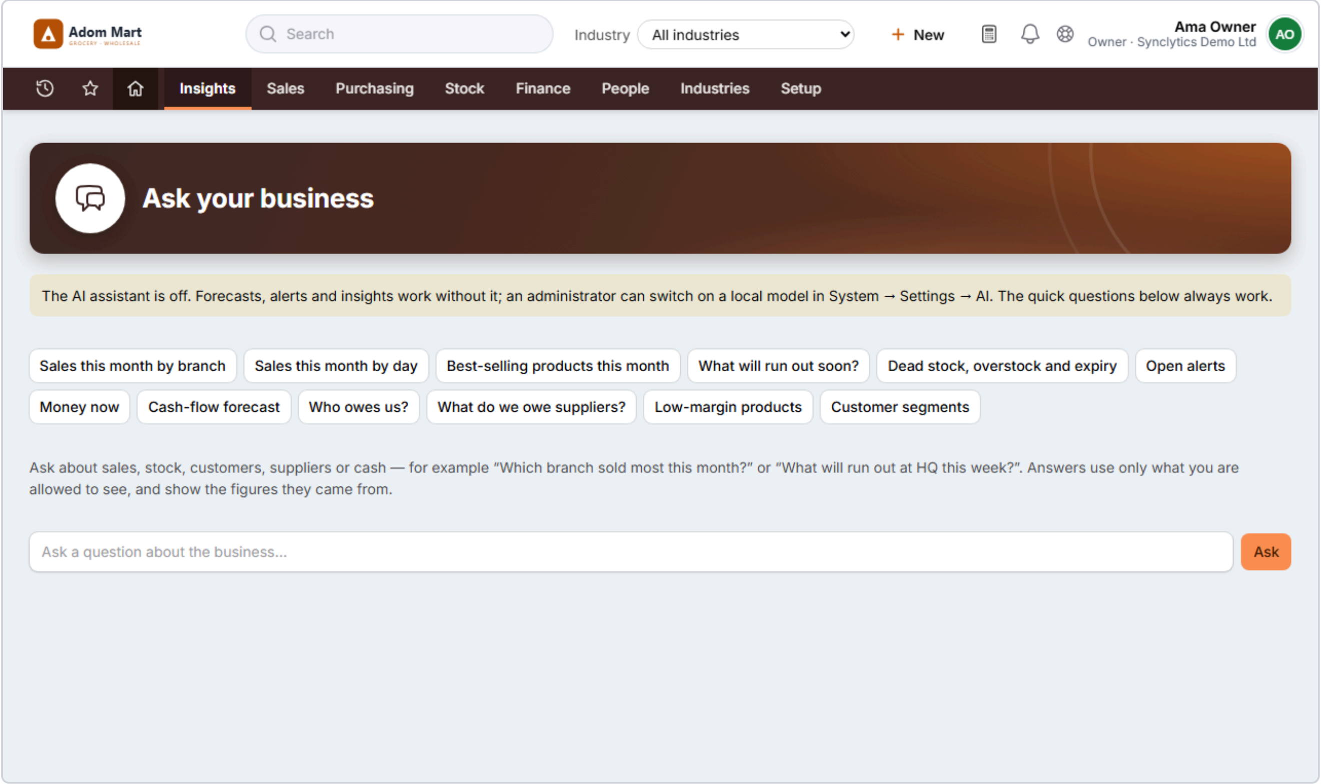
- **Ask your business** (the copilot): questions in plain language, answered from your reports.

- **Scan an invoice** on Supplier bills: a photo or PDF of a supplier invoice fills in the bill form. The AI also reads the order number on the invoice. It then matches each line to goods received (that order's deliveries first) or to a service confirmed on an order, and puts what matches nothing under *Other charges*. Above the form it lists what to look at:
    - the lines read do not add up to the invoice's total;
    - the invoice is already entered;
    - the supplier was not found;
    - the order number is not theirs;
    - some lines matched nothing.

  > **Why:** a missed or misread line is caught before the bill is entered, not at month end. Nothing is entered until a person presses *Create*.

- **Read with AI** on *Purchasing → Invoices from suppliers* (invoices sent on the supplier portal): the same, using the file the supplier sent. **Enter as bill** opens the bill form with the supplier, invoice number and date filled in, for when the AI is off. Either way, creating the bill marks the supplier's invoice *entered*, and the supplier sees this on their page.

- **Write description (AI)** on a product: a draft text for the online shop.


Adom Mart
Search
Industry
All industries
+ New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Ask your business
Sales this month by branch
Sales this month by day
Best-selling products this month
What will run out soon?
Dead stock, overstock and expiry
Open alerts
Money now
Cash-flow forecast
Who owes us?
What do we owe suppliers?
Low-margin products
Customer segments
Ask a question about the business...
Ask

The AI assistant is off. Forecasts, alerts and insights work without it; an administrator can switch on a local model in System → Settings → AI. The quick questions below always work.

Ask about sales, stock, customers, suppliers or cash — for example "Which branch sold most this month?" or "What will run out at HQ this week?". Answers use only what you are allowed to see, and show the figures they came from.

Without AI, *Ask your business* still offers **quick questions** (sales by branch, what will run out, who owes us...) answered straight from the reports.

You can switch on a **local model**: it runs on your own server, so nothing leaves your network and there is no monthly bill.

Requirements for a local model

	Minimum	Better
Memory (RAM)	16 GB	32 GB
Disk space	15 GB free	30 GB
Graphics card	Not required (slower)	NVIDIA with 8 GB or more

> **Why:** the model is a large file held in memory while it answers. On a small server answers take 20–60 seconds; with a graphics card a few seconds.

Installing Ollama (the program that runs local models)

On Windows:

1. Download the installer from https://ollama.com/download and run it (Next → Install). It starts by itself and sits next to the clock.
2. Open **Command Prompt** and download the two models (this takes a while, about 5 GB each):

   ```
   ollama pull qwen2.5:7b
   ollama pull qwen2.5vl:7b
   ```

   Why two: the first answers questions and writes text; the second reads photos of invoices.
3. Check it works: `ollama run qwen2.5:7b "Say hello"` — it should answer.

On Ubuntu:

```
curl -fsSL https://ollama.com/install.sh | sh
ollama pull qwen2.5:7b
ollama pull qwen2.5vl:7b
```

The installer sets Ollama to start with the server.

> **Note:** Ollama listens only on the server itself (`127.0.0.1:11434`). Keep it that way: do not open port 11434 in the firewall.

Switching the assistant on

1. **System** → **Settings** → **AI**.
2. **AI assistant:** *Local model (Ollama on your server)*.
3. Leave **Ollama address** as `http://127.0.0.1:11434` (Ollama on the same server). If Ollama runs on another computer in your office, put its address here.
4. The model names must match what you downloaded (`qwen2.5:7b` and `qwen2.5vl:7b`).
5. Save, then open **Insights** → **Ask your business** and ask "Which branch sold most this month?".

> **Check:** the answer appears with the table it came from. The question is listed in **Insights → AI usage log**.

If you later choose Claude (cloud AI)

Only with the owner's decision, because data then leaves your servers:

1. Get an API key from Anthropic (the owner's own account).
2. Put it in the server's `.env` as `ANTHROPIC_API_KEY=...`, then run `php artisan optimize`. *Why `.env`:* keys are never stored in the database or shown on screen.
3. **System → Settings → AI:** choose *Claude*, and set **Cloud AI budget per month (USD)**. With a budget of 0 Claude is never used; when the month's budget is spent it stops until the 1st.
4. Keep **Keep personal data on your server** on (recommended): questions about customers or staff are then answered by the local model only (or shown without AI wording if there is none). In every case names, phone numbers and e-mail addresses are replaced by placeholders such as *[Customer 1]* before anything is sent, and put back when the answer arrives.

If you choose another cloud model (some are free)

Instead of Claude you can use another cloud service. Google Gemini, Groq, OpenRouter (its *free* models) and Mistral have free plans that limit how many questions a minute or a day; OpenAI is paid. Data leaves your servers just as with Claude, so the owner decides.

1. **Get an API key** from the service, signed in with the business's own account:
   - Google Gemini: aistudio.google.com → **Get API key**;
   - Groq: console.groq.com → **API Keys**;
   - OpenRouter: openrouter.ai → **Keys** (choose models ending in *:free*);
   - Mistral: console.mistral.ai → **API Keys**.
2. Put it in the server's `.env` as `AI_CLOUD_API_KEY=...`, then run `php artisan optimize`. *Why `.env`:* keys are never stored in the database or shown on screen.
3. **System → Settings → AI:** choose **Another cloud model**, then the **service**. Leave the model names empty to use the service's usual model, or copy an exact name from the service's model list (services rename their models from time to time). For **Other OpenAI-compatible service**, also give its address ending in `/v1`.
4. Keep **Keep personal data on your server** on, as for Claude. Names, phones and e-mails are masked before anything is sent.

> **Check:** ask a question in **Insights → Ask your business**; **Insights → AI usage log** shows the service and model that answered.

> **Note:** the monthly cloud budget applies to Claude only. Free plans cost nothing; with a paid service (e.g. OpenAI) set a spending limit in that service's own account. Invoice scanning with these services reads photos (JPG, PNG), not PDFs.

15.3 Using the AI features safely

- **The AI never changes anything.** It reads reports; it cannot post, approve, order or pay. An invoice scan only fills in a form: you check each line against the paper and press **Create**, and the bill then follows the normal approval rules.
- **Answers show their figures.** Every copilot answer comes with the tables it used. If a figure looks wrong, the table shows where it came from.
- **People only see what they may see.** A branch manager's questions use only their branch; money and customer figures for the whole company need a company-wide role.
- **Everything is logged.** *Insights → AI usage log* lists every question, what was sent (as the model saw it, with names masked), the answer and the cost.
- Photos of invoices: the local model reads JPG and PNG photos (not PDFs — take a photo or screenshot). Claude can read PDFs.

15.4 Troubleshooting

Message	What to do
The AI assistant is off	Switch it on (15.2), or use the quick questions
The local AI model is not reachable	Start Ollama (Windows: open it from the Start menu; Ubuntu: `sudo systemctl start ollama`) and check the address in Settings → AI
The local AI model answered with an error: model not found	Download the model named in Settings (`ollama pull <name>`)
Answers are very slow	Normal without a graphics card; use a smaller model such as `qwen2.5:3b` (less accurate)
This month's cloud AI budget is used up	Wait for the 1st, raise the budget, or switch to the local model
No alerts ever appear	Check the scheduler is running (Chapter 4/5) or press **Refresh now**

16. Messages, reports and connecting other systems

16.1 E-mail (any mail server)

The system sends e-mail for scheduled reports, invoices to customers (when you switch that on) and digital receipts. It uses **any SMTP mail server** you already have: Google Workspace or Gmail, Zoho Mail, Microsoft 365, or your web host's mail.

1. Get the SMTP details from your mail provider: server name, port, user name and password.
    - **Gmail / Google Workspace:** server `smtp.gmail.com`, port `587`; the password must be an **app password** (Google account → Security → 2-Step Verification → App passwords), not your normal password.
    - **Zoho:** `smtp.zoho.com`, port `587`.
    - **Your web host:** see its "e-mail accounts" page (often `mail.yourdomain.com`, port `587` or `465`).
2. Open the server's `.env` file (Chapter 4/5) and set:

```
MAIL_MAILER=smtp
MAIL_HOST=smtp.gmail.com
MAIL_PORT=587
MAIL_USERNAME=reports@yourcompany.com
MAIL_PASSWORD=<the app password>
MAIL_FROM_ADDRESS=reports@yourcompany.com
MAIL_FROM_NAME="Your Company"
```

3. In the `server` folder run `php artisan optimize`.
4. **System** → **Messaging** → **Send a test e-mail** to your own address.

> **Why:** the password lives only in `.env` on the server — never in the database or on screen. `MAIL_MAILER=log` (the default) writes e-mails to the log file instead of sending them, which is safe for testing.

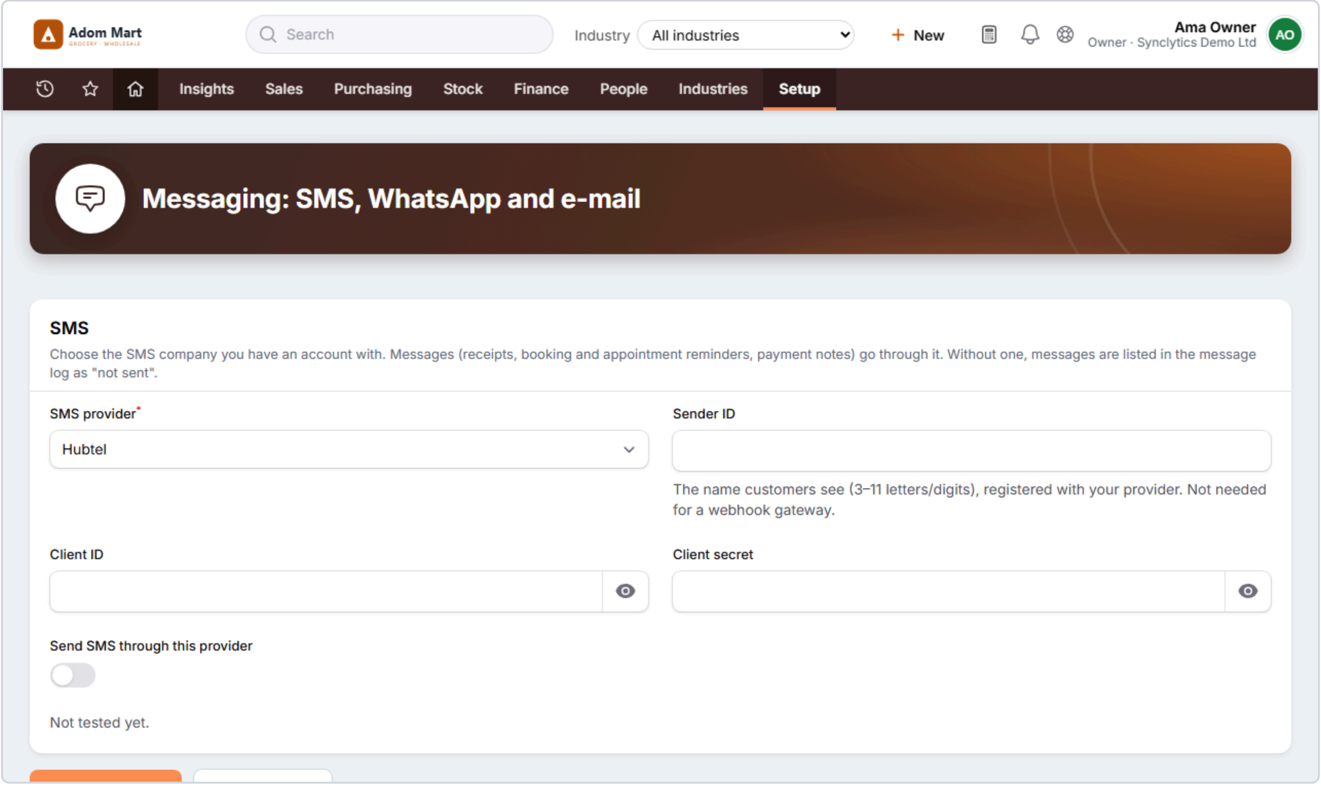
16.2 SMS — choose your provider

You can use **Hubtel**, **Arkesel**, **mNotify**, or **any other SMS company** through a small "webhook" bridge. Use the one you already have an account with.

1. Open an account with the provider (their website) and buy SMS credit.
2. Register a **sender ID** (the name customers see, 3–11 letters, e.g. `DemoShop`). In Ghana sender IDs must be approved (your provider handles it with the NCA); it can take a few days.
3. Copy the keys from the provider's dashboard:

Provider	What you need
Hubtel	Client ID and Client secret (Hubtel dashboard → Programmable SMS → API keys)

Provider	What you need
Arkesel	API key (Arkesel dashboard → SMS API)
mNotify	API key (mNotify dashboard → API)
Other (webhook)	The address of your gateway. It receives `{"to", "message", "reference"}` as JSON; an optional shared secret signs it (`X-Signature`).



Messaging: SMS, WhatsApp and e-mail

SMS

Choose the SMS company you have an account with. Messages (receipts, booking and appointment reminders, payment notes) go through it. Without one, messages are listed in the message log as "not sent".

SMS provider*: Hubtel

Sender ID:

The name customers see (3–11 letters/digits), registered with your provider. Not needed for a webhook gateway.

Client ID:

Client secret:

Send SMS through this provider ○

Not tested yet.

4. **System** → **Messaging** → **SMS**: choose the provider, type the sender ID and keys, tick **Send SMS through this provider**, **Save**.
5. **Send a test SMS** to your own phone.

> **Note:** keys are stored encrypted with this server's key and never shown again — an empty box keeps the saved key. Each server that sends messages (normally head office) has its own settings.

16.3 WhatsApp Business (optional)

WhatsApp messages need a **WhatsApp Business account on Meta's Cloud API**; Meta charges per conversation. Businesses may only start a conversation with a **template approved by Meta**.

1. In Meta Business Manager create a WhatsApp Business app, add your business phone number and create a **permanent access token** (system user).
2. Create templates with the same wording as in **Messages** → **Templates** (WhatsApp lines); the placeholders become `{{1}}`, `{{2}}` ... in order. Wait for approval.
3. **System** → **Messaging** → **WhatsApp Business**: Phone number ID and access token, tick **Send WhatsApp messages**, **Save**; **Send Meta's "hello_world" test**.

4. In **Messages** → **Templates**, put the approved template name on each WhatsApp line and switch it on.

When WhatsApp is on and a message has an active WhatsApp template, customers get WhatsApp instead of SMS.

16.3a Orders by WhatsApp (optional)

Customers send their list, or a photo of their prescription, to your WhatsApp Business number; staff turn it into an order and the customer is answered on WhatsApp.

1. In Meta's app settings (**App settings** → **Basic**) copy the **App secret** into **System** → **Messaging** → **WhatsApp Business** → **App secret**, and **Save**. **Why:** messages are accepted only when Meta signs them with this secret, so nobody else can post fake orders to your server.
2. The page then shows a **callback URL** and a **verify token**. In Meta (**WhatsApp** → **Configuration** → **Webhook**) enter both, press **Verify and save**, and subscribe to **messages**. **Why:** this tells Meta where to send your customers' messages.
3. **System** → **Settings** → **WhatsApp orders**: switch on **Take orders by WhatsApp**; change the **Automatic reply** if you like ({number} is the order's reference). **Why:** off by default, so nothing arrives until you are ready to answer.
4. Send a message from your own phone to the business number. It appears within seconds under **Online** → **WhatsApp orders** and you get the automatic reply.

Every day: open **Online** → **WhatsApp orders** (the number beside it is how many are waiting). Open one to read the message and see the photos (click a photo to see it full size). Then:

- **Reply** to ask something (which strength, a price);
- **Make the order**: the branch that fills it, picked up or delivered (with the address), and the items with quantities. Stock is held for it, the customer gets the items and the total, and it continues under **Online** → **Online orders** (**Mark ready** tells the customer it is ready; **Hand over** takes the payment);
- **Decline** with the reason the customer will read (e.g. *out of stock*, *a prescription is needed*).

> **Note:** WhatsApp lets you reply freely for 24 hours after the customer's last message; after that, the customer must write again. Prescription medicines still need the pharmacist's check when handed over, as at the counter.

16.4 Which messages are sent, and their wording

Messages → **Templates** lists every message per channel. Edit the text, use the placeholders shown, and switch each one **on or off**.

Message	Starts	Sent when
Receipt	On	The cashier presses **Send receipt** on the till (SMS, WhatsApp or e-mail)

Message	Starts	Sent when
Table booking / appointment confirmed and reminder	On	A booking or appointment is made, and before it starts
Invoice sent to the customer	**Off**	An invoice is issued (e-mail to the customer's address)
Payment received	**Off**	A customer pays on account
Scheduled report	On	A scheduled report runs (16.6)
Payment alert / Daily mobile money and card summary	On (sent only when *Payment alerts* are switched on, Chapter 7 Step 6)	A MoMo or card payment at a till, and the summary at its time. WhatsApp needs approved templates named `payment_alert` and `payment_summary`
Sign-in code (back office)	On	Someone whose sign-in method is *password and a code* signs in (WhatsApp needs an approved template named `signin_code` with one parameter, the code; otherwise e-mail). The code is never shown in the message log
Purchase order sent to a supplier	On	A buyer presses **Send to supplier** on an approved order (e-mail with the PDF, or WhatsApp with a download link; 14.6)

Messages → **Message log** shows every message: to whom, the text, and whether it was **sent**, is **waiting** (the provider was down — it is retried automatically for a few hours), **failed**, or **not sent** because no provider is set up. **Try again** resends one.

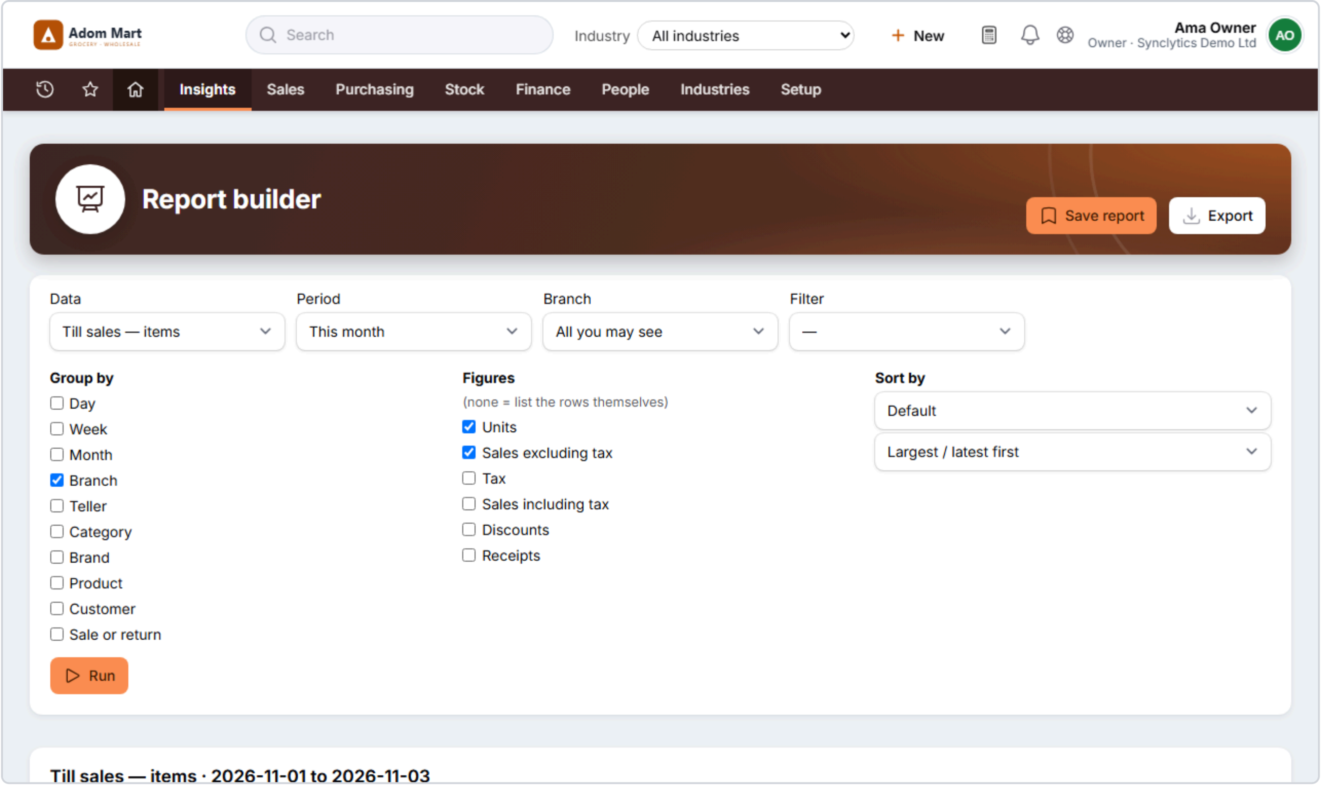
In the back office the bell (top right) tells approvers when something waits for them: purchase orders, supplier bills with price differences, stock adjustments, deliveries over the tolerance, stock counts, van settlements and expense claims.

16.5 Digital receipts at the till

After a sale, under the receipt: choose **SMS**, **WhatsApp** or **E-mail**, type the number or address, **Send receipt**. Only the channels switched on in Messages → Templates appear. A sale made offline can be sent once the till is back online.

16.6 Reports you build yourself

Reports → **Report builder**:


Adom Mart
GROCERY · WHOLESALE
Search
Industry
All industries
+ New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Report builder
Save report
Export
Run

Data: Till sales — items
Period: This month
Branch: All you may see
Filter: —

Group by
☐ Day
☐ Week
☐ Month
☑ Branch
☐ Teller
☐ Category
☐ Brand
☐ Product
☐ Customer
☐ Sale or return

Figures
(none = list the rows themselves)
☑ Units
☑ Sales excluding tax
☐ Tax
☐ Sales including tax
☐ Discounts
☐ Receipts

Sort by
Default
Largest / latest first

Till sales — items · 2026-11-01 to 2026-11-03

1. **Data**: till sales (items, receipts or payments), stock on hand, stock movements, customer invoices, supplier bills or the general ledger.
2. **Period** and **Branch**; optionally a **Filter** (e.g. Category contains "drinks").
3. **Group by** (branch, day, month, cashier, category, product, customer...) and **Figures** (units, sales, tax, discounts, receipts, value...). With no figures you get the rows themselves, each with an **Open** link to the document. For till sales, **Sales including tax** is what customers paid; **Sales excluding tax** is that amount with the tax (VAT, NHIL, GETFund...) taken off, the shop's own takings; **Tax** is the difference.
4. **Run**. Click any line to see the rows behind it (**Back to the summary** returns).
5. **Export** as Excel, CSV or PDF, or **Save report**.

Scheduling: in **Save report** choose *Every day / week / month*, the time, the format (PDF, Excel, CSV) and the recipients: e-mail addresses (the file is attached), WhatsApp numbers (a short note) or people in the system (bell). Reports are sent by head office every 15 minutes when due. **Reports** → **Saved reports** lists them with the next run and the last result.

> **Why:** a scheduled report runs *as the person who saved it*, so it never shows more than they are allowed to see.

16.7 Exporting to QuickBooks, Xero or Sage

If your accountant keeps the books in another program: **Finance** → **Export to QuickBooks / Xero / Sage**, choose the program and the period (normally last month), **Download the file**, and import it there:

Program	Import it under
QuickBooks Online	Settings → Import data → Journal entries

Program	Import it under
Xero	Accounting → Manual journals → Import
Sage 50 / Pastel	File → Import → Audit trail transactions

Before the first import, create the same account codes in the other program (or map them during import). The Xero file uses the tax rate "Tax Exempt" because tax is already in separate lines; rename it if your Xero uses another name.

16.8 Connecting other systems (the API)

For a developer connecting a website, a delivery partner or another program:

1. Optionally create a user just for that system with a narrow role (e.g. *Sales Rep*).
2. **System → API clients → New API client**: name, **scopes** (what it may do: read products, stock, customers, sales, invoices; create customers and orders; manage its webhooks), **Acts as** (that user), optionally **Only this branch**.
3. Copy the **key** shown once and give it to the developer securely. **New key** adds another (to rotate); **Revoke** stops every key at once.
4. The developer's documentation is at `https://<your server>/api/public/v1/openapi.json` (OpenAPI 3.1). Webhooks (product changed, sale, invoice, order confirmed or delivered) are signed with a secret shown once when they are added.

> **Why:** a key acts as a person, so it can never see more than that person may in the back office; everything it does is in the audit log under "API: ".

16.9 Importing from Excel

System → Import data now also takes **Excel (.xlsx)** files (the first worksheet). After choosing the file you see a **preview** of the first rows. If your columns have other names (e.g. "Item code" instead of `sku`), **Match columns** lets you say which column is which. Then **Check file** and **Import** as in Chapter 14.

> **Tip:** type dates in Excel as text like `2026-10-31` so they are read exactly.

17. Going live

This chapter takes a business from its old way of working (another till system, QuickBooks, Excel sheets or paper) to running on ZionEX, safely. Everything here is on one page in the back office: **System → Go-live** (owner and general manager).

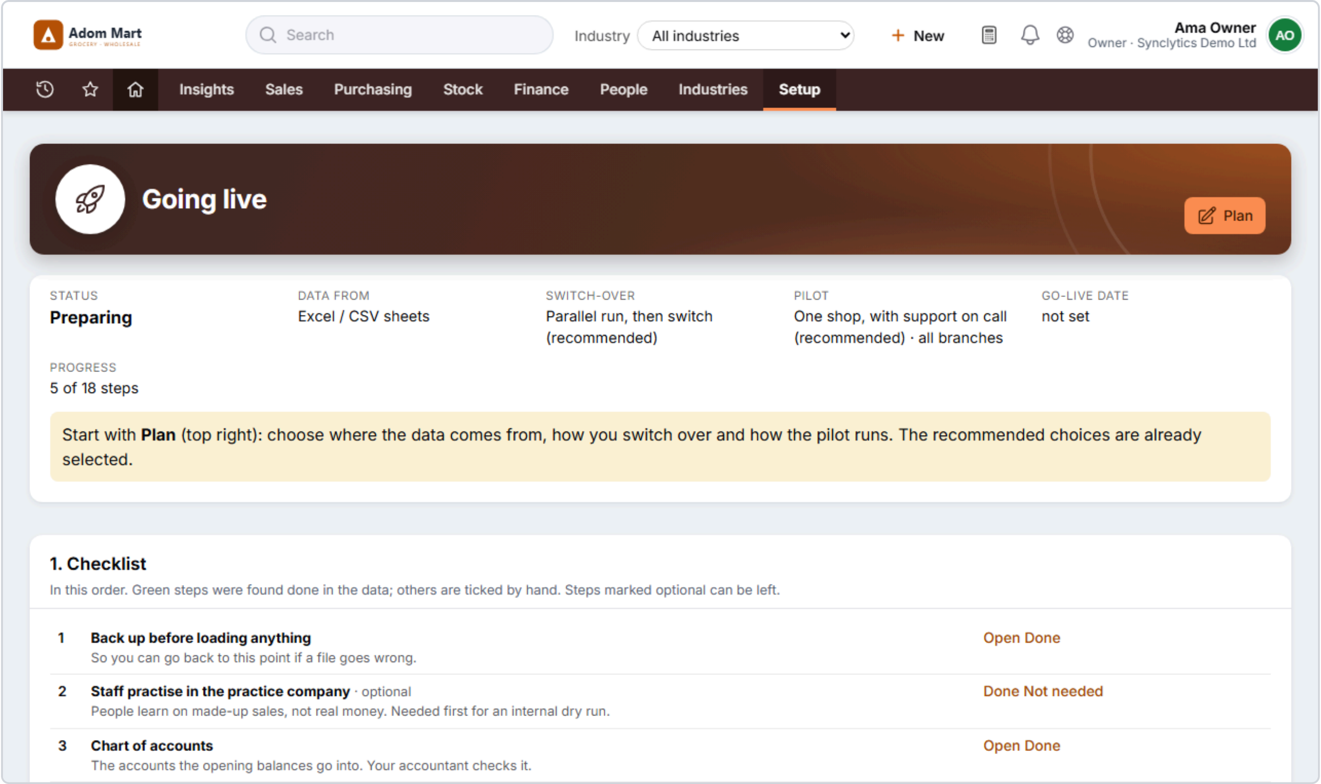
Why a plan: going live touches money, stock and every member of staff on the same day. A written plan, a checked set of figures and a way back make it a calm day instead of a risky one.

17.1 Make the plan

System → Go-live → Plan (top right). Choose:

1. **Where the data comes from** — tick every source you have:
    - **Excel / CSV sheets** — lists you keep yourself; use our templates.
    - **QuickBooks** — exports from QuickBooks Online or Desktop (lists and reports, saved as Excel or CSV).
    - **Another till (POS) system** — exports from Loyverse, Square, Lightspeed, Odoo and similar.
    - **Starting fresh** — nothing is brought over; you type in or import only products, prices and stock.
2. **How you switch over:**
    - **Parallel run, then switch (recommended)** — for one to two weeks both systems are used and each evening the day's totals are compared. **Why:** you see any difference while the old system still works.
    - **Switch fully on one day** — load the data on a quiet day (Saturday evening), use only the new system the next morning. Faster, but the go-live backup is your only way back.
    - **One till or branch first** — one till (or one branch) starts on the new system; the others follow after a week without problems.
3. **How the pilot runs:**
    - **One shop, with support on call (recommended)** — one real shop goes first. A **pilot report** is e-mailed every evening (17.6).
    - **Several shops at once** — two or three shops with different set-ups start together.
    - **Internal dry run first** — a whole simulated week in the **practice company** (17.3) before any real shop.
4. **Go-live date** — the opening balances are "as of" this date (normally the evening before the first day on the new system).
5. **Pilot branch(es)**, the **support e-mail address** (where problem reports and the pilot report go) and the **time of the pilot report** (default 19:00).

> **Check:** the top of the page shows your choices and the checklist below them.



Going live

STATUS
Preparing

DATA FROM
Excel / CSV sheets

SWITCH-OVER
Parallel run, then switch (recommended)

PILOT
One shop, with support on call (recommended) · all branches

GO-LIVE DATE
not set

PROGRESS
5 of 18 steps

Start with **Plan** (top right): choose where the data comes from, how you switch over and how the pilot runs. The recommended choices are already selected.

1. Checklist

In this order. Green steps were found done in the data; others are ticked by hand. Steps marked optional can be left.

1 **Back up before loading anything** — Open Done
So you can go back to this point if a file goes wrong.

2 **Staff practise in the practice company** · optional — Done Not needed
People learn on made-up sales, not real money. Needed first for an internal dry run.

3 **Chart of accounts** — Open Done
The accounts the opening balances go into. Your accountant checks it.

17.2 Bring the data over

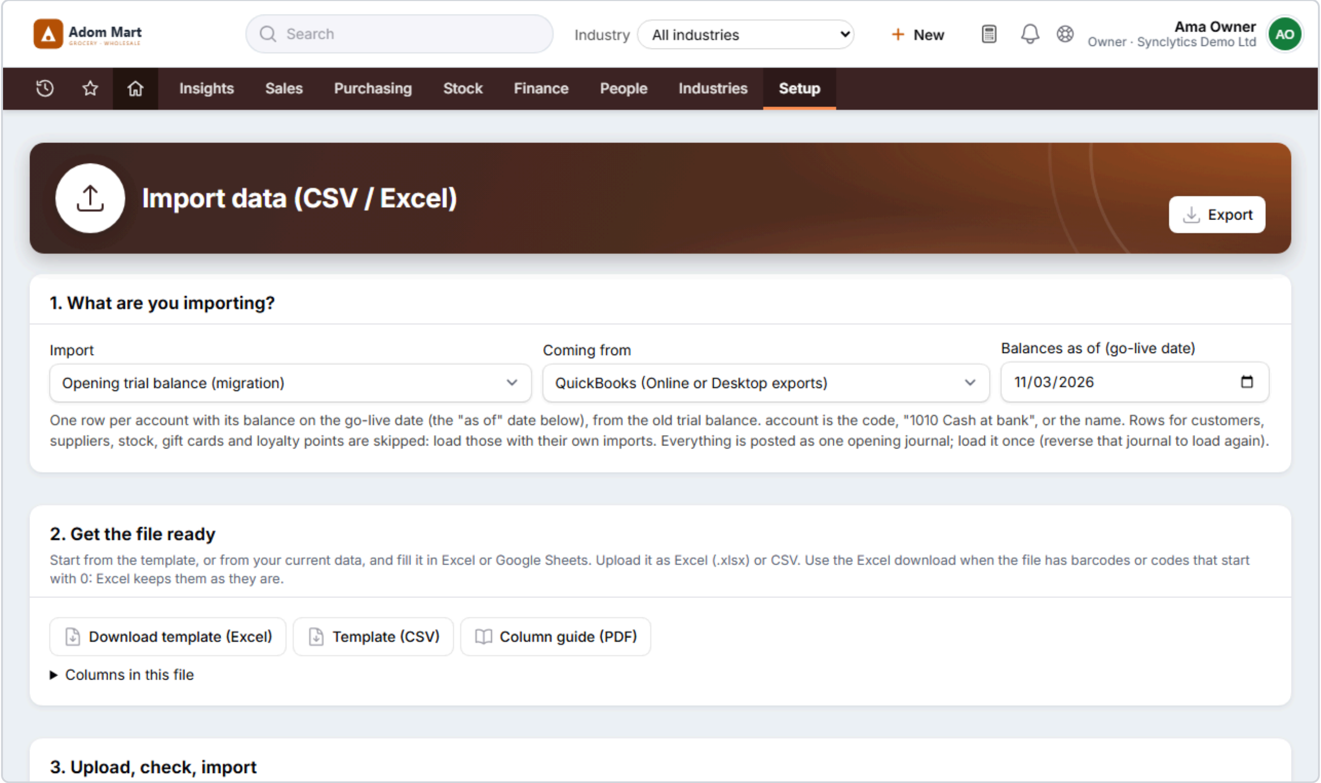
The **checklist** on the Go-live page lists each step in order. A step turns green by itself when the data is there; **Open** takes you to the right import, already set to your source. Do them in this order — later steps need the earlier ones.

Before you start: take a backup (**System** → **Backups** → **Back up now**). **Why:** if a file goes wrong, you can return to this point.

Step	From QuickBooks	From another till system	From Excel
Chart of accounts	*Account List* report	—	our template
Products	*Products and Services* list	the items export	our template
Prices	the same *Products and Services* file	the same items file	our template
Customers	*Customer Contact List*	the customers export	our template
Suppliers	*Vendor Contact List*	—	our template
Opening stock	*Products and Services* (Qty on hand) — or better, a fresh count	the stock / inventory export	our template

Step	From QuickBooks	From another till system	From Excel
Opening trial balance	*Trial Balance* report on the go-live date	—	our template
What customers owe	*Customer Balance Summary*	customer balances	our template
What you owe suppliers	*Vendor Balance Summary*	—	our template
Loyalty points	—	customers export with points	our template
Gift cards and store credit	—	gift card export	our template

For each file: **System** → **Import data**, choose the kind of import and **Coming from**. When you choose the file, the columns are matched for you (for example QuickBooks' *Product/Service Name* becomes *name*, *Sales Price / Rate* becomes *price*); check them, then **Check file**, fix what it reports, and **Import** (Chapter 14.2).


Adom Mart
Search
Industry
All industries
+ New
Ama Owner
Owner · Synclytics Demo Ltd
AO
Insights
Sales
Purchasing
Stock
Finance
People
Industries
Setup
Export
Download template (Excel)
Template (CSV)
Column guide (PDF)
▶ Columns in this file

Import data (CSV / Excel)

1. What are you importing?

Import: Opening trial balance (migration)

Coming from: QuickBooks (Online or Desktop exports)

Balances as of (go-live date): 11/03/2026

One row per account with its balance on the go-live date (the "as of" date below), from the old trial balance. account is the code, "1010 Cash at bank", or the name. Rows for customers, suppliers, stock, gift cards and loyalty points are skipped: load those with their own imports. Everything is posted as one opening journal; load it once (reverse that journal to load again).

2. Get the file ready

Start from the template, or from your current data, and fill it in Excel or Google Sheets. Upload it as Excel (.xlsx) or CSV. Use the Excel download when the file has barcodes or codes that start with 0: Excel keeps them as they are.

3. Upload, check, import

Good to know:

- **Codes from the old system are kept** as the products' SKUs, so the products, prices and stock files match; products added later in SyncLytics get system SKUs.
- **No SKU in the old system?** One is made from the name (*Malta Guinness 330ml* → `MALTA-GUINNESS-330ML`), the same way in the products, prices and stock files, so they match.

- **QuickBooks sub-items** (*Drinks:Malta 330ml*): the last part becomes the product name and the first its category.
- **Suppliers without a code** get one made from the name; customers are matched by code, phone number or exact name.
- **The opening trial balance skips** the lines for customers, suppliers, stock, gift cards and loyalty points: those come in through their own imports, with every customer's and supplier's detail. Whatever is left goes to *Opening balances (to clear)*.
- **Load the opening trial balance once.** To load it again, reverse the opening journal first (Finance → Journals).
- **A customer who is owed money** (a negative balance) gets it as **store credit** they can spend at the till.
- **Count the stock** rather than trusting the old system's quantities if you can: the opening stock is what everything else is measured against.

> **Warning:** only bring over customers who agreed to be on your records.

17.3 Let staff practise: the practice company

System → Go-live → Create the practice company. It copies your branches, tills, settings, products, prices and stock into a separate company with made-up customers. Choose the staff who get a practice login.

- In the back office, staff sign in at their e-mail address with **+practice** (*kofi+practice@yourshop.com*) and **their usual password** (and their usual two-step code).
- At a till, enrol a spare PC or tablet to the practice company (System → Devices, signed in as the practice owner); cashiers use **their usual username and PIN**.
- A purple **PRACTICE** bar shows on every screen and receipt. **Nothing leaves the practice company:** no GRA reporting, no SMS, WhatsApp or e-mail to customers, no updates to online stores or other systems. Its books never mix with yours.

> **Why:** people learn fastest by doing it, and mistakes there cost nothing.

Hand out the training material (17.7). When you want a clean start, **Archive the practice company** (its logins stop) and create a new one.

Clearing the test trading before go-live

Staff should practise in the **practice company** (17.3). If test sales, shifts or stock were entered in the real company instead, clear them the day before going live, **before loading the opening stock**:

1. On every till: make sure nothing is waiting to send (**Send now**), then close the till app.
2. On the server, in the `server` folder:

```
sudo -u www-data php artisan erp:clear-test-trading
```

This only **lists** what would be removed: till shifts and sales, cash counts, stock movements and stock levels, stock adjustments, counts and transfers, purchase orders, receipts, supplier bills, invoices and payments, prescriptions, loyalty and gift cards, and every accounting entry.

3. If the list is right, run it again with the company name exactly as it shows:

```
sudo -u www-data php artisan erp:clear-test-trading --apply --confirm="Your Company Name"
```

It takes a backup first, then clears. **Kept:** products, prices, barcodes, customers, suppliers, staff and roles, devices, branches and settings. Stock is then zero everywhere, and document numbers (invoices, receipts of payment, purchase orders …) start again.

4. Load the opening stock (Chapter 7, Step 10), then open the tills again.

Why the audit log keeps the test period: the audit log can never be edited or emptied, so it still shows the test sales, plus a line saying when the test trading was cleared and by whom. **Why it only works before go-live:** after go-live, every sale is real and is only ever corrected by a return or a reversal.

17.4 Check the figures

When everything is loaded: **System** → **Go-live** → **Check the figures**.

1. **Enter the old system's figures** on the go-live date: number of products, customers and suppliers; stock value at cost; what customers owe; what you owe suppliers; unspent gift cards; loyalty points. Take them from the old system's own reports (stock valuation, aged debtors, aged creditors).
2. Press **Post and check again**. The new system posts everything to the accounts and compares.

> **Check:** every difference shows ✓ and **Opening balances (to clear)** is **0.00**. If it is not zero, something from the old trial balance is missing or was loaded twice — your accountant compares the two trial balances.

> **Why:** a difference found now takes minutes to fix; found in a month, it takes days.

17.5 The switch-over

A parallel run (recommended)

1. **Start the parallel run** (Go-live page). From now, every sale is rung up in **both** systems.
2. Each evening, **Enter a day of the old system** for each pilot branch: its total sales, cash and number of receipts. The new system's figures for the same day are filled in for you.
3. A difference above 1.00 needs an **explanation** (a sale rung up in only one system, a price that differs). The checklist's *Parallel run is clean* step turns green after three days in a row that agree or are explained.

All on one day

Load everything on a quiet evening, check the figures (17.4), take the go-live backup and go live. Next morning only the new system is used.

One till or branch first

The first till or branch goes live as above. When it has run a week without problems, tick **First till ran a clean week** and move the others over the same way.

Going live

1. **Take the go-live backup** (Go-live page). **Why:** it is the exact point you can return to. It must be taken on the day you go live.
2. **Go live.** If a required step is still open, you are asked why you go live anyway; the answer is recorded.

From now on the new system holds the books. Keep the old system for **looking things up only** for at least three months.

17.6 During the pilot

- **Report a problem:** everyone has a **Report a problem** button (the life-buoy icon at the top of the back office; at the bottom of the till screen). It records what they wrote, the screen they were on and the system's recent errors, and e-mails support. Support and IT see the list under **System → Problem reports**, where they close each one with what was done.
- **The pilot report** is e-mailed every evening at the plan's time while the parallel run lasts and for 60 days after going live: sales per pilot branch, cash over/short, sales that reached the server late (a till was offline), the parallel-run difference, the problems reported today, and the health checks that are not green. **Show today's report** on the Go-live page shows it at any time.
- E-mail must work for both (Chapter 16.1). The support address can also be set in `.env` as `ERP_SUPPORT_EMAIL`.

17.7 Training material

Printable guides, in `docs/training/pdf/` of the release (rebuild them with `npm run training` in the `pos` folder):

File	For	What it covers
`teller-quick-card.pdf`	Tellers (people at a till)	One sheet to keep at the till: selling, payments, returns, offline, end of shift (with or without vaults)
`supervisor-guide.pdf`	Supervisors	Approvals, the review queue, lock-outs, end of day
`storekeeper-guide.pdf`	Storekeepers	Receiving, transfers, counts, adjustments
`accountant-guide.pdf`	Accountants	Opening balances, weekly checks, month-end
`manager-owner-guide.pdf`	Owners and managers	Access, daily checks, going live

File	For	What it covers
`cashier-and-income-controller-guide.pdf`	Cashiers and income controllers	Vaults: issuing floats, receiving trays, handing over, blind counts, differences, bank deposits
`video-scripts.pdf`	Whoever records training videos	Nine short videos, what to show and what to say

Each guide ends with exercises to do in the practice company.

17.8 The rollback plan: going back to the old system

Decide in advance **who** may decide to go back (normally the owner), and keep the old system ready until the pilot is over.

When to go back: the tills cannot sell for more than an hour and support cannot fix it; or the figures are wrong in a way nobody can explain. A slow screen or a wrong price is fixed, not rolled back.

1. **Decide and record it: System → Go-live → Go back to the old system**, with the reason.
2. **Keep selling:** switch the tills back to the old system straight away. During a parallel run the old system is already up to date — that is the point of it.
3. **Keep what happened on the new system:** export today's sales (**Sales** → **Till sales** → **Export**) and any other documents created since going live, so they can be entered in the old system.
4. **Only if the new system's books must be put back** to the go-live point (for a second attempt later): the technician restores the **go-live backup** (shown on the Go-live page) as in Chapter 12.3. Everything entered after it is replaced — that is why step 3 comes first.
5. **Find the cause** with support (the problem reports and the pilot reports help), fix it, practise again in the practice company, and plan a new date.

Check before the go-live day: the people involved know this plan; the old system still opens; the go-live backup was copied off the server (System → Backups shows its copies).

17.9 The independent security test

Before the first customer's server is reachable from the internet, SyncLytics has an outside security firm try to break into a test copy of ZionEX: a *penetration test*. The full pack is in `docs/pentest/`, with the request for quote, the rules for the testers, how to build the test machine and how findings are handled.

> **Why:** your sales, stock, payroll and accounts are in ZionEX. An independent firm finds what the people who built the software are used to not seeing. A bank, an insurer or a large customer may also ask for the firm's report.

For the technician building the test machine: use a spare PC or a virtual machine on a network of its own, never a server with real data. After installing, run `php artisan erp:pentest-setup` as described in `docs/pentest/03-test-environment.md`. It fills the empty installation with made-up companies and one sign-in per role, and writes the sign-ins to a private file instead of the screen. It refuses to run where a company already exists.

For the owner: the pilot does not start while a *Critical* or *High* finding is open. Line 21 of the launch checklist is ticked when the firm's retest letter confirms they are fixed.

18. Reference

18.1 Security levels

A person's **security level** is the highest level of role they can ever be given.

Level	Typical people
L6 · Owner	Owner / director (everything)
L5 · Executive	General Manager, Finance Manager
L4 · Manager	Branch Manager, Accountant, HR Manager, Warehouse Manager, Pharmacist, IT Admin
L3 · Supervisor	Supervisor, Stock Controller, Payroll Officer, Marketing
L2 · Staff	Cashier, Storekeeper, Waiter, Tailor, Repair Technician, Stylist / Therapist, Sales Rep
L1 · Restricted	Auditor (read-only), Kitchen Staff

IT Admin manages users, devices, modules and settings, but gets **no** access to sales, money or stock unless the owner also gives them a business role.

18.2 Roles provided

Owner · General Manager · Finance Manager · Accountant · Auditor · HR Manager · Payroll Officer · Purchasing Officer · Purchase Approver · Warehouse Manager · Storekeeper · Stock Controller (products and stock: create and edit products, categories and brands, stock adjustments, counts, receiving and transfers, reorder levels, bins, purchase requests; prices can be seen, not changed; no approvals, sales or money. It can be given for one branch: the person then works on stock at that branch and keeps the shared product list up to date for everyone; prices stay with managers) · Picker / Packer · Branch Manager · Supervisor · Teller (sells at a till; called Cashier before version 0.30) · Cashier (issues floats, receives the tellers' cash) · Income Controller (accounts: receives the day, decides differences, banks the cash) · Sales Rep · Van Sales Rep · Customer Service · Marketing · E-commerce Manager · Delivery Rider · Production Manager · Machine Operator · Quality Inspector · Pharmacist · Pharmacy Technician · Waiter / Bartender · Kitchen Staff · Repair Technician · Tailor · Stylist / Therapist · IT Admin.

You can change what a role allows under **People & access → Roles** (you can only add permissions you hold yourself).

18.3 Settings in `.env`

Key	Meaning
`APP_ENV`	`production` on a live system
`APP_DEBUG`	`false` on a live system (never `true`: it shows technical details)
`APP_URL`	The address people use to reach this server
`APP_KEY`	Encryption key (created once by `key:generate`). **Back it up, never change it.**

Key	Meaning
`DB_HOST`, `DB_PORT`, `DB_DATABASE`, `DB_USERNAME`, `DB_PASSWORD`	Database connection (`127.0.0.1`, `5432`, `erp`, `erp`, your password)
`ERP_NODE_ROLE`	`hq` (head office or single server) or `branch` (branch server)
`ERP_NODE_ID`	Short unique id of this server (`HQ`, `B01` ...), stamped on every record
`ERP_AUDIT_KEY`	Audit-log seal (created once by `erp:audit-key`). **Back it up, never change it.**
`REVERB_*`	Live updates (optional, 4.9 / 5.9)
`MAIL_MAILER`, `MAIL_HOST`, `MAIL_PORT`, `MAIL_USERNAME`, `MAIL_PASSWORD`, `MAIL_FROM_ADDRESS`, `MAIL_FROM_NAME`	Outgoing e-mail (Chapter 16.1). `MAIL_MAILER=log` only writes e-mails to the log
`BACKUP_PASSPHRASE`, `BACKUP_PG_BIN`, `BACKUP_TIME`, `BACKUP_KEEP_LOCAL`	Backups (Chapter 12.2). **Keep a written copy of the passphrase away from the server.**
`GRA_EVAT_URL`, `GRA_EVAT_COMPANY_REFERENCE`, `GRA_EVAT_KEY`, `GRA_EVAT_TIN`, `GRA_EVAT_ACCOUNTS`, `GRA_EVAT_TIMEOUT`	GRA E-VAT access from GRA (Chapter 11.5)
`REMOTE_RELAY_URL`	The remote support relay's address (Chapter 20); empty = no View screen / Take control buttons
`BRAND_NAME`, `BRAND_VENDOR`, `BRAND_WEBSITE`	The product's name, the company that publishes and licenses it (SyncLytics unless a reseller supplies it), and its website (normally left as delivered)
`ERP_SUPPORT_EMAIL`	Where problem reports and the evening pilot report go when the go-live plan names no address (Chapter 17.6); empty = SyncLytics support (info@synclyticsenterprisesystems.com)
`BRAND_SUPPORT_PHONE`, `BRAND_SUPPORT_EMAIL`	The publisher's hotline and e-mail on the Licence page and in licence messages; empty = SyncLytics' own (normally left as delivered)
`SESSION_SECURE_COOKIE`	`true` once the site uses HTTPS (the Ubuntu script sets it)
`ANTHROPIC_API_KEY`	Only if you choose Claude (cloud AI) in Chapter 15. Leave empty for no cloud AI.
`AI_CLOUD_API_KEY`	Only if you choose *Another cloud model* (Google Gemini, Groq, OpenRouter, Mistral, OpenAI ...) in Chapter 15.
`BROADCAST_CONNECTION`, `QUEUE_CONNECTION`, `CACHE_STORE`, `SESSION_DRIVER`	Leave as provided

After changing `.env`, run `php artisan optimize`.

18.4 Network ports

Port	Used by	Open to
443	The web server, Caddy (back office, tills, live updates)	Shop network, or the internet with an internet name
80	Sends visitors on to https; on a server without an internet name, also hands out its certificate (`/ca.crt`)	As 443
9001–9004	PHP workers behind the web server (Windows)	**This server only**
8080	Live updates (Reverb)	**This server only** (reached through 443)
5432	PostgreSQL	**This server only** — never open it to the network
22	SSH (Ubuntu remote administration)	Your administrators only

18.5 Useful commands (in the `server` folder)

Command	What it does
`php artisan erp:install`	Create the company and owner (once)
`php artisan erp:demo-catalog --as=<email>`	Sample products and stock — **test installations only**
`php artisan migrate --seed --force`	Create or update the database structure (installation and updates)
`php artisan optimize` / `optimize:clear`	Apply `.env` changes / clear cached settings
`php artisan schedule:work`	The scheduler (normally runs as a task or cron, not by hand)
`php artisan erp:sync`	Sync a branch server with head office now
`php artisan erp:node:join <hq-url> <hq-id> <company-id>`	Pair a branch server with head office
`php artisan erp:stock-check` after a large import	Confirms stock levels still match the ledger
`php artisan erp:bill-memberships`	Raise membership invoices that are due (runs daily by itself)
`php artisan erp:backup` / `erp:backup --verify`	Encrypted backup now / check the newest one (both also run by themselves)
`php artisan erp:backup:key`	Create the backup passphrase (once)
`php artisan erp:backup:restore <file>`	Put a backup back (replaces the data) — Chapter 12.3
`php artisan erp:health`	All health checks (System → Health); fails when something needs fixing
`php artisan erp:evat-batch --now`	Send the GRA E-VAT till sales waiting (daily mode)
`php artisan erp:load-test`	Speed test of sales and reports on a test system (leaves no data)

Command	What it does
`php artisan erp:pentest-setup --license=<file>`	Penetration test machine only: fill a **new, empty** installation with made-up companies and test sign-ins (Chapter 17.9)
`php artisan erp:reports`	Send scheduled reports that are due (runs every 15 minutes by itself)
`php artisan erp:pilot-report --show` / `--now`	Print / e-mail the pilot report now (sent every evening by itself during a pilot — Chapter 17.6)
`php artisan erp:insights`	Refresh forecasts and alerts now (runs daily at 04:30 by itself)
`php artisan license:status`	Licence state and limits
`php artisan erp:audit-verify`, `erp:stock-check`, `erp:gl-check`	Health checks (Chapter 13)

18.6 Till keyboard shortcuts

Key	Action
F2	Search / scan
F3	Today's route (vans)
F4	Choose customer
F6	Tables (Restaurant)
F7	Repairs (Electronics)
F8	Return
F9	Alterations (Fashion)
F10	Appointments (Services)
F12	Pay

18.7 Go-live checklist

The checklist lives in the software: **System** → **Go-live** (Chapter 17) shows each step in order and ticks most of them by itself. Before the go-live day also confirm:

- ☐ Licence **Active** with the right modules and limits; two-step sign-in for every manager, recovery codes stored safely.
- ☐ Branch names right (their codes are made from them and never change); tax groups checked by your accountant.
- ☐ Every user with the right role and scope; PINs set; IT Admin without business roles.
- ☐ Every till enrolled, printing, opening the drawer, and tested **offline**.
- ☐ Nightly backup running, copied off the server, and a **restore practised** (Chapter 12.3); **System** → **Health** all green.
- ☐ `.env`, the backup passphrase, the owner password and recovery codes stored safely, away from the server.
- ☐ Everyone knows the rollback plan (Chapter 17.8).

18.8 About this guide

This guide is generated from the files in `docs/manual/` and updated with every release. The software version and date are on the cover page.

19. Case study: a cloud server from start to finish

This chapter follows one real installation from start to finish: the first client's head office server, rented from **Contabo** and reached on the client's own internet name. It shows the order of the steps, what each screen asked for, and the problems met on the way. Use it beside Chapter 5, which explains each step in detail.

The client's real name, internet name and server address are replaced here by examples: `erp.clientdomain.com` and `203.0.113.10`.

19.1 What was decided first

Decision	Choice	Why
Where the server runs	Rented cloud server (setup D)	Two shops in different places; no server room; the tills keep selling if the internet drops.
One server or shared	**One server for this client**	The standard: own data, licence, backups and updates. A shared server (Chapter 5.0) is only for small clients starting out.
Provider and plan	Contabo **Cloud VPS 6**: 6 cores, 12 GB memory	12 GB lets the database keep the business in memory. The smaller plan saves very little and leaves no room to grow.
Disk	**100 GB NVMe** (not the 200 GB SSD at the same price)	The database works much faster on NVMe; 100 GB lasts two shops for years.
Region	**European Union** (Germany), free	Contabo serves Africa from Germany; its own test showed it as fast as the paid UK option from Ghana.
System	**Ubuntu 24.04**	What the install script is built and tested for.
Daily server copy	Contabo **Auto Backup**	A copy of the whole server kept off the server, on top of the system's own encrypted backups.
Term	24 months, paid up front	Cheapest monthly rate; matches the client's 24-month price lock.
Internet name	`erp.` in front of the client's own domain	Leaves the main name free for a website later.

19.2 Renting the server (Contabo)

1. On contabo.com: **VPS → Cloud VPS 6 → Get Started**.
2. In the order page, check every section, because some defaults are not what you want:
    - **Term:** 12 or 24 months (cheaper than monthly).
    - **Region:** European Union.
    - **Storage:** change the default 200 GB SSD to **100 GB NVMe**.
    - **Image:** Ubuntu; open its version list and choose **24.04**.
    - **Auto Backup:** the default is *No Data Protection*: choose **Auto Backup**.
    - **Password:** choose a long one (20+ characters) and keep it in a password manager. Never send it by WhatsApp or e-mail.
3. Pay. Contabo e-mails the server's **IP address** when it is ready (minutes to a few hours).

Lesson: the order page's defaults (SSD, no backup) are the cheaper choices, not the right ones for a business server. Go through it line by line.

19.3 Pointing the internet name at the server (Namecheap)

1. Namecheap → **Domain List** → **Manage** next to the domain.
2. **Domain** tab: *Nameservers* must say **Namecheap BasicDNS**.
3. **Advanced DNS** tab → **Add New Record**: type **A Record**, host `erp`, value = the server's IP address, TTL Automatic → green tick.
4. Check it from any PC (it took a few minutes here):

   ```
   nslookup erp.clientdomain.com 8.8.8.8
   ```

   It must answer with the server's IP address.

Why before installing: the install script asks Let's Encrypt for the HTTPS certificate, and Let's Encrypt checks that the name leads to this server.

Lesson: a newly bought domain does not answer at all until a record is added. That is normal.

19.4 Getting the installation files

The installation uses a **release package made for this client** (`release:build`, run by SyncLytics): the code stamped with the client's name and signed with the SyncLytics key, plus the till app. The Ubuntu `install.sh` is copied beside it.

Lesson (fixed on 14 October 2026): the first package built for this client showed that the signed manifest was never copied to servers, so tamper protection was silently off everywhere. Since the fix, the package is checked before sending (signature valid, protected files unchanged).

19.5 Copying the files to the server (from Windows)

Windows 10 and 11 already have the tools. In PowerShell:

```
scp "<folder>\ses-labs-<client>-<build>.zip" "<folder>\installer\ubuntu\install.sh" root@203.0.113.10:/root/
```

- **First time:** "*The authenticity of host … can't be established*" — type `yes`. Your PC remembers the server and will warn if it ever changes. (To be certain, compare the fingerprint in Contabo's **VNC console** with `ssh-keygen -lf /etc/ssh/ssh_host_ed25519_key.pub`.)
- **The password does not show** as you type. To paste it, **right-click once** (Ctrl+V may paste nothing).
- **"Connection closed"** right after the password question happened once here. The second try worked. Causes: the password typed too slowly or wrongly, or the new server still finishing its first start.

Done when both files show **100%**.

19.6 Installing

1. Connect: `ssh root@203.0.113.10` (same password). The prompt becomes `root@vmi…:~#`.
2. **Do not** accept the "*New release … available, run do-release-upgrade*" offer: stay on Ubuntu 24.04, which the install script is tested on.
3. Set the server's clock to Ghana time (Contabo's servers start on German time), so the nightly backup and the logs use Accra times:

   ```
   timedatectl set-timezone Africa/Accra
   ```

4. Run the install script with the internet name:

   ```
   sudo bash /root/install.sh --zip /root/ses-labs-<client>-<build>.zip --domain erp.clientdomain.com --email info@synclyticsenterprisesystems.com
   ```

   It shows steps **1/10 to 10/10** and ends with *Done: the server answers at https://erp.clientdomain.com/admin*. Here the packages, libraries and database took about 5 minutes on a new Cloud VPS 6.
5. **Write down the backup passphrase** it shows, and keep it with the SyncLytics key backup. Backups cannot be restored without it.
6. Answer its questions to create the **company and the owner account** (Chapter 7).

What happened here: the script ended at step **9/10** without a message, so the firewall, the final check, the passphrase display and the company questions did not run. The cause was the SSH port lookup, which fails on Ubuntu 24.04 when SSH is started on demand. Fixed on 14 October 2026: the script now carries on, and any unexpected stop names the line. Everything before step 9 had worked, so the remaining steps were done by hand:

```
ufw allow 22/tcp && ufw allow 80/tcp && ufw allow 443/tcp && ufw --force enable && ufw status
curl -fsS https://erp.clientdomain.com/up > /dev/null && echo "Site OK"
grep '^BACKUP_PASSPHRASE=' /var/www/seslabs/server/.env
cd /var/www/seslabs/server && php artisan erp:install && chown -R www-data:www-data storage bootstrap/cache
```

The site answered over HTTPS straight away, with a Let's Encrypt certificate valid for three months and renewed automatically.

19.7 Licence and first sign-in

1. Open `https://erp.clientdomain.com/admin` and sign in as the owner. The padlock shows a valid certificate.

2. **Licence** page: copy the **installation code** (`SYNREQ-…`) and send it to SyncLytics. The code names the release build, which confirms the signed release is installed.

3. SyncLytics issues the key for exactly what the client bought. Here that was:

    - 2 shops, 4 tills and devices, 10 users;
    - Pharmacy, Grocery (to start the retail shop) and Retail & Wholesale (for when it is released);
    - accounts, and HR and payroll (added free for this client);
    - valid to the end of the set-up period. The full-year key follows when the first year's subscription is paid at go-live.

   Paste it on the Licence page → **Activate**. A key re-issued with more features simply replaces the earlier one.

4. **Set up your company** wizard, *Branches & modules* page. It opens with a **demo layout**: head office plus Grocery, Restaurant and Pharmacy branches, 2 tills each. Change it to the client's real shops **before** clicking *Finish setup*, or the licence limits refuse it. Here:

    - delete the demo branches the client does not have;
    - the pharmacy, code `PH`, **head office on**, module **Pharmacy** only, 2 tills;
    - the retail shop, code `RT`, module **Grocery / Supermarket** only, 2 tills;
    - leave *Retail & Wholesale* unticked (not released yet).

5. **Books:** a client that is **one registered company** running both shops switches off *System* → *Settings* → *Finance* → *Separate books for each industry* straight away, before anything is posted.

6. Company details and TIN under *Company* → *Legal entities*; users and roles (Chapter 7).

Lesson: read the wizard's last page line by line. Its demo branches look finished but are only an example.

19.8 After installing (the same day)

Check	Where	Expected
Server health	System → Health	Scheduler OK, backups OK, *PHP code cache: On*, HTTPS OK
A backup outside Contabo	System → Backups → add a second destination (Chapter 12.2; cloud storage needs the off-site backup add-on)	Contabo's own backup is with the same provider; a second place protects against losing both
Firewall	Contabo panel and the server (`ufw status`)	Only SSH (22), 80 and 443 open; 443 for both TCP and UDP (HTTP/3)
Remote login	Your PC	Move from the password to an SSH key, and restrict port 22 to your office if possible
Tills	Each till device	Pair it (Chapter 9); it keeps selling if the internet drops. In the desktop till the server address is `https://erp.clientdomain.com`. Here the server's IP address was typed first, so the till could not connect; the settings were cleared (9.7) and the name entered. Newer till versions check the address before saving.
Speed	Outside trading hours: `php artisan erp:load-test --force` in `/var/www/seslabs/server`	No line marked *SLOW*

Speed from the shop

The back office felt slow at the shop. Measured: the server answered itself in 0.05 s, so the server was fine; the shop's line reached Google in about 50 ms with nothing lost, but the server in Europe in 150–230 ms with an occasional lost packet. Every click in the back office makes that trip. Two things help: **HTTP/3** (`sudo ufw allow 443/udp`; the installer now does it), and a **cable instead of Wi-Fi** at the till PC. Selling is not affected: the till keeps its products and sales on the PC and sends them in the background.

To measure it yourself, on the shop PC in PowerShell:

```
1..3 | % { curl.exe -o NUL -s -w "first reply %{time_starttransfer}s  total %{time_total}s`n" https://erp.clientdomain.com/up }; ping -n 20 erp.clientdomain.com
```

Under 1 s per reply and no *Request timed out* is good.

19.9 Time and cost (this installation)

Item	Amount
Contabo Cloud VPS 6, 24 months	about $7.20 a month, paid up front
Auto Backup	$4 a month
Domain	the client's own (bought separately)
SyncLytics time	renting and DNS 30 minutes; copying and installing 30 minutes; licence and first set-up 1 hour

20. Remote support relay on any VPS

The **remote support relay** lets SyncLytics support, and a client's own IT with the *Remote access* add-on, see or control a till, a back-office computer or a handheld from anywhere. Each device connects **out** to the relay, so nothing is opened on the shop's network or router. The relay is **MeshCentral** (free, open source); one relay serves every client, each in its own device group.

Why a separate server: the relay carries live screens for many clients. Keeping it apart from any client's business server means a client's server never holds another client's connections, and the relay can be updated or moved without touching a business server.

20.1 What you need

Item	Recommended	Why
A VPS	2 cores, 2–4 GB memory, 40 GB disk, from any provider (Contabo, Hetzner, DigitalOcean, Vultr, AWS Lightsail…) or a server in your office with a public address	The relay passes screens through; it needs little power. Recordings take the disk space.
System	**Ubuntu 24.04** or 22.04, or Debian 12	What the install script is built and tested for.
An internet name	e.g. `remote.yourdomain.com`	The free HTTPS certificate is issued for a name, not an IP address.
Ports	80 and 443 open to the internet	443 carries the connections; 80 is used once to issue the certificate and then redirects to 443.

20.2 Point the name at the server

At your domain provider, add an **A record**: name `remote`, value the VPS's public IP address. Wait until `ping remote.yourdomain.com` answers from that address (usually minutes, sometimes an hour). **Why:** the certificate is only issued once the name reaches this server.

20.3 Install the relay

From the PC, copy the script up and sign in (the address is an example):

```
scp "E:\…\Complete-ERP-POS\installer\remote-relay\install-relay.sh" root@203.0.113.20:/root/
ssh root@203.0.113.20
```

On the server:

```
sudo bash /root/install-relay.sh --domain remote.yourdomain.com --email you@yourdomain.com
```

It installs Node.js and MeshCentral, writes a safe configuration (nobody can create an account by themselves, two-factor sign-in is required, every session is recorded), starts it as a service that restarts by itself, and opens only ports 22, 80 and 443 in the firewall. At the end it prints the first account (**support**) and a **temporary password — shown once**; write it down.

Running the script again later **updates** the relay and keeps the configuration, accounts, device groups and recordings.

20.4 First sign-in

1. Open `https://remote.yourdomain.com` and sign in as **support** with the temporary password.
2. Change the password (12 characters or more) and set up **two-factor sign-in** with an authenticator app. **Why:** whoever signs in here can reach every client's tills.
3. **My Account** → **Users:** add one account per technician (never share an account), each with two-factor sign-in. Give technicians rights only to the device groups of the clients they look after.

20.5 One device group per client

For each client: **My Devices** → **Add Device Group**, named after the client (e.g. *Asante Pharmacy Ltd*).

- **User consent:** tick *Prompt for user consent* for desktop, terminal and files when the client wants every connection accepted at the device. Leave it off when the client chose unattended connections; the ERP still shows the person at the device what the branch rule is.
- **Session recording** is on for every group (the script set it); recordings stay in `/opt/meshcentral/meshcentral-recordings` on the relay.

Copy the group's **invitation link**: *Add Agent* → *Invite* → choose *link does not expire*, and copy it.

20.6 Link the client's ERP to the relay

On the client's **business server**:

1. Add the relay's address to the server's settings file (`.env`) — on Ubuntu `/var/www/<company>/server/.env` , on Windows `server\.env` in the install folder, in Docker the environment of the app container:

   ```
   REMOTE_RELAY_URL=https://remote.yourdomain.com
   ```

   Then run `php artisan config:cache` (Ubuntu and Windows) or restart the container. **Why:** until this is set, the ERP shows no remote buttons, so a client without the relay sees nothing new.
2. In the back office: **System** → **Settings** → **Remote access** → **Remote support agent install link**: paste the group's invitation link.
3. The owner chooses the **Remote connections** rule (Settings → Remote access, or per branch): unattended outside trading hours (the default), always unattended, or always ask.

20.7 Install the agent on each device

In the back office, **System** → **Devices** → **Install remote support** shows the link and these steps:

- **Windows tills and back-office PCs:** open the link on the computer, download the agent and choose **Install**. It runs in the background and starts with the computer.
- **Android handhelds and tablets:** install **MeshAgent** from Google Play, scan the QR code shown on the link page, and allow screen sharing when asked.
- Open the till once on each device, so the ERP learns its computer name.

The device then appears in the client's group on the relay, and on **System** → **Devices** with **View screen** and **Take control**.

20.8 Connecting

On **System** → **Devices**, **View screen** or **Take control** opens the device on the relay in a new tab:

- **Who may:** the client's own staff with the *Remote access* add-on and the *see* or *control devices* permission at that branch; **SyncLytics support** only while the owner's **Support access** is in force (System → Support access), and only at its level (see only, or see and control).
- The ERP first shows what the person at the device will see (asked to accept, or a notice), then opens the relay, which asks the technician to sign in.
- **Every connection** is written to the audit log (who, which device, when), and the relay records the session.

20.9 Looking after the relay

Task	How	Why
Update	Run the install script again	Security fixes; settings and recordings are kept
Status	`systemctl status meshcentral`	Should say *active (running)*
Back up	Copy `/opt/meshcentral/meshcentral-data` somewhere else weekly	It holds the accounts, groups and the relay's identity; without it every agent must be reinstalled
Recordings	Delete recordings older than the agreed period (e.g. 90 days)	They fill the disk; keep them only as long as agreed with the clients
A technician leaves	Remove their account on the relay the same day	The relay is the door to every client's devices

If a device does not appear: check the agent is running on it (Windows: *Services* → *SyncLyticsSupport*), that it can reach `https://remote.yourdomain.com`, and that it was installed from **this client's** link.